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O'Connor et al.

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(54) **METHODS FOR CONTROLLING
MERISTEM SIZE FOR CROP
IMPROVEMENT**

(56) **References Cited**

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(71) Applicant: **Pairwise Plants Services, Inc.**,
Durham, NC (US)

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(72) Inventors: **Devin O'Connor**, Hillsborough, NC
(US); **Nathaniel Graham**, Durham, NC
(US)

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(73) Assignee: **Pairwise Plants Services, Inc.**,
Durham, NC (US)

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Somssich et al. 2016 "Shared and distinct functions of the pseudokinase
Coryne (CRN) in shoot and root stem cell maintenance of *Arabidopsis*"
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Primary Examiner — Matthew R Keogh

Assistant Examiner — Rebecca Stephens

(74) *Attorney, Agent, or Firm* — Myers Bigel, P.A.

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C12N 15/82 (2006.01)

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(52) **U.S. Cl.**

CPC **C12N 15/8262** (2013.01); **C12N 9/22**
(2013.01); **C12N 15/11** (2013.01); **C12N**
15/8213 (2013.01); **C12N 2310/20** (2017.05);
C12N 2800/80 (2013.01)

(58) **Field of Classification Search**

CPC C12N 15/11; C12N 9/22; C12N 2310/20;
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See application file for complete search history.

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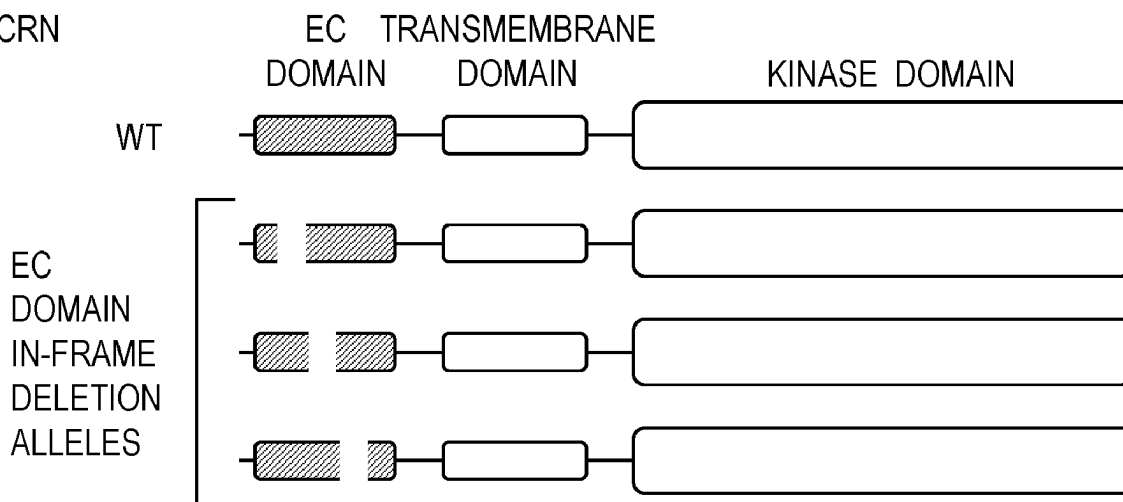
ABSTRACT

This invention relates to compositions and methods for
modifying CORYNE (CRN) genes in plants to optionally
increase kernel row number and/or improve one or more
yield traits. The invention further relates to plants having
increased kernel row number and/or one or more improved
yield traits produced using the methods and compositions of
the invention.

8 Claims, 3 Drawing Sheets

Specification includes a Sequence Listing.

ZmCRN



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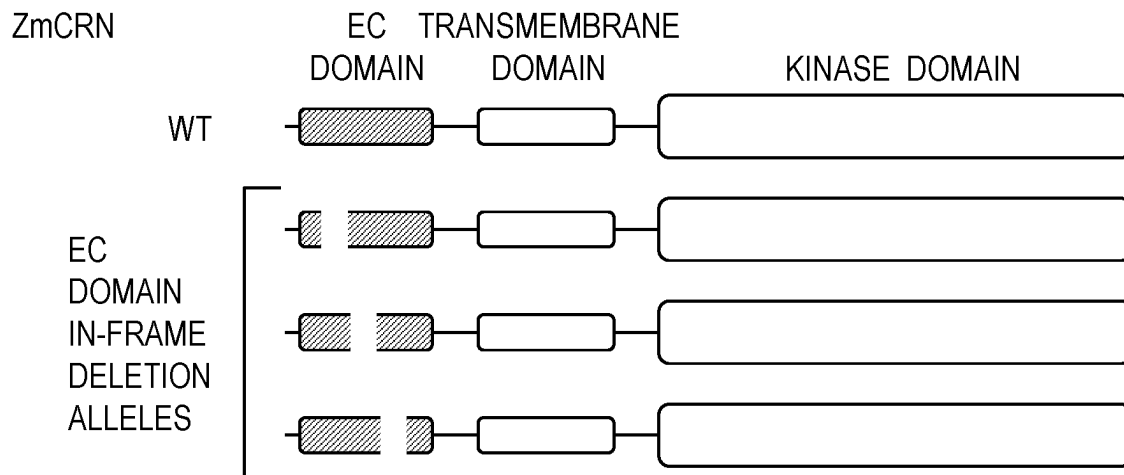
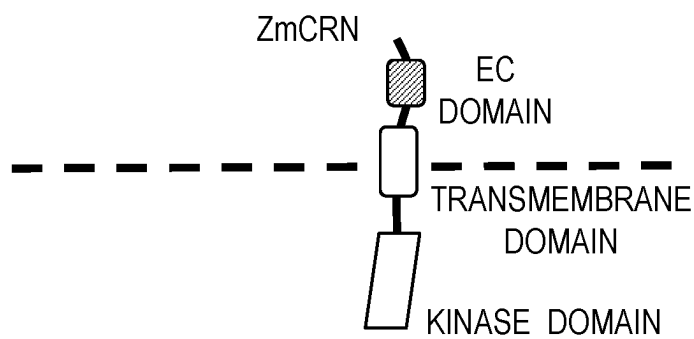
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1.	AT5G13290.2	ATHA1-401	MKQRRNRNGCSSNTI	SLLLL	FFLVFESRTSTSTSCRRRTVKH	LSTTSTSSTPL	-----	ESRITSKVIVL	
2.	ZM00001D042268	P001_ZMAY1-414	TMSRNPTKTLAA	-TLLV-	LEFLADLPCLASQPLHSEPM	ATQSPAPPLPPGSTIPRAQ	AGGAARLRRIAL		
3.	TRAESC33807G3475600	1_TAES1-427	MAPGAGTAAATKNPTKT	LAA--	LIILVV-	LIILADLPCLASQPLHSQLP	-----	ATQSFAAPLPPPOPSAPHARAGGAARLRRIAL	
4.	TRDC38G070150	1_TDIC1-427	MAPGAGTAAATKNPTKT	LAA--	IILLV-	LIILADLPCLASQPLHSQLP	-----	ATQSFAAPLPPPOPSAPHARAGGAARLRRIAL	
5.	TRAESC33002G434500	1_TAES1-427	MAPGAGTAAATKNPTKT	LAA--	TIILV-	LIILADLPCLASQPLHSQLP	-----	ATQSFAAPLPPPOPSAPHARAGGAARLRRIAL	
6.	AET3GV20979000	1_ATAU1-655	MAPGAGTAAATKNPTKT	LAA--	TLLV-	LIILADLPCLASQPLHSQLP	-----	ATQSFAAPLPPPOPSAPHARAGGAARLRRIAL	
7.	TRAESC3A023441700	1_TAES1-428	ATAGTAAAAAKNPTKT	LAT--	LILLV-	LIILADLPCLASQPLHSQLP	-----	ATQSFAAPLPPPOPSAPHARAGGAARLRRIAL	
8.	TRDC3AG062460	2_TDIC1-428	APGAGTAAAAAKNPTKT	LAT--	LILLV-	LIILADLPCLASQPLHSQLP	-----	ATQSFAAPLPPPOPSAPHARAGGAARLRRIAL	
9.	KQK11323	BDS1S1-420	MARGGGVVVGKNNPTKT	LAA--	IILLV-	LIILADLPCLASQPLHSEPM	-----	TQQSP-PPQQSQKI PHAQPGGAARLRRIAL	
10.	OBO1G51701	1_OBRA1-421	MEEGGMMAKNPTKT	LAI--	LILLV-	LIILVFCSLAASQPLHSEPMST	-----	AEYSPSPSPPOSKI PHAQAGGAARLRRIAL	
11.	OPUNCO1G41470	1_OPUN1-414	MEMMASKNPTKT	L--	AVHL-	LIILVFLSLASSQPLHSEPMST	-----	TQSPDPPOSK-----	IPHQAAGGAARLRRIAL
12.	OMERIO1G339550	3_OMER1-413	MEMMASKNPTK	-----	TILL-	LIILVFLSLACQPLHSEPMSTTTA	-----	TTQSAAPPPPPOSKI PHAEAGGAARLRRIAL	
13.	BGIOSGA005098-PA	OIND1-410	MASKNPTK	-----	KTILL-	LIILVFLSLACQPLHSEPMSTTTT	-----	TATQAPAPPPPPOSKI PHAEAGGAARLRRIAL	
14.	ONIVA01G48610	2_ONIV1-414	MEMMASKNPTK	-----	TILL-	LIILVFLSLACQPLHSEPMSTTT	-----	TATQAPAPPPPPOSKI PHAEAGGAARLRRIAL	
15.	ORUFJ01G45970	1_ORUF1-442	LCSPRAIASPLSMEMMASKNPTKT	LILL-	-----	LIILVFLSLACQPLHSEPMSTTTT	-----	TATQAPAPPPPPOSKI PHAEAGGAARLRRIAL	
16.	OBART01G42560	3_OBAR1-416	MEMMASKNPTKT	-----	HULL-	LIILVFLSLACQPLHSEPMSTTTT	-----	TATQTPTPPPPPOSKI PHAEAGGAARLRRIAL	
17.	OS01T0927500	02_OSAT1-414	MEMMASKNPTK	-----	TILL-	LIILVFLSLACQPLHSEPMSTTTT	-----	TATQAPAPPPPPOSKI PHAEAGGAARLRRIAL	
18.	OGLUM01G46830	2_OGLU1-414	MEMMASKNPTK	-----	TILL-	LIILVFLSLACQPLHSEPMSTTTT	-----	TATQAPAPPPPPOSKI PHAEAGGAARLRRIAL	
19.	KQL08435	SITAF1-414	IMEQVRMTSKNPTKT	PTPIIA-L	TILL-	LIILVFLSLACQPLHSEPMSTTTT	-----	TATQAPAPPPPPOSKI PHAEAGGAARLRRIAL	
20.	PAN26713	PHFI1-433	IMEQVRMTSKNPTKT	PTPIIA-L	TILL-	LIILVFLSLACQPLHSEPMSTTTT	-----	TATQAPAPPPPPOSKI PHAEAGGAARLRRIAL	
21.	PUZ53166	PHHA1-433	IMVGARMTSRNPTRL	PLIATTVLILL-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPPSPAPPOSIRIPRAQ	VGAARLRRIAL
22.	EES02007	SBIC1-417	MEQRARRRNINL	--SLLL-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
23.	BRAO23402	1-P_BRAP1-394	MEQRARRRNINL	--SLLL-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
24.	CX85658	BNAP1-394	MEQGRKRYCNTV	--YLLF-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
25.	CDV09782	BNAP1-398	MEQRRKRKYCNV	--YLLF-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
26.	CX86965	1-BNAP1-397	MEQRRKRKYCNV	--YLLF-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
27.	BRAO08839	1-P_BRAP1-395	MEQRRRRNRNGCSNTIS	LLLF-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
28.	AL_SCAFFOLD_0006	1229_ALYR1-402	MEQRRRRNRNGCSNTIS	LLLF-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
29.	CX78514	BNAP1-394	MEQRHRHTPCS	--LRL-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
30.	BD3G009090	1_BOLE1-394	MEQRHRHTPCS	--LRL-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
31.	CV53620	BNAP1-394	MEQRHRNTIY	--LRLF-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
32.	TP57577	JGAC_V2_MRNA4689_IPRA1-413	MEQRHRNTIY	--LRLF-	-----	LAFLALCAPASQPLHSEPM	-----	TQSPPTSPTPOSTIPRAQ	AGGAARLRRIAL
33.	KEH21855	INTRU1-413	GCGFVFCVLMFKRKQTFFY	--LAKELLVFOLVLL	FLFLSHHNTVQCGRLSKN	ISSEHSQS	-----	DYKDMPKRVIS	
34.	OWI7297	LANG1-392	GCGFVFCVLMFKRKQTFFY	--LAKELLVFOLVLL	FLFLSHHNTVQCGRLSKN	ISSEHSQS	-----	DYKDMPKRVIS	
35.	ESW11311	PVUL1-406	MEFRKHIASS	--LARELLALQPL	LIIFELSLHHHTVQCGRLSKH	VSSEPPSPRTSPSSS	-----	GYRDDPKKIILS	
36.	LRPD04G10110	1_VRAD1-429	MEFRKHIASS	--LARELLALQPL	LIIFELSLHHHTVQCGRLSKH	VSSEPPSPRTSPSSS	-----	GYRDDPKKIILS	
37.	KRH45205	GMAX1-408	MEFRKHILSS	--LARELLALQPL	LIIFELSLHHHTVQCGRLSKH	VSSEPPSPRTSPSSS	-----	GYDDPKKIILS	
38.	KRH01513	GMAX1-408	MEFRKHILSS	--LARELLALQPL	LIIFELSLHHHTVQCGRLSKH	VSSEPPSPRTSPSSS	-----	GYDDPKKIILS	
39.</									

FIG. 1



*FIG. 3**FIG. 4*

1

METHODS FOR CONTROLLING MERISTEM SIZE FOR CROP IMPROVEMENT

STATEMENT REGARDING ELECTRONIC FILING OF A SEQUENCE LISTING

A Sequence Listing in ASCII text format, submitted under 37 C.F.R. § 1.821, entitled 1499.29_ST25.txt, 596,676 bytes in size, generated on Jun. 1, 2021 and filed via EFS-Web, is provided in lieu of a paper copy. This Sequence Listing is hereby incorporated herein by reference into the specification for its disclosures.

STATEMENT OF PRIORITY

This application claims the benefit, under 35 U.S.C. § 119 (e), of U.S. Provisional Application No. 63/033,273 filed on Jun. 2, 2020, the entire contents of which is incorporated by reference herein.

FIELD OF THE INVENTION

This invention relates to compositions and methods for modifying CORYNE (CRN) genes in plants, optionally to increase kernel row number. The invention further relates to plants having a modified CRN gene and, for example, increased kernel row number produced using the methods and compositions of the invention.

BACKGROUND OF THE INVENTION

New plant organs are initiated at the growing tip of the plant called the meristem. In the meristem a population of undifferentiated stem cells is maintained. During growth, the meristem allocates stem-cells to newly formed organs, including seeds, while at the same time reserving some stem-cells to continually maintain the meristem. Several conserved molecular mechanisms have been described that control the size of the stem cell population to ensure organized growth and proper meristem size.

As a result of the modular nature of maize ear development, larger meristems tend to initiate more flowers, and thus, meristem size has a direct effect on kernel row number and yield. The number of flowers initiated during the development of the maize ear directly limits grain yield. An increased number of flowers initiated around the circumference of the ear (kernel row number or KRN) was a major trait selected during maize domestication. Significant advancements through breeding have resulted in dramatic increases in kernel row number, from 2 in teosinte, the ancestor of maize, to ~8-20 rows in modern elite maize varieties. In diverse maize lines kernel row number can get as high as 36.

In the canonical regulatory pathway described in the model plant *Arabidopsis*, CLAVATA3 (CLV3) peptide is secreted from cells in the meristem apex and moves through the apoplast into the central stem-cell domain where it interacts with several Leucine Rich Receptors (LRRs) including CLAVATA1 (CLV1) and CLAVATA2 (CLV2). This receptor-ligand interaction stimulates signaling that ultimately acts to reduce WUS expression and restrict the expansion of the stem cell population. One of the targets of WUS is the CLV3 gene itself, and in this way WUS acts to limit its own expression and maintain stem cell homeostasis (Fletcher, J. C., *Plants* 7:87 (2018)).

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Loss of function mutations in CLV1, CLV2, or CLV3 result in an expansion of the WUS domain and increased meristem size (Schoof et al., *Cell* 100:635-644 (2000)). Often this increase in meristem size results in aberrant plant growth because the meristem expands uncontrollably and becomes disorganized, a phenomenon called fasciation (Je et al., *Nat Genet* 48: ng.3567 (2016a)). Importantly, a larger meristem does not just make larger organs, but rather an increased number of organs around a larger area. Because of this relationship between meristem size and organ number, mutations in maize CLV-WUS signaling genes can lead to increased flower number and yield. CLV-WUS signaling is transduced downstream via CRN. While strong loss-of-function mutations in the maize CLV2 ortholog FACIATED EAR2 (FEA2) result in enlarged meristems and an increase in KRN, the ear is disordered and as a result there is no yield increase (Taguchi-Shiobara et al., *Gene Dev* 15:2755-2766 (2001)).

Improved strategies for modulating meristem size are needed to improve crop performance.

SUMMARY OF THE INVENTION

One aspect of the invention provides a plant or plant part thereof comprising at least one non-natural mutation in a short extracellular (EC) domain of an endogenous CORYNE (CRN) gene that encodes a CRN protein.

A second aspect of the invention provides a plant cell, comprising an editing system comprising: (a) a CRISPR-Cas effector protein; and (b) a guide nucleic acid (gRNA, gDNA, crRNA, crDNA, sgRNA, sgDNA) comprising a spacer sequence with complementarity to an endogenous target gene encoding an CRN protein.

A third aspect of the invention provides a corn plant cell comprising at least one non-naturally occurring mutation within a CRN gene, wherein the mutation is a substitution, insertion or a deletion that is introduced using an editing system that comprises a nucleic acid binding domain that binds to a target site in the CRN gene.

A fourth aspect of the invention provides a method of producing/breeding a transgene-free edited corn plant, comprising: crossing the corn plant of the invention with a transgene free corn plant, thereby introducing the at least one non-natural mutation into the corn plant that is transgene-free; and selecting a progeny corn plant that comprises the at least one non-natural mutation and is transgene-free, thereby producing a transgene free edited corn plant.

A fifth aspect of the invention provides a method of providing a plurality of corn plants having increased kernel number, the method comprising planting two or more plants of the invention in a growing area, thereby providing a plurality of corn plants having increased kernel number as compared to a plurality of control corn plants not comprising the mutation.

A sixth aspect of the invention provides a method of generating variation in a region of a corn CRN protein, comprising: introducing an editing system into a corn plant cell, wherein the editing system is targeted to a region of a corn CRN gene that encodes the region of the corn CRN protein, wherein the region comprises a sequence having at least 70% sequence identity (e.g., 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100%) to any one of the amino acid sequences of SEQ ID NO:119-121 or the region is encoded by a sequence having at least 70% identity to the nucleotide sequence of SEQ ID NOs: 123-125; and contacting the region of the CRN gene with the editing system,

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thereby introducing into the plant cell a mutation into the region of the CRN protein; and generating variation in the region of the CRN protein.

A seventh aspect of the invention provides a method for editing a specific site in the genome of a plant cell, the method comprising: cleaving, in a site specific manner, a target site within an endogenous CRN gene in the plant cell, the endogenous CRN gene comprising a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122, or encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118, thereby generating an edit in the endogenous CRN gene of the plant cell and producing a plant cell comprising the edit in the endogenous CRN gene.

An eighth aspect provides a method for making a corn plant, comprising: (a) contacting a population of corn plant cells comprising a wild-type endogenous CRN gene with a nuclease linked to a nucleic acid binding domain (e.g., DNA binding domain, e.g., editing system) that binds to a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122, to a sequence having at least 70% identity to the nucleotide sequence of SEQ ID NOs: 123-125, optionally SEQ ID NO:125, to a sequence encoding an amino acid sequence having at least 70% sequence identity to SEQ ID NO:118; or to a sequence encoding an amino acid sequence having at least 70% sequence identity to any one of the amino acid sequences of SEQ ID NOs: 112-114; (b) selecting a corn plant cell from the population in which at least one wild-type endogenous CRN gene has been mutated; and (c) growing the selected plant cell into a corn plant.

A ninth aspect provides a method for increasing kernel number in a corn plant, comprising (a) contacting a corn plant cell comprising an endogenous CRN gene with a nuclease targeting the endogenous CRN gene, wherein the nuclease is linked to a nucleic acid binding domain (e.g., DNA binding domain, e.g., editing system) that binds to a target site in the endogenous CRN gene, wherein the endogenous CRN gene: (i) encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (ii) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122; (iii) comprises a region having a sequence with at least 70% sequence identity to the nucleotide sequence of SEQ ID NOs: 123-125, optionally SEQ ID NO: 125; and/or (iv) comprises a region encoding a sequence having at least 70% sequence identity to any one of the amino acid sequences of SEQ ID NOs: 119-121 to produce a corn plant cell comprising a mutation in the endogenous CRN gene, thereby producing the corn plant comprising at least one cell having a mutation in the endogenous CRN gene; and (b) growing the corn plant cell into a corn plant comprising the mutation in the endogenous CRN gene, thereby producing a corn plant have a mutated endogenous CRN gene and an increased kernel number.

A tenth aspect provides method for producing a corn plant or part thereof comprising at least one cell having a mutated endogenous CRN gene, the method comprising contacting a target site in an endogenous CRN gene in the corn plant or plant part with a nuclease comprising a cleavage domain and a nucleic acid binding domain, wherein the nucleic acid binding domain binds to a target site in the endogenous CRN gene, wherein the endogenous CRN gene (a) encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO: 118; (b) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprises a region having

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a sequence with at least 70% sequence identity to the nucleotide sequence of SEQ ID NOs: 123-125, optionally SEQ ID NO: 125; and/or (d) comprises a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121, to produce a plant cell comprising a mutation in the endogenous CRN gene, thereby producing the corn plant or part thereof comprising at least one cell having a mutation in the endogenous CRN gene.

An eleventh aspect of the invention provides a method for producing a corn plant or part thereof comprising a mutated endogenous CRN gene and exhibiting increased kernel number, the method comprising contacting a target site in an endogenous CRN gene in the corn plant or plant part with a nuclease comprising a cleavage domain and a nucleic acid binding domain, wherein the nucleic acid binding domain binds to a target site in the endogenous CRN gene, wherein the endogenous CRN gene: (a) encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO: 118; (b) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprises a region having a sequence with at least 70% sequence identity to the nucleotide sequence of SEQ ID NOs: 123-125, optionally SEQ ID NO:125; and/or (d) comprises a region encoding a sequence having at least 70% sequence identity to any one of the amino acid sequence of SEQ ID NOs: 119-121, thereby producing the corn plant or part thereof comprising an endogenous CRN gene having a mutation and exhibiting increased kernel number.

A twelfth aspect provides a guide nucleic acid that binds to a target site in a CRN gene, the target site comprising a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122, at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 123-125, optionally SEQ ID NO:125; encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118, or encoding a sequence having at least 70% sequence identity to any one of the amino acid sequences of SEQ ID NO: 119-121.

In a thirteenth aspect, a system is provided comprising a guide nucleic acid of the invention and a CRISPR-Cas effector protein that associates with the guide nucleic acid.

A fourteenth aspect provides a gene editing system comprising a CRISPR-Cas effector protein in association with a guide nucleic acid, wherein the guide nucleic acid comprises a spacer sequence that binds to an endogenous CRN gene.

In a fifteenth aspect, a complex comprising a CRISPR-Cas effector protein comprising a cleavage domain and a guide nucleic acid, wherein the guide nucleic acid binds to a target site in an endogenous CRN gene, wherein the endogenous CRN gene, wherein the endogenous CRN gene: (a) encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (b) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122; (c) comprises a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NO: 123-125, optionally SEQ ID NO:125; and/or (d) comprises a region encoding a sequence having at least 70% sequence identity to any one of the amino acid sequences of SEQ ID NOs: 119-121, wherein the cleavage domain cleaves a target strand in the CRN gene.

In a sixteenth aspect, an expression cassette is provided, the expression cassette comprising (a) a polynucleotide encoding CRISPR-Cas effector protein comprising a cleavage domain and (b) a guide nucleic acid that binds to a target site

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in an endogenous CRN gene, wherein the guide nucleic acid comprises a spacer sequence that is complementary to and binds to (i) a portion of a nucleic acid encoding an amino acid sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (ii) a portion of a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (iii) a portion of a sequence having at least 70% sequence identity to any one of the nucleotide sequences of SEQ ID NO: 123-125, optionally SEQ ID NO:125; and/or (iv) a portion of a sequence having at least 70% sequence identity to a sequence encoding any one of the amino acid sequences of SEQ ID NO: 119-121.

In an additional aspect, a method of creating a mutation in an endogenous CRN gene in a plant provided, comprising: (a) targeting a gene editing system to a portion of the CRN gene, the portion comprising (i) a sequence having at least 70% sequence identity to any one of the nucleotide sequences of SEQ ID NOs: 123-125, optionally SEQ ID NO:125; and/or (ii) a sequence having at least 70% sequence identity to a sequence encoding any one of the amino acid sequences of SEQ ID NO:118-121, and (b) selecting a plant that comprises a substitution of an amino acid residue in the EC domain of the CRN gene, optionally an alternative amino acid in amino acid residue at position 477.

A further aspect of the invention provides a nucleic acid encoding a dominant negative mutation, a semi-dominant mutation, a hypomorphic mutation, or a weak loss-of-function mutation of a corn CRN protein.

In an additional aspect, a corn plant or part thereof is provided comprising a nucleic acid of the invention. In a further aspect, a corn plant or part thereof is provided that exhibits increased kernel number. In some aspects a corn plant is provided that also exhibits increased yield, and improved disease resistance as well as exhibits larger meristems and root meristems that are maintained. Further provided are plants comprising in their genome one or more mutated CORYNE (CRN) genes produced by the methods of the invention as well as polypeptides, polynucleotides, nucleic acid constructs, expression cassettes and vectors for making a plant of this invention.

These and other aspects of the invention are set forth in more detail in the description of the invention below.

BRIEF DESCRIPTION OF THE SEQUENCES

SEQ ID NOs: 1-17 are exemplary Cas12a amino acid sequences useful with this invention.

SEQ ID NOs: 18-20 are exemplary Cas12a nucleotide sequences useful with this invention.

SEQ ID NO:21-22 are exemplary regulatory sequences encoding a promoter and intron.

SEQ ID NOs: 23-29 are exemplary cytosine deaminase sequences useful with this invention.

SEQ ID NOs: 30-40 are exemplary adenine deaminase amino acid sequences useful with this invention.

SEQ ID NO:41 is an exemplary uracil-DNA glycosylase inhibitor (UGI) sequences useful with this invention.

SEQ ID NOs: 42-44 provides an example of a protospacer adjacent motif position for a Type V CRISPR-Cas12a nucle-ase.

SEQ ID NOs: 45-47 provide example peptide tags and affinity polypeptides useful with this invention.

SEQ ID NOs: 48-58 provide example RNA recruiting motifs and corresponding affinity polypeptides useful with this invention.

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SEQ ID NOs: 59-60 are exemplary Cas9 polypeptide sequences useful with this invention.

SEQ ID NOs: 61-71 are exemplary Cas9 polynucleotide sequences useful with this invention.

SEQ ID NOs: 72-118 are example CRN polypeptide sequences.

SEQ ID NO:119 is an example EC domain amino acid sequence from a maize CRN polypeptide.

SEQ ID NO:120 and SEQ ID NO:121 are example target regions of a CRN polypeptide.

SEQ ID NO:122 is an example CRN genomic sequence.

SEQ ID NOs: 123-125 are example target regions in the CRN genomic sequence.

SEQ ID NOs: 126-130 are example spacer sequences for nucleic acid guides useful with this invention.

SEQ ID NOs: 131-178 are the sequences shown in FIG. 1 from top to bottom.

SEQ ID NO:179 is a portion of the wildtype CRN sequence as shown in FIG. 2, top line.

SEQ ID NOs: 180-191 show example edits of a CRN nucleotide sequence as shown in FIG. 2, second line to bottom line.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 provides an alignment of various CRN polypeptides showing the EC domain (from top to bottom: SEQ ID NOs: 131-178).

FIG. 2 provides an alignment of mutated sequences with the wild type CRN sequence (from top to bottom: SEQ ID NO:179-191).

FIG. 3 provides a cartoon exemplifying a structure of wild type (WT) maize CRN protein and edited maize CRN proteins showing in-frame deletions.

FIG. 4 provides a cartoon exemplifying a maize CRN polypeptide functioning in planta.

DETAILED DESCRIPTION

The present invention now will be described hereinafter with reference to the accompanying drawings and examples, in which embodiments of the invention are shown. This description is not intended to be a detailed catalog of all the different ways in which the invention may be implemented, or all the features that may be added to the instant invention. For example, features illustrated with respect to one embodiment may be incorporated into other embodiments, and features illustrated with respect to a particular embodiment may be deleted from that embodiment. Thus, the invention contemplates that in some embodiments of the invention, any feature or combination of features set forth herein can be excluded or omitted. In addition, numerous variations and additions to the various embodiments suggested herein will be apparent to those skilled in the art in light of the instant disclosure, which do not depart from the instant invention. Hence, the following descriptions are intended to illustrate some particular embodiments of the invention, and not to exhaustively specify all permutations, combinations and variations thereof.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. The terminology used in the description of the invention herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention.

All publications, patent applications, patents and other references cited herein are incorporated by reference in their entirety for the teachings relevant to the sentence and/or paragraph in which the reference is presented.

Unless the context indicates otherwise, it is specifically intended that the various features of the invention described herein can be used in any combination. Moreover, the present invention also contemplates that in some embodiments of the invention, any feature or combination of features set forth herein can be excluded or omitted. To illustrate, if the specification states that a composition comprises components A, B and C, it is specifically intended that any of A, B or C, or a combination thereof, can be omitted and disclaimed singularly or in any combination.

As used in the description of the invention and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

Also as used herein, “and/or” refers to and encompasses any and all possible combinations of one or more of the associated listed items, as well as the lack of combinations when interpreted in the alternative (“or”).

The term “about,” as used herein when referring to a measurable value such as an amount or concentration and the like, is meant to encompass variations of $\pm 10\%$, $\pm 5\%$, $\pm 1\%$, $\pm 0.5\%$, or even $\pm 0.1\%$ of the specified value as well as the specified value. For example, “about X” where X is the measurable value, is meant to include X as well as variations of $\pm 10\%$, $\pm 5\%$, $\pm 1\%$, $\pm 0.5\%$, or even $\pm 0.1\%$ of X. A range provided herein for a measurable value may include any other range and/or individual value therein.

As used herein, phrases such as “between X and Y” and “between about X and Y” should be interpreted to include X and Y. As used herein, phrases such as “between about X and Y” mean “between about X and about Y” and phrases such as “from about X to Y” mean “from about X to about Y.”

Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. For example, if the range 10 to 15 is disclosed, then 11, 12, 13, and 14 are also disclosed.

The term “comprise,” “comprises” and “comprising” as used herein, specify the presence of the stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

As used herein, the transitional phrase “consisting essentially of” means that the scope of a claim is to be interpreted to encompass the specified materials or steps recited in the claim and those that do not materially affect the basic and novel characteristic(s) of the claimed invention. Thus, the term “consisting essentially of” when used in a claim of this invention is not intended to be interpreted to be equivalent to “comprising.”

As used herein, the terms “increase,” “increasing,” “increased,” “enhance,” “enhanced,” “enhancing,” and “enhancement” (and grammatical variations thereof) describe an elevation of at least about 5%, 10%, 15%, 20%, 25%, 50%, 75%, 100%, 150%, 200%, 300%, 400%, 500% or more as compared to a control. For example, a plant comprising a mutation in a CRN gene as described herein can exhibit increased kernel row number (e.g., producing ears having increased kernel row number) that is at least about 5% or greater than that of a control plant not com-

prising the same mutation, optionally wherein the length of the ears comprising increased kernel row number is not substantially decreased (e.g., a decrease in length of less than 30% as compared to an ear of a plant not comprising the same FEA2 mutation). A control plant is typically the same plant as the edited plant but the control plant has not been similarly edited and therefore does not comprise the mutation. A control plant may be an isogenic plant and/or a wild type plant. Thus, a control plant can be the same breeding line, variety, or cultivar as the subject plant into which a mutation as described herein is introgressed, but the control breeding line, variety, or cultivar is free of the mutation. In some embodiments, a comparison between a plant of the invention and a control plant is made under the same growth conditions, e.g., the same environmental conditions (soil, hydration, light, heat, nutrients and the like).

A “control” plant may be an isogenic plant and/or a wild type plant. Thus, a control plant can be the same breeding line, variety, or cultivar as the subject plant into which a mutation as described herein is introgressed, but the control breeding line, variety, or cultivar is free of the mutation. In some embodiments, a comparison between a plant of the invention and a control plant is made under the same growth conditions, e.g., the same environmental conditions (soil, hydration, light, heat, nutrients and the like).

As used herein, the terms “reduce,” “reduced,” “reducing,” “reduction,” “diminish,” and “decrease” (and grammatical variations thereof), describe, for example, a decrease of at least about 5%, 10%, 15%, 20%, 25%, 35%, 50%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% as compared to a control. In particular embodiments, the reduction can result in no or essentially no (i.e., an insignificant amount, e.g., less than about 10% or even 5%) detectable activity or amount.

As used herein, the terms “express,” “expresses,” “expressed” or “expression,” and the like, with respect to a nucleic acid molecule and/or a nucleotide sequence (e.g., RNA or DNA) indicates that the nucleic acid molecule and/or a nucleotide sequence is transcribed and, optionally, translated. Thus, a nucleic acid molecule and/or a nucleotide sequence may express a polypeptide of interest or, for example, a functional untranslated RNA.

A “heterologous” or a “recombinant” nucleotide sequence is a nucleotide sequence not naturally associated with a host cell into which it is introduced, including non-naturally occurring multiple copies of a naturally occurring nucleotide sequence. A “heterologous” nucleotide/polypeptide may originate from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention.

A “native” or “wild type” nucleic acid, nucleotide sequence, polypeptide or amino acid sequence refers to a naturally occurring or endogenous nucleic acid, nucleotide sequence, polypeptide or amino acid sequence. In some contexts, a “wild type” nucleic acid is a nucleic acid that is not edited as described herein and can differ from an “endogenous” gene that may be edited as described herein (e.g., a mutated endogenous gene). In some contexts, a “wild type” nucleic acid (e.g., unedited) may be heterologous to the organism in which the wild type nucleic acid is found (e.g., a transgenic organism). As an example, a “wild type endogenous CORYNE (CRN) gene” is an CRN gene that is naturally occurring in or endogenous to the reference organism, e.g., a plant, e.g., a maize plant, and may be subject to modification as described herein, after which, such a modified endogenous gene is no longer wild type.

As used herein, the term “heterozygous” refers to a genetic status wherein different alleles reside at corresponding loci on homologous chromosomes.

As used herein, the term “homozygous” refers to a genetic status wherein identical alleles reside at corresponding loci on homologous chromosomes.

As used herein, the term “allele” refers to one of two or more different nucleotides or nucleotide sequences that occur at a specific locus.

A “null allele” is a nonfunctional allele caused by a genetic mutation that results in a complete lack of production of the corresponding protein or produces a protein that is nonfunctional.

A “dominant negative mutation” is a mutation that produces an altered gene product (e.g., having an aberrant function relative to wild type), which gene product adversely affects the function of the wild-type allele or gene product. For example, a “dominant negative mutation” may block a function of the wild type gene product. A dominant negative mutation may also be referred to as an “antimorphic mutation.”

A “semi-dominant mutation” refers to a mutation in which the penetrance of the phenotype in a heterozygous organism is less than that observed for a homozygous organism.

A “weak loss-of-function mutation” is a mutation that results in a gene product having partial function or reduced function (partially inactivated) as compared to the wildtype gene product.

A “hypomorphic mutation” is a mutation that results in a partial loss of gene function, which may occur through reduced expression (e.g., reduced protein and/or reduced RNA) or reduced functional performance (e.g., reduced activity), but not a complete loss of function/activity. A “hypomorphic” allele is a semi-functional allele caused by a genetic mutation that results in production of the corresponding protein that functions at anywhere between 1% and 99% of normal efficiency.

A “hypermorphic mutation” is a mutation that results in increased expression of the gene product and/or increased activity of the gene product.

A “locus” is a position on a chromosome where a gene or marker or allele is located. In some embodiments, a locus may encompass one or more nucleotides.

As used herein, the terms “desired allele,” “target allele” and/or “allele of interest” are used interchangeably to refer to an allele associated with a desired trait. In some embodiments, a desired allele may be associated with either an increase or a decrease (relative to a control) of or in a given trait, depending on the nature of the desired phenotype.

A marker is “associated with” a trait when said trait is linked to it and when the presence of the marker is an indicator of whether and/or to what extent the desired trait or trait form will occur in a plant/germplasm comprising the marker. Similarly, a marker is “associated with” an allele or chromosome interval when it is linked to it and when the presence of the marker is an indicator of whether the allele or chromosome interval is present in a plant/germplasm comprising the marker.

As used herein, the terms “backcross” and “backcrossing” refer to the process whereby a progeny plant is crossed back to one of its parents one or more times (e.g., 1, 2, 3, 4, 5, 6, 7, 8, etc.). In a backcrossing scheme, the “donor” parent refers to the parental plant with the desired gene or locus to be introgressed. The “recipient” parent (used one or more times) or “recurrent” parent (used two or more times) refers to the parental plant into which the gene or locus is being introgressed. For example, see Ragot, M. et al. *Marker-*

assisted Backcrossing: A Practical Example, in *TECHNIQUES ET UTILISATIONS DES MARQUEURS MOLECULAIRES LES COLLOQUES*, Vol. 72, pp. 45-56 (1995); and Openshaw et al., *Marker-assisted Selection in Backcross Breeding*, in *PROCEEDINGS OF THE SYMPOSIUM “ANALYSIS OF MOLECULAR MARKER DATA,”* pp. 41-43 (1994). The initial cross gives rise to the F1 generation. The term “BC1” refers to the second use of the recurrent parent, “BC2” refers to the third use of the recurrent parent, and so on.

As used herein, the terms “cross” or “crossed” refer to the fusion of gametes via pollination to produce progeny (e.g., cells, seeds or plants). The term encompasses both sexual crosses (the pollination of one plant by another) and selfing (self-pollination, e.g., when the pollen and ovule are from the same plant). The term “crossing” refers to the act of fusing gametes via pollination to produce progeny.

As used herein, the terms “introgression,” “introgressing” and “introgressed” refer to both the natural and artificial transmission of a desired allele or combination of desired alleles of a genetic locus or genetic loci from one genetic background to another. For example, a desired allele at a specified locus can be transmitted to at least one progeny via a sexual cross between two parents of the same species, where at least one of the parents has the desired allele in its genome. Alternatively, for example, transmission of an allele can occur by recombination between two donor genomes, e.g., in a fused protoplast, where at least one of the donor protoplasts has the desired allele in its genome. The desired allele may be a selected allele of a marker, a QTL, a transgene, or the like. Offspring comprising the desired allele can be backcrossed one or more times (e.g., 1, 2, 3, 4, or more times) to a line having a desired genetic background, selecting for the desired allele, with the result being that the desired allele becomes fixed in the desired genetic background. For example, a marker associated with increased yield under non-water stress conditions may be introgressed from a donor into a recurrent parent that does not comprise the marker and does not exhibit increased yield under non-water stress conditions. The resulting offspring could then be backcrossed one or more times and selected until the progeny possess the genetic marker(s) associated with increased yield under non-water stress conditions in the recurrent parent background.

A “genetic map” is a description of genetic linkage relationships among loci on one or more chromosomes within a given species, generally depicted in a diagrammatic or tabular form. For each genetic map, distances between loci are measured by the recombination frequencies between them. Recombination between loci can be detected using a variety of markers. A genetic map is a product of the mapping population, types of markers used, and the polymorphic potential of each marker between different populations. The order and genetic distances between loci can differ from one genetic map to another.

As used herein, the term “genotype” refers to the genetic constitution of an individual (or group of individuals) at one or more genetic loci, as contrasted with the observable and/or detectable and/or manifested trait (the phenotype). Genotype is defined by the allele(s) of one or more known loci that the individual has inherited from its parents. The term genotype can be used to refer to an individual’s genetic constitution at a single locus, at multiple loci, or more generally, the term genotype can be used to refer to an individual’s genetic make-up for all the genes in its genome. Genotypes can be indirectly characterized, e.g., using markers and/or directly characterized by nucleic acid sequencing.

As used herein, the term “germplasm” refers to genetic material of or from an individual (e.g., a plant), a group of individuals (e.g., a plant line, variety or family), or a clone derived from a line, variety, species, or culture. The germplasm can be part of an organism or cell, or can be separate from the organism or cell. In general, germplasm provides genetic material with a specific genetic makeup that provides a foundation for some or all of the hereditary qualities of an organism or cell culture. As used herein, germplasm includes cells, seed or tissues from which new plants may be grown, as well as plant parts that can be cultured into a whole plant (e.g., leaves, stems, buds, roots, pollen, cells, etc.).

As used herein, the terms “cultivar” and “variety” refer to a group of similar plants that by structural or genetic features and/or performance can be distinguished from other varieties within the same species.

As used herein, the terms “exotic,” “exotic line” and “exotic germplasm” refer to any plant, line or germplasm that is not elite. In general, exotic plants/germplasms are not derived from any known elite plant or germplasm, but rather are selected to introduce one or more desired genetic elements into a breeding program (e.g., to introduce novel alleles into a breeding program).

As used herein, the term “hybrid” in the context of plant breeding refers to a plant that is the offspring of genetically dissimilar parents produced by crossing plants of different lines or breeds or species, including but not limited to the cross between two inbred lines.

As used herein, the term “inbred” refers to a substantially homozygous plant or variety. The term may refer to a plant or plant variety that is substantially homozygous throughout the entire genome or that is substantially homozygous with respect to a portion of the genome that is of particular interest.

A “haplotype” is the genotype of an individual at a plurality of genetic loci, i.e., a combination of alleles. Typically, the genetic loci that define a haplotype are physically and genetically linked, i.e., on the same chromosome segment. The term “haplotype” can refer to polymorphisms at a particular locus, such as a single marker locus, or polymorphisms at multiple loci along a chromosomal segment.

As used herein, the term “heterologous” refers to a nucleotide/polypeptide that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention.

A plant in which at least one orthologous IPA1 gene encoding an SPL transcription factor is modified as described herein (e.g., comprises a modification as described herein) may have improved yield traits as compared to a plant that does not comprise the modification in the at least one orthologous IPA1 gene. As used herein, “improved yield traits” refers to any plant trait associated with growth, for example, biomass, yield, nitrogen use efficiency (NUE), inflorescence size/weight, fruit yield, fruit quality, fruit size, seed size, seed number, foliar tissue weight, nodulation number, nodulation mass, nodulation activity, number of seed heads, number of tillers, number of branches, number of flowers, number of tubers, tuber mass, bulb mass, number of seeds, total seed mass, rate of leaf emergence, rate of tiller/branch emergence, rate of seedling emergence, length of roots, number of roots, size and/or weight of root mass, or any combination thereof. Thus, in some aspects, “improved yield traits” may include, but is not limited to, increased inflorescence production, increased fruit produc-

tion (e.g., increased number, weight and/or size of fruit; e.g., increase number, weight, and/or size of ears for, e.g., maize), increased fruit quality, increased number, size and/or weight of roots, increased meristem size, increased seed size, increased biomass, increased leaf size, increased nitrogen use efficiency, increased height, increased internode number and/or increased internode length as compared to a control plant or part thereof (e.g., a plant that does not comprise a mutated endogenous IPA1 nucleic acid (e.g., a mutated IPA1 gene)). Improved yield traits can also result from increased planting density of plants of the invention. Thus, in some aspects, a plant of the invention is capable of being planted at an increased density (as a consequence of altered plant architecture resulting from the endogenous mutation), which results in improved yield traits as compared to a control plant that is planted at the same density. In some aspects, improved yield traits can be expressed as quantity of grain produced per area of land (e.g., bushels per acre of land).

As used herein a “control plant” means a plant that does not contain an edited IPA1 gene or genes as described herein that imparts an enhanced/improved trait (e.g., yield trait) or altered phenotype. A control plant is used to identify and select a plant edited as described herein and that has an enhanced trait or altered phenotype as compared to the control plant. A suitable control plant can be a plant of the parental line used to generate a plant comprising a mutated IPA1 gene(s), for example, a wild type plant devoid of an edit in an endogenous IPA1 gene as described herein. A suitable control plant can also be a plant that contains recombinant nucleic acids that impart other traits, for example, a transgenic plant having enhanced herbicide tolerance. A suitable control plant can in some cases be a progeny of a heterozygous or hemizygous transgenic plant line that is devoid of the mutated IPA1 gene as described herein, known as a negative segregant, or a negative isogenic line.

An enhanced trait may be, for example, decreased days from planting to maturity, increased stalk size, increased number of leaves, increased plant height growth rate in vegetative stage, increased ear size, increased ear dry weight per plant, increased number of kernels per ear, increased weight per kernel, increased number of kernels per plant, decreased ear void, extended grain fill period, reduced plant height, increased number of root branches, increased total root length, increased yield, increased nitrogen use efficiency, and increased water use efficiency as compared to a control plant. An altered phenotype may be, for example, plant height, biomass, canopy area, anthocyanin content, chlorophyll content, water applied, water content, and water use efficiency.

As used herein a “trait” is a physiological, morphological, biochemical, or physical characteristic of a plant or particular plant material or cell. In some instances, this characteristic is visible to the human eye and can be measured mechanically, such as seed or plant size, weight, shape, form, length, height, growth rate and development stage, or can be measured by biochemical techniques, such as detecting the protein, starch, certain metabolites, or oil content of seed or leaves, or by observation of a metabolic or physiological process, for example, by measuring tolerance to water deprivation or particular salt or sugar concentrations, or by the measurement of the expression level of a gene or genes, for example, by employing Northern analysis, RT-PCR, microarray gene expression assays, or reporter gene expression systems, or by agricultural observations such as hyperosmotic stress tolerance or yield. However, any technique can be used to measure the amount of, the comparative

level of, or the difference in any selected chemical compound or macromolecule in the transgenic plants.

As used herein an “enhanced trait” means a characteristic of a plant resulting from mutations in an IPA1 gene(s) as described herein. Such traits include, but are not limited to, an enhanced agronomic trait characterized by enhanced plant morphology, physiology, growth and development, yield, nutritional enhancement, disease or pest resistance, or environmental or chemical tolerance. In some embodiments, an enhanced trait/alterd phenotype may be, for example, decreased days from planting to maturity, increased stalk size, increased number of leaves, increased plant height growth rate in vegetative stage, increased ear size, increased ear dry weight per plant, increased number of kernels per ear, increased weight per kernel, increased number of kernels per plant, decreased ear void, extended grain fill period, reduced plant height, increased number of root branches, increased total root length, drought tolerance, increased water use efficiency, cold tolerance, increased nitrogen use efficiency, and increased yield. In some embodiments, a trait is increased yield under nonstress conditions or increased yield under environmental stress conditions. Stress conditions can include both biotic and abiotic stress, for example, drought, shade, fungal disease, viral disease, bacterial disease, insect infestation, nematode infestation, cold temperature exposure, heat exposure, osmotic stress, reduced nitrogen nutrient availability, reduced phosphorus nutrient availability and high plant density. “Yield” can be affected by many properties including without limitation, plant height, plant biomass, pod number, pod position on the plant, number of internodes, incidence of pod shatter, grain size, ear size, ear tip filling, kernel abortion, efficiency of nodulation and nitrogen fixation, efficiency of nutrient assimilation, resistance to biotic and abiotic stress, carbon assimilation, plant architecture, resistance to lodging, percent seed germination, seedling vigor, and juvenile traits. Yield can also be affected by efficiency of germination (including germination in stressed conditions), growth rate (including growth rate in stressed conditions), flowering time and duration, ear number, ear size, ear weight, seed number per ear or pod, seed size, composition of seed (starch, oil, protein) and characteristics of seed fill.

Also used herein, the term “trait modification” encompasses altering the naturally occurring trait by producing a detectable difference in a characteristic in a plant comprising a mutation in an endogenous IPA1 gene encoding an SPL transcription factor as described herein relative to a plant not comprising the mutation, such as a wild-type plant, or a negative segregant. In some cases, the trait modification can be evaluated quantitatively. For example, the trait modification can entail an increase or decrease in an observed trait characteristics or phenotype as compared to a control plant. It is known that there can be natural variations in a modified trait. Therefore, the trait modification observed entails a change of the normal distribution and magnitude of the trait characteristics or phenotype in the plants as compared to a control plant.

The present disclosure relates to a plant with improved economically important characteristics, more specifically increased yield. More specifically the present disclosure relates to a plant comprising a mutation(s) in an IPA1 gene(s) as described herein, wherein the plant has increased yield as compared to a control plant devoid of said mutation(s). In some embodiments, plants produced as described herein exhibit increased yield or improved yield trait components as compared to a control plant. In some embodiments, a plant of the present disclosure exhibits an improved

trait that is related to yield, including but not limited to increased nitrogen use efficiency, increased nitrogen stress tolerance, increased water use efficiency and increased drought tolerance, as defined and discussed infra.

Yield can be defined as the measurable produce of economic value from a crop. Yield can be defined in the scope of quantity and/or quality. Yield can be directly dependent on several factors, for example, the number and size of organs, plant architecture (such as the number of branches, plant biomass, e.g., increased root biomass, steeper root angle and/or longer roots, and the like), flowering time and duration, grain fill period. Root architecture and development, photosynthetic efficiency, nutrient uptake, stress tolerance, early vigor, delayed senescence and functional stay green phenotypes may be factors in determining yield. Optimizing the above-mentioned factors can therefore contribute to increasing crop yield.

Reference herein to an increase/improvement in yield-related traits can also be taken to mean an increase in biomass (weight) of one or more parts of a plant, which can include above ground and/or below ground (harvestable) plant parts. In particular, such harvestable parts are seeds, and performance of the methods of the disclosure results in plants with increased yield and in particular increased seed yield relative to the seed yield of suitable control plants. The term “yield” of a plant can relate to vegetative biomass (root and/or shoot biomass), to reproductive organs, and/or to propagules (such as seeds) of that plant.

Increased yield of a plant of the present disclosure can be measured in a number of ways, including test weight, seed number per plant, seed weight, seed number per unit area (for example, seeds, or weight of seeds, per acre), bushels per acre, tons per acre, or kilo per hectare. Increased yield can result from improved utilization of key biochemical compounds, such as nitrogen, phosphorous and carbohydrate, or from improved responses to environmental stresses, such as cold, heat, drought, salt, shade, high plant density, and attack by pests or pathogens.

“Increased yield” can manifest as one or more of the following: (i) increased plant biomass (weight) of one or more parts of a plant, particularly aboveground (harvestable) parts, of a plant, increased root biomass (increased number of roots, increased root thickness, increased root length) or increased biomass of any other harvestable part; or (ii) increased early vigor, defined herein as an improved seedling aboveground area approximately three weeks post-germination.

“Early vigor” refers to active healthy plant growth especially during early stages of plant growth, and can result from increased plant fitness due to, for example, the plants being better adapted to their environment (for example, optimizing the use of energy resources, uptake of nutrients and partitioning carbon allocation between shoot and root). Early vigor, for example, can be a combination of the ability of seeds to germinate and emerge after planting and the ability of the young plants to grow and develop after emergence. Plants having early vigor also show increased seedling survival and better establishment of the crop, which often results in highly uniform fields with the majority of the plants reaching the various stages of development at substantially the same time, which often results in increased yield. Therefore, early vigor can be determined by measuring various factors, such as kernel weight, percentage germination, percentage emergence, seedling growth, seedling height, root length, root and shoot biomass, canopy size and color and others.

Further, increased yield can also manifest as increased total seed yield, which may result from one or more of an increase in seed biomass (seed weight) due to an increase in the seed weight on a per plant and/or on an individual seed basis an increased number of, for example, flowers/panicles per plant; an increased number of pods; an increased number of nodes; an increased number of flowers (“florets”) per panicle/plant; increased seed fill rate; an increased number of filled seeds; increased seed size (length, width, area, perimeter), which can also influence the composition of seeds; and/or increased seed volume, which can also influence the composition of seeds. In one embodiment, increased yield can be increased seed yield, for example, increased seed weight; increased number of filled seeds; and increased harvest index.

Increased yield can also result in modified architecture, or can occur because of modified plant architecture.

Increased yield can also manifest as increased harvest index, which is expressed as a ratio of the yield of harvestable parts, such as seeds, over the total biomass

The disclosure also extends to harvestable parts of a plant such as, but not limited to, seeds, leaves, fruits, flowers, bolls, pods, siliques, nuts, stems, rhizomes, tubers and bulbs. The disclosure furthermore relates to products derived from a harvestable part of such a plant, such as dry pellets, powders, oil, fat and fatty acids, starch or proteins.

The present disclosure provides a method for increasing “yield” of a plant or “broad acre yield” of a plant or plant part defined as the harvestable plant parts per unit area, for example seeds, or weight of seeds, per acre, pounds per acre, bushels per acre, tones per acre, tons per acre, kilo per hectare.

As used herein “nitrogen use efficiency” refers to the processes which lead to an increase in the plant’s yield, biomass, vigor, and growth rate per nitrogen unit applied. The processes can include the uptake, assimilation, accumulation, signaling, sensing, retranslocation (within the plant) and use of nitrogen by the plant.

As used herein “increased nitrogen use efficiency” refers to the ability of plants to grow, develop, or yield faster or better than normal when subjected to the same amount of available/applied nitrogen as under normal or standard conditions; ability of plants to grow, develop, or yield normally, or grow, develop, or yield faster or better when subjected to less than optimal amounts of available/applied nitrogen, or under nitrogen limiting conditions.

As used herein “nitrogen limiting conditions” refers to growth conditions or environments that provide less than optimal amounts of nitrogen needed for adequate or successful plant metabolism, growth, reproductive success and/or viability.

As used herein the “increased nitrogen stress tolerance” refers to the ability of plants to grow, develop, or yield normally, or grow, develop, or yield faster or better when subjected to less than optimal amounts of available/applied nitrogen, or under nitrogen limiting conditions.

Increased plant nitrogen use efficiency can be translated in the field into either harvesting similar quantities of yield, while supplying less nitrogen, or increased yield gained by supplying optimal/sufficient amounts of nitrogen. The increased nitrogen use efficiency can improve plant nitrogen stress tolerance and can also improve crop quality and biochemical constituents of the seed such as protein yield and oil yield. The terms “increased nitrogen use efficiency”, “enhanced nitrogen use efficiency”, and “nitrogen stress

tolerance” are used inter-changeably in the present disclosure to refer to plants with improved productivity under nitrogen limiting conditions.

As used herein “water use efficiency” refers to the amount of carbon dioxide assimilated by leaves per unit of water vapor transpired. It constitutes one of the most important traits controlling plant productivity in dry environments. “Drought tolerance” refers to the degree to which a plant is adapted to arid or drought conditions. The physiological responses of plants to a deficit of water include leaf wilting, a reduction in leaf area, leaf abscission, and the stimulation of root growth by directing nutrients to the underground parts of the plants. Typically, plants are more susceptible to drought during flowering and seed development (the reproductive stages), as plant’s resources are deviated to support root growth. In addition, abscisic acid (ABA), a plant stress hormone, induces the closure of leaf stomata (microscopic pores involved in gas exchange), thereby reducing water loss through transpiration, and decreasing the rate of photosynthesis. These responses improve the water-use efficiency of the plant on the short term. The terms “increased water use efficiency”, “enhanced water use efficiency”, and “increased drought tolerance” are used inter-changeably in the present disclosure to refer to plants with improved productivity under water-limiting conditions.

As used herein “increased water use efficiency” refers to the ability of plants to grow, develop, or yield faster or better than normal when subjected to the same amount of available/applied water as under normal or standard conditions; ability of plants to grow, develop, or yield normally, or grow, develop, or yield faster or better when subjected to reduced amounts of available/applied water (water input) or under conditions of water stress or water deficit stress.

As used herein “increased drought tolerance” refers to the ability of plants to grow, develop, or yield normally, or grow, develop, or yield faster or better than normal when subjected to reduced amounts of available/applied water and/or under conditions of acute or chronic drought; ability of plants to grow, develop, or yield normally when subjected to reduced amounts of available/applied water (water input) or under conditions of water deficit stress or under conditions of acute or chronic drought.

As used herein, “drought stress” refers to a period of dryness (acute or chronic/prolonged) that results in water deficit and subjects plants to stress and/or damage to plant tissues and/or negatively affects grain/crop yield; a period of dryness (acute or chronic/prolonged) that results in water deficit and/or higher temperatures and subjects plants to stress and/or damage to plant tissues and/or negatively affects grain/crop yield.

As used herein, “water deficit” refers to the conditions or environments that provide less than optimal amounts of water needed for adequate/successful growth and development of plants.

As used herein, “water stress” refers to the conditions or environments that provide improper (either less/insufficient or more/excessive) amounts of water than that needed for adequate/successful growth and development of plants/crops thereby subjecting the plants to stress and/or damage to plant tissues and/or negatively affecting grain/crop yield.

As used herein “water deficit stress” refers to the conditions or environments that provide less/insufficient amounts of water than that needed for adequate/successful growth and development of plants/crops thereby subjecting the plants to stress and/or damage to plant tissues and/or negatively affecting grain yield.

As used herein, the terms “nucleic acid,” “nucleic acid molecule,” “nucleotide sequence” and “polynucleotide” refer to RNA or DNA that is linear or branched, single or double stranded, or a hybrid thereof. The term also encompasses RNA/DNA hybrids. When dsRNA is produced synthetically, less common bases, such as inosine, 5-methylcytosine, 6-methyladenine, hypoxanthine and others can also be used for antisense, dsRNA, and ribozyme pairing. For example, polynucleotides that contain C-5 propyne analogues of uridine and cytidine have been shown to bind RNA with high affinity and to be potent antisense inhibitors of gene expression. Other modifications, such as modification to the phosphodiester backbone, or the 2'-hydroxy in the ribose sugar group of the RNA can also be made.

As used herein, the term “nucleotide sequence” refers to a heteropolymer of nucleotides or the sequence of these nucleotides from the 5' to 3' end of a nucleic acid molecule and includes DNA or RNA molecules, including cDNA, a DNA fragment or portion, genomic DNA, synthetic (e.g., chemically synthesized) DNA, plasmid DNA, mRNA, and anti-sense RNA, any of which can be single stranded or double stranded. The terms “nucleotide sequence” “nucleic acid,” “nucleic acid molecule,” “nucleic acid construct,” “oligonucleotide” and “polynucleotide” are also used interchangeably herein to refer to a heteropolymer of nucleotides. Nucleic acid molecules and/or nucleotide sequences provided herein are presented herein in the 5' to 3' direction, from left to right and are represented using the standard code for representing the nucleotide characters as set forth in the U.S. sequence rules, 37 CFR §§ 1.821-1.825 and the World Intellectual Property Organization (WIPO) Standard ST.25. A “5' region” as used herein can mean the region of a polynucleotide that is nearest the 5' end of the polynucleotide. Thus, for example, an element in the 5' region of a polynucleotide can be located anywhere from the first nucleotide located at the 5' end of the polynucleotide to the nucleotide located halfway through the polynucleotide. A “3' region” as used herein can mean the region of a polynucleotide that is nearest the 3' end of the polynucleotide. Thus, for example, an element in the 3' region of a polynucleotide can be located anywhere from the first nucleotide located at the 3' end of the polynucleotide to the nucleotide located halfway through the polynucleotide.

As used herein with respect to nucleic acids, the term “fragment” or “portion” refers to a nucleic acid that is reduced in length relative (e.g., reduced by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 20, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 310, 320, 330, 340, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, or 900 or more nucleotides or any range or value therein) to a reference nucleic acid and that comprises, consists essentially of and/or consists of a nucleotide sequence of contiguous nucleotides identical or almost identical (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% identical) to a corresponding portion of the reference nucleic acid. Such a nucleic acid fragment may be, where appropriate, included in a larger polynucleotide of which it is a constituent. As an example, a repeat sequence of guide nucleic acid of this invention may comprise a “portion” of a wild type CRISPR-Cas repeat sequence (e.g., a wild Type CRISPR-Cas repeat, e.g., a repeat from the CRISPR Cas system of, for example, a Cas9, Cas12a (Cpf1), Cas12b, Cas12c (C2c3), Cas12d (CasY), Cas12e (CasX),

Cas12g, Cas12h, Cas12i, C2c4, C2c5, C2c6, C2c7, C2c8, C2c9, C2c10, Cas14a, Cas14b, and/or a Cas14c, and the like).

In some embodiments, a fragment or a portion of a CRN nucleic acid may comprise, consist essentially of or consist of about 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 120, 130, 140, 150, 160, 170, 180, 190, 200, 220, 240, 260, 280, 300, 320, 340, 360, 380, 400, 420, 440, 460, 480, 500, 520, 540, 550, 560, 565, 570, 575, 580, 585, 590, 595, 600, 605, 610, 615, 620, 625, 630, 635, 640, 645, 650, 655, 660, 665, 670, 680, 690, 700, 720, 740, 760, 780, 800, 820, 840, 860, 880, 900, 920, 940, 960, 980, 1000, 1020, 1040, 1060, 1080, 1100, 1120, 1140, 1160, 1180, 1100, 1120, 1140, 1160, 1180, 1200, 1220, 1240, 1260, or 1280 consecutive nucleotides or any range or value therein of a nucleic acid encoding a CRN polypeptide, optionally a fragment or a portion of a CRN gene may be about 10, 20, 30, 40, 50, 90, 100 to about 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, or about 1280 consecutive nucleotides in length, about 100 to about 1150 consecutive nucleotides in length, about 400 to about 800 consecutive nucleotides in length, about 500 to about 700 consecutive nucleotides in length, about 550 to about 600 consecutive nucleotides in length, about 300 to about 400 consecutive nucleotides in length, about 200 to about 300 consecutive nucleotides in length, about 100 to about 200 consecutive nucleotides in length, about 100 to about 150 consecutive nucleotides in length, about 50 to about 100 consecutive nucleotides in length, about 10 to about 50 consecutive nucleotides in length, or any range or value therein.

In some embodiments, a “sequence-specific nucleic acid binding domain” (e.g., sequence-specific DNA binding domain, sequence-specific RNA binding domain) may bind to one or more fragments or portions of nucleotide sequences encoding CRN polypeptides as described herein.

As used herein with respect to polypeptides, the term “fragment” or “portion” may refer to a polypeptide that is reduced in length relative to a reference polypeptide and that comprises, consists essentially of and/or consists of an amino acid sequence of contiguous amino acids identical or almost identical (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% identical) to a corresponding portion of the reference polypeptide. Such a polypeptide fragment may be, where appropriate, included in a larger polypeptide of which it is a constituent. In some embodiments, the polypeptide fragment comprises, consists essentially of or consists of at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 125, 150, 175, 200, 225, 250, 260, 270, 280, 290, or more consecutive amino acids of a reference polypeptide. In some embodiments, a fragment or portion of a CRN polypeptide may comprise, consist essentially of or consist of about 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 22, 223, 224, 225, 226, 227, 228, 229, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 290, 295, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350, 355, 360, 365, 370, 375, 380, 385, 390, 395, 400, 410, or 420 consecutive amino acid residues, or any range or value therein, (e.g., a fragment or a portion of any one of SEQ ID NOs: 66-111 (e.g., SEQ ID NOs: 112-114)).

As used herein with respect to nucleic acids, the term “functional fragment” refers to nucleic acid that encodes a functional fragment of a polypeptide.

The term “gene,” as used herein, refers to a nucleic acid molecule capable of being used to produce mRNA, antisense RNA, miRNA, anti-microRNA antisense oligodeoxyribonucleotide (AMO) and the like. Genes may or may not be capable of being used to produce a functional protein or gene product. Genes can include both coding and non-coding regions (e.g., introns, regulatory elements, promoters, enhancers, termination sequences and/or 5' and 3' untranslated regions). A gene may be “isolated” by which is meant a nucleic acid that is substantially or essentially free from components normally found in association with the nucleic acid in its natural state. Such components include other cellular material, culture medium from recombinant production, and/or various chemicals used in chemically synthesizing the nucleic acid.

The term “mutation” refers to point mutations (e.g., missense, or nonsense, or insertions or deletions of single base pairs that result in frame shifts), insertions, deletions, and/or truncations. When the mutation is a substitution of a residue within an amino acid sequence with another residue, or a deletion or insertion of one or more residues within a sequence, the mutations are typically described by identifying the original residue followed by the position of the residue within the sequence and by the identity of the newly substituted residue. A truncation can include a truncation at the C-terminal end of a polypeptide or at the N-terminal end of a polypeptide. A truncation of a polypeptide can be the result of a deletion of the corresponding 5' end or 3' end of the gene encoding the polypeptide. A frameshift mutation can occur when deletions or insertions of one or more base pairs are introduced into a gene. Frameshift mutations in a gene can result in the production of a polypeptide that is longer, shorter or the same length as the wild type polypeptide depending on when the first stop codon occurs following the mutated region of the gene.

In some embodiments, a deletion useful with this invention may result in an in-frame mutation. In some embodiments, such a deletion may be a dominant negative mutation, a semi-dominant mutation, a hypomorphic mutation, a weak loss-of-function mutation or a null allele, which when comprised in a plant can result in the plant exhibiting increased kernel number as compared to a plant not comprising said deletion. In some embodiments, a deletion may be a deletion of 1 base pair to about 160 base pairs (e.g., about 3 base pairs to about 10 base pairs, about 3 base pairs to about 20 base pairs, about 3 base pairs to about 30 base pairs, about 3 base pairs to about 35 base pairs, about 3 base pairs to about 40 base pairs, about 3 base pairs to about 45 base pairs, about 3 base pairs to about 50 base pairs, about 3 base pairs to about 80 base pairs, about 3 base pairs to about 100 base pairs, about 3 base pairs to about 160 base pairs, about 10 base pairs to about 160 base pairs, about 15 base pairs to about 160 base pairs, about 20 base pairs to about 160 base pairs, about 30 base pairs to about 160 base pairs, about 50 base pairs to about 160 base pairs, about 75 base pairs to about 160 base pairs, or about 96 base pairs to about 160 base pairs; e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or 15 base pairs to about 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 110, 120, 130, 140, 150, or 160 base pairs). An CRN gene may be edited in more than one location, thereby providing a CRN gene comprising more than one mutation. In some embodiments, such a plant may also exhibit larger meristems, maintenance of root meristems and increased yield and increased disease resistance. Additional advantages of the present invention include, but are not limited to regulating growth, regulating meristem size, regulating vascular development, controlling

organ number, facilitating increased plant regeneration from tissue culture, increasing fruit size and or controlling plant secondary growth.

The terms “complementary” or “complementarity,” as used herein, refer to the natural binding of polynucleotides under permissive salt and temperature conditions by base-pairing. For example, the sequence “A-G-T” (5' to 3') binds to the complementary sequence “T-C-A” (3' to 5'). Complementarity between two single-stranded molecules may be “partial,” in which only some of the nucleotides bind, or it may be complete when total complementarity exists between the single stranded molecules. The degree of complementarity between nucleic acid strands has significant effects on the efficiency and strength of hybridization between nucleic acid strands.

“Complement,” as used herein, can mean 100% complementarity with the comparator nucleotide sequence or it can mean less than 100% complementarity (e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, and the like, complementarity) to the comparator nucleotide sequence.

Different nucleic acids or proteins having homology are referred to herein as “homologues.” The term homologue includes homologous sequences from the same and from other species and orthologous sequences from the same and other species. “Homology” refers to the level of similarity between two or more nucleic acid and/or amino acid sequences in terms of percent of positional identity (i.e., sequence similarity or identity). Homology also refers to the concept of similar functional properties among different nucleic acids or proteins. Thus, the compositions and methods of the invention further comprise homologues to the nucleotide sequences and polypeptide sequences of this invention. “Orthologous,” as used herein, refers to homologous nucleotide sequences and/or amino acid sequences in different species that arose from a common ancestral gene during speciation. A homologue of a nucleotide sequence of this invention has a substantial sequence identity (e.g., at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% or 100%) to said nucleotide sequence of the invention.

As used herein “sequence identity” refers to the extent to which two optimally aligned polynucleotide or polypeptide sequences are invariant throughout a window of alignment of components, e.g., nucleotides or amino acids. “Identity” can be readily calculated by known methods including, but not limited to, those described in: *Computational Molecular Biology* (Lesk, A. M., ed.) Oxford University Press, New York (1988); *Biocomputing: Informatics and Genome Projects* (Smith, D. W., ed.) Academic Press, New York (1993); *Computer Analysis of Sequence Data, Part I* (Griffin, A. M., and Griffin, H. G., eds.) Humana Press, New Jersey (1994); *Sequence Analysis in Molecular Biology* (von Heinje, G., ed.) Academic Press (1987); and *Sequence Analysis Primer* (Gribskov, M. and Devereux, J., eds.) Stockton Press, New York (1991).

As used herein, the term “percent sequence identity” or “percent identity” refers to the percentage of identical nucleotides in a linear polynucleotide sequence of a reference (“query”) polynucleotide molecule (or its complementary strand) as compared to a test (“subject”) polynucleotide molecule (or its complementary strand) when the two sequences are optimally aligned. In some embodiments,

“percent sequence identity” can refer to the percentage of identical amino acids in an amino acid sequence as compared to a reference polypeptide.

As used herein, the phrase “substantially identical,” or “substantial identity” in the context of two nucleic acid molecules, nucleotide sequences, or polypeptide sequences, refers to two or more sequences or subsequences that have at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% or 100% nucleotide or amino acid residue identity, when compared and aligned for maximum correspondence, as measured using one of the following sequence comparison algorithms or by visual inspection. In some embodiments of the invention, the substantial identity exists over a region of consecutive nucleotides of a nucleotide sequence of the invention that is about 10 nucleotides to about 20 nucleotides, about 10 nucleotides to about 25 nucleotides, about 10 nucleotides to about 30 nucleotides, about 15 nucleotides to about 25 nucleotides, about 30 nucleotides to about 40 nucleotides, about 50 nucleotides to about 60 nucleotides, about 70 nucleotides to about 80 nucleotides, about 90 nucleotides to about 100 nucleotides, about 100 nucleotides to about 200 nucleotides, about 100 nucleotides to about 300 nucleotides, about 100 nucleotides to about 400 nucleotides, about 100 nucleotides to about 500 nucleotides, about 100 nucleotides to about 600 nucleotides, about 100 nucleotides to about 800 nucleotides, about 100 nucleotides to about 900 nucleotides, about 100, 200, 300, 400 consecutive nucleotides to about 500, 600, 700, 800, 1000, 1100, 1200 consecutive nucleotides or more, or any range therein, up to the full length of the sequence. In some embodiments, nucleotide sequences can be substantially identical over at least about 20 consecutive nucleotides (e.g., about 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 50, 60, 70, or 80 or more consecutive nucleotides).

In some embodiments of the invention, the substantial identity exists over a region of consecutive amino acid residues of a polypeptide of the invention that is about 3 amino acid residues to about 20 amino acid residues, about 5 amino acid residues to about 25 amino acid residues, about 7 amino acid residues to about 30 amino acid residues, about 10 amino acid residues to about 25 amino acid residues, about 15 amino acid residues to about 30 amino acid residues, about 20 amino acid residues to about 40 amino acid residues, about 25 amino acid residues to about 40 amino acid residues, about 25 amino acid residues to about 50 amino acid residues, about 30 amino acid residues to about 50 amino acid residues, about 40 amino acid residues to about 50 amino acid residues, about 40 amino acid residues to about 70 amino acid residues, about 50 amino acid residues to about 70 amino acid residues, about 60 amino acid residues to about 80 amino acid residues, about 70 amino acid residues to about 80 amino acid residues, about 90 amino acid residues to about 100 amino acid residues, or more amino acid residues in length, and any range therein, up to the full length of the sequence. In some embodiments, polypeptide sequences can be substantially identical to one another over at least about 8 consecutive amino acid residues (e.g., about 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108,

109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 130, 140, 150, 175, 200, 225, 250, 300, 350 or more amino acids in length or more consecutive amino acid residues). In some embodiments, two or more CRN polypeptides may be identical or substantially identical (e.g., at least 70% to 99.9% identical, e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9% identical or any range or value therein).

For sequence comparison, typically one sequence acts as a reference sequence to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are entered into a computer, subsequence coordinates are designated if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.

Optimal alignment of sequences for aligning a comparison window are well known to those skilled in the art and may be conducted by tools such as the local homology algorithm of Smith and Waterman, the homology alignment algorithm of Needleman and Wunsch, the search for similarity method of Pearson and Lipman, and optionally by computerized implementations of these algorithms such as GAP, BESTFIT, FASTA, and TFASTA available as part of the GCG® Wisconsin Package® (Accelrys Inc., San Diego, CA). An “identity fraction” for aligned segments of a test sequence and a reference sequence is the number of identical components which are shared by the two aligned sequences divided by the total number of components in the reference sequence segment, e.g., the entire reference sequence or a smaller defined part of the reference sequence. Percent sequence identity is represented as the identity fraction multiplied by 100. The comparison of one or more polynucleotide sequences may be to a full-length polynucleotide sequence or a portion thereof, or to a longer polynucleotide sequence. For purposes of this invention “percent identity” may also be determined using BLASTX version 2.0 for translated nucleotide sequences and BLASTN version 2.0 for polynucleotide sequences.

Two nucleotide sequences may also be considered substantially complementary when the two sequences hybridize to each other under stringent conditions. In some embodiments, two nucleotide sequences considered to be substantially complementary hybridize to each other under highly stringent conditions.

“Stringent hybridization conditions” and “stringent hybridization wash conditions” in the context of nucleic acid hybridization experiments such as Southern and Northern hybridizations are sequence dependent and are different under different environmental parameters. An extensive guide to the hybridization of nucleic acids is found in Tijssen *Laboratory Techniques in Biochemistry and Molecular Biology-Hybridization with Nucleic Acid Probes* part I chapter 2 “Overview of principles of hybridization and the strategy of nucleic acid probe assays” Elsevier, New York (1993). Generally, highly stringent hybridization and wash conditions are selected to be about 5° C. lower than the thermal melting point (T_m) for the specific sequence at a defined ionic strength and pH.

The T_m is the temperature (under defined ionic strength and pH) at which 50% of the target sequence hybridizes to a perfectly matched probe. Very stringent conditions are selected to be equal to the T_m for a particular probe. An

example of stringent hybridization conditions for hybridization of complementary nucleotide sequences which have more than 100 complementary residues on a filter in a Southern or northern blot is 50% formamide with 1 mg of heparin at 42° C., with the hybridization being carried out overnight. An example of highly stringent wash conditions is 0.1 M NaCl at 72° C. for about 15 minutes. An example of stringent wash conditions is a 0.2×SSC wash at 65° C. for 15 minutes (see, Sambrook, *infra*, for a description of SSC buffer). Often, a high stringency wash is preceded by a low stringency wash to remove background probe signal. An example of a medium stringency wash for a duplex of, e.g., more than 100 nucleotides, is 1×SSC at 45° C. for 15 minutes. An example of a low stringency wash for a duplex of, e.g., more than 100 nucleotides, is 4-6×SSC at 40° C. for 15 minutes. For short probes (e.g., about 10 to 50 nucleotides), stringent conditions typically involve salt concentrations of less than about 1.0 M Na ion, typically about 0.01 to 1.0 M Na ion concentration (or other salts) at pH 7.0 to 8.3, and the temperature is typically at least about 30° C. Stringent conditions can also be achieved with the addition of destabilizing agents such as formamide. In general, a signal to noise ratio of 2× (or higher) than that observed for an unrelated probe in the particular hybridization assay indicates detection of a specific hybridization. Nucleotide sequences that do not hybridize to each other under stringent conditions are still substantially identical if the proteins that they encode are substantially identical. This can occur, for example, when a copy of a nucleotide sequence is created using the maximum codon degeneracy permitted by the genetic code.

A polynucleotide and/or recombinant nucleic acid construct of this invention (e.g., expression cassettes and/or vectors) may be codon optimized for expression. In some embodiments, the polynucleotides, nucleic acid constructs, expression cassettes, and/or vectors of the editing systems of the invention (e.g., comprising/encoding a sequence-specific nucleic acid binding domain (e.g., a sequence-specific nucleic acid binding domain from a polynucleotide-guided endonuclease, a zinc finger nuclease, a transcription activator-like effector nuclease (TALEN), an Argonaute protein, and/or a CRISPR-Cas endonuclease (e.g., CRISPR-Cas effector protein) (e.g., a Type I CRISPR-Cas effector protein, a Type II CRISPR-Cas effector protein, a Type III CRISPR-Cas effector protein, a Type IV CRISPR-Cas effector protein, a Type V CRISPR-Cas effector protein or a Type VI CRISPR-Cas effector protein)), a nuclease (e.g., an endonuclease (e.g., FokI), a polynucleotide-guided endonuclease, a CRISPR-Cas endonuclease (e.g., CRISPR-Cas effector protein), a zinc finger nuclease, and/or a transcription activator-like effector nuclease (TALEN)), deaminase proteins/domains (e.g., adenine deaminase, cytosine deaminase), a polynucleotide encoding a reverse transcriptase protein or domain, a polynucleotide encoding a 5'-3' exonuclease polypeptide, and/or affinity polypeptides, peptide tags, etc.) may be codon optimized for expression in a plant. In some embodiments, the codon optimized nucleic acids, polynucleotides, expression cassettes, and/or vectors of the invention have about 70% to about 99.9% (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9% or 100%) identity or more to the reference nucleic acids, polynucleotides, expression cassettes, and/or vectors that have not been codon optimized.

In any of the embodiments described herein, a polynucleotide or nucleic acid construct of the invention may be

operatively associated with a variety of promoters and/or other regulatory elements for expression in a plant and/or a cell of a plant. Thus, in some embodiments, a polynucleotide or nucleic acid construct of this invention may further comprise one or more promoters, introns, enhancers, and/or terminators operably linked to one or more nucleotide sequences. In some embodiments, a promoter may be operably associated with an intron (e.g., Ubi1 promoter and intron). In some embodiments, a promoter associated with an intron may be referred to as a "promoter region" (e.g., Ubi1 promoter and intron).

By "operably linked" or "operably associated" as used herein in reference to polynucleotides, it is meant that the indicated elements are functionally related to each other and are also generally physically related. Thus, the term "operably linked" or "operably associated" as used herein, refers to nucleotide sequences on a single nucleic acid molecule that are functionally associated. Thus, a first nucleotide sequence that is operably linked to a second nucleotide sequence means a situation when the first nucleotide sequence is placed in a functional relationship with the second nucleotide sequence. For instance, a promoter is operably associated with a nucleotide sequence if the promoter effects the transcription or expression of said nucleotide sequence. Those skilled in the art will appreciate that the control sequences (e.g., promoter) need not be contiguous with the nucleotide sequence to which it is operably associated, as long as the control sequences function to direct the expression thereof. Thus, for example, intervening untranslated, yet transcribed, nucleic acid sequences can be present between a promoter and the nucleotide sequence, and the promoter can still be considered "operably linked" to the nucleotide sequence.

As used herein, the term "linked," in reference to polypeptides, refers to the attachment of one polypeptide to another. A polypeptide may be linked to another polypeptide (at the N-terminus or the C-terminus) directly (e.g., via a peptide bond) or through a linker.

The term "linker" is art-recognized and refers to a chemical group, or a molecule linking two molecules or moieties, e.g., two domains of a fusion protein, such as, for example, a nucleic acid binding polypeptide or domain and peptide tag and/or a reverse transcriptase and an affinity polypeptide that binds to the peptide tag; or a DNA endonuclease polypeptide or domain and peptide tag and/or a reverse transcriptase and an affinity polypeptide that binds to the peptide tag. A linker may be comprised of a single linking molecule or may comprise more than one linking molecule. In some embodiments, the linker can be an organic molecule, group, polymer, or chemical moiety such as a bivalent organic moiety. In some embodiments, the linker may be an amino acid, or it may be a peptide. In some embodiments, the linker is a peptide.

In some embodiments, a peptide linker useful with this invention may be about 2 to about 100 or more amino acids in length, for example, about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100 or more amino acids in length (e.g., about 2 to about 40, about 2 to about 50, about 2 to about 60, about 4 to about 40, about 4 to about 50, about 4 to about 60, about 5 to about 40, about 5 to about 50, about 5 to about 60, about 9 to about 40, about 9 to about 50, about 9 to about 60, about 10 to about 40, about 10 to about 50,

about 10 to about 60, or about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 amino acids to about 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100 or more amino acids in length (e.g., about 105, 110, 115, 120, 130, 140 150 or more amino acids in length). In some embodiments, a peptide linker may be a GS linker.

As used herein, the term “linked,” or “fused” in reference to polynucleotides, refers to the attachment of one polynucleotide to another. In some embodiments, two or more polynucleotide molecules may be linked by a linker that can be an organic molecule, group, polymer, or chemical moiety such as a bivalent organic moiety. A polynucleotide may be linked or fused to another polynucleotide (at the 5' end or the 3' end) via a covalent or non-covalent linkage or binding, including e.g., Watson-Crick base-pairing, or through one or more linking nucleotides. In some embodiments, a polynucleotide motif of a certain structure may be inserted within another polynucleotide sequence (e.g. extension of the hairpin structure in the guide RNA). In some embodiments, the linking nucleotides may be naturally occurring nucleotides. In some embodiments, the linking nucleotides may be non-naturally occurring nucleotides.

A “promoter” is a nucleotide sequence that controls or regulates the transcription of a nucleotide sequence (e.g., a coding sequence) that is operably associated with the promoter. The coding sequence controlled or regulated by a promoter may encode a polypeptide and/or a functional RNA. Typically, a “promoter” refers to a nucleotide sequence that contains a binding site for RNA polymerase II and directs the initiation of transcription. In general, promoters are found 5', or upstream, relative to the start of the coding region of the corresponding coding sequence. A promoter may comprise other elements that act as regulators of gene expression; e.g., a promoter region. These include a TATA box consensus sequence, and often a CAAT box consensus sequence (Breathnach and Chambon, (1981) *Annu. Rev. Biochem.* 50:349). In plants, the CAAT box may be substituted by the AGGA box (Messing et al., (1983) in *Genetic Engineering of Plants*, T. Kosuge, C. Meredith and A. Hollaender (eds.), Plenum Press, pp. 211-227).

Promoters useful with this invention can include, for example, constitutive, inducible, temporally regulated, developmentally regulated, chemically regulated, tissue-preferred and/or tissue-specific promoters for use in the preparation of recombinant nucleic acid molecules, e.g., “synthetic nucleic acid constructs” or “protein-RNA complex.” These various types of promoters are known in the art.

The choice of promoter may vary depending on the temporal and spatial requirements for expression, and also may vary based on the host cell to be transformed. Promoters for many different organisms are well known in the art. Based on the extensive knowledge present in the art, the appropriate promoter can be selected for the particular host organism of interest. Thus, for example, much is known about promoters upstream of highly constitutively expressed genes in model organisms and such knowledge can be readily accessed and implemented in other systems as appropriate.

In some embodiments, a promoter functional in a plant may be used with the constructs of this invention. Non-limiting examples of a promoter useful for driving expression in a plant include the promoter of the RubisCo small subunit gene 1 (PrbcS1), the promoter of the actin gene

(Pactin), the promoter of the nitrate reductase gene (Pnr) and the promoter of duplicated carbonic anhydrase gene 1 (Pdca1) (See, Walker et al. *Plant Cell Rep.* 23:727-735 (2005); Li et al. *Gene* 403:132-142 (2007); Li et al. *Mol Biol. Rep.* 37:1143-1154 (2010)). PrbcS1 and Pactin are constitutive promoters and Pnr and Pdca1 are inducible promoters. Pnr is induced by nitrate and repressed by ammonium (Li et al. *Gene* 403:132-142 (2007)) and Pdca1 is induced by salt (Li et al. *Mol Biol. Rep.* 37:1143-1154 (2010)). In some embodiments, a promoter useful with this invention is RNA polymerase II (Pol II) promoter. In some embodiments, a U6 promoter or a 7SL promoter from *Zea mays* may be useful with constructs of this invention. In some embodiments, the U6c promoter and/or 7SL promoter from *Zea mays* may be useful for driving expression of a guide nucleic acid. In some embodiments, a U6c promoter, U6i promoter and/or 7SL promoter from *Glycine max* may be useful with constructs of this invention. In some embodiments, the U6c promoter, U6i promoter and/or 7SL promoter from *Glycine max* may be useful for driving expression of a guide nucleic acid.

Examples of constitutive promoters useful for plants include, but are not limited to, cestrum virus promoter (cmp) (U.S. Pat. No. 7,166,770), the rice actin 1 promoter (Wang et al. (1992) *Mol. Cell. Biol.* 12:3399-3406; as well as U.S. Pat. No. 5,641,876), CaMV 35S promoter (Odell et al. (1985) *Nature* 313:810-812), CaMV 19S promoter (Lawton et al. (1987) *Plant Mol. Biol.* 9:315-324), nos promoter (Ebert et al. (1987) *Proc. Natl. Acad. Sci USA* 84:5745-5749), Adh promoter (Walker et al. (1987) *Proc. Natl. Acad. Sci. USA* 84:6624-6629), sucrose synthase promoter (Yang & Russell (1990) *Proc. Natl. Acad. Sci. USA* 87:4144-4148), and the ubiquitin promoter. The constitutive promoter derived from ubiquitin accumulates in many cell types. Ubiquitin promoters have been cloned from several plant species for use in transgenic plants, for example, sunflower (Binet et al., 1991. *Plant Science* 79:87-94), maize (Christensen et al., 1989. *Plant Molec. Biol.* 12:619-632), and *Arabidopsis* (Norris et al. 1993. *Plant Molec. Biol.* 21:895-906). The maize ubiquitin promoter (UbiP) has been developed in transgenic monocot systems and its sequence and vectors constructed for monocot transformation are disclosed in the patent publication EP 0 342 926. The ubiquitin promoter is suitable for the expression of the nucleotide sequences of the invention in transgenic plants, especially monocotyledons. Further, the promoter expression cassettes described by McElroy et al. (*Mol. Gen. Genet.* 231:150-160 (1991)) can be easily modified for the expression of the nucleotide sequences of the invention and are particularly suitable for use in monocotyledonous hosts.

In some embodiments, tissue specific/tissue preferred promoters can be used for expression of a heterologous polynucleotide in a plant cell. Tissue specific or preferred expression patterns include, but are not limited to, green tissue specific or preferred, root specific or preferred, stem specific or preferred, flower specific or preferred or pollen specific or preferred. Promoters suitable for expression in green tissue include many that regulate genes involved in photosynthesis and many of these have been cloned from both monocotyledons and dicotyledons. In one embodiment, a promoter useful with the invention is the maize PEPC promoter from the phosphoenol carboxylase gene (Hudspeth & Grula, *Plant Molec. Biol.* 12:579-589 (1989)). Non-limiting examples of tissue-specific promoters include those associated with genes encoding the seed storage proteins (such as β -conglycinin, cruciferin, napin and phaseolin), zein or oil body proteins (such as oleosin), or proteins

involved in fatty acid biosynthesis (including acyl carrier protein, stearoyl-ACP desaturase and fatty acid desaturases (fad 2-1)), and other nucleic acids expressed during embryo development (such as Bce4, see, e.g., Kridl et al. (1991) *Seed Sci. Res.* 1:209-219; as well as EP U.S. Pat. No. 255,378). Tissue-specific or tissue-preferential promoters useful for the expression of the nucleotide sequences of the invention in plants, particularly maize, include but are not limited to those that direct expression in root, pith, leaf or pollen. Such promoters are disclosed, for example, in WO 93/07278, herein incorporated by reference in its entirety. Other non-limiting examples of tissue specific or tissue preferred promoters useful with the invention the cotton rubisco promoter disclosed in U.S. Pat. No. 6,040,504; the rice sucrose synthase promoter disclosed in U.S. Pat. No. 5,604,121; the root specific promoter described by de Fromond (FEBS 290:103-106 (1991); EP 0 452 269 to Ciba-Geigy); the stem specific promoter described in U.S. Pat. No. 5,625,136 (to Ciba-Geigy) and which drives expression of the maize trpA gene; the cestrum yellow leaf curling virus promoter disclosed in WO 01/73087; and pollen specific or preferred promoters including, but not limited to, ProOsLPS10 and ProOsLPS11 from rice (Nguyen et al. *Plant Biotechnol. Reports* 9 (5): 297-306 (2015)), ZmSTK2_USP from maize (Wang et al. *Genome* 60 (6): 485-495 (2017)), LAT52 and LAT59 from tomato (Twell et al. *Development* 109 (3): 705-713 (1990)), Zm13 (U.S. Pat. No. 10,421,972), PLA₂-δ promoter from *Arabidopsis* (U.S. Pat. No. 7,141,424), and/or the ZmC5 promoter from maize (International PCT Publication No. WO1999/042587).

Additional examples of plant tissue-specific/tissue preferred promoters include, but are not limited to, the root hair-specific cis-elements (RHEs) (Kim et al. *The Plant Cell* 18:2958-2970 (2006)), the root-specific promoters RCc3 (Jeong et al. *Plant Physiol.* 153:185-197 (2010)) and RB7 (U.S. Pat. No. 5,459,252), the lectin promoter (Lindstrom et al. (1990) *Dev. Genet.* 11:160-167; and Vodkin (1983) *Prog. Clin. Biol. Res.* 138:87-98), corn alcohol dehydrogenase 1 promoter (Dennis et al. (1984) *Nucleic Acids Res.* 12:3983-4000), S-adenosyl-L-methionine synthetase (SAMS) (Vander Mijnsbrugge et al. (1996) *Plant and Cell Physiology*, 37 (8): 1108-1115), corn light harvesting complex promoter (Bansal et al. (1992) *Proc. Natl. Acad. Sci. USA* 89:3654-3658), corn heat shock protein promoter (O'Dell et al. (1985) *EMBO J.* 5:451-458; and Rochester et al. (1986) *EMBO J.* 5:451-458), pea small subunit RuBP carboxylase promoter (Cashmore, "Nuclear genes encoding the small subunit of ribulose-1,5-bisphosphate carboxylase" pp. 29-39 In: *Genetic Engineering of Plants* (Hollaender ed., Plenum Press 1983; and Poulsen et al. (1986) *Mol. Gen. Genet.* 205:193-200), Ti plasmid mannopine synthase promoter (Langridge et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:3219-3223), Ti plasmid nopaline synthase promoter (Langridge et al. (1989), supra), *petunia* chalcone isomerase promoter (van Tunen et al. (1988) *EMBO J.* 7:1257-1263), bean glycine rich protein 1 promoter (Keller et al. (1989) *Genes Dev.* 3:1639-1646), truncated CaMV 35S promoter (O'Dell et al. (1985) *Nature* 313:810-812), potato patatin promoter (Wenzler et al. (1989) *Plant Mol. Biol.* 13:347-354), root cell promoter (Yamamoto et al. (1990) *Nucleic Acids Res.* 18:7449), maize zein promoter (Kriz et al. (1987) *Mol. Gen. Genet.* 207:90-98; Langridge et al. (1983) *Cell* 34:1015-1022; Reina et al. (1990) *Nucleic Acids Res.* 18:6425; Reina et al. (1990) *Nucleic Acids Res.* 18:7449; and Wandelt et al. (1989) *Nucleic Acids Res.* 17:2354), globulin-1 promoter (Belanger et al. (1991) *Genetics* 129:863-872), α-tubulin cab promoter (Sullivan et al. (1989) *Mol. Gen. Genet.*

215:431-440), PEPCase promoter (Hudspeth & Grula (1989) *Plant Mol. Biol.* 12:579-589), R gene complex-associated promoters (Chandler et al. (1989) *Plant Cell* 1:1175-1183), and chalcone synthase promoters (Franken et al. (1991) *EMBO J.* 10:2605-2612).

Useful for seed-specific expression is the pea vicilin promoter (Czako et al. (1992) *Mol. Gen. Genet.* 235:33-40; as well as the seed-specific promoters disclosed in U.S. Pat. No. 5,625,136. Useful promoters for expression in mature leaves are those that are switched at the onset of senescence, such as the SAG promoter from *Arabidopsis* (Gan et al. (1995) *Science* 270:1986-1988).

In addition, promoters functional in chloroplasts can be used. Non-limiting examples of such promoters include the bacteriophage T3 gene 9 5' UTR and other promoters disclosed in U.S. Pat. No. 7,579,516. Other promoters useful with the invention include but are not limited to the S-E9 small subunit RuBP carboxylase promoter and the Kunitz trypsin inhibitor gene promoter (Kti3).

Additional regulatory elements useful with this invention include, but are not limited to, introns, enhancers, termination sequences and/or 5' and 3' untranslated regions.

An intron useful with this invention can be an intron identified in and isolated from a plant and then inserted into an expression cassette to be used in transformation of a plant. As would be understood by those of skill in the art, introns can comprise the sequences required for self-excision and are incorporated into nucleic acid constructs/ expression cassettes in frame. An intron can be used either as a spacer to separate multiple protein-coding sequences in one nucleic acid construct, or an intron can be used inside one protein-coding sequence to, for example, stabilize the mRNA. If they are used within a protein-coding sequence, they are inserted "in-frame" with the excision sites included. Introns may also be associated with promoters to improve or modify expression. As an example, a promoter/intron combination useful with this invention includes but is not limited to that of the maize Ubi1 promoter and intron (see, e.g., SEQ ID NO:21 and SEQ ID NO:22).

Non-limiting examples of introns useful with the present invention include introns from the ADHI gene (e.g., Adh1-S introns 1, 2 and 6), the ubiquitin gene (Ubi1), the RuBisCO small subunit (rbcS) gene, the RuBisCO large subunit (rbcL) gene, the actin gene (e.g., actin-1 intron), the pyruvate dehydrogenase kinase gene (pdk), the nitrate reductase gene (nr), the duplicated carbonic anhydrase gene 1 (Tdca1), the psbA gene, the atpA gene, or any combination thereof.

In some embodiments, a polynucleotide and/or a nucleic acid construct of the invention can be an "expression cassette" or can be comprised within an expression cassette. As used herein, "expression cassette" means a recombinant nucleic acid molecule comprising, for example, a one or more polynucleotides of the invention (e.g., a polynucleotide encoding a sequence-specific nucleic acid binding domain, a polynucleotide encoding a deaminase protein or domain, a polynucleotide encoding a reverse transcriptase protein or domain, a polynucleotide encoding a 5'-3' exonuclease polypeptide or domain, a guide nucleic acid and/or reverse transcriptase (RT) template), wherein polynucleotide(s) is/are operably associated with one or more control sequences (e.g., a promoter, terminator and the like). Thus, in some embodiments, one or more expression cassettes may be provided, which are designed to express, for example, a nucleic acid construct of the invention (e.g., a polynucleotide encoding a sequence-specific nucleic acid binding domain (e.g., a sequence-specific DNA binding domain), a polynucleotide encoding a nuclease polypeptide/domain, a

polynucleotide encoding a deaminase protein/domain, a polynucleotide encoding a reverse transcriptase protein/domain, a polynucleotide encoding a 5'-3' exonuclease polypeptide/domain, a polynucleotide encoding a peptide tag, and/or a polynucleotide encoding an affinity polypeptide, and the like, or comprising a guide nucleic acid, an extended guide nucleic acid, and/or RT template, and the like). When an expression cassette of the present invention comprises more than one polynucleotide, the polynucleotides may be operably linked to a single promoter that drives expression of all of the polynucleotides or the polynucleotides may be operably linked to one or more separate promoters (e.g., three polynucleotides may be driven by one, two or three promoters in any combination). When two or more separate promoters are used, the promoters may be the same promoter, or they may be different promoters. Thus, a polynucleotide encoding a sequence specific nucleic acid binding domain, a polynucleotide encoding a nuclease protein/domain, a polynucleotide encoding a CRISPR-Cas effector protein/domain, a polynucleotide encoding an deaminase protein/domain, a polynucleotide encoding a reverse transcriptase polypeptide/domain (e.g., RNA-dependent DNA polymerase), and/or a polynucleotide encoding a 5'-3' exonuclease polypeptide/domain, a guide nucleic acid, an extended guide nucleic acid and/or RT template when comprised in a single expression cassette may each be operably linked to a single promoter, or separate promoters in any combination.

An expression cassette comprising a nucleic acid construct of the invention may be chimeric, meaning that at least one (e.g., one or more) of its components is heterologous with respect to at least one of its other components (e.g., a promoter from the host organism operably linked to a polynucleotide of interest to be expressed in the host organism, wherein the polynucleotide of interest is from a different organism than the host or is not normally found in association with that promoter). An expression cassette may also be one that is naturally occurring but has been obtained in a recombinant form useful for heterologous expression.

An expression cassette can optionally include a transcriptional and/or translational termination region (i.e., termination region) and/or an enhancer region that is functional in the selected host cell. A variety of transcriptional terminators and enhancers are known in the art and are available for use in expression cassettes. Transcriptional terminators are responsible for the termination of transcription and correct mRNA polyadenylation. A termination region and/or the enhancer region may be native to the transcriptional initiation region, may be native to, for example, a gene encoding a sequence-specific nucleic acid binding protein, a gene encoding a nuclease, a gene encoding a reverse transcriptase, a gene encoding a deaminase, and the like, or may be native to a host cell, or may be native to another source (e.g., foreign or heterologous to, for example, to a promoter, to a gene encoding a sequence-specific nucleic acid binding protein, a gene encoding a nuclease, a gene encoding a reverse transcriptase, a gene encoding a deaminase, and the like, or to the host cell, or any combination thereof).

An expression cassette of the invention also can include a polynucleotide encoding a selectable marker, which can be used to select a transformed host cell. As used herein, "selectable marker" means a polynucleotide sequence that when expressed imparts a distinct phenotype to the host cell expressing the marker and thus allows such transformed cells to be distinguished from those that do not have the marker. Such a polynucleotide sequence may encode either a selectable or screenable marker, depending on whether the

marker confers a trait that can be selected for by chemical means, such as by using a selective agent (e.g., an antibiotic and the like), or on whether the marker is simply a trait that one can identify through observation or testing, such as by screening (e.g., fluorescence). Many examples of suitable selectable markers are known in the art and can be used in the expression cassettes described herein.

In addition to expression cassettes, the nucleic acid molecules/constructs and polynucleotide sequences described herein can be used in connection with vectors. The term "vector" refers to a composition for transferring, delivering or introducing a nucleic acid (or nucleic acids) into a cell. A vector comprises a nucleic acid construct (e.g., expression cassette(s)) comprising the nucleotide sequence(s) to be transferred, delivered or introduced. Vectors for use in transformation of host organisms are well known in the art. Non-limiting examples of general classes of vectors include viral vectors, plasmid vectors, phage vectors, phagemid vectors, cosmid vectors, fosmid vectors, bacteriophages, artificial chromosomes, minicircles, or *Agrobacterium* binary vectors in double or single stranded linear or circular form which may or may not be self-transmissible or mobilizable. In some embodiments, a viral vector can include, but is not limited, to a retroviral, lentiviral, adenoviral, adeno-associated, or herpes simplex viral vector. A vector as defined herein can transform a prokaryotic or eukaryotic host either by integration into the cellular genome or exist extrachromosomally (e.g., autonomous replicating plasmid with an origin of replication). Additionally, included are shuttle vectors by which is meant a DNA vehicle capable, naturally or by design, of replication in two different host organisms, which may be selected from actinomycetes and related species, bacteria and eukaryotic (e.g., higher plant, mammalian, yeast or fungal cells). In some embodiments, the nucleic acid in the vector is under the control of, and operably linked to, an appropriate promoter or other regulatory elements for transcription in a host cell. The vector may be a bi-functional expression vector which functions in multiple hosts. In the case of genomic DNA, this may contain its own promoter and/or other regulatory elements and in the case of cDNA this may be under the control of an appropriate promoter and/or other regulatory elements for expression in the host cell. Accordingly, a nucleic acid or polynucleotide of this invention and/or expression cassettes comprising the same may be comprised in vectors as described herein and as known in the art.

As used herein, "contact," "contacting," "contacted," and grammatical variations thereof, refer to placing the components of a desired reaction together under conditions suitable for carrying out the desired reaction (e.g., transformation, transcriptional control, genome editing, nicking, and/or cleavage). As an example, a target nucleic acid may be contacted with a sequence-specific nucleic acid binding protein (e.g., polynucleotide-guided endonuclease, a CRISPR-Cas endonuclease (e.g., CRISPR-Cas effector protein), a zinc finger nuclease, a transcription activator-like effector nuclease (TALEN) and/or an Argonaute protein)) and a deaminase or a nucleic acid construct encoding the same, under conditions whereby the sequence-specific nucleic acid binding protein, the reverse transcriptase and/or the deaminase are expressed and the sequence-specific nucleic acid binding protein binds to the target nucleic acid, and the reverse transcriptase and/or deaminase may be fused to either the sequence-specific nucleic acid binding protein or recruited to the sequence-specific nucleic acid binding protein (via, for example, a peptide tag fused to the sequence-specific nucleic acid binding protein and an affin-

ity tag fused to the reverse transcriptase and/or deaminase) and thus, the deaminase and/or reverse transcriptase is positioned in the vicinity of the target nucleic acid, thereby modifying the target nucleic acid. Other methods for recruiting reverse transcriptase and/or deaminase may be used that take advantage of other protein-protein interactions, and also RNA-protein interactions and chemical interactions may be used for protein-protein and protein-nucleic acid recruitment.

As used herein, “modifying” or “modification” in reference to a target nucleic acid includes editing (e.g., mutating), covalent modification, exchanging/substituting nucleic acids/nucleotide bases, deleting, cleaving, nicking, and/or altering transcriptional control of a target nucleic acid. In some embodiments, a modification may include one or more single base changes (SNPs) of any type.

“Introducing,” “introduce,” “introduced” (and grammatical variations thereof) in the context of a polynucleotide of interest means presenting a nucleotide sequence of interest (e.g., polynucleotide, RT template, a nucleic acid construct, and/or a guide nucleic acid) to a plant, plant part thereof, or cell thereof, in such a manner that the nucleotide sequence gains access to the interior of a cell.

The terms “transformation” or transfection” may be used interchangeably and as used herein refer to the introduction of a heterologous nucleic acid into a cell. Transformation of a cell may be stable or transient. Thus, in some embodiments, a host cell or host organism (e.g., a plant) may be stably transformed with a polynucleotide/nucleic acid molecule of the invention. In some embodiments, a host cell or host organism may be transiently transformed with a polynucleotide/nucleic acid molecule of the invention.

“Transient transformation” in the context of a polynucleotide means that a polynucleotide is introduced into the cell and does not integrate into the genome of the cell.

By “stably introducing” or “stably introduced” in the context of a polynucleotide introduced into a cell is intended that the introduced polynucleotide is stably incorporated into the genome of the cell, and thus the cell is stably transformed with the polynucleotide.

“Stable transformation” or “stably transformed” as used herein means that a nucleic acid molecule is introduced into a cell and integrates into the genome of the cell. As such, the integrated nucleic acid molecule is capable of being inherited by the progeny thereof, more particularly, by the progeny of multiple successive generations. “Genome” as used herein includes the nuclear and the plastid genome, and therefore includes integration of the nucleic acid into, for example, the chloroplast or mitochondrial genome. Stable transformation as used herein can also refer to a transgene that is maintained extrachromosomally, for example, as a minichromosome or a plasmid.

Transient transformation may be detected by, for example, an enzyme-linked immunosorbent assay (ELISA) or Western blot, which can detect the presence of a peptide or polypeptide encoded by one or more transgene introduced into an organism. Stable transformation of a cell can be detected by, for example, a Southern blot hybridization assay of genomic DNA of the cell with nucleic acid sequences which specifically hybridize with a nucleotide sequence of a transgene introduced into an organism (e.g., a plant). Stable transformation of a cell can be detected by, for example, a Northern blot hybridization assay of RNA of the cell with nucleic acid sequences which specifically hybridize with a nucleotide sequence of a transgene introduced into a host organism. Stable transformation of a cell can also be detected by, e.g., a polymerase chain reaction (PCR) or other

amplification reactions as are well known in the art, employing specific primer sequences that hybridize with target sequence(s) of a transgene, resulting in amplification of the transgene sequence, which can be detected according to standard methods. Transformation can also be detected by direct sequencing and/or hybridization protocols well known in the art.

Accordingly, in some embodiments, nucleotide sequences, polynucleotides, nucleic acid constructs, and/or expression cassettes of the invention may be expressed transiently and/or they can be stably incorporated into the genome of the host organism. Thus, in some embodiments, a nucleic acid construct of the invention (e.g., one or more expression cassettes comprising polynucleotides for editing as described herein) may be transiently introduced into a cell with a guide nucleic acid and as such, no DNA is maintained in the cell.

A nucleic acid construct of the invention may be introduced into a plant cell by any method known to those of skill in the art. Non-limiting examples of transformation methods include transformation via bacterial-mediated nucleic acid delivery (e.g., via *Agrobacteria*), viral-mediated nucleic acid delivery, silicon carbide or nucleic acid whisker-mediated nucleic acid delivery, liposome mediated nucleic acid delivery, microinjection, microparticle bombardment, calcium-phosphate-mediated transformation, cyclodextrin-mediated transformation, electroporation, nanoparticle-mediated transformation, sonication, infiltration, PEG-mediated nucleic acid uptake, as well as any other electrical, chemical, physical (mechanical) and/or biological mechanism that results in the introduction of nucleic acid into the plant cell, including any combination thereof. Procedures for transforming both eukaryotic and prokaryotic organisms are well known and routine in the art and are described throughout the literature (See, for example, Jiang et al. 2013. *Nat. Biotechnol.* 31:233-239; Ran et al. *Nature Protocols* 8:2281-2308 (2013)). General guides to various plant transformation methods known in the art include Miki et al. (“Procedures for Introducing Foreign DNA into Plants” in *Methods in Plant Molecular Biology and Biotechnology*, Glick, B. R. and Thompson, J. E., Eds. (CRC Press, Inc., Boca Raton, 1993), pages 67-88) and Rakowoczy-Trojanowska (*Cell. Mol. Biol. Lett.* 7:849-858 (2002)).

In some embodiments of the invention, transformation of a cell may comprise nuclear transformation. In other embodiments, transformation of a cell may comprise plastid transformation (e.g., chloroplast transformation). In still further embodiments, nucleic acids of the invention may be introduced into a cell via conventional breeding techniques. In some embodiments, one or more of the polynucleotides, expression cassettes and/or vectors may be introduced into a plant cell via *Agrobacterium* transformation.

A polynucleotide therefore can be introduced into a plant, plant part, plant cell in any number of ways that are well known in the art. The methods of the invention do not depend on a particular method for introducing one or more nucleotide sequences into a plant, only that they gain access to the interior the cell. Where more than polynucleotide is to be introduced, they can be assembled as part of a single nucleic acid construct, or as separate nucleic acid constructs, and can be located on the same or different nucleic acid constructs. Accordingly, the polynucleotide can be introduced into the cell of interest in a single transformation event, or in separate transformation events, or, alternatively, a polynucleotide can be incorporated into a plant as part of a breeding protocol.

The present invention provides methods and compositions for reducing the influence of genes that normally act to restrict meristem size to generate plants with larger meristems, to maintain the root meristem, to increase kernel row number (optionally without substantially decreasing ear length (e.g., without decreasing ear length more than 30% as compared to an ear of a plant not comprising the same CRN mutation)) and/or to improve one or more yield traits, as well as to improve disease resistance.

The CLV1, CLV2, and CLV3 genes are part of several overlapping signaling pathways that utilize a similar mechanism to regulate meristem size. Plants express many CLV3-like (CLE) peptides which are sensed by many leucine-rich repeat (LRR) domains, and the downstream signaling regulates a myriad of growth processes in the plant (Fletcher, J. C., *Plants* 7:87 (2018)). The role of CRN in conveying downstream signaling appears to be conserved between *Arabidopsis* and Maize (Nimchuk et al., *Plant Cell* 23:851-854 (2011); Guo et al., *Plant J.* 63:889-900 (2010); Zhu et al., *Plant J.* 61:223-233 (2009); Müller et al., *The Plant Cell* Online 20:934-946 (2008); Miwa et al., *Plant Cell Physiol.* 49:1752-1757 (2008); Je et al., *Nat. Genet.* 48: ng.3567 (2018)). Both *Arabidopsis* and Maize CRN are predicted to encode inactive pseudokinases suggesting CRN may act as a protein scaffold for complex formation (Nimchuk et al., *Plant Cell* 23:851-854 (2011)). Indeed, the current model for CRN function in *Arabidopsis* hypothesizes that CRN/CLV2 dimerization facilitates transit of both proteins from the endoplasmic reticulum (ER) to the plasma membrane (PM) where CLE-mediated signaling takes place. When CLV2 and CRN are separated as monomers, charged amino acids prevent PM transiting and the proteins are held in the ER. When CRN and CLV2 pair, these charged amino acids are masked and export of both proteins from the ER to PM takes place, a phenomenon called charge masking (Bleckmann et al. *Plant Phys.* 152:166-176 (2010)). The short extracellular domain (EC) of CRN is required for membrane transiting but is not required for dimerization with CLV2, suggesting the EC domain may perform a charge masking function. See, e.g., FIG. 4.

As is understood in the art, different amino acids can carry a negative, positive, or neutral charge. The combined action of a sequence of amino acids determines the charge of a region of a particular protein. In the present invention, by changing the amino acid composition along the CRN EC domain, the charge of the protein region that is involved in charge masking FEA2 can be changed and in-turn the transiting of the FEA2/CRN protein complex to the membrane can be modified. Without intending to be bound by any particular theory, it is believed that the positive charge in CRN masks the negative charge of FEA2, which is required for transiting. Thus, as an example, deletions of a CRN polypeptide may provide a reduction in positive charge, and thus, less charge masking of FEA2 and greater complex retention in the ER. Increased ER retention in this model would antagonize signaling mediated by the CRN/FEA2 complex resulting in increased meristem size, increased kernel row number and improved yield.

Accordingly, as described herein, editing technology is used to target CRN genes in plants to generate plants with larger meristems, having increased kernel number (optionally without substantially reducing ear length, e.g., exhibits a decrease in ear length of no more than 30% as compared to an ear of a plant not comprising the same CRN mutation), increased yield, and improved disease resistance as well as to generate plants that maintain their root meristem. Mutations that may be useful for production of plants exhibiting

increased kernel number include, for example, substitutions, deletions and insertions. In some aspects, a mutation generated by the editing technology can be a point mutation, an in-frame deletion, a mutation that generates a premature stop codon, a null mutation, a dominant negative mutation, a semi-dominant mutation, hypomorphic mutation, or a weak loss-of-function mutation.

In some embodiments, the invention provides a plant or plant part thereof, the plant or plant part comprising at least one non-natural mutation (e.g., 1, 2, 3, 4, 5, or more mutations) in an endogenous CORYNE (CRN) gene that encodes a CRN protein. In some embodiments, the at least one non-natural mutation results in a null mutation, a dominant negative mutation, a semi-dominant mutation, a hypomorphic mutation, and/or a weak loss-of-function mutation.

In some embodiments, a plant cell is provided, the plant cell comprising an editing system comprising: (a) a CRISPR-Cas effector protein; and (b) a guide nucleic acid (gRNA, gDNA, crRNA, crDNA, sgRNA, sgDNA) comprising a spacer sequence with complementarity to an endogenous target gene encoding a CRN protein. The editing system may be used to generate a mutation in the endogenous target gene encoding a CRN protein. In some embodiments, the mutation is a non-natural mutation. In some embodiments, a guide nucleic acid of an editing system may comprise the nucleotide sequence (a spacer sequence; guide) of any one of SEQ ID NOs: 126-130.

The mutation in a CRN gene of the plant, plant part thereof or the plant cell may be any type of mutation, including a base substitution, a base deletion and/or a base insertion. In some embodiments, a non-natural mutation may comprise a base substitution to an A, a T, a G, or a C. In some embodiments, the at least one non-natural mutation may be a base substitution from a C to a T (C>T), a G to an A (G>A), an A to a G (A>G) or a T to a C (T>C). In some embodiments, a non-natural mutation may be a deletion of at least one base pair (e.g., 1 base pair to about 200 base pairs) or an insertion of at least one base pair (e.g., 1 base pair to about 50 base pairs). In some embodiments, a deletion may comprise 1 base pair to about 10 consecutive base pairs (e.g., 1, 2 bp to about 3, 4, 5, 6, 7, 8, 9, or 10 bp; e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 consecutive base pairs), 1 base pair to about 20 consecutive base pairs (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 consecutive base pairs; e.g., 1, 2, 3, 4, 5, 6 bp to about 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 consecutive base pairs), 1 base pair to about 30 consecutive base pairs, 1 base pair to about 40 consecutive base pairs, 1 base pair to about 50 consecutive base pairs (e.g., 1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 consecutive base pairs to about 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50 consecutive base pairs), or about 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50 consecutive base pairs to about 60, 70, 80, 90, 100, 110, 120, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 145, 150, 155, or 160 consecutive base pairs or any value or range therein of a CRN gene. In some embodiments, a deletion may be a deletion that includes the entire EC region in a CRN gene (e.g., SEQ ID NO: 119 in maize). In some embodiments, a deletion is an in-frame deletion. In some embodiments, a deletion provides a premature stop codon and a truncated CRN gene and/or protein.

In some embodiments, a non-natural mutation useful with this invention may result in at least one (e.g., one or more) amino acid substitution in the EC region of a CRN gene. In

some embodiments, a non-natural mutation resulting in one or more substituted amino acid residues in a CRN gene may be located: (a) at position 23 to position 66 with reference amino acid position numbering of SEQ ID NO:118; (b) at position 29 to position 75 with reference amino acid position numbering of SEQ ID NOs: 72-77; (c) at position 29 to position 74 with reference amino acid position numbering of SEQ ID NO:78; (d) at position 29 to position 76 with reference amino acid position numbering of SEQ ID NO:79; (e) at position 22 to position 65 with reference amino acid position numbering of SEQ ID NO:80; (f) at position 22 to position 68 with reference amino acid position numbering of SEQ ID NO:81; (g) at position 16 to position 65 with reference amino acid position numbering of SEQ ID NO:82; (h) at position 20 to position 69 with reference amino acid position numbering of SEQ ID NOs: 83, 86, or 87; (i) at position 48 to position 97 with reference amino acid position numbering of SEQ ID NO:84; (j) at position 21 to position 71 with reference amino acid position numbering of SEQ ID NO: 85; (k) at position 41 to position 86 with reference amino acid position numbering of SEQ ID NO: 88 or SEQ ID NO:89; (l) at position 31 to position 71 with reference amino acid position numbering of SEQ ID NO:90; (m) at position 21 to position 56 with reference amino acid position numbering of SEQ ID NO:91; (n) at position 22 to position 56 with reference amino acid position numbering of SEQ ID NO:92; (o) at position 22 to position 59 with reference amino acid position numbering of SEQ ID NO:93 or SEQ ID NO:94; (p) at position 20 to position 57 with reference amino acid position numbering of SEQ ID NOs: 95, 98, 99 or 100; (q) at position 25 to position 62 with reference amino acid position numbering of SEQ ID NO: 96; (r) at position 26 to position 63 with reference amino acid position numbering of SEQ ID NO: 97; (s) at position 40 to position 75 with reference amino acid position numbering of SEQ ID NO:101; (t) at position 40 to position 76 with reference amino acid position numbering of SEQ ID NO: 102; (u) at position 19 to position 54 with reference amino acid position numbering of SEQ ID NO:103; (v) at position 25 to position 68 with reference amino acid position numbering of SEQ ID NO: 104; (w) at position 47 to position 90 with reference amino acid position numbering of SEQ ID NO:105; (x) at position 25 to position 70 with reference amino acid position numbering of SEQ ID NO:106 or SEQ ID NO:107; (y) at position 22 to position 60 with reference amino acid position numbering of SEQ ID NO: 108 or SEQ ID NO: 109; (z) at position 23 to position 53 with reference amino acid position numbering of SEQ ID NO: 110; (aa) at position 15 to position 53 with reference amino acid position numbering of SEQ ID NO:111 or SEQ ID NO:113; (bb) at position 17 to position 55 with reference amino acid position numbering of SEQ ID NO: 112; (cc) at position 25 to position 63 with reference amino acid position numbering of SEQ ID NO:114; (dd) at position 21 to position 61 with reference amino acid position numbering of SEQ ID NO: 115; or (ee) at position 23 to position 57 with reference amino acid position numbering of SEQ ID NO:116 or SEQ ID NO:117. In some embodiments, the mutation results in one or more substituted amino acid residues located at position 23 to position 66 with reference amino acid position numbering of SEQ ID NO:118

An endogenous CRN gene useful with this invention may (a) encode a sequence having at least 95% sequence identity to any one of the amino acid sequences of SEQ ID NO: 72-118, optionally SEQ ID NO:118; (b) comprise a region having a sequence with at least 90% sequence identity to the nucleotide sequence of SEQ ID NO:122-125; and/or (c)

encode a sequence having at least 90% sequence identity to any one of the amino acid sequences of SEQ ID NOs: 119-121. In some embodiments, a mutated CRN gene comprises at least about 90% sequence identity to any one of the nucleic acid sequences of SEQ ID NOs: 180-191.

In some embodiments, a plant comprising at least one mutation (e.g., one or more mutations) in an endogenous CRN gene exhibits increased maintenance of meristems and/or increased kernel number as compared to a plant without the at least one non-natural mutation. In some embodiments, the plant comprising at least one mutation in an endogenous CRN gene is a corn plant which exhibits increased kernel number. In some embodiments, the plant comprising at least one mutation in an endogenous CRN gene is a corn plant which exhibits increased yield and/or increased disease resistance. In some embodiments, a plant (e.g., a corn plant) may be regenerated from a plant part and/or plant cell of the invention, wherein the regenerated plant (e.g., regenerated corn plant) comprises the mutation in the endogenous (CRN gene and a phenotype of increased kernel number as compared to a plant (e.g., a corn plant) not comprising the mutation.

In some embodiments, a corn plant cell is provided, the corn plant cell comprising at least one non-natural mutation within a CRN gene, wherein the mutation is a substitution, insertion or a deletion that is introduced using an editing system that comprises a nucleic acid binding domain that binds to a target site in the CRN gene. In some embodiments, the substitution, insertion or a deletion results in a null allele, a dominant negative allele, a semi-dominant allele, hypomorphic mutation, or a weak loss-of-function allele. In some embodiments, a deletion results in an in-frame deletion allele. In some embodiments, the target site is within a region of the CRN gene, the region comprising a sequence having at least 90% sequence identity (e.g., about 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 99.5, 99.6, 99.7, 99.8, 99.9, or 100% sequence identity) to the nucleotide sequence of any one of SEQ ID NOs: 123-125 and/or encoding a sequence having at least 95% sequence identity (e.g., about 95, 96, 97, 98, 99, 99.5, 99.6, 99.7, 99.8, 99.9, or 100% sequence identity) to any one of the amino acid sequences of SEQ ID NO: 119-121.

In some embodiments, a method of producing/breeding a transgene-free edited corn plant is provided, the method comprising: crossing a corn plant of the present invention (e.g., a corn plant comprising a mutation in a CRN gene and having increased kernel number) with a transgene free corn plant, thereby introducing the at least one non-natural mutation (e.g., one or more non-natural mutations) into the corn plant that is transgene-free; and selecting a progeny corn plant that comprises the at least one non-natural mutation and is transgene-free, thereby producing a transgene free edited corn plant.

Also provided herein is a method of providing a plurality of corn plants having increased kernel number, the method comprising planting two or more corn plants of the invention (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10 or more corn plants comprising a mutation in a CRN polypeptide and having increased kernel number) in a growing area (e.g., a field (e.g., a cultivated field, an agricultural field), a growth chamber, a greenhouse, a recreational area, a lawn, and/or a roadside and the like), thereby providing a plurality of corn plants having increased kernel number as compared to a plurality of control corn plants not comprising the mutation. In some embodiments, the plurality of plants may also exhibit larger meristems, increased yield, increased disease resistance and maintenance of their root meristems. The invention further

provides a method of generating variation in a region of a CRN protein, comprising: introducing an editing system into a corn plant cell, wherein the editing system is targeted to a region of a CRN gene that encodes the region of the CRN protein, wherein the region comprises a sequence having at least 70% sequence identity to any one of the amino acid sequences of SEQ ID NO: 119-121 or the region is encoded by a sequence having at least 70% identity to the nucleotide sequence of SEQ ID NOs: 123-125; and contacting the region of the CRN gene with the editing system, thereby introducing into the plant cell a mutation into the region of the CRN protein; and generating variation in the region of the CRN protein.

In some embodiments, a method for editing a specific site in the genome of a corn plant cell, the method comprising: cleaving, in a site specific manner, a target site within an endogenous CRN gene in the plant cell, the endogenous CRN gene comprising a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122, or encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO: 118, thereby generating an edit in the endogenous CRN gene of the plant cell and producing a plant cell comprising the edit in the endogenous CRN gene. The endogenous CRN gene encodes a CRN protein and the edit results in variation of amino acids in the coding region of the CRN protein. In some embodiments, the edit results in a non-naturally occurring mutation, including but not limited to a deletion, substitution, or insertion, wherein the edit may result in a null mutation, a dominant negative mutation, a semi-dominant mutation, a hypomorphic mutation, or a weak loss-of-function mutation. In some embodiments, the non-naturally occurring mutation is a deletion, optionally wherein the deletion is in the EC domain of a CRN gene (See, FIG. 3—representation of various deletions within the EC domain). In some embodiments, the entire EC domain of the CRN gene may be deleted. In some embodiments, the deletion may result in an edited CRN nucleic acid having at least 90% sequence identity to any one of SEQ ID NOs: 180-191. In some embodiments, a deletion results in an in-frame deletion. In some embodiments, a deletion results in the generation of a premature stop codon. In some embodiments, a non-natural mutation produces variability in a region (e.g., EC domain) of a CRN polypeptide (e.g., amino acid residues located at position 23 to position 66 with reference amino acid position numbering of SEQ ID NO:118 and/or in the EC region of any one of the amino acid sequences of SEQ ID NOs: 72-117 as described herein). In some embodiments, a method of editing may further comprise regenerating a corn plant from the corn plant cell comprising the edit in the endogenous CRN gene, thereby producing a corn plant comprising the edit in its endogenous CRN gene and having a phenotype of increased kernel number when compared to a control corn plant (e.g., an isogenic corn plant) that does not comprise the edit.

In some embodiments, a method for making a corn plant, comprising: (a) contacting a population of corn plant cells comprising a wild-type endogenous CRN gene with a nuclease linked to a nucleic acid binding domain (e.g., DNA binding domain; e.g., editing system) that binds to a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122, to a sequence having at least 70% identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125, to a sequence encoding an amino acid sequence having at least 70% sequence identity to SEQ ID NO:118; or to a sequence encoding an amino acid sequence having at least 70% sequence identity

to any one of SEQ ID NOs: 119-121; (b) selecting a corn plant cell from the population in which at least one endogenous CRN gene has been mutated; and (c) growing the selected plant cell into a corn plant.

In some embodiments, a method increasing kernel number in a corn plant, comprising (a) contacting a corn plant cell comprising an endogenous CRN gene with a nuclease targeting the endogenous CRN gene, wherein the nuclease is linked to a nucleic acid binding domain (e.g., editing system) that binds to a target site in the endogenous CRN gene, wherein the endogenous CRN gene: (i) encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (ii) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122; (iii) comprises a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (iv) comprises a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121 to produce a corn plant cell comprising a mutation in the endogenous CRN gene, thereby producing the corn plant comprising at least one cell (e.g., one or more cells) having a mutation in the endogenous CRN gene; and (b) growing the corn plant cell into a corn plant comprising the mutation in the endogenous CRN gene, thereby producing a corn plant have a mutated endogenous CRN gene and an increased kernel number.

In some embodiments, a method for producing a corn plant or part thereof comprising at least one cell (e.g., one or more cells) having a mutated endogenous CRN gene, the method comprising contacting a target site in an endogenous CRN gene in the corn plant or plant part with a nuclease comprising a cleavage domain and a nucleic acid binding domain, wherein the nucleic acid binding domain binds to a target site in the endogenous CRN gene, wherein the endogenous CRN gene (a) encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (b) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122; (c) comprises a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprises a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121, to produce a plant cell comprising a mutation in the endogenous CRN gene, thereby producing the corn plant or part thereof comprising at least one cell having a mutation in the endogenous CRN gene.

Also provided herein is a method for producing a corn plant or part thereof comprising a mutated endogenous CRN gene and exhibiting increased kernel number, the method comprising contacting a target site in an endogenous CRN gene in the corn plant or plant part with a nuclease comprising a cleavage domain and a nucleic acid binding domain, wherein the nucleic acid binding domain binds to a target site in the endogenous CRN gene, wherein the endogenous CRN gene: (a) encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (b) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprises a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprises a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121,

thereby producing the corn plant or part thereof comprising an endogenous CRN gene having a mutation and exhibiting increased kernel number.

In some embodiments, a corn plant or part thereof comprising at least one cell (e.g., one or more cells) having a mutation in the endogenous CRN gene as described herein, comprises a sequence having at least 90% identity to any one of the sequences of SEQ ID NOs: 180-191.

In some embodiments, a nuclease may cleave an endogenous CRN gene, thereby introducing the mutation into the endogenous CRN gene. A nuclease useful with the invention may be any nuclease that can be utilized to edit/modify a target nucleic acid. Such nucleases include, but are not limited to a zinc finger nuclease, transcription activator-like effector nucleases (TALEN), endonuclease (e.g., FokI) and/or a CRISPR-Cas effector protein. Likewise, a nucleic acid binding domain useful with the invention may be any nucleic acid binding domain (e.g., DNA binding domain) that can be utilized to edit/modify a target nucleic acid. Such nucleic acid binding domains include, but are not limited to, a zinc finger, transcription activator-like DNA binding domain (TAL), an argonaute and/or a CRISPR-Cas effector DNA binding domain.

In some embodiments, a method of editing an endogenous CRN gene in a corn plant or plant part is provided, the method comprising contacting a target site in an CRN gene in the corn plant or plant part with a cytosine base editing system comprising a cytosine deaminase and a nucleic acid binding domain that binds to a target site in the CRN gene, the CRN gene (a) encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO: 118; (b) comprising a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprising a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprising a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121, thereby editing the endogenous CRN gene in the corn plant or part thereof and producing a corn plant or part thereof comprising at least one cell (e.g., one or more cells) having a mutation in the endogenous CRN gene.

In some embodiments, a method of editing an endogenous CRN gene in a corn plant or plant part is provided, the method comprising contacting a target site in an CRN gene in the corn plant or plant part with an adenosine base editing system comprising an adenosine deaminase and a nucleic acid binding domain that binds to a target site in the CRN gene, the CRN gene (a) encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO: 118; (b) comprising a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprising a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprising a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121, thereby editing the endogenous CRN gene in the corn plant or part thereof and producing a plant or part thereof comprising at least one cell having a mutation in the endogenous CRN gene.

In some embodiments, a mutation in an edited endogenous CRN gene as described herein produces a sequence having at least 90% identity to any one of the sequences of SEQ ID NOs: 180-191.

In some embodiments, a method of detecting a mutant CRN gene (a mutation in an endogenous CRN gene) is

provide, the method comprising detecting in the genome of a plant a deletion in a nucleic acid encoding the amino acid sequence of SEQ ID NO:118, wherein the amino acid sequence of SEQ ID NO:118 comprises a mutation in one or more amino acid residue(s) located from position 23 to position 66 with reference amino acid position numbering of SEQ ID NO:118. In some embodiments, the mutation is the result of a nucleotide substitution of C>T, G>A, A>G or T>C.

In some embodiments, the present invention provides a method of detecting a mutation in an endogenous CRN gene, comprising detecting in the genome of a plant a mutated CRN gene. In some embodiments, the mutated CRN gene comprises a sequence having at least 90% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 180-191.

In some embodiments, the present invention provides a method of producing a plant comprising a mutation in an endogenous CRN gene and at least one polynucleotide of interest (e.g., one or more polynucleotides of interest), the method comprising crossing a plant of the invention comprising at least one mutation in an endogenous CRN gene (a first plant) with a second plant that comprises the at least one polynucleotide of interest to produce progeny plants; and selecting progeny plants comprising at least one mutation in the CRN gene and the at least one polynucleotide of interest, thereby producing the plant comprising a mutation in an endogenous CRN gene and at least one polynucleotide of interest.

The present invention further provides a method of producing a plant comprising a mutation in an endogenous CRN gene and at least one polynucleotide of interest, the method comprising introducing at least one polynucleotide of interest into a plant of the present invention comprising at least one mutation in a CRN gene, thereby producing a plant comprising at least one mutation in a CRN gene and at least one polynucleotide of interest. In some embodiments, the plant is a corn plant.

In some embodiments, the present invention provides a method of producing a plant comprising a mutation in an endogenous CRN gene and at least one polynucleotide of interest, the method comprising introducing at least one polynucleotide of interest into a plant of the invention comprising at least one mutation in an endogenous CRN gene, thereby producing a plant comprising at least one mutation in a CRN gene and at least one polynucleotide of interest. In some embodiments, the plant is a corn plant.

A polynucleotide of interest may be any polynucleotide that can confer a desirable phenotype or otherwise modify the phenotype or genotype of a plant. In some embodiments, a polynucleotide of interest may be polynucleotide that confers herbicide tolerance, insect resistance, disease resistance, increased yield, increased nutrient use efficiency or abiotic stress resistance.

A CORYNE (CRN) gene useful with this invention includes any CRN gene in which a mutation as described herein can confer increased kernel number in a plant or part thereof comprising the mutation (e.g., SEQ ID NOs: 72-118). In some embodiments, a CRN polypeptide comprises an amino acid sequence having at least 70% identity (e.g., about 70, 71, 72, 73, 74, 75, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 99.5, 100% sequence identity) to SEQ ID NO: 118 or comprising any one of the amino acid sequences of SEQ ID NOs: 119-121 (e.g., the CRN polypeptide comprises an EC domain comprising the amino acid sequence of, for example, SEQ ID NO:119, SEQ ID NO:120, or SEQ ID NO:121) within the

CRN polypeptide). In some embodiments, a CRN polypeptide comprises an amino acid sequence having at least 70% identity (e.g., about 70, 71, 72, 73, 74, 75, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 99.5, 100% sequence identity) to SEQ ID NO:72-118. In some embodiments, a CRN gene may comprise a sequence having at least about 70% sequence identity (e.g., about 70, 71, 72, 73, 74, 75, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 99.5, 100% sequence identity) to the nucleotide sequence of SEQ ID NO:122, or the CRN gene comprises within it a sequence having at least 70% identity to any one of the nucleotide sequences of any one of SEQ ID NOs: 123-125.

In some embodiments, the at least one non-natural mutation (e.g., one or more non-natural mutations) in an endogenous CRN gene in a corn plant may be a substitution, a deletion and/or an insertion. In some embodiments, the at least one non-natural mutation in an endogenous CRN gene in a corn plant may be a substitution, a deletion and/or an insertion that results in a null mutation, a dominant negative mutation, a semi-dominant mutation, hypomorphic mutation, or a weak loss-of-function mutation and a plant having the phenotype of increased kernel number as compared to a control corn plant (e.g., isogenic plant) not comprising the edit/mutation. For example, the mutation may be a substitution, a deletion and/or an insertion of one or more amino acid residues (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more amino acids of the CRN polypeptide) or the mutation may be a substitution, a deletion and/or an insertion of at least 1 nucleotide to about 160 consecutive nucleotides (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 consecutive nucleotides, or any range or value therein) (e.g., a base substitution, deletion and/or insertion) from the gene encoding the CRN polypeptide. In some embodiments, the non-natural mutation may be a deletion. In some embodiments, the mutation may be a deletion of at least about 18 consecutive base pairs to about 160 consecutive base pairs (e.g., about 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50 consecutive base pairs to about 60, 70, 80, 90, 100, 110, 120, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 145, 150, 155, or 160 consecutive base pairs or any value or range therein of a CRN gene. In some embodiments, a deletion results in an in-frame deletion allele. In some embodiments, the at least one non-natural mutation may be a base substitution to an A, a T, a G, or a C. In some embodiments, the at least one non-natural mutation may be a base substitution to from a C to a T (C>T), a G to an A (G>A), an A to a G (A>G) or a T to a C (T>C). In some embodiments, the non-natural mutation is in the EC region of a CRN.

In some embodiments, a mutation in an endogenous CRN gene may be made following cleavage by an editing system that comprises a nuclease and a nucleic acid binding domain that binds to a target site within a target nucleic acid (e.g., a CRN gene), the target nucleic acid (e.g., CRN gene) (a) encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (b) comprising a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprising a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprising a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121. In some embodi-

ments, the nuclease cleaves the endogenous CRN gene, and a mutation is introduced into the endogenous CRN gene. In some embodiments, the cleavage results in a mutation in an endogenous CRN gene comprising a sequence having at least 90% identity to any one of the sequences of SEQ ID NOs: 180-191.

Further provided herein are guide nucleic acids (e.g., gRNA, gDNA, crRNA, crDNA) that bind to a target site in a CRN gene, wherein the endogenous CRN gene comprises: a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO:122 or any one of the nucleotide sequences of any one of SEQ ID NOs: 123-125; encodes a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118, or any one of the amino acid sequences of SEQ ID NO:119-121. In some embodiments, a guide nucleic acid comprises a spacer having the nucleotide sequence of any one of SEQ ID NOs: 126-130.

In some embodiments, a guide nucleic acid is provided that binds to a target nucleic acid in a CORYNE (CRN) gene in a corn plant, the CRN gene having the gene identification number (gene ID) of Zm00001d042268.

With regard to corn (*Zea mays*), markers of the present invention are described herein with respect to the positions of marker loci in the B73 corn genome, version 4, "B73 RefGen_v4" (assembly aka B73 RefGen_v4, AGPv4) at the MaizeGDB internet resource (maizegdb.org/assembly).

In some embodiments, a system is provided comprising a guide nucleic acid comprising a spacer having the nucleotide sequence of any one of SEQ ID NOs: 126-130 and a CRISPR-Cas effector protein that associates with the guide nucleic acid. In some embodiments, the system may further comprise a tracr nucleic acid that associates with the guide nucleic acid and a CRISPR-Cas effector protein, optionally wherein the tracr nucleic acid and the guide nucleic acid are covalently linked.

The invention further provides a gene editing system comprising a CRISPR-Cas effector protein in association with a guide nucleic acid and the guide nucleic acid comprises a spacer sequence that binds to a CRN gene, the CRN gene (a) encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (b) comprising a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprising a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprising a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121. In some embodiments, a spacer sequence of the guide nucleic acid may comprise the nucleotide sequence of any one of SEQ ID NOs: 126-130. In some embodiments, the gene editing system may further comprise a tracr nucleic acid that associates with the guide nucleic acid and a CRISPR-Cas effector protein, optionally wherein the tracr nucleic acid and the guide nucleic acid are covalently linked. As used herein, "a CRISPR-Cas effector protein in association with a guide nucleic acid" refers to the complex that is formed between a CRISPR-Cas effector protein and a guide nucleic acid in order to direct the CRISPR-Cas effector protein to a target site in a gene.

The present invention further provides a complex comprising a CRISPR-Cas effector protein comprising a cleavage domain and a guide nucleic acid, wherein the guide nucleic acid binds to a target site in an endogenous CRN gene, wherein the endogenous CRN gene, wherein the endogenous CRN gene: (a) encodes a sequence having at

least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (b) comprises a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprises a region having a sequence with at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprises a region encoding a sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NOs: 119-121, wherein the cleavage domain cleaves a target strand in the CRN gene. In some embodiments, the cleavage domain cleaves a target strand in the CRN gene such that it results in a mutation in an endogenous CRN gene, the mutated CRN gene comprising a sequence having at least 90% identity to any one of the sequences of SEQ ID NOs: 180-191.

In some embodiments, expression cassettes are provided that comprise (a) a polynucleotide encoding CRISPR-Cas effector protein comprising a cleavage domain and (b) a guide nucleic acid that binds to a target site in an endogenous CRN gene, wherein the guide nucleic acid comprises a spacer sequence that is complementary to and binds to (i) a portion of a nucleic acid encoding an amino acid sequence having at least 70% sequence identity to the amino acid sequence of SEQ ID NO:118; (ii) a portion of a sequence having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (iii) a portion of a sequence having at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (iv) a portion of sequence having at least 70% sequence identity to a sequence encoding the amino acid sequence of SEQ ID NO: 119-121.

Also provided herein are nucleic acids encoding a mutated CRN gene that when present in a corn plant or plant part results in the corn plant comprising a phenotype of increased kernel number as compared to a corn plant or plant part not comprising the CRN mutation. In some embodiments, a mutated CRN gene may comprise a sequence having at least 90% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 180-191.

Nucleic acid constructs of the invention (e.g., a construct comprising a sequence specific nucleic acid binding domain, a CRISPR-Cas effector domain, a deaminase domain, reverse transcriptase (RT), RT template and/or a guide nucleic acid, etc.) and expression cassettes/vectors comprising the same may be used as an editing system of this invention for modifying target nucleic acids (e.g., endogenous CRN genes) and/or their expression.

Any corn plant comprising an endogenous CRN gene that is capable of conferring increased kernel number when modified as described herein may be modified (e.g., mutated, e.g., base edited, cleaved, nicked, etc.) as described herein (e.g., using the polypeptides, polynucleotides, RNPs, nucleic acid constructs, expression cassettes, and/or vectors of the invention) to increase kernel number in the corn plant.

A plant exhibiting increased kernel number (e.g., a corn plant) may have an increase in kernel number by about 5% to about 100% (e.g., about 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% or more or any range or value therein; e.g., about 5% to about 10%, about 5% to about 15%, about 5% to about 20%, about 10% to about 50%, about 10% to about 80%, about 10% to about 90%, about 10% to about 100%, about 20% to about 50%, about 20% to

about 80%, about 20% to about 90%, about 20% to about 100%, about 30% to about 50%, about 30% to about 80%, about 30% to about 90%, about 30% to about 100%, about 50% to about 100%, about 75% to about 100% or more, and any range or value therein) as compared to a plant or part thereof that does not comprise the mutated endogenous CRN gene.

In some embodiments, plants exhibiting increased kernel row number as described herein (e.g., a plant that produces ears having increased kernel row number) produce ears that are also not substantially decreased in length. As used herein, an ear of a plant comprising a mutation as described herein that is "not substantially decreased in length" has a length that is reduced by less than 30% (e.g., reduced by 0% or reduced by about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30%) as compared to a plant that does not comprise the same CRN mutation.

In some embodiments, a corn plant or plant part thereof is provided comprising at least one non-natural mutation in at least one endogenous CORYNE (CRN) gene having the gene identification number (gene ID) of Zm00001d042268.

The term "plant part," as used herein, includes but is not limited to reproductive tissues (e.g., petals, sepals, stamens, pistils, receptacles, anthers, pollen, flowers, fruits, flower bud, ovules, seeds, and embryos); vegetative tissues (e.g., petioles, stems, roots, root hairs, root tips, pith, coleoptiles, stalks, shoots, branches, bark, apical meristem, axillary bud, cotyledon, hypocotyls, and leaves); vascular tissues (e.g., phloem and xylem); specialized cells such as epidermal cells, parenchyma cells, chlorenchyma cells, schlerenchyma cells, stomates, guard cells, cuticle, mesophyll cells; callus tissue; and cuttings. The term "plant part" also includes plant cells, including plant cells that are intact in plants and/or parts of plants, plant protoplasts, plant tissues, plant organs, plant cell tissue cultures, plant calli, plant clumps, and the like. As used herein, "shoot" refers to the above ground parts including the leaves and stems. As used herein, the term "tissue culture" encompasses cultures of tissue, cells, protoplasts and callus.

As used herein, "plant cell" refers to a structural and physiological unit of the plant, which typically comprise a cell wall but also includes protoplasts. A plant cell of the present invention can be in the form of an isolated single cell or can be a cultured cell or can be a part of a higher-organized unit such as, for example, a plant tissue (including callus) or a plant organ. A "protoplast" is an isolated plant cell without a cell wall or with only parts of the cell wall. Thus, in some embodiments of the invention, a transgenic cell comprising a nucleic acid molecule and/or nucleotide sequence of the invention is a cell of any plant or plant part including, but not limited to, a root cell, a leaf cell, a tissue culture cell, a seed cell, a flower cell, a fruit cell, a pollen cell, and the like. In some aspects of the invention, the plant part can be a plant germplasm. In some aspects, a plant cell can be non-propagating plant cell that does not regenerate into a plant.

"Plant cell culture" means cultures of plant units such as, for example, protoplasts, cell culture cells, cells in plant tissues, pollen, pollen tubes, ovules, embryo sacs, zygotes and embryos at various stages of development.

As used herein, a "plant organ" is a distinct and visibly structured and differentiated part of a plant such as a root, stem, leaf, flower bud, or embryo.

"Plant tissue" as used herein means a group of plant cells organized into a structural and functional unit. Any tissue of a plant in planta or in culture is included. This term includes,

but is not limited to, whole plants, plant organs, plant seeds, tissue culture and any groups of plant cells organized into structural and/or functional units. The use of this term in conjunction with, or in the absence of, any specific type of plant tissue as listed above or otherwise embraced by this definition is not intended to be exclusive of any other type of plant tissue.

In some embodiments of the invention, a transgenic tissue culture or transgenic plant cell culture is provided, wherein the transgenic tissue or cell culture comprises a nucleic acid molecule/nucleotide sequence of the invention. In some 10 embodiments, transgenes may be eliminated from a plant developed from the transgenic tissue or cell by breeding of the transgenic plant with a non-transgenic plant and selecting among the progeny for the plants comprising the desired gene edit and not the transgenes used in producing the edit.

An editing system useful with this invention can be any site-specific (sequence-specific) genome editing system now known or later developed, which system can introduce mutations in target specific manner. For example, an editing system (e.g., site- or sequence-specific editing system) can include, but is not limited to, a CRISPR-Cas editing system, a meganuclease editing system, a zinc finger nuclease (ZFN) editing system, a transcription activator-like effector nuclease (TALEN) editing system, a base editing system and/or a prime editing system, each of which can comprise one or more polypeptides and/or one or more polynucleotides that when expressed as a system in a cell can modify (mutate) a target nucleic acid in a sequence specific manner. In some 20 embodiments, an editing system (e.g., site- or sequence-specific editing system) can comprise one or more polynucleotides and/or one or more polypeptides, including but not limited to a nucleic acid binding domain (e.g., DNA binding domain), a nuclease, and/or other polypeptide, and/or a polynucleotide.

In some embodiments, an editing system can comprise one or more sequence-specific nucleic acid binding domains (e.g., DNA binding domains) that can be from, for example, a polynucleotide-guided endonuclease, a CRISPR-Cas endonuclease (e.g., CRISPR-Cas effector protein), a zinc 40 finger nuclease, a transcription activator-like effector nuclease (TALEN) and/or an Argonaute protein. In some embodiments, an editing system can comprise one or more cleavage domains (e.g., nucleases) including, but not limited to, an endonuclease (e.g., FokI), a polynucleotide-guided endonuclease, a CRISPR-Cas endonuclease (e.g., CRISPR-Cas effector protein), a zinc finger nuclease, and/or a transcription activator-like effector nuclease (TALEN). In some 50 embodiments, an editing system can comprise one or more polypeptides that include, but are not limited to, a deaminase (e.g., a cytosine deaminase, an adenine deaminase), a reverse transcriptase, a Dna2 polypeptide, and/or a 5' flap endonuclease (FEN). In some embodiments, an editing system can comprise one or more polynucleotides, including, but is not limited to, a CRISPR array (CRISPR guide) 55 nucleic acid, extended guide nucleic acid, and/or a reverse transcriptase template.

In some embodiments, a method of modifying or editing an CRN gene may comprise contacting a target nucleic acid (e.g., a nucleic acid encoding an CRN protein) with a base-editing fusion protein (e.g., a sequence specific DNA binding protein (e.g., a CRISPR-Cas effector protein or domain) fused to a deaminase domain (e.g., an adenine 60 deaminase and/or a cytosine deaminase) and a guide nucleic acid, wherein the guide nucleic acid is capable of guiding/targeting the base editing fusion protein to the target nucleic acid, thereby editing a locus within the target nucleic acid.

In some embodiments, a base editing fusion protein and guide nucleic acid may be comprised in one or more expression cassettes. In some embodiments, the target nucleic acid may be contacted with a base editing fusion protein and an expression cassette comprising a guide 5 nucleic acid. In some embodiments, the sequence-specific nucleic acid binding fusion proteins and guides may be provided as ribonucleoproteins (RNPs). In some embodiments, a cell may be contacted with more than one base-editing fusion protein and/or one or more guide nucleic acids that may target one or more target nucleic acids in the cell.

In some embodiments, a method of modifying or editing an CRN gene may comprise contacting a target nucleic acid (e.g., a nucleic acid encoding a CRN) with a sequence-specific nucleic acid binding fusion protein (e.g., a sequence-specific DNA binding protein (e.g., a CRISPR-Cas effector protein or domain)) fused to a peptide tag, a deaminase fusion protein comprising a deaminase domain (e.g., an adenine deaminase and/or a cytosine deaminase) 20 fused to an affinity polypeptide that is capable of binding to the peptide tag, and a guide nucleic acid, wherein the guide nucleic acid is capable of guiding/targeting the sequence-specific nucleic acid binding fusion protein to the target nucleic acid and the sequence-specific nucleic acid binding fusion protein is capable of recruiting the deaminase fusion protein to the target nucleic acid via the peptide tag-affinity polypeptide interaction, thereby editing a locus within the target nucleic acid. In some embodiments, the sequence-specific nucleic acid binding fusion protein may be fused to the affinity polypeptide that binds the peptide tag and the 30 deaminase may be fused to the peptide tag, thereby recruiting the deaminase to the sequence-specific nucleic acid binding fusion protein and to the target nucleic acid. In some embodiments, the sequence-specific binding fusion protein, deaminase fusion protein, and guide nucleic acid may be comprised in one or more expression cassettes. In some 35 embodiments, the target nucleic acid may be contacted with a sequence-specific binding fusion protein, deaminase fusion protein, and an expression cassette comprising a guide nucleic acid. In some embodiments, the sequence-specific nucleic acid binding fusion proteins, deaminase fusion proteins and guides may be provided as ribonucleoproteins (RNPs).

In some embodiments, methods such as prime editing may be used to generate a mutation in an endogenous CRN gene. In prime editing, RNA-dependent DNA polymerase (reverse transcriptase, RT) and reverse transcriptase templates (RT template) are used in combination with sequence specific nucleic acid binding domains that confer the ability 45 to recognize and bind the target in a sequence-specific manner, and which can also cause a nick of the PAM-containing strand within the target. The nucleic acid binding domain may be a CRISPR-Cas effector protein and in this case, the CRISPR array or guide RNA may be an extended guide that comprises an extended portion comprising a primer binding site (PSB) and the edit to be incorporated into the genome (the template). Similar to base editing, prime editing can take advantage of the various methods of recruiting proteins for use in the editing to the target site, such methods including both non-covalent and covalent interactions between the proteins and nucleic acids used in the selected process of genome editing.

As used herein, a "CRISPR-Cas effector protein" is a protein or polypeptide or domain thereof that cleaves or cuts a nucleic acid, binds a nucleic acid (e.g., a target nucleic acid and/or a guide nucleic acid), and/or that identifies, recognizes, or binds a guide nucleic acid as defined herein. In

some embodiments, a CRISPR-Cas effector protein may be an enzyme (e.g., a nuclease, endonuclease, nickase, etc.) or portion thereof and/or may function as an enzyme. In some embodiments, a CRISPR-Cas effector protein refers to a CRISPR-Cas nuclease polypeptide or domain thereof that comprises nuclease activity or in which the nuclease activity has been reduced or eliminated, and/or comprises nickase activity or in which the nickase has been reduced or eliminated, and/or comprises single stranded DNA cleavage activity (ss DNase activity) or in which the ss DNase activity has been reduced or eliminated, and/or comprises self-processing RNase activity or in which the self-processing RNase activity has been reduced or eliminated. A CRISPR-Cas effector protein may bind to a target nucleic acid.

In some embodiments, a sequence-specific nucleic acid binding domain may be a CRISPR-Cas effector protein. In some embodiments, a CRISPR-Cas effector protein may be from a Type I CRISPR-Cas system, a Type II CRISPR-Cas system, a Type III CRISPR-Cas system, a Type IV CRISPR-Cas system, a Type V CRISPR-Cas system, or a Type VI CRISPR-Cas system. In some embodiments, a CRISPR-Cas effector protein of the invention may be from a Type II CRISPR-Cas system or a Type V CRISPR-Cas system. In some embodiments, a CRISPR-Cas effector protein may be Type II CRISPR-Cas effector protein, for example, a Cas9 effector protein. In some embodiments, a CRISPR-Cas effector protein may be Type V CRISPR-Cas effector protein, for example, a Cas12 effector protein.

In some embodiments, a CRISPR-Cas effector protein may include, but is not limited to, a Cas9, C2c1, C2c3, Cas12a (also referred to as Cpf1), Cas12b, Cas12c, Cas12d, Cas12e, Cas13a, Cas13b, Cas13c, Cas13d, Cas1, Cas1B, Cas2, Cas3, Cas3', Cas3'', Cas4, Cas5, Cas6, Cas7, Cas8, Cas9 (also known as Csn1 and Csx12), Cas10, Csy1, Csy2, Csy3, Cse1, Cse2, Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4 (dinG), and/or Csf5 nuclease, optionally wherein the CRISPR-Cas effector protein may be a Cas9, Cas12a (Cpf1), Cas12b, Cas12c (C2c3), Cas12d (CasY), Cas12e (CasX), Cas12g, Cas12h, Cas12i, C2c4, C2c5, C2c8, C2c9, C2c10, Cas14a, Cas14b, and/or Cas14c effector protein.

In some embodiments, a CRISPR-Cas effector protein useful with the invention may comprise a mutation in its nuclease active site (e.g., RuvC, HNH, e.g., RuvC site of a Cas12a nuclease domain; e.g., RuvC site and/or HNH site of a Cas9 nuclease domain). A CRISPR-Cas effector protein having a mutation in its nuclease active site, and therefore, no longer comprising nuclease activity, is commonly referred to as “dead,” e.g., dCas. In some embodiments, a CRISPR-Cas effector protein domain or polypeptide having a mutation in its nuclease active site may have impaired activity or reduced activity as compared to the same CRISPR-Cas effector protein without the mutation, e.g., a nickase, e.g., Cas9 nickase, Cas12a nickase.

A CRISPR Cas9 effector protein or CRISPR Cas9 effector domain useful with this invention may be any known or later identified Cas9 nuclease. In some embodiments, a CRISPR Cas9 polypeptide can be a Cas9 polypeptide from, for example, *Streptococcus* spp. (e.g., *S. pyogenes*, *S. thermophilus*), *Lactobacillus* spp., *Bifidobacterium* spp., *Kandleria* spp., *Leuconostoc* spp., *Oenococcus* spp., *Pediococcus* spp., *Weissella* spp., and/or *Olsenella* spp. Example Cas9 sequences include, but are not limited to, the amino acid

sequences of SEQ ID NO: 59 and SEQ ID NO:60 or the nucleotide sequences of SEQ ID NOs: 61-71.

In some embodiments, the CRISPR-Cas effector protein may be a Cas9 polypeptide derived from *Streptococcus pyogenes* and recognizes the PAM sequence motif NGG, NAG, NGA (Mali et al, Science 2013; 339 (6121): 823-826). In some embodiments, the CRISPR-Cas effector protein may be a Cas9 polypeptide derived from *Streptococcus thermophilus* and recognizes the PAM sequence motif NGGNG and/or NNAGAAW (W=A or T) (See, e.g., Horvath et al, Science, 2010; 327 (5962): 167-170, and Deveau et al, J Bacteriol 2008; 190 (4): 1390-1400). In some embodiments, the CRISPR-Cas effector protein may be a Cas9 polypeptide derived from *Streptococcus mutans* and recognizes the PAM sequence motif NGG and/or NAAR (R=A or G) (See, e.g., Deveau et al, J BACTERIOL 2008; 190 (4): 1390-1400). In some embodiments, the CRISPR-Cas effector protein may be a Cas9 polypeptide derived from *Streptococcus aureus* and recognizes the PAM sequence motif NNGRR (R=A or G). In some embodiments, the CRISPR-Cas effector protein may be a Cas9 protein derived from *S. aureus*, which recognizes the PAM sequence motif N GRRT (R=A or G). In some embodiments, the CRISPR-Cas effector protein may be a Cas9 polypeptide derived from *S. aureus*, which recognizes the PAM sequence motif N GRRV (R=A or G). In some embodiments, the CRISPR-Cas effector protein may be a Cas9 polypeptide that is derived from *Neisseria meningitidis* and recognizes the PAM sequence motif N GATT or N GCTT (R=A or G, V=A, G or C) (See, e.g., Hou et al, PNAS 2013, 1-6). In the aforementioned embodiments, N can be any nucleotide residue, e.g., any of A, G, C or T. In some embodiments, the CRISPR-Cas effector protein may be a Cas13a protein derived from *Leptotrichia shahii*, which recognizes a protospacer flanking sequence (PFS) (or RNA PAM (rPAM)) sequence motif of a single 3' A, U, or C, which may be located within the target nucleic acid.

In some embodiments, the CRISPR-Cas effector protein may be derived from Cas12a, which is a Type V Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)-Cas nuclease see, e.g., SEQ ID NOs: 1-20). Cas12a differs in several respects from the more well-known Type II CRISPR Cas9 nuclease. For example, Cas9 recognizes a G-rich protospacer-adjacent motif (PAM) that is 3' to its guide RNA (gRNA, sgRNA, crRNA, crDNA, CRISPR array) binding site (protospacer, target nucleic acid, target DNA) (3'-NGG), while Cas12a recognizes a T-rich PAM that is located 5' to the target nucleic acid (5'-TTN, 5'-TTTN. In fact, the orientations in which Cas9 and Cas12a bind their guide RNAs are very nearly reversed in relation to their N and C termini. Furthermore, Cas12a enzymes use a single guide RNA (gRNA, CRISPR array, crRNA) rather than the dual guide RNA (sgRNA (e.g., crRNA and tracrRNA)) found in natural Cas9 systems, and Cas12a processes its own gRNAs. Additionally, Cas12a nuclease activity produces staggered DNA double stranded breaks instead of blunt ends produced by Cas9 nuclease activity, and Cas12a relies on a single RuvC domain to cleave both DNA strands, whereas Cas9 utilizes an HNH domain and a RuvC domain for cleavage.

A CRISPR Cas12a effector protein/domain useful with this invention may be any known or later identified Cas12a polypeptide (previously known as Cpf1) (see, e.g., U.S. Pat. No. 9,790,490, which is incorporated by reference for its disclosures of Cpf1 (Cas12a) sequences). The term “Cas12a”, “Cas12a polypeptide” or “Cas12a domain” refers to an RNA-guided nuclease comprising a Cas12a polypep-

tide, or a fragment thereof, which comprises the guide nucleic acid binding domain of Cas12a and/or an active, inactive, or partially active DNA cleavage domain of Cas12a. In some embodiments, a Cas12a useful with the invention may comprise a mutation in the nuclease active site (e.g., RuvC site of the Cas12a domain). A Cas12a domain or Cas12a polypeptide having a mutation in its nuclease active site, and therefore, no longer comprising nuclease activity, is commonly referred to as deadCas12a (e.g., dCas12a). In some embodiments, a Cas12a domain or Cas12a polypeptide having a mutation in its nuclease active site may have impaired activity, e.g., may have nickase activity.

Any deaminase domain/polypeptide useful for base editing may be used with this invention. In some embodiments, the deaminase domain may be a cytosine deaminase domain or an adenine deaminase domain. A cytosine deaminase (or cytidine deaminase) useful with this invention may be any known or later identified cytosine deaminase from any organism (see, e.g., U.S. Pat. No. 10,167,457 and Thuronyi et al. *Nat. Biotechnol.* 37:1070-1079 (2019), each of which is incorporated by reference herein for its disclosure of cytosine deaminases). Cytosine deaminases can catalyze the hydrolytic deamination of cytidine or deoxycytidine to uridine or deoxyuridine, respectively. Thus, in some embodiments, a deaminase or deaminase domain useful with this invention may be a cytidine deaminase domain, catalyzing the hydrolytic deamination of cytosine to uracil. In some embodiments, a cytosine deaminase may be a variant of a naturally-occurring cytosine deaminase, including but not limited to a primate (e.g., a human, monkey, chimpanzee, gorilla), a dog, a cow, a rat or a mouse. Thus, in some embodiments, a cytosine deaminase useful with the invention may be about 70% to about 100% identical to a wild type cytosine deaminase (e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical, and any range or value therein, to a naturally occurring cytosine deaminase).

In some embodiments, a cytosine deaminase useful with the invention may be an apolipoprotein B mRNA-editing complex (APOBEC) family deaminase. In some embodiments, the cytosine deaminase may be an APOBEC1 deaminase, an APOBEC2 deaminase, an APOBEC3A deaminase, an APOBEC3B deaminase, an APOBEC3C deaminase, an APOBEC3D deaminase, an APOBEC3F deaminase, an APOBEC3G deaminase, an APOBEC3H deaminase, an APOBEC4 deaminase, a human activation induced deaminase (hAID), an rAPOBEC1, FERNY, and/or a CDA1, optionally a pmCDA1, an atCDA1 (e.g., At2g19570), and evolved versions of the same (e.g., SEQ ID NO:27, SEQ ID NO:28 or SEQ ID NO: 29). In some embodiments, the cytosine deaminase may be an APOBEC1 deaminase having the amino acid sequence of SEQ ID NO:23. In some embodiments, the cytosine deaminase may be an APOBEC3A deaminase having the amino acid sequence of SEQ ID NO: 24. In some embodiments, the cytosine deaminase may be an CDA1 deaminase, optionally a CDA1 having the amino acid sequence of SEQ ID NO:25. In some embodiments, the cytosine deaminase may be a FERNY deaminase, optionally a FERNY having the amino acid sequence of SEQ ID NO:26. In some embodiments, a cytosine deaminase useful with the invention may be about 70% to about 100% identical (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%,

94%, 95%, 96%, 97%, 98%, 99%, 99.5% or 100% identical) to the amino acid sequence of a naturally occurring cytosine deaminase (e.g., an evolved deaminase). In some embodiments, a cytosine deaminase useful with the invention may be about 70% to about 99.5% identical (e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.5% identical) to the amino acid sequence of SEQ ID NO:23, SEQ ID NO:24, SEQ ID NO: 25 or SEQ ID NO:26 (e.g., at least 80%, at least 85%, at least 90%, at least 92%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5% identical to the amino acid sequence of SEQ ID NO:23, SEQ ID NO:24, SEQ ID NO:25, SEQ ID NO:26, SEQ ID NO:27, SEQ ID NO:28 or SEQ ID NO:29). In some embodiments, a polynucleotide encoding a cytosine deaminase may be codon optimized for expression in a plant and the codon optimized polypeptide may be about 70% to 99.5% identical to the reference polynucleotide.

In some embodiments, a nucleic acid construct of this invention may further encode a uracil glycosylase inhibitor (UGI) (e.g., uracil-DNA glycosylase inhibitor) polypeptide/domain. Thus, in some embodiments, a nucleic acid construct encoding a CRISPR-Cas effector protein and a cytosine deaminase domain (e.g., encoding a fusion protein comprising a CRISPR-Cas effector protein domain fused to a cytosine deaminase domain, and/or a CRISPR-Cas effector protein domain fused to a peptide tag or to an affinity polypeptide capable of binding a peptide tag and/or a deaminase protein domain fused to a peptide tag or to an affinity polypeptide capable of binding a peptide tag) may further encode a uracil-DNA glycosylase inhibitor (UGI), optionally wherein the UGI may be codon optimized for expression in a plant. In some embodiments, the invention provides fusion proteins comprising a CRISPR-Cas effector polypeptide, a deaminase domain, and a UGI and/or one or more polynucleotides encoding the same, optionally wherein the one or more polynucleotides may be codon optimized for expression in a plant. In some embodiments, the invention provides fusion proteins, wherein a CRISPR-Cas effector polypeptide, a deaminase domain, and a UGI may be fused to any combination of peptide tags and affinity polypeptides as described herein, thereby recruiting the deaminase domain and UGI to the CRISPR-Cas effector polypeptide and a target nucleic acid. In some embodiments, a guide nucleic acid may be linked to a recruiting RNA motif and one or more of the deaminase domain and/or UGI may be fused to an affinity polypeptide that is capable of interacting with the recruiting RNA motif, thereby recruiting the deaminase domain and UGI to a target nucleic acid.

A "uracil glycosylase inhibitor" useful with the invention may be any protein that is capable of inhibiting a uracil-DNA glycosylase base-excision repair enzyme. In some embodiments, a UGI domain comprises a wild type UGI or a fragment thereof. In some embodiments, a UGI domain useful with the invention may be about 70% to about 100% identical (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% or 100% identical and any range or value therein) to the amino acid sequence of a naturally occurring UGI domain. In some embodiments, a UGI domain may comprise the amino acid sequence of SEQ ID NO:41 or a polypeptide having about 70% to about 99.5% sequence identity to the amino acid sequence of SEQ ID NO:41 (e.g., at least 80%, at least 85%, at least 90%, at least 92%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%,

or at least 99.5% identical to the amino acid sequence of SEQ ID NO:41). For example, in some embodiments, a UGI domain may comprise a fragment of the amino acid sequence of SEQ ID NO:41 that is 100% identical to a portion of consecutive nucleotides (e.g., 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80 consecutive nucleotides; e.g., about 10, 15, 20, 25, 30, 35, 40, 45, to about 50, 55, 60, 65, 70, 75, 80 consecutive nucleotides) of the amino acid sequence of SEQ ID NO:41. In some embodiments, a UGI domain may be a variant of a known UGI (e.g., SEQ ID NO:41) having about 70% to about 99.5% sequence identity (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% sequence identity, and any range or value therein) to the known UGI. In some embodiments, a polynucleotide encoding a UGI may be codon optimized for expression in a plant (e.g., a plant) and the codon optimized polypeptide may be about 70% to about 99.5% identical to the reference polynucleotide.

An adenine deaminase (or adenosine deaminase) useful with this invention may be any known or later identified adenine deaminase from any organism (see, e.g., U.S. Pat. No. 10,113,163, which is incorporated by reference herein for its disclosure of adenine deaminases). An adenine deaminase can catalyze the hydrolytic deamination of adenine or adenosine. In some embodiments, the adenine deaminase may catalyze the hydrolytic deamination of adenosine or deoxyadenosine to inosine or deoxyinosine, respectively. In some embodiments, the adenosine deaminase may catalyze the hydrolytic deamination of adenine or adenosine in DNA. In some embodiments, an adenine deaminase encoded by a nucleic acid construct of the invention may generate an A→G conversion in the sense (e.g., “+”; template) strand of the target nucleic acid or a T→C conversion in the antisense (e.g., “-”, complementary) strand of the target nucleic acid.

In some embodiments, an adenosine deaminase may be a variant of a naturally occurring adenine deaminase. Thus, in some embodiments, an adenosine deaminase may be about 70% to 100% identical to a wild type adenine deaminase (e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8% or 99.9% identical, and any range or value therein, to a naturally occurring adenine deaminase). In some embodiments, the deaminase or deaminase does not occur in nature and may be referred to as an engineered, mutated or evolved adenosine deaminase. Thus, for example, an engineered, mutated or evolved adenine deaminase polypeptide or an adenine deaminase domain may be about 70% to 99.9% identical to a naturally occurring adenine deaminase polypeptide/domain (e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8% or 99.9% identical, and any range or value therein, to a naturally occurring adenine deaminase polypeptide or adenine deaminase domain). In some embodiments, the adenosine deaminase may be from a bacterium, (e.g., *Escherichia coli*, *Staphylococcus aureus*, *Haemophilus influenzae*, *Caulobacter crescentus*, and the like). In some embodiments, a polynucleotide encoding an adenine deaminase polypeptide/domain may be codon optimized for expression in a plant.

In some embodiments, an adenine deaminase domain may be a wild type tRNA-specific adenosine deaminase domain,

e.g., a tRNA-specific adenosine deaminase (TadA) and/or a mutated/evolved adenosine deaminase domain, e.g., mutated/evolved tRNA-specific adenosine deaminase domain (TadA*). In some embodiments, a TadA domain may be from *E. coli*. In some embodiments, the TadA may be modified, e.g., truncated, missing one or more N-terminal and/or C-terminal amino acids relative to a full-length TadA (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 6, 17, 18, 19, or 20 N-terminal and/or C terminal amino acid residues may be missing relative to a full length TadA. In some embodiments, a TadA polypeptide or TadA domain does not comprise an N-terminal methionine. In some embodiments, a wild type *E. coli* TadA comprises the amino acid sequence of SEQ ID NO:30. In some embodiments, a mutated/evolved *E. coli* TadA* comprises the amino acid sequence of SEQ ID NOs: 31-40 (e.g., SEQ ID NOs: 31, 32, 33, 34, 35, 36, 37, 38, 39 or 40). In some embodiments, a polynucleotide encoding a TadA/TadA* may be codon optimized for expression in a plant.

A cytosine deaminase catalyzes cytosine deamination and results in a thymidine (through a uracil intermediate), causing a C to T conversion, or a G to A conversion in the complementary strand in the genome. Thus, in some embodiments, the cytosine deaminase encoded by the polynucleotide of the invention generates a C→T conversion in the sense (e.g., “+”; template) strand of the target nucleic acid or a G→A conversion in antisense (e.g., “-”, complementary) strand of the target nucleic acid.

In some embodiments, the adenine deaminase encoded by the nucleic acid construct of the invention generates an A→G conversion in the sense (e.g., “+”; template) strand of the target nucleic acid or a T→C conversion in the antisense (e.g., “-”, complementary) strand of the target nucleic acid.

The nucleic acid constructs of the invention encoding a base editor comprising a sequence-specific DNA binding protein and a cytosine deaminase polypeptide, and nucleic acid constructs/expression cassettes/vectors encoding the same, may be used in combination with guide nucleic acids for modifying target nucleic acid including, but not limited to, generation of C→T or G→A mutations in a target nucleic acid including, but not limited to, a plasmid sequence; generation of C→T or G→A mutations in a coding sequence to alter an amino acid identity; generation of C→T or G→A mutations in a coding sequence to generate a stop codon; generation of C→T or G→A mutations in a coding sequence to disrupt a start codon; generation of point mutations in genomic DNA to disrupt function; and/or generation of point mutations in genomic DNA to disrupt splice junctions.

The nucleic acid constructs of the invention encoding a base editor comprising a sequence-specific DNA binding protein and an adenine deaminase polypeptide, and expression cassettes and/or vectors encoding the same may be used in combination with guide nucleic acids for modifying a target nucleic acid including, but not limited to, generation of A→G or T→C mutations in a target nucleic acid including, but not limited to, a plasmid sequence; generation of A→G or T→C mutations in a coding sequence to alter an amino acid identity; generation of A→G or T→C mutations in a coding sequence to generate a stop codon; generation of A→G or T→C mutations in a coding sequence to disrupt a start codon; generation of point mutations in genomic DNA to disrupt function; and/or generation of point mutations in genomic DNA to disrupt splice junctions.

The nucleic acid constructs of the invention comprising a CRISPR-Cas effector protein or a fusion protein thereof may be used in combination with a guide RNA (gRNA, CRISPR array, CRISPR RNA, crRNA), designed to function with the

encoded CRISPR-Cas effector protein or domain, to modify a target nucleic acid. A guide nucleic acid useful with this invention comprises at least one spacer sequence and at least one repeat sequence. The guide nucleic acid is capable of forming a complex with the CRISPR-Cas nuclease domain encoded and expressed by a nucleic acid construct of the invention and the spacer sequence is capable of hybridizing to a target nucleic acid, thereby guiding the complex (e.g., a CRISPR-Cas effector fusion protein (e.g., CRISPR-Cas effector domain fused to a deaminase domain and/or a CRISPR-Cas effector domain fused to a peptide tag or an affinity polypeptide to recruit a deaminase domain and optionally, a UGI) to the target nucleic acid, wherein the target nucleic acid may be modified (e.g., cleaved or edited) or modulated (e.g., modulating transcription) by the deaminase domain.

As an example, a nucleic acid construct encoding a Cas9 domain linked to a cytosine deaminase domain (e.g., fusion protein) may be used in combination with a Cas9 guide nucleic acid to modify a target nucleic acid, wherein the cytosine deaminase domain of the fusion protein deaminates a cytosine base in the target nucleic acid, thereby editing the target nucleic acid. In a further example, a nucleic acid construct encoding a Cas9 domain linked to an adenine deaminase domain (e.g., fusion protein) may be used in combination with a Cas9 guide nucleic acid to modify a target nucleic acid, wherein the adenine deaminase domain of the fusion protein deaminates an adenosine base in the target nucleic acid, thereby editing the target nucleic acid.

Likewise, a nucleic acid construct encoding a Cas12a domain (or other selected CRISPR-Cas nuclease, e.g., C2c1, C2c3, Cas12b, Cas12c, Cas12d, Cas12e, Cas13a, Cas13b, Cas13c, Cas13d, Cas1, Cas1B, Cas2, Cas3, Cas3', Cas3'', Cas4, Cas5, Cas6, Cas7, Cas8, Cas9 (also known as Csn1 and Csx12), Cas10, Csy1, Csy2, Csy3, Cse1, Cse2, Cse1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4 (dinG), and/or Csf5) linked to a cytosine deaminase domain or adenine deaminase domain (e.g., fusion protein) may be used in combination with a Cas12a guide nucleic acid (or the guide nucleic acid for the other selected CRISPR-Cas nuclease) to modify a target nucleic acid, wherein the cytosine deaminase domain or adenine deaminase domain of the fusion protein deaminates a cytosine base in the target nucleic acid, thereby editing the target nucleic acid.

A “guide nucleic acid,” “guide RNA,” “gRNA,” “CRISPR RNA/DNA” “crRNA” or “crDNA” as used herein means a nucleic acid that comprises at least one spacer sequence, which is complementary to (and hybridizes to) a target DNA (e.g., protospacer), and at least one repeat sequence (e.g., a repeat of a Type V Cas12a CRISPR-Cas system, or a fragment or portion thereof; a repeat of a Type II Cas9 CRISPR-Cas system, or fragment thereof; a repeat of a Type V C2c1 CRISPR Cas system, or a fragment thereof; a repeat of a CRISPR-Cas system of, for example, C2c3, Cas12a (also referred to as Cpf1), Cas12b, Cas12c, Cas12d, Cas12e, Cas13a, Cas13b, Cas13c, Cas13d, Cas1, Cas1B, Cas2, Cas3, Cas3', Cas3'', Cas4, Cas5, Cas6, Cas7, Cas8, Cas9 (also known as Csn1 and Csx12), Cas10, Csy1, Csy2, Csy3, Cse1, Cse2, Cse1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4 (dinG), and/or Csf5, or a fragment thereof), wherein the repeat sequence may be linked to the 5' end and/or the 3' end of the

spacer sequence. The design of a gRNA of this invention may be based on a Type I, Type II, Type III, Type IV, Type V, or Type VI CRISPR-Cas system.

In some embodiments, a Cas12a gRNA may comprise, from 5' to 3', a repeat sequence (full length or portion thereof (“handle”); e.g., pseudoknot-like structure) and a spacer sequence.

In some embodiments, a guide nucleic acid may comprise more than one repeat sequence-spacer sequence (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, or more repeat-spacer sequences) (e.g., repeat-spacer-repeat, e.g., repeat-spacer-repeat-spacer-repeat-spacer-repeat-spacer-repeat-spacer-repeat-spacer, and the like). The guide nucleic acids of this invention are synthetic, human-made and not found in nature. A gRNA can be quite long and may be used as an aptamer (like in the MS2 recruitment strategy) or other RNA structures hanging off the spacer.

A “repeat sequence” as used herein, refers to, for example, any repeat sequence of a wild-type CRISPR Cas locus (e.g., a Cas9 locus, a Cas12a locus, a C2c1 locus, etc.) or a repeat sequence of a synthetic crRNA that is functional with the CRISPR-Cas effector protein encoded by the nucleic acid constructs of the invention. A repeat sequence useful with this invention can be any known or later identified repeat sequence of a CRISPR-Cas locus (e.g., Type I, Type II, Type III, Type IV, Type V or Type VI) or it can be a synthetic repeat designed to function in a Type I, II, III, IV, V or VI CRISPR-Cas system. A repeat sequence may comprise a hairpin structure and/or a stem loop structure. In some embodiments, a repeat sequence may form a pseudoknot-like structure at its 5' end (i.e., “handle”). Thus, in some embodiments, a repeat sequence can be identical to or substantially identical to a repeat sequence from wild-type Type I CRISPR-Cas loci, Type II, CRISPR-Cas loci, Type III, CRISPR-Cas loci, Type IV CRISPR-Cas loci, Type V CRISPR-Cas loci and/or Type VI CRISPR-Cas loci. A repeat sequence from a wild-type CRISPR-Cas locus may be determined through established algorithms, such as using the CRISPRfinder offered through CRISPRdb (see, Grissa et al. *Nucleic Acids Res.* 35 (Web Server issue): W52-7). In some embodiments, a repeat sequence or portion thereof is linked at its 3' end to the 5' end of a spacer sequence, thereby forming a repeat-spacer sequence (e.g., guide nucleic acid, guide RNA/DNA, crRNA, crDNA).

In some embodiments, a repeat sequence comprises, consists essentially of, or consists of at least 10 nucleotides depending on the particular repeat and whether the guide nucleic acid comprising the repeat is processed or unprocessed (e.g., about 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50 to 100 or more nucleotides, or any range or value therein). In some embodiments, a repeat sequence comprises, consists essentially of, or consists of about 10 to about 20, about 10 to about 30, about 10 to about 45, about 10 to about 50, about 15 to about 30, about 15 to about 40, about 15 to about 45, about 15 to about 50, about 20 to about 30, about 20 to about 40, about 20 to about 50, about 30 to about 40, about 40 to about 80, about 50 to about 100 or more nucleotides.

A repeat sequence linked to the 5' end of a spacer sequence can comprise a portion of a repeat sequence (e.g., 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35 or more contiguous nucleotides of a wild type repeat sequence). In some embodiments, a portion of a repeat sequence linked to the 5' end of a spacer sequence can be about five to about ten consecutive nucleotides in length (e.g., about 5, 6, 7, 8, 9, 10 nucleotides) and have at least 90% sequence identity (e.g.,

at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more (e.g., 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8, 99.9, or 100%)) to the same region (e.g., 5' end) of a wild type CRISPR Cas repeat nucleotide sequence. In some embodiments, a portion of a repeat sequence may comprise a pseudoknot-like structure at its 5' end (e.g., "handle").

A "spacer sequence" as used herein is a nucleotide sequence that is complementary to a target nucleic acid (e.g., target DNA) (e.g., protospacer) (e.g., consecutive nucleotides of a sequence (a) encoding an amino acid sequence having at least 70% sequence identity to SEQ ID NO: 118; (b) having at least 70% sequence identity to the nucleotide sequence of SEQ ID NO: 122; (c) comprising a region having at least 70% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 123-125; and/or (d) comprising a region encoding an amino acid sequence having at least 70% sequence identity to any one of SEQ ID NOs: 119-121). In some embodiments, a spacer sequence may include, but is not limited to, the nucleotide sequences of any one of SEQ ID NOs: 126-130. The spacer sequence can be fully complementary or substantially complementary (e.g., at least about 70% complementary (e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more (e.g., 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8, 99.9, or 100%)) to a target nucleic acid. Thus, in some embodiments, the spacer sequence can have one, two, three, four, or five mismatches as compared to the target nucleic acid, which mismatches can be contiguous or noncontiguous. In some embodiments, the spacer sequence can have 70% complementarity to a target nucleic acid. In other embodiments, the spacer nucleotide sequence can have 80% complementarity to a target nucleic acid. In still other embodiments, the spacer nucleotide sequence can have 85%, 90%, 95%, 96%, 97%, 98%, 99% or 99.5% complementarity, and the like, to the target nucleic acid (protospacer). In some embodiments, the spacer sequence is 100% complementary to the target nucleic acid. A spacer sequence may have a length from about 15 nucleotides to about 30 nucleotides (e.g., 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides, or any range or value therein). Thus, in some embodiments, a spacer sequence may have complete complementarity or substantial complementarity over a region of a target nucleic acid (e.g., protospacer) that is at least about 15 nucleotides to about 30 nucleotides in length. In some embodiments, the spacer is about 20 nucleotides in length. In some embodiments, the spacer is about 21, 22, or 23 nucleotides in length.

In some embodiments, the 5' region of a spacer sequence of a guide nucleic acid may be identical to a target DNA, while the 3' region of the spacer may be substantially complementary to the target DNA (e.g., Type V CRISPR-Cas), or the 3' region of a spacer sequence of a guide nucleic acid may be identical to a target DNA, while the 5' region of the spacer may be substantially complementary to the target DNA (e.g., Type II CRISPR-Cas), and therefore, the overall complementarity of the spacer sequence to the target DNA may be less than 100%. Thus, for example, in a guide for a Type V CRISPR-Cas system, the first 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 nucleotides in the 5' region (i.e., seed region) of,

for example, a 20 nucleotide spacer sequence may be 100% complementary to the target DNA, while the remaining nucleotides in the 3' region of the spacer sequence are substantially complementary (e.g., at least about 70% complementary) to the target DNA. In some embodiments, the first 1 to 8 nucleotides (e.g., the first 1, 2, 3, 4, 5, 6, 7, 8, nucleotides, and any range therein) of the 5' end of the spacer sequence may be 100% complementary to the target DNA, while the remaining nucleotides in the 3' region of the spacer sequence are substantially complementary (e.g., at least about 50% complementary (e.g., 50%, 55%, 60%, 65%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more)) to the target DNA.

As a further example, in a guide for a Type II CRISPR-Cas system, the first 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 nucleotides in the 3' region (i.e., seed region) of, for example, a 20 nucleotide spacer sequence may be 100% complementary to the target DNA, while the remaining nucleotides in the 5' region of the spacer sequence are substantially complementary (e.g., at least about 70% complementary) to the target DNA. In some embodiments, the first 1 to 10 nucleotides (e.g., the first 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 nucleotides, and any range therein) of the 3' end of the spacer sequence may be 100% complementary to the target DNA, while the remaining nucleotides in the 5' region of the spacer sequence are substantially complementary (e.g., at least about 50% complementary (e.g., at least about 50%, 55%, 60%, 65%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more or any range or value therein)) to the target DNA.

In some embodiments, a seed region of a spacer may be about 8 to about 10 nucleotides in length, about 5 to about 6 nucleotides in length, or about 6 nucleotides in length.

As used herein, a "target nucleic acid", "target DNA," "target nucleotide sequence," "target region," or a "target region in the genome" refers to a region of a plant's genome that is fully complementary (100% complementary) or substantially complementary (e.g., at least 70% complementary (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more)) to a spacer sequence in a guide nucleic acid of this invention. A target region useful for a CRISPR-Cas system may be located immediately 3' (e.g., Type V CRISPR-Cas system) or immediately 5' (e.g., Type II CRISPR-Cas system) to a PAM sequence in the genome of the organism (e.g., a plant genome). A target region may be selected from any region of at least 15 consecutive nucleotides (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 nucleotides, and the like) located immediately adjacent to a PAM sequence.

A "protospacer sequence" refers to the target double stranded DNA and specifically to the portion of the target DNA (e.g., or target region in the genome) that is fully or substantially complementary (and hybridizes) to the spacer sequence of the CRISPR repeat-spacer sequences (e.g., guide nucleic acids, CRISPR arrays, crRNAs).

In the case of Type V CRISPR-Cas (e.g., Cas12a) systems and Type II CRISPR-Cas (Cas9) systems, the protospacer sequence is flanked by (e.g., immediately adjacent to) a protospacer adjacent motif (PAM). For Type IV CRISPR-Cas systems, the PAM is located at the 5' end on the non-target strand and at the 3' end of the target strand (see below, as an example).

5'-NNNNNNNNNNNNNNNNNN-3' RNA Spacer (SEQ ID NO: 42)
 |||||
 3'-AAANNNNNNNNNNNNNNNN-5' Target strand (SEQ ID NO: 43)
 |||||
 5'-TTNNNNNNNNNNNNNNNNNN-3' Non-target strand (SEQ ID NO: 44)

In the case of Type II CRISPR-Cas (e.g., Cas9) systems, the PAM is located immediately 3' of the target region. The PAM for Type I CRISPR-Cas systems is located 5' of the target strand. There is no known PAM for Type III CRISPR-Cas systems. Makarova et al. describes the nomenclature for all the classes, types and subtypes of CRISPR systems (*Nature Reviews Microbiology* 13:722-736 (2015)). Guide structures and PAMs are described in by R. Barrangou (*Genome Biol.* 16:247 (2015)).

Canonical Cas12a PAMs are T rich. In some embodiments, a canonical Cas12a PAM sequence may be 5'-TTN, 5'-TTTN, or 5'-TTTV. In some embodiments, canonical Cas9 (e.g., *S. pyogenes*) PAMs may be 5'-NGG-3'. In some embodiments, non-canonical PAMs may be used but may be less efficient.

Additional PAM sequences may be determined by those skilled in the art through established experimental and computational approaches. Thus, for example, experimental approaches include targeting a sequence flanked by all possible nucleotide sequences and identifying sequence members that do not undergo targeting, such as through the transformation of target plasmid DNA (Esvelt et al. 2013. *Nat. Methods* 10:1116-1121; Jiang et al. 2013. *Nat. Biotechnol.* 31:233-239). In some aspects, a computational approach can include performing BLAST searches of natural spacers to identify the original target DNA sequences in bacteriophages or plasmids and aligning these sequences to determine conserved sequences adjacent to the target sequence (Briner and Barrangou. 2014. *Appl. Environ. Microbiol.* 80:994-1001; Mojica et al. 2009. *Microbiology* 155:733-740).

In some embodiments, the present invention provides expression cassettes and/or vectors comprising the nucleic acid constructs of the invention (e.g., one or more components of an editing system of the invention). In some embodiments, expression cassettes and/or vectors comprising the nucleic acid constructs of the invention and/or one or more guide nucleic acids may be provided. In some embodiments, a nucleic acid construct of the invention encoding a base editor (e.g., a construct comprising a CRISPR-Cas effector protein and a deaminase domain (e.g., a fusion protein)) or the components for base editing (e.g., a CRISPR-Cas effector protein fused to a peptide tag or an affinity polypeptide, a deaminase domain fused to a peptide tag or an affinity polypeptide, and/or a UGI fused to a peptide tag or an affinity polypeptide), may be comprised on the same or on a separate expression cassette or vector from that comprising the one or more guide nucleic acids. When the nucleic acid construct encoding a base editor or the components for base editing is/are comprised on separate expression cassette(s) or vector(s) from that comprising the guide nucleic acid, a target nucleic acid may be contacted with (e.g., provided with) the expression cassette(s) or vector(s) encoding the base editor or components for base editing in any order from one another and the guide nucleic acid, e.g., prior to, concurrently with, or after the expression cassette comprising the guide nucleic acid is provided (e.g., contacted with the target nucleic acid).

Fusion proteins of the invention may comprise sequence-specific nucleic acid binding domains, CRISPR-Cas poly-

peptides, and/or deaminase domains fused to peptide tags or affinity polypeptides that interact with the peptide tags, as known in the art, for use in recruiting the deaminase to the target nucleic acid. Methods of recruiting may also comprise guide nucleic acids linked to RNA recruiting motifs and deaminases fused to affinity polypeptides capable of interacting with RNA recruiting motifs, thereby recruiting the deaminase to the target nucleic acid. Alternatively, chemical interactions may be used to recruit polypeptides (e.g., deaminases) to a target nucleic acid.

A peptide tag (e.g., epitope) useful with this invention may include, but is not limited to, a GCN4 peptide tag (e.g., Sun-Tag), a c-Myc affinity tag, an HA affinity tag, a His affinity tag, an S affinity tag, a methionine-His affinity tag, an RGD-His affinity tag, a FLAG octapeptide, a strep tag or strep tag II, a V5 tag, and/or a VSV-G epitope. Any epitope that may be linked to a polypeptide and for which there is a corresponding affinity polypeptide that may be linked to another polypeptide may be used with this invention as a peptide tag. In some embodiments, a peptide tag may comprise 1 or 2 or more copies of a peptide tag (e.g., repeat unit, multimerized epitope (e.g., tandem repeats)) (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 or more repeat units. In some embodiments, an affinity polypeptide that interacts with/binds to a peptide tag may be an antibody. In some embodiments, the antibody may be a scFv antibody. In some embodiments, an affinity polypeptide that binds to a peptide tag may be synthetic (e.g., evolved for affinity interaction) including, but not limited to, an affibody, an anticalin, a monobody and/or a DARPin (see, e.g., Sha et al., *Protein Sci.* 26 (5): 910-924 (2017)); Gilbreth (*Curr Opin Struc Biol* 22 (4): 413-420 (2013)), U.S. Pat. No. 9,982,053, each of which are incorporated by reference in their entireties for the teachings relevant to affibodies, anticalins, monobodies and/or DARPins. Example peptide tag sequences and their affinity polypeptides include, but are not limited to, the amino acid sequences of SEQ ID NOs: 45-47.

In some embodiments, a guide nucleic acid may be linked to an RNA recruiting motif, and a polypeptide to be recruited (e.g., a deaminase) may be fused to an affinity polypeptide that binds to the RNA recruiting motif, wherein the guide binds to the target nucleic acid and the RNA recruiting motif binds to the affinity polypeptide, thereby recruiting the polypeptide to the guide and contacting the target nucleic acid with the polypeptide (e.g., deaminase). In some embodiments, two or more polypeptides may be recruited to a guide nucleic acid, thereby contacting the target nucleic acid with two or more polypeptides (e.g., deaminases). Example RNA recruiting motifs and their affinity polypeptides include, but are not limited to, the sequences of SEQ ID NOs: 48-58.

In some embodiments, a polypeptide fused to an affinity polypeptide may be a reverse transcriptase and the guide nucleic acid may be an extended guide nucleic acid linked to an RNA recruiting motif. In some embodiments, an RNA recruiting motif may be located on the 3' end of the extended portion of an extended guide nucleic acid (e.g., 5'-3', repeat-spacer-extended portion (RT template-primer binding site)-

RNA recruiting motif). In some embodiments, an RNA recruiting motif may be embedded in the extended portion.

In some embodiments of the invention, an extended guide RNA and/or guide RNA may be linked to one or two or more RNA recruiting motifs (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more motifs, e.g., at least 10 to about 25 motifs, e.g., about 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 motifs), optionally wherein when two or more RNA recruiting motifs are present they may be the same RNA recruiting motif or different RNA recruiting motifs. In some embodiments, an RNA recruiting motif and corresponding affinity polypeptide may include, but is not limited, to a telomerase Ku binding motif (e.g., Ku binding hairpin) and the corresponding affinity polypeptide Ku (e.g., Ku heterodimer), a telomerase Sm7 binding motif and the corresponding affinity polypeptide Sm7, an MS2 phage operator stem-loop and the corresponding affinity polypeptide MS2 Coat Protein (MCP), a PP7 phage operator stem-loop and the corresponding affinity polypeptide PP7 Coat Protein (PCP), an SfMu phage Com stem-loop and the corresponding affinity polypeptide Com RNA binding protein, a PUF binding site (PBS) and the affinity polypeptide Pumilio/fem-3 mRNA binding factor (PUF), and/or a synthetic RNA-aptamer and the aptamer ligand as the corresponding affinity polypeptide. In some embodiments, the RNA recruiting motif and corresponding affinity polypeptide may be an MS2 phage operator stem-loop and the affinity polypeptide MS2 Coat Protein (MCP). In some embodiments, the RNA recruiting motif and corresponding affinity polypeptide may be a PUF binding site (PBS) and the affinity polypeptide Pumilio/fem-3 mRNA binding factor (PUF).

In some embodiments, the components for recruiting polypeptides and nucleic acids may those that function through chemical interactions that may include, but are not limited to, rapamycin-inducible dimerization of FRB-FKBP; Biotin-streptavidin; SNAP tag; Halo tag; CLIP tag; DmrA-DmrC heterodimer induced by a compound; bifunctional ligand (e.g., fusion of two protein-binding chemicals together, e.g., dihydrofolate reductase (DHFR)).

In some embodiments, the nucleic acid constructs, expression cassettes or vectors of the invention that are optimized for expression in a plant may be about 70% to 100% identical (e.g., about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% or 100%) to the nucleic acid constructs, expression cassettes or vectors comprising the same polynucleotide(s) but which have not been codon optimized for expression in a plant.

Further provided herein are cells comprising one or more polynucleotides, guide nucleic acids, nucleic acid constructs, expression cassettes or vectors of the invention.

The nucleic acid constructs of the invention (e.g., a construct comprising a sequence specific nucleic acid binding domain, a CRISPR-Cas effector domain, a deaminase domain, reverse transcriptase (RT), RT template and/or a guide nucleic acid, etc.) and expression cassettes/vectors comprising the same may be used as an editing system of this invention for modifying target nucleic acids and/or their expression.

Accordingly, plants or plant cultivars which are to be treated with preference in accordance with the invention include all plants which, through genetic modification, received genetic material, which imparts particular advantageous useful properties ("traits") to these plants. Examples of such properties are better plant growth, vigor, stress

tolerance, standability, lodging resistance, nutrient uptake, plant nutrition, and/or yield, in particular improved growth, increased tolerance to high or low temperatures, increased tolerance to drought or to levels of water or soil salinity, enhanced flowering performance, easier harvesting, accelerated ripening, higher yields, higher quality and/or a higher nutritional value of the harvested products, better storage life and/or processability of the harvested products.

Further examples of such properties are an increased resistance against animal and microbial pests, such as against insects, arachnids, nematodes, mites, slugs and snails owing, for example, to toxins formed in the plants. Among DNA sequences encoding proteins which confer properties of tolerance to such animal and microbial pests, in particular insects, mention will particularly be made of the genetic material from *Bacillus thuringiensis* encoding the Bt proteins widely described in the literature and well known to those skilled in the art. Mention will also be made of proteins extracted from bacteria such as *Photobacterium* (WO97/17432 and WO98/08932). In particular, mention will be made of the Bt Cry or VIP proteins which include the CryIA, CryIAb, CryIAc, CryIIA, CryIIIA, CryIIIB2, Cry9c, Cry2Ab, Cry3Bb and CryIF proteins or toxic fragments thereof and also hybrids or combinations thereof, especially the CryIF protein or hybrids derived from a CryIF protein (e.g. hybrid CryIA-CryIF proteins or toxic fragments thereof), the CryIA-type proteins or toxic fragments thereof, preferably the CryIAc protein or hybrids derived from the CryIAc protein (e.g. hybrid CryIAb-CryIAc proteins) or the CryIAb or Bt2 protein or toxic fragments thereof, the Cry2Ae, Cry2Af or Cry2Ag proteins or toxic fragments thereof, the CryIA, 105 protein or a toxic fragment thereof, the VIP3Aa19 protein, the VIP3Aa20 protein, the VIP3A proteins produced in the COT202 or COT203 cotton events, the VIP3Aa protein or a toxic fragment thereof as described in Estruch et al. (1996), Proc Natl Acad Sci US A. 28; 93 (11): 5389-94, the Cry proteins as described in WO2001/47952, the insecticidal proteins from *Xenorhabdus* (as described in WO98/50427), *Serratia* (particularly from *S. entomophila*) or *Photobacterium* species strains, such as Tc-proteins from *Photobacterium* as described in WO98/08932. Also any variants or mutants of any one of these proteins differing in some amino acids (1-10, preferably 1-5) from any of the above named sequences, particularly the sequence of their toxic fragment, or which are fused to a transit peptide, such as a plastid transit peptide, or another protein or peptide, is included herein.

Another and particularly emphasized example of such properties is conferred tolerance to one or more herbicides, for example imidazolinones, sulphonylureas, glyphosate or phosphinothricin. Among DNA sequences encoding proteins (i.e., polynucleotides of interest) which confer properties of tolerance to certain herbicides on the transformed plant cells and plants, mention will be particularly be made to the bar or PAT gene or the *Streptomyces coelicolor* gene described in WO2009/152359 which confers tolerance to glufosinate herbicides, a gene encoding a suitable EPSPS (5-Enolpyruvylshikimate-3-phosphat-Synthase) which confers tolerance to herbicides having EPSPS as a target, especially herbicides such as glyphosate and its salts, a gene encoding glyphosate-n-acetyltransferase, or a gene encoding glyphosate oxidoreductase. Further suitable herbicide tolerance traits include at least one ALS (acetolactate synthase) inhibitor (e.g. WO2007/024782), a mutated *Arabidopsis* ALS/AHAS gene (e.g. U.S. Pat. No. 6,855,533), genes encoding 2,4-D-monooxygenases conferring tolerance to 2,4-D (2,4-dichlorophenoxyacetic acid) and genes encoding

Dicamba monooxygenases conferring tolerance to dicamba (3,6-dichloro-2-methoxybenzoic acid).

Further examples of such properties are increased resistance against phytopathogenic fungi, bacteria and/or viruses owing, for example, to systemic acquired resistance (SAR), systemin, phytoalexins, elicitors and also resistance genes and correspondingly expressed proteins and toxins.

Particularly useful transgenic events in transgenic plants or plant cultivars which can be treated with preference in accordance with the invention include Event 531/PV-GHBK04 (cotton, insect control, described in WO2002/040677), Event 1143-14A (cotton, insect control, not deposited, described in WO2006/128569); Event 1143-51B (cotton, insect control, not deposited, described in WO2006/128570); Event 1445 (cotton, herbicide tolerance, not deposited, described in US-A 2002-120964 or WO2002/034946); Event 17053 (rice, herbicide tolerance, deposited as PTA-9843, described in WO2010/117737); Event 17314 (rice, herbicide tolerance, deposited as PTA-9844, described in WO2010/117735); Event 281-24-236 (cotton, insect control-herbicide tolerance, deposited as PTA-6233, described in WO2005/103266 or US-A 2005-216969); Event 3006-210-23 (cotton, insect control-herbicide tolerance, deposited as PTA-6233, described in US-A 2007-143876 or WO2005/103266); Event 3272 (corn, quality trait, deposited as PTA-9972, described in WO2006/098952 or US-A 2006-230473); Event 33391 (wheat, herbicide tolerance, deposited as PTA-2347, described in WO2002/027004), Event 40416 (corn, insect control-herbicide tolerance, deposited as ATCC PTA-11508, described in WO 11/075593); Event 43A47 (corn, insect control-herbicide tolerance, deposited as ATCC PTA-11509, described in WO2011/075595); Event 5307 (corn, insect control, deposited as ATCC PTA-9561, described in WO2010/077816); Event ASR-368 (bent grass, herbicide tolerance, deposited as ATCC PTA-4816, described in US-A 2006-162007 or WO2004/053062); Event B16 (corn, herbicide tolerance, not deposited, described in US-A 2003-126634); Event BPS-CV127-9 (soybean, herbicide tolerance, deposited as NCIMB No. 41603, described in WO2010/080829); Event BLR1 (oilseed rape, restoration of male sterility, deposited as NCIMB 41193, described in WO2005/074671), Event CE43-67B (cotton, insect control, deposited as DSM ACC2724, described in US-A 2009-217423 or WO2006/128573); Event CE44-69D (cotton, insect control, not deposited, described in US-A 2010-0024077); Event CE44-69D (cotton, insect control, not deposited, described in WO2006/128571); Event CE46-02A (cotton, insect control, not deposited, described in WO2006/128572); Event COT102 (cotton, insect control, not deposited, described in US-A 2006-130175 or WO2004/039986); Event COT202 (cotton, insect control, not deposited, described in US-A 2007-067868 or WO2005/054479); Event COT203 (cotton, insect control, not deposited, described in WO2005/054480); Event DAS21606-3/1606 (soybean, herbicide tolerance, deposited as PTA-11028, described in WO2012/033794), Event DAS40278 (corn, herbicide tolerance, deposited as ATCC PTA-10244, described in WO2011/022469); Event DAS-44406-6/pDAB8264.44.06.1 (soybean, herbicide tolerance, deposited as PTA-11336, described in WO2012/075426), Event DAS-14536-7/pDAB8291.45.36.2 (soybean, herbicide tolerance, deposited as PTA-11335, described in WO2012/075429), Event DAS-59122-7 (corn, insect control-herbicide tolerance, deposited as ATCC PTA 11384, described in US-A 2006-070139); Event DAS-59132 (corn, insect control-herbicide tolerance, not deposited, described in WO2009/100188); Event

DAS68416 (soybean, herbicide tolerance, deposited as ATCC PTA-10442, described in WO2011/066384 or WO2011/066360); Event DP-098140-6 (corn, herbicide tolerance, deposited as ATCC PTA-8296, described in US-A 2009-137395 or WO 08/112019); Event DP-305423-1 (soybean, quality trait, not deposited, described in US-A 2008-312082 or WO2008/054747); Event DP-32138-1 (corn, hybridization system, deposited as ATCC PTA-9158, described in US-A 2009-0210970 or WO2009/103049); Event DP-356043-5 (soybean, herbicide tolerance, deposited as ATCC PTA-8287, described in US-A 2010-0184079 or WO2008/002872); Event EE-I (brinjal, insect control, not deposited, described in WO 07/091277); Event Fil 17 (corn, herbicide tolerance, deposited as ATCC 209031, described in US-A 2006-059581 or WO 98/044140); Event FG72 (soybean, herbicide tolerance, deposited as PTA-11041, described in WO2011/063413), Event GA21 (corn, herbicide tolerance, deposited as ATCC 209033, described in US-A 2005-086719 or WO 98/044140); Event GG25 (corn, herbicide tolerance, deposited as ATCC 209032, described in US-A 2005-188434 or WO98/044140); Event GHB119 (cotton, insect control-herbicide tolerance, deposited as ATCC PTA-8398, described in WO2008/151780); Event GHB614 (cotton, herbicide tolerance, deposited as ATCC PTA-6878, described in US-A 2010-050282 or WO2007/017186); Event GJ11 (corn, herbicide tolerance, deposited as ATCC 209030, described in US-A 2005-188434 or WO98/044140); Event GM RZ13 (sugar beet, virus resistance, deposited as NCIMB-41601, described in WO2010/076212); Event H7-1 (sugar beet, herbicide tolerance, deposited as NCIMB 41158 or NCIMB 41159, described in US-A 2004-172669 or WO 2004/074492); Event JOPLIN1 (wheat, disease tolerance, not deposited, described in US-A 2008-064032); Event LL27 (soybean, herbicide tolerance, deposited as NCIMB41658, described in WO2006/108674 or US-A 2008-320616); Event LL55 (soybean, herbicide tolerance, deposited as NCIMB 41660, described in WO 2006/108675 or US-A 2008-196127); Event LLcotton25 (cotton, herbicide tolerance, deposited as ATCC PTA-3343, described in WO2003/013224 or US-A 2003-097687); Event LLRICE06 (rice, herbicide tolerance, deposited as ATCC 203353, described in U.S. Pat. No. 6,468,747 or WO2000/026345); Event LLRice62 (rice, herbicide tolerance, deposited as ATCC 203352, described in WO2000/026345), Event LLRICE601 (rice, herbicide tolerance, deposited as ATCC PTA-2600, described in US-A 2008-2289060 or WO2000/026356); Event LY038 (corn, quality trait, deposited as ATCC PTA-5623, described in US-A 2007-028322 or WO2005/061720); Event MIR162 (corn, insect control, deposited as PTA-8166, described in US-A 2009-300784 or WO2007/142840); Event MIR604 (corn, insect control, not deposited, described in US-A 2008-167456 or WO2005/103301); Event MON15985 (cotton, insect control, deposited as ATCC PTA-2516, described in US-A 2004-250317 or WO2002/100163); Event MON810 (corn, insect control, not deposited, described in US-A 2002-102582); Event MON863 (corn, insect control, deposited as ATCC PTA-2605, described in WO2004/011601 or US-A 2006-095986); Event MON87427 (corn, pollination control, deposited as ATCC PTA-7899, described in WO2011/062904); Event MON87460 (corn, stress tolerance, deposited as ATCC PTA-8910, described in WO2009/111263 or US-A 2011-0138504); Event MON87701 (soybean, insect control, deposited as ATCC PTA-8194, described in US-A 2009-130071 or WO2009/064652); Event MON87705 (soybean, quality trait-herbicide tolerance, deposited as ATCC PTA-9241, described in US-A

2010-0080887 or WO2010/037016); Event MON87708 (soybean, herbicide tolerance, deposited as ATCC PTA-9670, described in WO2011/034704); Event MON87712 (soybean, yield, deposited as PTA-10296, described in WO2012/051199), Event MON87754 (soybean, quality trait, deposited as ATCC PTA-9385, described in WO2010/024976); Event MON87769 (soybean, quality trait, deposited as ATCC PTA-8911, described in US-A 2011-0067141 or WO2009/102873); Event MON88017 (corn, insect control herbicide tolerance, deposited as ATCC PTA-5582, described in US-A 2008-028482 or WO2005/059103); Event MON88913 (cotton, herbicide tolerance, deposited as ATCC PTA-4854, described in WO2004/072235 or US-A 2006-059590); Event MON88302 (oilseed rape, herbicide tolerance, deposited as PTA-10955, described in WO2011/153186), Event MON88701 (cotton, herbicide tolerance, deposited as PTA-11754, described in WO2012/134808), Event MON89034 (corn, insect control, deposited as ATCC PTA-7455, described in WO 07/140256 or US-A 2008-260932); Event MON89788 (soybean, herbicide tolerance, deposited as ATCC PTA-6708, described in US-A 2006-282915 or WO2006/130436); Event MS1 1 (oilseed rape, pollination control-herbicide tolerance, deposited as ATCC PTA-850 or PTA-2485, described in WO2001/031042); Event MS8 (oilseed rape, pollination control-herbicide tolerance, deposited as ATCC PTA-730, described in WO2001/041558 or US-A 2003-188347); Event NK603 (corn, herbicide tolerance, deposited as ATCC PTA-2478, described in US-A 2007-292854); Event PE-7 (rice, insect control, not deposited, described in WO2008/114282); Event RF3 (oilseed rape, pollination control-herbicide tolerance, deposited as ATCC PTA-730, described in WO2001/041558 or US-A 2003-188347); Event RT73 (oilseed rape, herbicide tolerance, not deposited, described in WO2002/036831 or US-A 2008-070260); Event SYHT0H2/SYN-000H2-5 (soybean, herbicide tolerance, deposited as PTA-11226, described in WO2012/082548), Event T227-1 (sugar beet, herbicide tolerance, not deposited, described in WO2002/44407 or US-A 2009-265817); Event T25 (corn, herbicide tolerance, not deposited, described in US-A 2001-029014 or WO2001/051654); Event T304-40 (cotton, insect control-herbicide tolerance, deposited as ATCC PTA-8171, described in US-A 2010-077501 or WO2008/122406); Event T342-142 (cotton, insect control, not deposited, described in WO2006/128568); Event TC1507 (corn, insect control-herbicide tolerance, not deposited, described in US-A 2005-039226 or WO2004/099447); Event VIP1034 (corn, insect control-herbicide tolerance, deposited as ATCC PTA-3925, described in WO2003/052073), Event 32316 (corn, insect control-herbicide tolerance, deposited as PTA-11507, described in WO2011/084632), Event 4114 (corn, insect control-herbicide tolerance, deposited as PTA-11506, described in WO2011/084621), event EE-GM3/FG72 (soybean, herbicide tolerance, ATCC Accession N° PTA-11041) optionally stacked with event EE-GM1/LL27 or event EE-GM2/LL55 (WO2011/063413A2), event DAS-68416-4 (soybean, herbicide tolerance, ATCC Accession N° PTA-10442, WO2011/066360A1), event DAS-68416-4 (soybean, herbicide tolerance, ATCC Accession N° PTA-10442, WO2011/066384A1), event DP-040416-8 (corn, insect control, ATCC Accession N° PTA-11508, WO2011/075593A1), event DP-043A47-3 (corn, insect control, ATCC Accession N° PTA-11509, WO2011/075595A1), event DP-004114-3 (corn, insect control, ATCC Accession N° PTA-11506, WO2011/084621A1), event DP-032316-8 (corn, insect control, ATCC Accession N° PTA-11507, WO2011/084632A1), event MON-88302-9 (oilseed rape, herbicide tolerance,

ATCC Accession N° PTA-10955, WO2011/153186A1), event DAS-21606-3 (soybean, herbicide tolerance, ATCC Accession No. PTA-11028, WO2012/033794A2), event MON-87712-4 (soybean, quality trait, ATCC Accession N° PTA-10296, WO2012/051199A2), event DAS-44406-6 (soybean, stacked herbicide tolerance, ATCC Accession N° PTA-11336, WO2012/075426A1), event DAS-14536-7 (soybean, stacked herbicide tolerance, ATCC Accession N° PTA-11335, WO2012/075429A1), event SYN-000H2-5 (soybean, herbicide tolerance, ATCC Accession N° PTA-11226, WO2012/082548A2), event DP-061061-7 (oilseed rape, herbicide tolerance, no deposit N° available, WO2012071039A1), event DP-073496-4 (oilseed rape, herbicide tolerance, no deposit N° available, US2012131692), event 8264.44.06.1 (soybean, stacked herbicide tolerance, Accession N° PTA-11336, WO2012075426A2), event 8291.45.36.2 (soybean, stacked herbicide tolerance, Accession N° PTA-11335, WO2012075429A2), event SYHT0H2 (soybean, ATCC Accession N° PTA-11226, WO2012/082548A2), event MON88701 (cotton, ATCC Accession N° PTA-11754, WO2012/134808A1), event KK179-2 (alfalfa, ATCC Accession N° PTA-11833, WO2013/003558A1), event pDAB8264.42.32.1 (soybean, stacked herbicide tolerance, ATCC Accession N° PTA-11993, WO2013/010094A1), event MZDT09Y (corn, ATCC Accession N° PTA-13025, WO2013/012775A1).

The genes/events (e.g., polynucleotides of interest), which impart the desired traits in question, may also be present in combinations with one another in the transgenic plants. Examples of transgenic plants which may be mentioned are the important crop plants, such as cereals (wheat, rice, triticale, barley, rye, oats), maize, soya beans, potatoes, sugar beet, sugar cane, tomatoes, peas and other types of vegetable, cotton, tobacco, oilseed rape and also fruit plants (with the fruits apples, pears, citrus fruits and grapes), with particular emphasis being given to maize, soya beans, wheat, rice, potatoes, cotton, sugar cane, tobacco and oilseed rape. Traits which are particularly emphasized are the increased resistance of the plants to insects, arachnids, nematodes and slugs and snails, as well as the increased resistance of the plants to one or more herbicides.

Commercially available examples of such plants, plant parts or plant seeds that may be treated with preference in accordance with the invention include commercial products, such as plant seeds, sold or distributed under the GENUITY®, DROUGHTGARD®, SMARTSTAX®, RIB COMPLETE®, ROUNDUP READY®, VT DOUBLE PRO®, VT TRIPLE PRO®, BOLLGARD II®, ROUNDUP READY 2 YIELD®, YIELDGARD®, ROUNDUP READY® 2 XTEND™, INTACTA RR2 PRO®, VISTIVE GOLD®, and/or XTENDFLEX™ trade names.

The invention will now be described with reference to the following examples. It should be appreciated that these examples are not intended to limit the scope of the claims to the invention but are rather intended to be exemplary of certain embodiments. Any variations in the exemplified methods that occur to the skilled artisan are intended to fall within the scope of the invention.

EXAMPLES

Example 1

Gene editing tools are used to create alleles of the CRN gene that reduce signaling and increase kernel row number (KRN) without compromising ear organization (e.g., without substantially reducing ear length). The CRN gene

encodes an inactive pseudo-kinase with a short extracellular (EC) domain. See FIG. 1. The EC of CRN is required for plasma membrane transiting but is not required for dimerization with CLV2, suggesting that the EC domain may perform a charge masking function. To alter the charge masking of FEA2, CRN genes are modified to produce CRN polypeptide mutations that may provide a reduction in positive charge, and thus, less charge masking of FEA2 and greater complex retention in the ER.

Example 2. Design of the Editing Constructs for Fea2 Editing

The genomic sequence of the CRN gene was identified in a proprietary maize line. From this reference sequence, spacer sequences (SEQ ID NOs: 126-130) were designed for use in editing constructs. The editing constructs contained a CRISPR-Cas effector and a spacer sequence designed to target the EC domain of the CRN protein encoded by the CRN gene. Spacers were deployed with either a Cas-effector cutting enzyme or a Cas-effector base editing complex.

Example 3. Transformation and Selection of Edited E0 Plants

A vector encoding a selected spacer as well as the chosen CRISPR-Cas effector were introduced into dried excised maize embryos using *Agrobacterium*. Transformed tissue was maintained in vitro with antibiotic selection to regenerate positive transformants. Healthy non-chimeric plants (E0) were selected and plugged in growth trays. Tissue was collected from regenerating plants (E0 generation) for DNA extraction and subsequent molecular screening was employed to identify edits in the CRN gene. FIG. 2 shows some of the edits obtained and their position within the EC domain of the CRN gene. Plants identified to be (1) healthy, non-chimeric and fertile, with (2) low transgene copy and

(3) a deletion in the EC domain were advanced to the next generation. E0 plants that satisfied all the above criteria were selfed to produce the E1 generation. Selected E1s were selfed to generate the E2 generation. We identified several families with deletions in and around the EC domain.

Example 4. Phenotypic Assessment of Trait Activity

Seeds for E1 and E2 material were sown in flats and later transferred to pots after seedlings were established. All materials were cultivated under standard greenhouse conditions and grown to reproductive maturity. Following standard practices, emerging ears were covered with small paper bags prior to the emergence of silk and tassels were covered during anthesis for the capture of pollen on a plant-by-plant basis. All ears were removed from the plants after dry-down.

After ear harvest and dry-down, kernel row number was manually counted for all ears. Data represent the average of three row counts per ear taken from the mid-section of the ear where row lineages were most defined. In order to prevent double counting of rows, a marker (e.g., paper clip) was inserted in between the rows where the counts initiated and to designate where row counting should cease. Unedited germplasm, and lines transformed with a Gus plasmid were used as wild-type controls for phenotyping. Table 1 shows kernel row number (KRN) for E1 and E2 families derived from selfed E0 and selfed E1 ears, respectively. BCF1 ears are a cross of edited E1 plants with a WT inbred different than the original WT transformation line.

In Table 1, the edit is provided with reference to the CRN gene represented by SEQ ID NO: 122. Thus, the full-length sequence for each edited gene can be determined by locating within SEQ ID NO:122 the named base pair position under column labeled "allele" and delete the number of base pairs (nucleotides) provided (e.g., 964:9D is a 9 base pair deletion starting at position 964 of SEQ ID NO:122).

TABLE 1

CRN gene edits					
Generation	Spacer SEQ ID	Allele (Coordinates with respect to SEQ ID NO. 122)	Allele Description	KRN Average	Sample Size
WT Control	NA	WT	WT Transformation line	15.6	10
WT Control	NA	WT	WT Inbred used for hybrid cross	12.6	7
E1	126	964:9D, 960:17D compound het	3AA in-frame deletion, premature stop compound het	18	1
E1	126	964:3D, 964:9D compound het	1AA, 3AA in-frame deletion compound het	16	1
E1	126	959:27D, 964:9D compound het	9AA, 3AA in-frame deletion compound het	16	1
E1	126	959:27D, 964:3D compound het	9AA, 1AA in-frame deletion compound het	18	1
E1	127	983:11D het	premature stop het	17.8	9
E1	127	983:11D homo	premature stop homo	18.2	3
E1	127	987:6D homo	2AA in-frame deletion homo	18	2
E1	128	977:42D homo	14AA in-frame deletion homo	17.4	12
E1	128	964:6D homo	2AA in-frame deletion homo	16.4	5
E1	128	998:9D het	3AA in-frame deletion het	16	4
E1	128	964:6D-964:6D, 1000:6D compound het	2x 2AA in-frame deletion with >V compound het	17	4
E1	128	998:9D homo	3AA in-frame deletion homo	16	2
E1	128	964:6D,1000:6D homo	2x 2AA in-frame deletion with >V homo	17	1
E1	128	964:6D het	2AA in-frame deletion het	16	1
WT Control	NA	WT	WT Transformation line	16	1
WT Control	NA	WT	WT Inbred used for hybrid cross	14	1
E2	126	964:9D homo	3AA in-frame deletion homo	20	1
E2	126	964:3D homo	1AA in-frame deletion homo	16	1
E2	126	960:17D homo	premature stop homo	18	1
E2	126	959:27D homo	9AA in-frame deletion homo	18	1
E2	127	987:6D homo	2AA in-frame deletion homo	16	1

TABLE 1-continued

CRN gene edits					
Generation	Spacer SEQ ID	Allele (Coordinates with respect to SEQ ID NO. 122)	Allele Description	KRN Average	Sample Size
E2	127	983:11D homo	premature stop homo	26	1
E2	128	964:6D, 1000:6D homo	2x 2AA in-frame deletion with >V homo	22	1
E2	128	964:6D homo	2AA in-frame deletion homo	20	1
BCF1 Hybrid	127	983:11D hybrid het	premature stop F1 hybrid het	18	1
BCF1 Hybrid	128	977:42D hybrid het	14AA in-frame deletion F1 hybrid het	16	1

WT = wild-type

Homo = homozygous;

Het = heterozygous (one WT copy, one edited copy);

Compound het = one copy has one edit, the other copy has the other edit

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The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. The invention is

defined by the following claims, with equivalents of the claims to be included therein.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 191

<210> SEQ ID NO 1

<211> LENGTH: 1228

<212> TYPE: PRT

<213> ORGANISM: Lachnospiraceae sp.

<400> SEQUENCE: 1

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1          5          10          15

Leu Arg Phe Lys Ala Ile Pro Val Gly Lys Thr Gln Glu Asn Ile Asp
          20          25          30

Asn Lys Arg Leu Leu Val Glu Asp Glu Lys Arg Ala Glu Asp Tyr Lys
          35          40          45

Gly Val Lys Lys Leu Leu Asp Arg Tyr Tyr Leu Ser Phe Ile Asn Asp
          50          55          60

Val Leu His Ser Ile Lys Leu Lys Asn Leu Asn Asn Tyr Ile Ser Leu
          65          70          75          80

Phe Arg Lys Lys Thr Arg Thr Glu Lys Glu Asn Lys Glu Leu Glu Asn
          85          90          95

Leu Glu Ile Asn Leu Arg Lys Glu Ile Ala Lys Ala Phe Lys Gly Asn
          100          105          110

Glu Gly Tyr Lys Ser Leu Phe Lys Lys Asp Ile Ile Glu Thr Ile Leu
          115          120          125

Pro Glu Phe Leu Asp Asp Lys Asp Glu Ile Ala Leu Val Asn Ser Phe
          130          135          140

Asn Gly Phe Thr Thr Ala Phe Thr Gly Phe Phe Asp Asn Arg Glu Asn
          145          150          155          160

Met Phe Ser Glu Glu Ala Lys Ser Thr Ser Ile Ala Phe Arg Cys Ile
          165          170          175

Asn Glu Asn Leu Thr Arg Tyr Ile Ser Asn Met Asp Ile Phe Glu Lys
          180          185          190

Val Asp Ala Ile Phe Asp Lys His Glu Val Gln Glu Ile Lys Glu Lys
          195          200          205

Ile Leu Asn Ser Asp Tyr Asp Val Glu Asp Phe Phe Glu Gly Glu Phe
          210          215          220

Phe Asn Phe Val Leu Thr Gln Glu Gly Ile Asp Val Tyr Asn Ala Ile
          225          230          235          240

Ile Gly Gly Phe Val Thr Glu Ser Gly Glu Lys Ile Lys Gly Leu Asn

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-continued

245								250				255			
Glu	Tyr	Ile	Asn	Leu	Tyr	Asn	Gln	Lys	Thr	Lys	Gln	Lys	Leu	Pro	Lys
			260				265						270		
Phe	Lys	Pro	Leu	Tyr	Lys	Gln	Val	Leu	Ser	Asp	Arg	Glu	Ser	Leu	Ser
			275				280						285		
Phe	Tyr	Gly	Glu	Gly	Tyr	Thr	Ser	Asp	Glu	Glu	Val	Leu	Glu	Val	Phe
			290				295						300		
Arg	Asn	Thr	Leu	Asn	Lys	Asn	Ser	Glu	Ile	Phe	Ser	Ser	Ile	Lys	Lys
			305				310						315		
Leu	Glu	Lys	Leu	Phe	Lys	Asn	Phe	Asp	Glu	Tyr	Ser	Ser	Ala	Gly	Ile
			325				330						335		
Phe	Val	Lys	Asn	Gly	Pro	Ala	Ile	Ser	Thr	Ile	Ser	Lys	Asp	Ile	Phe
			340				345						350		
Gly	Glu	Trp	Asn	Val	Ile	Arg	Asp	Lys	Trp	Asn	Ala	Glu	Tyr	Asp	Asp
			355				360						365		
Ile	His	Leu	Lys	Lys	Lys	Ala	Val	Val	Thr	Glu	Lys	Tyr	Glu	Asp	Asp
			370				375						380		
Arg	Arg	Lys	Ser	Phe	Lys	Lys	Ile	Gly	Ser	Phe	Ser	Leu	Glu	Gln	Leu
			385				390						395		
Gln	Glu	Tyr	Ala	Asp	Ala	Asp	Leu	Ser	Val	Val	Glu	Lys	Leu	Lys	Glu
			405				410						415		
Ile	Ile	Ile	Gln	Lys	Val	Asp	Glu	Ile	Tyr	Lys	Val	Tyr	Gly	Ser	Ser
			420				425						430		
Glu	Lys	Leu	Phe	Asp	Ala	Asp	Phe	Val	Leu	Glu	Lys	Ser	Leu	Lys	Lys
			435				440						445		
Asn	Asp	Ala	Val	Val	Ala	Ile	Met	Lys	Asp	Leu	Leu	Asp	Ser	Val	Lys
			450				455						460		
Ser	Phe	Glu	Asn	Tyr	Ile	Lys	Ala	Phe	Phe	Gly	Glu	Gly	Lys	Glu	Thr
			465				470						475		
Asn	Arg	Asp	Glu	Ser	Phe	Tyr	Gly	Asp	Phe	Val	Leu	Ala	Tyr	Asp	Ile
			485				490						495		
Leu	Leu	Lys	Val	Asp	His	Ile	Tyr	Asp	Ala	Ile	Arg	Asn	Tyr	Val	Thr
			500				505						510		
Gln	Lys	Pro	Tyr	Ser	Lys	Asp	Lys	Phe	Lys	Leu	Tyr	Phe	Gln	Asn	Pro
			515				520						525		
Gln	Phe	Met	Gly	Gly	Trp	Asp	Lys	Asp	Lys	Glu	Thr	Asp	Tyr	Arg	Ala
			530				535						540		
Thr	Ile	Leu	Arg	Tyr	Gly	Ser	Lys	Tyr	Tyr	Leu	Ala	Ile	Met	Asp	Lys
			545				550						555		
Lys	Tyr	Ala	Lys	Cys	Leu	Gln	Lys	Ile	Asp	Lys	Asp	Asp	Val	Asn	Gly
			565				570						575		
Asn	Tyr	Glu	Lys	Ile	Asn	Tyr	Lys	Leu	Leu	Pro	Gly	Pro	Asn	Lys	Met
			580				585						590		
Leu	Pro	Lys	Val	Phe	Phe	Ser	Lys	Lys	Trp	Met	Ala	Tyr	Tyr	Asn	Pro
			595				600						605		
Ser	Glu	Asp	Ile	Gln	Lys	Ile	Tyr	Lys	Asn	Gly	Thr	Phe	Lys	Lys	Gly
			610				615						620		
Asp	Met	Phe	Asn	Leu	Asn	Asp	Cys	His	Lys	Leu	Ile	Asp	Phe	Phe	Lys
			625				630						635		
Asp	Ser	Ile	Ser	Arg	Tyr	Pro	Lys	Trp	Ser	Asn	Ala	Tyr	Asp	Phe	Asn
			645				650						655		
Phe	Ser	Glu	Thr	Glu	Lys	Tyr	Lys	Asp	Ile	Ala	Gly	Phe	Tyr	Arg	Glu
			660				665						670		

Val 675	Glu 675	Gln 675	Gly 675	Tyr 675	Lys 675	Val 680	Ser 680	Phe 680	Glu 680	Ser 685	Ala 685	Ser 685	Lys 685	Lys 685	
Glu 690	Val 690	Asp 690	Lys 690	Leu 690	Val 695	Glu 695	Glu 695	Gly 695	Lys 695	Leu 700	Tyr 700	Met 700	Phe 700	Gln 700	Ile 700
Tyr 705	Asn 705	Lys 705	Asp 705	Phe 705	Ser 710	Asp 710	Lys 710	Ser 710	His 710	Gly 715	Thr 715	Pro 715	Asn 715	Leu 720	His 720
Thr 725	Met 725	Tyr 725	Phe 725	Lys 725	Leu 725	Leu 730	Phe 730	Asp 730	Glu 730	Asn 730	Asn 730	His 730	Gly 735	Gln 735	Ile 735
Arg 740	Leu 740	Ser 740	Gly 740	Gly 740	Ala 740	Glu 745	Leu 745	Phe 745	Met 745	Arg 745	Arg 745	Ala 750	Ser 750	Leu 750	Lys 750
Lys 755	Glu 755	Glu 755	Leu 755	Val 755	Val 755	His 760	Pro 760	Ala 760	Asn 760	Ser 760	Pro 765	Ile 765	Ala 765	Asn 765	Lys 765
Asn 770	Pro 770	Asp 770	Asn 770	Pro 770	Lys 775	Lys 775	Thr 775	Thr 775	Thr 775	Leu 780	Ser 780	Tyr 780	Asp 780	Val 780	Tyr 780
Lys 785	Asp 785	Lys 785	Arg 785	Phe 785	Ser 790	Glu 790	Asp 790	Gln 790	Tyr 790	Glu 795	Leu 795	His 795	Ile 795	Pro 800	Ile 800
Ala 805	Ile 805	Asn 805	Lys 805	Cys 805	Pro 805	Lys 810	Asn 810	Ile 810	Phe 810	Lys 810	Ile 810	Asn 810	Thr 810	Glu 815	Val 815
Arg 820	Val 820	Leu 820	Leu 820	Lys 820	His 820	Asp 825	Asp 825	Asn 825	Pro 825	Tyr 825	Val 825	Ile 825	Gly 830	Ile 830	Asp 830
Arg 835	Gly 835	Glu 835	Arg 835	Asn 835	Leu 835	Leu 840	Tyr 840	Ile 840	Val 840	Val 840	Val 845	Asp 845	Gly 845	Lys 845	Gly 845
Asn 850	Ile 850	Val 850	Glu 850	Gln 850	Tyr 850	Ser 855	Leu 855	Asn 855	Glu 855	Ile 855	Ile 860	Asn 860	Asn 860	Phe 860	Asn 860
Gly 865	Ile 865	Arg 865	Ile 865	Lys 865	Thr 870	Asp 870	Tyr 870	His 870	Ser 870	Leu 875	Leu 875	Asp 875	Lys 875	Lys 875	Glu 880
Lys 885	Glu 885	Arg 885	Phe 885	Glu 885	Ala 885	Arg 885	Gln 885	Asn 885	Trp 890	Thr 890	Ser 890	Ile 890	Glu 895	Asn 895	Ile 895
Lys 900	Glu 900	Leu 900	Lys 900	Ala 900	Gly 900	Tyr 900	Ile 900	Ser 905	Gln 905	Val 905	Val 905	His 905	Lys 910	Ile 910	Cys 910
Glu 915	Leu 915	Val 915	Glu 915	Lys 915	Tyr 915	Asp 915	Ala 920	Val 920	Ile 920	Ala 920	Leu 920	Glu 925	Asp 925	Leu 925	Asn 925
Ser 930	Gly 930	Phe 930	Lys 930	Asn 930	Ser 930	Arg 935	Val 935	Lys 935	Val 935	Glu 935	Lys 940	Gln 940	Val 940	Tyr 940	Gln 940
Lys 945	Phe 945	Glu 945	Lys 945	Met 945	Leu 945	Ile 945	Asp 945	Lys 945	Leu 945	Asn 945	Tyr 945	Met 945	Val 945	Asp 945	Lys 945
Lys 950	Ser 950	Asn 950	Pro 950	Cys 950	Ala 950	Thr 950	Gly 950	Gly 950	Ala 950	Leu 950	Lys 950	Gly 950	Tyr 950	Gln 950	Ile 950
Thr 955	Asn 955	Lys 955	Phe 955	Glu 955	Ser 955	Phe 955	Lys 955	Ser 955	Met 955	Ser 955	Thr 955	Gln 955	Asn 955	Gly 955	Phe 955
Ile 960	Phe 960	Tyr 960	Ile 960	Pro 960	Ala 960	Trp 960	Leu 960	Thr 960	Ser 960	Lys 960	Ile 960	Asp 960	Pro 960	Ser 960	Thr 960
Gly 965	Phe 965	Val 965	Asn 965	Leu 965	Leu 965	Lys 965	Thr 965	Lys 965	Tyr 965	Thr 965	Ser 965	Ile 965	Ala 965	Asp 965	
Ser 970	Lys 970	Lys 970	Phe 970	Ile 970	Ser 970	Ser 970	Phe 970	Asp 970	Arg 970	Ile 970	Met 970	Tyr 970	Val 970	Pro 970	
Glu 975	Glu 975	Asp 975	Leu 975	Phe 975	Glu 975	Phe 975	Ala 975	Leu 975	Asp 975	Tyr 975	Lys 975	Asn 975	Phe 975	Ser 975	
Arg 980	Thr 980	Asp 980	Ala 980	Asp 980	Tyr 980	Ile 980	Lys 980	Lys							

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Phe	Asp	Trp	Glu	Glu	Val	Cys	Leu	Thr	Ser	Ala	Tyr	Lys	Glu	Leu
1085						1090					1095			
Phe	Asn	Lys	Tyr	Gly	Ile	Asn	Tyr	Gln	Gln	Gly	Asp	Ile	Arg	Ala
1100						1105					1110			
Leu	Leu	Cys	Glu	Gln	Ser	Asp	Lys	Ala	Phe	Tyr	Ser	Ser	Phe	Met
1115						1120					1125			
Ala	Leu	Met	Ser	Leu	Met	Leu	Gln	Met	Arg	Asn	Ser	Ile	Thr	Gly
1130						1135					1140			
Arg	Thr	Asp	Val	Asp	Phe	Leu	Ile	Ser	Pro	Val	Lys	Asn	Ser	Asp
1145						1150					1155			
Gly	Ile	Phe	Tyr	Asp	Ser	Arg	Asn	Tyr	Glu	Ala	Gln	Glu	Asn	Ala
1160						1165					1170			
Ile	Leu	Pro	Lys	Asn	Ala	Asp	Ala	Asn	Gly	Ala	Tyr	Asn	Ile	Ala
1175						1180					1185			
Arg	Lys	Val	Leu	Trp	Ala	Ile	Gly	Gln	Phe	Lys	Lys	Ala	Glu	Asp
1190						1195					1200			
Glu	Lys	Leu	Asp	Lys	Val	Lys	Ile	Ala	Ile	Ser	Asn	Lys	Glu	Trp
1205						1210					1215			
Leu	Glu	Tyr	Ala	Gln	Thr	Ser	Val	Lys	His					
1220						1225								

<210> SEQ ID NO 2
 <211> LENGTH: 1307
 <212> TYPE: PRT
 <213> ORGANISM: Acidaminococcus sp.

<400> SEQUENCE: 2

Met	Thr	Gln	Phe	Glu	Gly	Phe	Thr	Asn	Leu	Tyr	Gln	Val	Ser	Lys	Thr
1				5					10					15	
Leu	Arg	Phe	Glu	Leu	Ile	Pro	Gln	Gly	Lys	Thr	Leu	Lys	His	Ile	Gln
			20					25					30		
Glu	Gln	Gly	Phe	Ile	Glu	Glu	Asp	Lys	Ala	Arg	Asn	Asp	His	Tyr	Lys
		35					40					45			
Glu	Leu	Lys	Pro	Ile	Ile	Asp	Arg	Ile	Tyr	Lys	Thr	Tyr	Ala	Asp	Gln
	50					55					60				
Cys	Leu	Gln	Leu	Val	Gln	Leu	Asp	Trp	Glu	Asn	Leu	Ser	Ala	Ala	Ile
65				70					75						80
Asp	Ser	Tyr	Arg	Lys	Glu	Lys	Thr	Glu	Glu	Thr	Arg	Asn	Ala	Leu	Ile
			85					90						95	
Glu	Glu	Gln	Ala	Thr	Tyr	Arg	Asn	Ala	Ile	His	Asp	Tyr	Phe	Ile	Gly
		100					105					110			
Arg	Thr	Asp	Asn	Leu	Thr	Asp	Ala	Ile	Asn	Lys	Arg	His	Ala	Glu	Ile
	115					120						125			
Tyr	Lys	Gly	Leu	Phe	Lys	Ala	Glu	Leu	Phe	Asn	Gly	Lys	Val	Leu	Lys
	130				135						140				
Gln	Leu	Gly	Thr	Val	Thr	Thr	Glu	His	Glu	Asn	Ala	Leu	Leu	Arg	
145				150					155					160	
Ser	Phe	Asp	Lys	Phe	Thr	Thr	Tyr	Phe	Ser	Gly	Phe	Tyr	Glu	Asn	Arg
			165					170						175	
Lys	Asn	Val	Phe	Ser	Ala	Glu	Asp	Ile	Ser	Thr	Ala	Ile	Pro	His	Arg
			180					185					190		
Ile	Val	Gln	Asp	Asn	Phe	Pro	Lys	Phe	Lys	Glu	Asn	Cys	His	Ile	Phe
	195						200					205			
Thr	Arg	Leu	Ile	Thr	Ala	Val	Pro	Ser	Leu	Arg	Glu	His	Phe	Glu	Asn
	210					215					220				

Val 225	Lys	Lys	Ala	Ile	Gly 230	Ile	Phe	Val	Ser	Thr 235	Ser	Ile	Glu	Glu	Val 240
Phe	Ser	Phe	Pro	Phe 245	Tyr	Asn	Gln	Leu	Leu 250	Thr	Gln	Thr	Gln	Ile 255	Asp
Leu	Tyr	Asn	Gln	Leu 260	Leu	Gly	Gly	Ile	Ser 265	Arg	Glu	Ala	Gly	Thr	Glu
Lys	Ile	Lys	Gly	Leu 275	Asn	Glu	Val	Leu	Asn 280	Leu	Ala	Ile	Gln	Lys	Asn
Asp	Glu	Thr	Ala	His 290	Ile	Ile	Ala	Ser	Leu 295	Pro	His	Arg	Phe	Ile	Pro
Leu	Phe	Lys	Gln	Ile 305	Leu	Ser	Asp	Arg	Asn 310	Thr	Leu	Ser	Phe	Ile	Leu 320
Glu	Glu	Phe	Lys	Ser 325	Asp	Glu	Glu	Val	Ile 330	Gln	Ser	Phe	Cys	Lys	Tyr
Lys	Thr	Leu	Leu	Arg 340	Asn	Glu	Asn	Val	Leu 345	Glu	Thr	Ala	Glu	Ala	Leu
Phe	Asn	Glu	Leu	Asn 355	Ser	Ile	Asp	Leu	Thr 360	His	Ile	Phe	Ile	Ser	His
Lys	Lys	Leu	Glu	Thr 370	Ile	Ser	Ser	Ala	Leu 375	Cys	Asp	His	Trp	Asp	Thr
Leu	Arg	Asn	Ala	Leu 385	Tyr	Glu	Arg	Arg	Ile 390	Ser	Glu	Leu	Thr	Gly	Lys 400
Ile	Thr	Lys	Ser	Ala 405	Lys	Glu	Lys	Val	Gln 410	Arg	Ser	Leu	Lys	His	Glu
Asp	Ile	Asn	Leu	Gln 420	Glu	Ile	Ile	Ser	Ala 425	Ala	Gly	Lys	Glu	Leu	Ser
Glu	Ala	Phe	Lys	Gln 435	Lys	Thr	Ser	Glu	Ile 440	Leu	Ser	His	Ala	His	Ala
Ala	Leu	Asp	Gln	Pro 450	Leu	Pro	Thr	Thr	Leu 455	Lys	Lys	Gln	Glu	Glu	Lys
Glu	Ile	Leu	Lys	Ser 465	Gln	Leu	Asp	Ser	Leu 470	Leu	Gly	Leu	Tyr	His	Leu 480
Leu	Asp	Trp	Phe	Ala 485	Val	Asp	Glu	Ser	Asn 490	Glu	Val	Asp	Pro	Glu	Phe
Ser	Ala	Arg	Leu	Thr 500	Gly	Ile	Lys	Leu	Glu 505	Met	Glu	Pro	Ser	Leu	Ser
Phe	Tyr	Asn	Lys	Ala 515	Arg	Asn	Tyr	Ala	Thr 520	Lys	Lys	Pro	Tyr	Ser	Val
Glu	Lys	Phe	Lys	Leu 530	Asn	Phe	Gln	Met	Pro 535	Thr	Leu	Ala	Ser	Gly	Trp
Asp	Val	Asn	Lys	Glu 545	Lys	Asn	Asn	Gly	Ala 550	Ile	Leu	Phe	Val	Lys	Asn 560
Gly	Leu	Tyr	Tyr	Leu 565	Gly	Ile	Met	Pro	Lys 570	Gln	Lys	Gly	Arg	Tyr	Lys
Ala	Leu	Ser	Phe	Glu 580	Pro	Thr	Glu	Lys	Thr 585	Ser	Glu	Gly	Phe	Asp	Lys
Met	Tyr	Tyr	Asp	Tyr 595	Phe	Pro	Asp	Ala	Ala 600	Lys	Met	Ile	Pro	Lys	Cys
Ser	Thr	Gln	Leu	Lys 610	Ala	Val	Thr	Ala	His 615	Phe	Gln	Thr	His	Thr	Thr
Pro	Ile	Leu	Leu	Ser 625	Asn	Asn	Phe	Ile	Glu 630	Pro	Leu	Glu	Ile	Thr	Lys 640

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Glu Ile Tyr Asp Leu Asn Asn Pro Glu Lys Glu Pro Lys Lys Phe Gln
 645 650 655
 Thr Ala Tyr Ala Lys Lys Thr Gly Asp Gln Lys Gly Tyr Arg Glu Ala
 660 665 670
 Leu Cys Lys Trp Ile Asp Phe Thr Arg Asp Phe Leu Ser Lys Tyr Thr
 675 680 685
 Lys Thr Thr Ser Ile Asp Leu Ser Ser Leu Arg Pro Ser Ser Gln Tyr
 690 695 700
 Lys Asp Leu Gly Glu Tyr Tyr Ala Glu Leu Asn Pro Leu Leu Tyr His
 705 710 715 720
 Ile Ser Phe Gln Arg Ile Ala Glu Lys Glu Ile Met Asp Ala Val Glu
 725 730 735
 Thr Gly Lys Leu Tyr Leu Phe Gln Ile Tyr Asn Lys Asp Phe Ala Lys
 740 745 750
 Gly His His Gly Lys Pro Asn Leu His Thr Leu Tyr Trp Thr Gly Leu
 755 760 765
 Phe Ser Pro Glu Asn Leu Ala Lys Thr Ser Ile Lys Leu Asn Gly Gln
 770 775 780
 Ala Glu Leu Phe Tyr Arg Pro Lys Ser Arg Met Lys Arg Met Ala His
 785 790 795 800
 Arg Leu Gly Glu Lys Met Leu Asn Lys Lys Leu Lys Asp Gln Lys Thr
 805 810 815
 Pro Ile Pro Asp Thr Leu Tyr Gln Glu Leu Tyr Asp Tyr Val Asn His
 820 825 830
 Arg Leu Ser His Asp Leu Ser Asp Glu Ala Arg Ala Leu Leu Pro Asn
 835 840 845
 Val Ile Thr Lys Glu Val Ser His Glu Ile Ile Lys Asp Arg Arg Phe
 850 855 860
 Thr Ser Asp Lys Phe Phe Phe His Val Pro Ile Thr Leu Asn Tyr Gln
 865 870 875 880
 Ala Ala Asn Ser Pro Ser Lys Phe Asn Gln Arg Val Asn Ala Tyr Leu
 885 890 895
 Lys Glu His Pro Glu Thr Pro Ile Ile Gly Ile Asp Arg Gly Glu Arg
 900 905 910
 Asn Leu Ile Tyr Ile Thr Val Ile Asp Ser Thr Gly Lys Ile Leu Glu
 915 920 925
 Gln Arg Ser Leu Asn Thr Ile Gln Gln Phe Asp Tyr Gln Lys Lys Leu
 930 935 940
 Asp Asn Arg Glu Lys Glu Arg Val Ala Ala Arg Gln Ala Trp Ser Val
 945 950 955 960
 Val Gly Thr Ile Lys Asp Leu Lys Gln Gly Tyr Leu Ser Gln Val Ile
 965 970 975
 His Glu Ile Val Asp Leu Met Ile His Tyr Gln Ala Val Val Val Leu
 980 985 990
 Glu Asn Leu Asn Phe Gly Phe Lys Ser Lys Arg Thr Gly Ile Ala Glu
 995 1000 1005
 Lys Ala Val Tyr Gln Gln Phe Glu Lys Met Leu Ile Asp Lys Leu
 1010 1015 1020
 Asn Cys Leu Val Leu Lys Asp Tyr Pro Ala Glu Lys Val Gly Gly
 1025 1030 1035
 Val Leu Asn Pro Tyr Gln Leu Thr Asp Gln Phe Thr Ser Phe Ala
 1040 1045 1050
 Lys Met Gly Thr Gln Ser Gly Phe Leu Phe Tyr Val Pro Ala Pro

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1055	1060	1065
Tyr Thr Ser Lys Ile Asp Pro Leu Thr Gly Phe Val Asp Pro Phe		
1070	1075	1080
Val Trp Lys Thr Ile Lys Asn His Glu Ser Arg Lys His Phe Leu		
1085	1090	1095
Glu Gly Phe Asp Phe Leu His Tyr Asp Val Lys Thr Gly Asp Phe		
1100	1105	1110
Ile Leu His Phe Lys Met Asn Arg Asn Leu Ser Phe Gln Arg Gly		
1115	1120	1125
Leu Pro Gly Phe Met Pro Ala Trp Asp Ile Val Phe Glu Lys Asn		
1130	1135	1140
Glu Thr Gln Phe Asp Ala Lys Gly Thr Pro Phe Ile Ala Gly Lys		
1145	1150	1155
Arg Ile Val Pro Val Ile Glu Asn His Arg Phe Thr Gly Arg Tyr		
1160	1165	1170
Arg Asp Leu Tyr Pro Ala Asn Glu Leu Ile Ala Leu Leu Glu Glu		
1175	1180	1185
Lys Gly Ile Val Phe Arg Asp Gly Ser Asn Ile Leu Pro Lys Leu		
1190	1195	1200
Leu Glu Asn Asp Asp Ser His Ala Ile Asp Thr Met Val Ala Leu		
1205	1210	1215
Ile Arg Ser Val Leu Gln Met Arg Asn Ser Asn Ala Ala Thr Gly		
1220	1225	1230
Glu Asp Tyr Ile Asn Ser Pro Val Arg Asp Leu Asn Gly Val Cys		
1235	1240	1245
Phe Asp Ser Arg Phe Gln Asn Pro Glu Trp Pro Met Asp Ala Asp		
1250	1255	1260
Ala Asn Gly Ala Tyr His Ile Ala Leu Lys Gly Gln Leu Leu Leu		
1265	1270	1275
Asn His Leu Lys Glu Ser Lys Asp Leu Lys Leu Gln Asn Gly Ile		
1280	1285	1290
Ser Asn Gln Asp Trp Leu Ala Tyr Ile Gln Glu Leu Arg Asn		
1295	1300	1305

<210> SEQ ID NO 3

<211> LENGTH: 1241

<212> TYPE: PRT

<213> ORGANISM: Butyrivibrio proteoclasticus

<400> SEQUENCE: 3

Met Leu Leu Tyr Glu Asn Tyr Thr Lys Arg Asn Gln Ile Thr Lys Ser		
1	5	10
Leu Arg Leu Glu Leu Arg Pro Gln Gly Lys Thr Leu Arg Asn Ile Lys		
20	25	30
Glu Leu Asn Leu Leu Glu Gln Asp Lys Ala Ile Tyr Ala Leu Leu Glu		
35	40	45
Arg Leu Lys Pro Val Ile Asp Glu Gly Ile Lys Asp Ile Ala Arg Asp		
50	55	60
Thr Leu Lys Asn Cys Glu Leu Ser Phe Glu Lys Leu Tyr Glu His Phe		
65	70	75
Leu Ser Gly Asp Lys Lys Ala Tyr Ala Lys Glu Ser Glu Arg Leu Lys		
85	90	95
Lys Glu Ile Val Lys Thr Leu Ile Lys Asn Leu Pro Glu Gly Ile Gly		
100	105	110

Lys	Ile	Ser	Glu	Ile	Asn	Ser	Ala	Lys	Tyr	Leu	Asn	Gly	Val	Leu	Tyr
115						120						125			
Asp	Phe	Ile	Asp	Lys	Thr	His	Lys	Asp	Ser	Glu	Glu	Lys	Gln	Asn	Ile
130						135				140					
Leu	Ser	Asp	Ile	Leu	Glu	Thr	Lys	Gly	Tyr	Leu	Ala	Leu	Phe	Ser	Lys
145				150				155						160	
Phe	Leu	Thr	Ser	Arg	Ile	Thr	Thr	Leu	Glu	Gln	Ser	Met	Pro	Lys	Arg
				165				170						175	
Val	Ile	Glu	Asn	Phe	Glu	Ile	Tyr	Ala	Ala	Asn	Ile	Pro	Lys	Met	Gln
				180				185				190			
Asp	Ala	Leu	Glu	Arg	Gly	Ala	Val	Ser	Phe	Ala	Ile	Glu	Tyr	Glu	Ser
195						200						205			
Ile	Cys	Ser	Val	Asp	Tyr	Tyr	Asn	Gln	Ile	Leu	Ser	Gln	Glu	Asp	Ile
210						215				220					
Asp	Ser	Tyr	Asn	Arg	Leu	Ile	Ser	Gly	Ile	Met	Asp	Glu	Asp	Gly	Ala
225				230						235				240	
Lys	Glu	Lys	Gly	Ile	Asn	Gln	Thr	Ile	Ser	Glu	Lys	Asn	Ile	Lys	Ile
				245				250						255	
Lys	Ser	Glu	His	Leu	Glu	Glu	Lys	Pro	Phe	Arg	Ile	Leu	Lys	Gln	Leu
				260				265				270			
His	Lys	Gln	Ile	Leu	Glu	Glu	Arg	Glu	Lys	Ala	Phe	Thr	Ile	Asp	His
275						280						285			
Ile	Asp	Ser	Asp	Glu	Glu	Val	Val	Gln	Val	Thr	Lys	Glu	Ala	Phe	Glu
290						295				300					
Gln	Thr	Lys	Glu	Gln	Trp	Glu	Asn	Ile	Lys	Lys	Ile	Asn	Gly	Phe	Tyr
305				310						315				320	
Ala	Lys	Asp	Pro	Gly	Asp	Ile	Thr	Leu	Phe	Ile	Val	Val	Gly	Pro	Asn
				325				330						335	
Gln	Thr	His	Val	Leu	Ser	Gln	Leu	Ile	Tyr	Gly	Glu	His	Asp	Arg	Ile
				340				345				350			
Arg	Leu	Leu	Leu	Glu	Glu	Tyr	Glu	Lys	Asn	Thr	Leu	Glu	Val	Leu	Pro
355						360						365			
Arg	Arg	Thr	Lys	Ser	Glu	Asp	Ala	Arg	Tyr	Asp	Lys	Phe	Val	Asn	Ala
370						375				380					
Val	Pro	Lys	Lys	Val	Ala	Lys	Glu	Ser	His	Thr	Phe	Asp	Gly	Leu	Gln
385				390						395				400	
Lys	Met	Thr	Gly	Asp	Asp	Arg	Leu	Phe	Ile	Leu	Tyr	Arg	Asp	Glu	Leu
				405				410						415	
Ala	Arg	Asn	Tyr	Met	Arg	Ile	Lys	Glu	Ala	Tyr	Gly	Thr	Phe	Glu	Arg
				420				425				430			
Asp	Ile	Leu	Lys	Ser	Arg	Arg	Gly	Ile	Lys	Gly	Asn	Arg	Asp	Val	Gln
435						440						445			
Glu	Ser	Leu	Val	Ser	Phe	Tyr	Asp	Glu	Leu	Thr	Lys	Phe	Arg	Ser	Ala
450						455				460					
Leu	Arg	Ile	Ile	Asn	Ser	Gly	Asn	Asp	Glu	Lys	Ala	Asp	Pro	Ile	Phe
465				470						475				480	
Tyr	Asn	Thr	P												

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530	535	540
Ile Arg Arg Gly Asp Glu Tyr Tyr Tyr Phe Val Leu Thr Pro Asp Val 545 550 555 560		
Lys Pro Val Asp Leu Lys Thr Lys Asp Glu Thr Asp Ala Gln Ile Phe 565 570 575		
Val Gln Arg Lys Gly Ala Lys Ser Phe Leu Gly Leu Pro Lys Ala Leu 580 585 590		
Phe Lys Cys Ile Leu Glu Pro Tyr Phe Glu Ser Pro Glu His Lys Asn 595 600 605		
Asp Lys Asn Cys Val Ile Glu Glu Tyr Val Ser Lys Pro Leu Thr Ile 610 615 620		
Asp Arg Arg Ala Tyr Asp Ile Phe Lys Asn Gly Thr Phe Lys Lys Thr 625 630 635 640		
Asn Ile Gly Ile Asp Gly Leu Thr Glu Glu Lys Phe Lys Asp Asp Cys 645 650 655		
Arg Tyr Leu Ile Asp Val Tyr Lys Glu Phe Ile Ala Val Tyr Thr Arg 660 665 670		
Tyr Ser Cys Phe Asn Met Ser Gly Leu Lys Arg Ala Asp Glu Tyr Asn 675 680 685		
Asp Ile Gly Glu Phe Phe Ser Asp Val Asp Thr Arg Leu Cys Thr Met 690 695 700		
Glu Trp Ile Pro Val Ser Phe Glu Arg Ile Asn Asp Met Val Asp Lys 705 710 715 720		
Lys Glu Gly Leu Leu Phe Leu Val Arg Ser Met Phe Leu Tyr Asn Arg 725 730 735		
Pro Arg Lys Pro Tyr Glu Arg Thr Phe Ile Gln Leu Phe Ser Asp Ser 740 745 750		
Asn Met Glu His Thr Ser Met Leu Leu Asn Ser Arg Ala Met Ile Gln 755 760 765		
Tyr Arg Ala Ala Ser Leu Pro Arg Arg Val Thr His Lys Lys Gly Ser 770 775 780		
Ile Leu Val Ala Leu Arg Asp Ser Asn Gly Glu His Ile Pro Met His 785 790 795 800		
Ile Arg Glu Ala Ile Tyr Lys Met Lys Asn Asn Phe Asp Ile Ser Ser 805 810 815		
Glu Asp Phe Ile Met Ala Lys Ala Tyr Leu Ala Glu His Asp Val Ala 820 825 830		
Ile Lys Lys Ala Asn Glu Asp Ile Ile Arg Asn Arg Arg Tyr Thr Glu 835 840 845		
Asp Lys Phe Phe Leu Ser Leu Ser Tyr Thr Lys Asn Ala Asp Ile Ser 850 855 860		
Ala Arg Thr Leu Asp Tyr Ile Asn Asp Lys Val Glu Glu Asp Thr Gln 865 870 875 880		
Asp Ser Arg Met Ala Val Ile Val Thr Arg Asn Leu Lys Asp Leu Thr 885 890 895		
Tyr Val Ala Val Val Asp Glu Lys Asn Asn Val Leu Glu Glu Lys Ser 900 905 910		
Leu Asn Glu Ile Asp Gly Val Asn Tyr Arg Glu Leu Leu Lys Glu Arg 915 920 925		
Thr Lys Ile Lys Tyr His Asp Lys Thr Arg Leu Trp Gln Tyr Asp Val 930 935 940		
Ser Ser Lys Gly Leu Lys Glu Ala Tyr Val Glu Leu Ala Val Thr Gln 945 950 955 960		

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Ile Ser Lys Leu Ala Thr Lys Tyr Asn Ala Val Val Val Val Glu Ser
      965                      970                      975

Met Ser Ser Thr Phe Lys Asp Lys Phe Ser Phe Leu Asp Glu Gln Ile
      980                      985                      990

Phe Lys Ala Phe Glu Ala Arg Leu Cys Ala Arg Met Ser Asp Leu Ser
      995                      1000                      1005

Phe Asn Thr Ile Lys Glu Gly Glu Ala Gly Ser Ile Ser Asn Pro
      1010                      1015                      1020

Ile Gln Val Ser Asn Asn Asn Gly Asn Ser Tyr Gln Asp Gly Val
      1025                      1030                      1035

Ile Tyr Phe Leu Asn Asn Ala Tyr Thr Arg Thr Leu Cys Pro Asp
      1040                      1045                      1050

Thr Gly Phe Val Asp Val Phe Asp Lys Thr Arg Leu Ile Thr Met
      1055                      1060                      1065

Gln Ser Lys Arg Gln Phe Phe Ala Lys Met Lys Asp Ile Arg Ile
      1070                      1075                      1080

Asp Asp Gly Glu Met Leu Phe Thr Phe Asn Leu Glu Glu Tyr Pro
      1085                      1090                      1095

Thr Lys Arg Leu Leu Asp Arg Lys Glu Trp Thr Val Lys Ile Ala
      1100                      1105                      1110

Gly Asp Gly Ser Tyr Phe Asp Lys Asp Lys Gly Glu Tyr Val Tyr
      1115                      1120                      1125

Val Asn Asp Ile Val Arg Glu Gln Ile Ile Pro Ala Leu Leu Glu
      1130                      1135                      1140

Asp Lys Ala Val Phe Asp Gly Asn Met Ala Glu Lys Phe Leu Asp
      1145                      1150                      1155

Lys Thr Ala Ile Ser Gly Lys Ser Val Glu Leu Ile Tyr Lys Trp
      1160                      1165                      1170

Phe Ala Asn Ala Leu Tyr Gly Ile Ile Thr Lys Lys Asp Gly Glu
      1175                      1180                      1185

Lys Ile Tyr Arg Ser Pro Ile Thr Gly Thr Glu Ile Asp Val Ser
      1190                      1195                      1200

Lys Asn Thr Thr Tyr Asn Phe Gly Lys Lys Phe Met Phe Lys Gln
      1205                      1210                      1215

Glu Tyr Arg Gly Asp Gly Asp Phe Leu Asp Ala Phe Leu Asn Tyr
      1220                      1225                      1230

Met Gln Ala Gln Asp Ile Ala Val
      1235                      1240

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<210> SEQ ID NO 4
<211> LENGTH: 1238
<212> TYPE: PRT
<213> ORGANISM: Candidatus Methanoplasma termitum

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<400> SEQUENCE: 4

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Met Asn Asn Tyr Asp Glu Phe Thr Lys Leu Tyr Pro Ile Gln Lys Thr
1          5          10          15

Ile Arg Phe Glu Leu Lys Pro Gln Gly Arg Thr Met Glu His Leu Glu
20         25         30

Thr Phe Asn Phe Phe Glu Glu Asp Arg Asp Arg Ala Glu Lys Tyr Lys
35         40         45

Ile Leu Lys Glu Ala Ile Asp Glu Tyr His Lys Lys Phe Ile Asp Glu
50         55         60

His Leu Thr Asn Met Ser Leu Asp Trp Asn Ser Leu Lys Gln Ile Ser

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65	70	75	80
Glu Lys Tyr Tyr Lys Ser Arg Glu Glu Lys Asp Lys Lys Val Phe Leu	85	90	95
Ser Glu Gln Lys Arg Met Arg Gln Glu Ile Val Ser Glu Phe Lys Lys	100	105	110
Asp Asp Arg Phe Lys Asp Leu Phe Ser Lys Lys Leu Phe Ser Glu Leu	115	120	125
Leu Lys Glu Glu Ile Tyr Lys Lys Gly Asn His Gln Glu Ile Asp Ala	130	135	140
Leu Lys Ser Phe Asp Lys Phe Ser Gly Tyr Phe Ile Gly Leu His Glu	145	150	155
Asn Arg Lys Asn Met Tyr Ser Asp Gly Asp Glu Ile Thr Ala Ile Ser	165	170	175
Asn Arg Ile Val Asn Glu Asn Phe Pro Lys Phe Leu Asp Asn Leu Gln	180	185	190
Lys Tyr Gln Glu Ala Arg Lys Lys Tyr Pro Glu Trp Ile Ile Lys Ala	195	200	205
Glu Ser Ala Leu Val Ala His Asn Ile Lys Met Asp Ile Val Phe Ser	210	215	220
Leu Glu Tyr Phe Asn Lys Val Leu Asn Gln Glu Gly Ile Gln Arg Tyr	225	230	235
Asn Leu Ala Leu Gly Gly Tyr Val Thr Lys Ser Gly Glu Lys Met Met	245	250	255
Gly Leu Asn Asp Ala Leu Asn Leu Ala His Gln Ser Glu Lys Ser Ser	260	265	270
Lys Gly Arg Ile His Met Thr Pro Leu Phe Lys Gln Ile Leu Ser Glu	275	280	285
Lys Glu Ser Phe Ser Tyr Ile Pro Asp Val Phe Thr Glu Asp Ser Gln	290	295	300
Leu Leu Pro Ser Ile Gly Gly Phe Phe Ala Gln Ile Glu Asn Asp Lys	305	310	315
Asp Gly Asn Ile Phe Asp Arg Ala Leu Glu Leu Ile Ser Ser Tyr Ala	325	330	335
Glu Tyr Asp Thr Glu Arg Ile Tyr Ile Arg Gln Ala Asp Ile Asn Arg	340	345	350
Val Ser Asn Val Ile Phe Gly Glu Trp Gly Thr Leu Gly Gly Leu Met	355	360	365
Arg Glu Tyr Lys Ala Asp Ser Ile Asn Asp Ile Asn Leu Glu Arg Thr	370	375	380
Cys Lys Lys Val Asp Lys Trp Leu Asp Ser Lys Glu Phe Ala Leu Ser	385	390	395
Asp Val Leu Glu Ala Ile Asp Arg Thr Gly Asn Asn Asp Ala Phe Asn	405	410	415
Glu Tyr Ile Ser Lys Met Arg Thr Ala Arg Glu Lys Ile Asp Ala Ala	420	425	430
Arg Lys Glu Met Lys Phe Ile Ser Glu Lys Ile Ser Gly Asp Glu Glu	435	440	445
Ser Ile His Ile Ile Lys Thr Leu Leu Asp Ser Val Gln Gln Phe Leu	450	455	460
His Phe Phe Asn Leu Phe Lys Ala Arg Gln Asp Ile Pro Leu Asp Gly	465	470	475
Ala Phe Tyr Ala Glu Phe Asp Glu Val His Ser Lys Leu Phe Ala Ile	485	490	495

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Val	Pro	Leu	Tyr	Asn	Lys	Val	Arg	Asn	Tyr	Leu	Thr	Lys	Asn	Asn	Leu
				500				505					510		
Asn	Thr	Lys	Lys	Ile	Lys	Leu	Asn	Phe	Lys	Asn	Pro	Thr	Leu	Ala	Asn
		515						520				525			
Gly	Trp	Asp	Gln	Asn	Lys	Val	Tyr	Asp	Tyr	Ala	Ser	Leu	Ile	Phe	Leu
	530					535					540				
Arg	Asp	Gly	Asn	Tyr	Tyr	Leu	Gly	Ile	Ile	Asn	Pro	Lys	Arg	Lys	Lys
545					550					555					560
Asn	Ile	Lys	Phe	Glu	Gln	Gly	Ser	Gly	Asn	Gly	Pro	Phe	Tyr	Arg	Lys
				565					570					575	
Met	Val	Tyr	Lys	Gln	Ile	Pro	Gly	Pro	Asn	Lys	Asn	Leu	Arg	Pro	Val
			580					585					590		
Phe	Leu	Thr	Ser	Thr	Lys	Gly	Lys	Lys	Glu	Tyr	Lys	Pro	Ser	Lys	Glu
		595					600					605			
Ile	Ile	Glu	Gly	Tyr	Glu	Ala	Asp	Lys	His	Ile	Arg	Gly	Asp	Lys	Phe
	610					615					620				
Asp	Leu	Asp	Phe	Cys	His	Lys	Leu	Ile	Asp	Phe	Phe	Lys	Glu	Ser	Ile
625					630					635					640
Glu	Lys	His	Lys	Asp	Trp	Ser	Lys	Phe	Asn	Phe	Tyr	Phe	Ser	Pro	Thr
				645					650					655	
Glu	Ser	Tyr	Gly	Asp	Ile	Ser	Glu	Phe	Tyr	Leu	Asp	Val	Glu	Lys	Gln
			660					665				670			
Gly	Tyr	Arg	Met	His	Phe	Glu	Asn	Ile	Ser	Ala	Glu	Thr	Ile	Asp	Glu
		675					680					685			
Tyr	Val	Glu	Lys	Gly	Asp	Leu	Phe	Leu	Phe	Gln	Ile	Tyr	Asn	Lys	Asp
	690					695					700				
Phe	Val	Lys	Ala	Ala	Thr	Gly	Lys	Lys	Asp	Met	His	Thr	Ile	Tyr	Trp
705					710					715					720
Asn	Ala	Ala	Phe	Ser	Pro	Glu	Asn	Leu	Gln	Asp	Val	Val	Val	Lys	Leu
				725					730					735	
Asn	Gly	Glu	Ala	Glu	Leu	Phe	Tyr	Arg	Asp	Lys	Ser	Asp	Ile	Lys	Glu
			740					745					750		
Ile	Val	His	Arg	Glu	Gly	Glu	Ile	Leu	Val	Asn	Arg	Thr	Tyr	Asn	Gly
		755					760					765			
Arg	Thr	Pro	Val	Pro	Asp	Lys	Ile	His	Lys	Lys	Leu	Thr	Asp	Tyr	His
	770					775						780			
Asn	Gly	Arg	Thr	Lys	Asp	Leu	Gly	Glu	Ala	Lys	Glu	Tyr	Leu	Asp	Lys
785					790					795					800
Val	Arg	Tyr	Phe	Lys	Ala	His	Tyr	Asp	Ile	Thr	Lys	Asp	Arg	Arg	Tyr
				805					810					815	
Leu	Asn	Asp	Lys	Ile	Tyr	Phe	His	Val	Pro	Leu	Thr	Leu	Asn	Phe	Lys
			820					825					830		
Ala	Asn	Gly	Lys	Lys	Asn	Leu	Asn	Lys	Met	Val	Ile	Glu	Lys	Phe	Leu
			835				840					845			
Ser	Asp	Glu	Lys	Ala	His	Ile	Ile	Gly	Ile	Asp	Arg	Gly	Glu	Arg	Asn
	850					855					860				
Leu	Leu	Tyr	Tyr	Ser	Ile	Ile	Asp	Arg	Ser	Gly	Lys	Ile	Ile	Asp	Gln
865					870					875					880
Gln	Ser	Leu	Asn	Val	Ile	Asp	Gly	Phe	Asp	Tyr	Arg	Glu	Lys	Leu	Asn
				885				890						895	
Gln	Arg	Glu	Ile	Glu	Met	Lys	Asp	Ala	Arg	Gln	Ser	Trp	Asn	Ala	Ile
				900				905						910	

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Gly Lys Ile Lys Asp Leu Lys Glu Gly Tyr Leu Ser Lys Ala Val His
 915 920 925
 Glu Ile Thr Lys Met Ala Ile Gln Tyr Asn Ala Ile Val Val Met Glu
 930 935 940
 Glu Leu Asn Tyr Gly Phe Lys Arg Gly Arg Phe Lys Val Glu Lys Gln
 945 950 955 960
 Ile Tyr Gln Lys Phe Glu Asn Met Leu Ile Asp Lys Met Asn Tyr Leu
 965 970 975
 Val Phe Lys Asp Ala Pro Asp Glu Ser Pro Gly Gly Val Leu Asn Ala
 980 985 990
 Tyr Gln Leu Thr Asn Pro Leu Glu Ser Phe Ala Lys Leu Gly Lys Gln
 995 1000 1005
 Thr Gly Ile Leu Phe Tyr Val Pro Ala Ala Tyr Thr Ser Lys Ile
 1010 1015 1020
 Asp Pro Thr Thr Gly Phe Val Asn Leu Phe Asn Thr Ser Ser Lys
 1025 1030 1035
 Thr Asn Ala Gln Glu Arg Lys Glu Phe Leu Gln Lys Phe Glu Ser
 1040 1045 1050
 Ile Ser Tyr Ser Ala Lys Asp Gly Gly Ile Phe Ala Phe Ala Phe
 1055 1060 1065
 Asp Tyr Arg Lys Phe Gly Thr Ser Lys Thr Asp His Lys Asn Val
 1070 1075 1080
 Trp Thr Ala Tyr Thr Asn Gly Glu Arg Met Arg Tyr Ile Lys Glu
 1085 1090 1095
 Lys Lys Arg Asn Glu Leu Phe Asp Pro Ser Lys Glu Ile Lys Glu
 1100 1105 1110
 Ala Leu Thr Ser Ser Gly Ile Lys Tyr Asp Gly Gly Gln Asn Ile
 1115 1120 1125
 Leu Pro Asp Ile Leu Arg Ser Asn Asn Asn Gly Leu Ile Tyr Thr
 1130 1135 1140
 Met Tyr Ser Ser Phe Ile Ala Ala Ile Gln Met Arg Val Tyr Asp
 1145 1150 1155
 Gly Lys Glu Asp Tyr Ile Ile Ser Pro Ile Lys Asn Ser Lys Gly
 1160 1165 1170
 Glu Phe Phe Arg Thr Asp Pro Lys Arg Arg Glu Leu Pro Ile Asp
 1175 1180 1185
 Ala Asp Ala Asn Gly Ala Tyr Asn Ile Ala Leu Arg Gly Glu Leu
 1190 1195 1200
 Thr Met Arg Ala Ile Ala Glu Lys Phe Asp Pro Asp Ser Glu Lys
 1205 1210 1215
 Met Ala Lys Leu Glu Leu Lys His Lys Asp Trp Phe Glu Phe Met
 1220 1225 1230
 Gln Thr Arg Gly Asp
 1235

<210> SEQ ID NO 5

<211> LENGTH: 1281

<212> TYPE: PRT

<213> ORGANISM: Eubacterium eligens

<400> SEQUENCE: 5

Met Asn Gly Asn Arg Ser Ile Val Tyr Arg Glu Phe Val Gly Val Ile
 1 5 10 15

Pro Val Ala Lys Thr Leu Arg Asn Glu Leu Arg Pro Val Gly His Thr
 20 25 30

Gln	Glu	His	Ile	Ile	Gln	Asn	Gly	Leu	Ile	Gln	Glu	Asp	Glu	Leu	Arg
35															
Gln	Glu	Lys	Ser	Thr	Glu	Leu	Lys	Asn	Ile	Met	Asp	Asp	Tyr	Tyr	Arg
50															
Glu	Tyr	Ile	Asp	Lys	Ser	Leu	Ser	Gly	Val	Thr	Asp	Leu	Asp	Phe	Thr
65															
Leu	Leu	Phe	Glu	Leu	Met	Asn	Leu	Val	Gln	Ser	Ser	Pro	Ser	Lys	Asp
85															
Asn	Lys	Lys	Ala	Leu	Glu	Lys	Glu	Gln	Ser	Lys	Met	Arg	Glu	Gln	Ile
100															
Cys	Thr	His	Leu	Gln	Ser	Asp	Ser	Asn	Tyr	Lys	Asn	Ile	Phe	Asn	Ala
115															
Lys	Leu	Leu	Lys	Glu	Ile	Leu	Pro	Asp	Phe	Ile	Lys	Asn	Tyr	Asn	Gln
130															
Tyr	Asp	Val	Lys	Asp	Lys	Ala	Gly	Lys	Leu	Glu	Thr	Leu	Ala	Leu	Phe
145															
Asn	Gly	Phe	Ser	Thr	Tyr	Phe	Thr	Asp	Phe	Phe	Glu	Lys	Arg	Lys	Asn
165															
Val	Phe	Thr	Lys	Glu	Ala	Val	Ser	Thr	Ser	Ile	Ala	Tyr	Arg	Ile	Val
180															
His	Glu	Asn	Ser	Leu	Ile	Phe	Leu	Ala	Asn	Met	Thr	Ser	Tyr	Lys	Lys
195															
Ile	Ser	Glu	Lys	Ala	Leu	Asp	Glu	Ile	Glu	Val	Ile	Glu	Lys	Asn	Asn
210															
Gln	Asp	Lys	Met	Gly	Asp	Trp	Glu	Leu	Asn	Gln	Ile	Phe	Asn	Pro	Asp
225															
Phe	Tyr	Asn	Met	Val	Leu	Ile	Gln	Ser	Gly	Ile	Asp	Phe	Tyr	Asn	Glu
245															
Ile	Cys	Gly	Val	Val	Asn	Ala	His	Met	Asn	Leu	Tyr	Cys	Gln	Gln	Thr
260															
Lys	Asn	Asn	Tyr	Asn	Leu	Phe	Lys	Met	Arg	Lys	Leu	His	Lys	Gln	Ile
275															
Leu	Ala	Tyr	Thr	Ser	Thr	Ser	Phe	Glu	Val	Pro	Lys	Met	Phe	Glu	Asp
290															
Asp	Met	Ser	Val	Tyr	Asn	Ala	Val	Asn	Ala	Phe	Ile	Asp	Glu	Thr	Glu
305															
Lys	Gly	Asn	Ile	Ile	Gly	Lys	Leu	Lys	Asp	Ile	Val	Asn	Lys	Tyr	Asp
325															
Glu	Leu	Asp	Glu	Lys	Arg	Ile	Tyr	Ile	Ser	Lys	Asp	Phe	Tyr	Glu	Thr
340															
Leu	Ser	Cys	Phe	Met	Ser	Gly	Asn	Trp	Asn	Leu	Ile	Thr	Gly	Cys	Val
355															
Glu	Asn	Phe	Tyr	Asp	Glu	Asn	Ile	His	Ala	Lys	Gly	Lys	Ser	Lys	Glu
370															
Glu	Lys	Val	Lys	Lys	Ala	Val	Lys	Glu	Asp	Lys	Tyr	Lys	Ser	Ile	Asn
385															
Asp	Val	Asn	Asp	Leu	Val	Glu	Lys	Tyr	Ile	Asp	Glu	Lys	Glu	Arg	Asn
405															
Glu	Phe	Lys	Asn	Ser	Asn	Ala	Lys	Gln	Tyr	Ile	Arg	Glu	Ile	Ser	Asn
420															
Ile	Ile	Thr	Asp	Thr	Glu	Thr	Ala	His	Leu	Glu	Tyr	Asp	Asp	His	Ile
435															
440															
445															

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Ser	Leu	Ile	Glu	Ser	Glu	Glu	Lys	Ala	Asp	Glu	Met	Lys	Lys	Arg	Leu
450						455				460					
Asp	Met	Tyr	Met	Asn	Met	Tyr	His	Trp	Ala	Lys	Ala	Phe	Ile	Val	Asp
465					470					475					480
Glu	Val	Leu	Asp	Arg	Asp	Glu	Met	Phe	Tyr	Ser	Asp	Ile	Asp	Asp	Ile
			485						490					495	
Tyr	Asn	Ile	Leu	Glu	Asn	Ile	Val	Pro	Leu	Tyr	Asn	Arg	Val	Arg	Asn
			500					505					510		
Tyr	Val	Thr	Gln	Lys	Pro	Tyr	Asn	Ser	Lys	Lys	Ile	Lys	Leu	Asn	Phe
		515					520					525			
Gln	Ser	Pro	Thr	Leu	Ala	Asn	Gly	Trp	Ser	Gln	Ser	Lys	Glu	Phe	Asp
	530					535					540				
Asn	Asn	Ala	Ile	Ile	Leu	Ile	Arg	Asp	Asn	Lys	Tyr	Tyr	Leu	Ala	Ile
545					550					555					560
Phe	Asn	Ala	Lys	Asn	Lys	Pro	Asp	Lys	Lys	Ile	Ile	Gln	Gly	Asn	Ser
			565						570					575	
Asp	Lys	Lys	Asn	Asp	Asn	Asp	Tyr	Lys	Lys	Met	Val	Tyr	Asn	Leu	Leu
			580					585					590		
Pro	Gly	Ala	Asn	Lys	Met	Leu	Pro	Lys	Val	Phe	Leu	Ser	Lys	Lys	Gly
		595					600					605			
Ile	Glu	Thr	Phe	Lys	Pro	Ser	Asp	Tyr	Ile	Ile	Ser	Gly	Tyr	Asn	Ala
	610					615					620				
His	Lys	His	Ile	Lys	Thr	Ser	Glu	Asn	Phe	Asp	Ile	Ser	Phe	Cys	Arg
625					630					635					640
Asp	Leu	Ile	Asp	Tyr	Phe	Lys	Asn	Ser	Ile	Glu	Lys	His	Ala	Glu	Trp
			645						650					655	
Arg	Lys	Tyr	Glu	Phe	Lys	Phe	Ser	Ala	Thr	Asp	Ser	Tyr	Ser	Asp	Ile
			660					665					670		
Ser	Glu	Phe	Tyr	Arg	Glu	Val	Glu	Met	Gln	Gly	Tyr	Arg	Ile	Asp	Trp
		675					680					685			
Thr	Tyr	Ile	Ser	Glu	Ala	Asp	Ile	Asn	Lys	Leu	Asp	Glu	Glu	Gly	Lys
	690					695					700				
Ile	Tyr	Leu	Phe	Gln	Ile	Tyr	Asn	Lys	Asp	Phe	Ala	Glu	Asn	Ser	Thr
705					710					715					720
Gly	Lys	Glu	Asn	Leu	His	Thr	Met	Tyr	Phe	Lys	Asn	Ile	Phe	Ser	Glu
			725						730					735	
Glu	Asn	Leu	Asp	Lys	Ile	Ile	Lys	Leu	Asn	Gly	Gln	Ala	Glu	Leu	Phe
			740					745					750		
Tyr	Arg	Arg	Ala	Ser	Val	Lys	Asn	Pro	Val	Lys	His	Lys	Lys	Asp	Ser
		755					760					765			
Val	Leu	Val	Asn	Lys	Thr	Tyr	Lys	Asn	Gln	Leu	Asp	Asn	Gly	Asp	Val
	770					775					780				
Val	Arg	Ile	Pro	Ile	Pro	Asp	Asp	Ile	Tyr	Asn	Glu	Ile	Tyr	Lys	Met
785					790					795					800
Tyr	Asn	Gly	Tyr	Ile	Lys	Glu	Ser	Asp	Leu	Ser	Glu	Ala	Ala	Lys	Glu
			805						810					815	
Tyr	Leu	Asp	Lys	Val	Glu	Val	Arg	Thr	Ala	Gln	Lys	Asp	Ile	Val	Lys
			820					825					830		
Asp	Tyr	Arg	Tyr	Thr	Val	Asp	Lys	Tyr	Phe	Ile	His	Thr	Pro	Ile	Thr
		835				840						845			
Ile	Asn	Tyr	Lys	Val	Thr	Ala	Arg	Asn	Asn	Val	Asn	Asp	Met	Val	Val
850					855					860					
Lys	Tyr	Ile	Ala	Gln	Asn	Asp	Asp	Ile	His	Val	Ile	Gly	Ile	Asp	Arg

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865	870	875	880
Gly Glu Arg Asn Leu Ile Tyr Ile Ser Val Ile Asp Ser His Gly Asn			
	885	890	895
Ile Val Lys Gln Lys Ser Tyr Asn Ile Leu Asn Asn Tyr Asp Tyr Lys			
	900	905	910
Lys Lys Leu Val Glu Lys Glu Lys Thr Arg Glu Tyr Ala Arg Lys Asn			
	915	920	925
Trp Lys Ser Ile Gly Asn Ile Lys Glu Leu Lys Glu Gly Tyr Ile Ser			
	930	935	940
Gly Val Val His Glu Ile Ala Met Leu Ile Val Glu Tyr Asn Ala Ile			
	945	950	955
Ile Ala Met Glu Asp Leu Asn Tyr Gly Phe Lys Arg Gly Arg Phe Lys			
	965	970	975
Val Glu Arg Gln Val Tyr Gln Lys Phe Glu Ser Met Leu Ile Asn Lys			
	980	985	990
Leu Asn Tyr Phe Ala Ser Lys Glu Lys Ser Val Asp Glu Pro Gly Gly			
	995	1000	1005
Leu Leu Lys Gly Tyr Gln Leu Thr Tyr Val Pro Asp Asn Ile Lys			
	1010	1015	1020
Asn Leu Gly Lys Gln Cys Gly Val Ile Phe Tyr Val Pro Ala Ala			
	1025	1030	1035
Phe Thr Ser Lys Ile Asp Pro Ser Thr Gly Phe Ile Ser Ala Phe			
	1040	1045	1050
Asn Phe Lys Ser Ile Ser Thr Asn Ala Ser Arg Lys Gln Phe Phe			
	1055	1060	1065
Met Gln Phe Asp Glu Ile Arg Tyr Cys Ala Glu Lys Asp Met Phe			
	1070	1075	1080
Ser Phe Gly Phe Asp Tyr Asn Asn Phe Asp Thr Tyr Asn Ile Thr			
	1085	1090	1095
Met Gly Lys Thr Gln Trp Thr Val Tyr Thr Asn Gly Glu Arg Leu			
	1100	1105	1110
Gln Ser Glu Phe Asn Asn Ala Arg Arg Thr Gly Lys Thr Lys Ser			
	1115	1120	1125
Ile Asn Leu Thr Glu Thr Ile Lys Leu Leu Leu Glu Asp Asn Glu			
	1130	1135	1140
Ile Asn Tyr Ala Asp Gly His Asp Ile Arg Ile Asp Met Glu Lys			
	1145	1150	1155
Met Asp Glu Asp Lys Lys Ser Glu Phe Phe Ala Gln Leu Leu Ser			
	1160	1165	1170
Leu Tyr Lys Leu Thr Val Gln Met Arg Asn Ser Tyr Thr Glu Ala			
	1175	1180	1185
Glu Glu Gln Glu Asn Gly Ile Ser Tyr Asp Lys Ile Ile Ser Pro			
	1190	1195	1200
Val Ile Asn Asp Glu Gly Glu Phe Phe Asp Ser Asp Asn Tyr Lys			
	1205	1210	1215
Glu Ser Asp Asp Lys Glu Cys Lys Met Pro Lys Asp Ala Asp Ala			
	1220	1225	1230
Asn Gly Ala Tyr Cys Ile Ala Leu Lys Gly Leu Tyr Glu Val Leu			
	1235	1240	1245
Lys Ile Lys Ser Glu Trp Thr Glu Asp Gly Phe Asp Arg Asn Cys			
	1250	1255	1260
Leu Lys Leu Pro His Ala Glu Trp Leu Asp Phe Ile Gln Asn Lys			
	1265	1270	1275

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Arg Tyr Glu
1280

<210> SEQ ID NO 6
<211> LENGTH: 1300
<212> TYPE: PRT
<213> ORGANISM: Francisella novicida

<400> SEQUENCE: 6

Met Ser Ile Tyr Gln Glu Phe Val Asn Lys Tyr Ser Leu Ser Lys Thr
1 5 10 15

Leu Arg Phe Glu Leu Ile Pro Gln Gly Lys Thr Leu Glu Asn Ile Lys
20 25 30

Ala Arg Gly Leu Ile Leu Asp Asp Glu Lys Arg Ala Lys Asp Tyr Lys
35 40 45

Lys Ala Lys Gln Ile Ile Asp Lys Tyr His Gln Phe Phe Ile Glu Glu
50 55 60

Ile Leu Ser Ser Val Cys Ile Ser Glu Asp Leu Leu Gln Asn Tyr Ser
65 70 75 80

Asp Val Tyr Phe Lys Leu Lys Lys Ser Asp Asp Asp Asn Leu Gln Lys
85 90 95

Asp Phe Lys Ser Ala Lys Asp Thr Ile Lys Lys Gln Ile Ser Glu Tyr
100 105 110

Ile Lys Asp Ser Glu Lys Phe Lys Asn Leu Phe Asn Gln Asn Leu Ile
115 120 125

Asp Ala Lys Lys Gly Gln Glu Ser Asp Leu Ile Leu Trp Leu Lys Gln
130 135 140

Ser Lys Asp Asn Gly Ile Glu Leu Phe Lys Ala Asn Ser Asp Ile Thr
145 150 155 160

Asp Ile Asp Glu Ala Leu Glu Ile Ile Lys Ser Phe Lys Gly Trp Thr
165 170 175

Thr Tyr Phe Lys Gly Phe His Glu Asn Arg Lys Val Asn Tyr Ser Ser
180 185 190

Asn Asp Ile Pro Thr Ser Ile Ile Tyr Arg Ile Val Asp Asp Asn Leu
195 200 205

Pro Lys Phe Leu Glu Asn Lys Ala Lys Tyr Glu Ser Leu Lys Asp Lys
210 215 220

Ala Pro Glu Ala Ile Asn Tyr Glu Gln Ile Lys Lys Asp Leu Ala Glu
225 230 235 240

Glu Leu Thr Phe Asp Ile Asp Tyr Lys Thr Ser Glu Val Asn Gln Arg
245 250 255

Val Phe Ser Leu Asp Glu Val Phe Glu Ile Ala Asn Phe Asn Asn Tyr
260 265 270

Leu Asn Gln Ser Gly Ile Thr Lys Phe Asn Thr Ile Ile Gly Gly Lys
275 280 285

Phe Val Asn Gly Glu Asn Thr Lys Arg Lys Gly Ile Asn Glu Tyr Ile
290 295 300

Asn Leu Tyr Ser Gln Gln Ile Asn Asp Lys Thr Leu Lys Lys Tyr Lys
305 310 315 320

Met Ser Val Leu Phe Lys Gln Ile Leu Ser Asp Thr Glu Ser Lys Ser
325 330 335

Phe Val Ile Asp Lys Leu Glu Asp Asp Ser Asp Val Val Thr Thr Met
340 345 350

Gln Ser Phe Tyr Glu Gln Ile Ala Ala Phe Lys Thr Val Glu Glu Lys

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355	360	365
Ser Ile Lys Glu Thr Leu	Ser Leu Leu Phe Asp Asp	Leu Lys Ala Gln
370	375	380
Lys Leu Asp Leu Ser Lys	Ile Tyr Phe Lys Asn Asp	Lys Ser Leu Thr
385	390	395
400		
Asp Leu Ser Gln Gln Val	Phe Asp Asp Tyr Ser Val	Ile Gly Thr Ala
405	410	415
Val Leu Glu Tyr Ile Thr	Gln Gln Ile Ala Pro Lys	Asn Leu Asp Asn
420	425	430
Pro Ser Lys Lys Glu Gln	Glu Leu Ile Ala Lys Lys	Thr Glu Lys Ala
435	440	445
Lys Tyr Leu Ser Leu Glu	Thr Ile Lys Leu Ala Leu	Glu Glu Phe Asn
450	455	460
Lys His Arg Asp Ile Asp	Lys Gln Cys Arg Phe Glu	Glu Ile Leu Ala
465	470	475
480		
Asn Phe Ala Ala Ile Pro	Met Ile Phe Asp Glu Ile	Ala Gln Asn Lys
485	490	495
Asp Asn Leu Ala Gln Ile	Ser Ile Lys Tyr Gln Asn	Gln Gly Lys Lys
500	505	510
Asp Leu Leu Gln Ala Ser	Ala Glu Asp Asp Val Lys	Ala Ile Lys Asp
515	520	525
Leu Leu Asp Gln Thr Asn	Asn Leu Leu His Lys Leu	Lys Ile Phe His
530	535	540
Ile Ser Gln Ser Glu Asp	Lys Ala Asn Ile Leu Asp	Lys Asp Glu His
545	550	555
560		
Phe Tyr Leu Val Phe Glu	Glu Cys Tyr Phe Glu Leu	Ala Asn Ile Val
565	570	575
Pro Leu Tyr Asn Lys Ile	Arg Asn Tyr Ile Thr Gln	Lys Pro Tyr Ser
580	585	590
Asp Glu Lys Phe Lys Leu	Asn Phe Glu Asn Ser Thr	Leu Ala Asn Gly
595	600	605
Trp Asp Lys Asn Lys Glu	Pro Asp Asn Thr Ala Ile	Leu Phe Ile Lys
610	615	620
Asp Asp Lys Tyr Tyr Leu	Gly Val Met Asn Lys Lys	Asn Asn Lys Ile
625	630	635
640		
Phe Asp Asp Lys Ala Ile	Lys Glu Asn Lys Gly Glu	Gly Tyr Lys Lys
645	650	655
Ile Val Tyr Lys Leu Leu	Pro Gly Ala Asn Lys Met	Leu Pro Lys Val
660	665	670
Phe Phe Ser Ala Lys Ser	Ile Lys Phe Tyr Asn Pro	Ser Glu Asp Ile
675	680	685
Leu Arg Ile Arg Asn His	Ser Thr His Thr Lys Asn	Gly Ser Pro Gln
690	695	700
Lys Gly Tyr Glu Lys Phe	Glu Phe Asn Ile Glu Asp	Cys Arg Lys Phe
705	710	715
720		
Ile Asp Phe Tyr Lys Gln	Ser Ile Ser Lys His Pro	Glu Trp Lys Asp
725	730	735
Phe Gly Phe Arg Phe Ser	Asp Thr Gln Arg Tyr Asn	Ser Ile Asp Glu
740	745	750
Phe Tyr Arg Glu Val Glu	Asn Gln Gly Tyr Lys Leu	Thr Phe Glu Asn
755	760	765
Ile Ser Glu Ser Tyr Ile	Asp Ser Val Val Asn Gln	Gly Lys Leu Tyr
770	775	780

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Leu Phe Gln Ile Tyr Asn Lys Asp Phe Ser Ala Tyr Ser Lys Gly Arg
 785 790 795 800
 Pro Asn Leu His Thr Leu Tyr Trp Lys Ala Leu Phe Asp Glu Arg Asn
 805 810 815
 Leu Gln Asp Val Val Tyr Lys Leu Asn Gly Glu Ala Glu Leu Phe Tyr
 820 825 830
 Arg Lys Gln Ser Ile Pro Lys Lys Ile Thr His Pro Ala Lys Glu Ala
 835 840 845
 Ile Ala Asn Lys Asn Lys Asp Asn Pro Lys Lys Glu Ser Val Phe Glu
 850 855 860
 Tyr Asp Leu Ile Lys Asp Lys Arg Phe Thr Glu Asp Lys Phe Phe Phe
 865 870 875 880
 His Cys Pro Ile Thr Ile Asn Phe Lys Ser Ser Gly Ala Asn Lys Phe
 885 890 895
 Asn Asp Glu Ile Asn Leu Leu Leu Lys Glu Lys Ala Asn Asp Val His
 900 905 910
 Ile Leu Ser Ile Asp Arg Gly Glu Arg His Leu Ala Tyr Tyr Thr Leu
 915 920 925
 Val Asp Gly Lys Gly Asn Ile Ile Lys Gln Asp Thr Phe Asn Ile Ile
 930 935 940
 Gly Asn Asp Arg Met Lys Thr Asn Tyr His Asp Lys Leu Ala Ala Ile
 945 950 955 960
 Glu Lys Asp Arg Asp Ser Ala Arg Lys Asp Trp Lys Lys Ile Asn Asn
 965 970 975
 Ile Lys Glu Met Lys Glu Gly Tyr Leu Ser Gln Val Val His Glu Ile
 980 985 990
 Ala Lys Leu Val Ile Glu Tyr Asn Ala Ile Val Val Phe Glu Asp Leu
 995 1000 1005
 Asn Phe Gly Phe Lys Arg Gly Arg Phe Lys Val Glu Lys Gln Val
 1010 1015 1020
 Tyr Gln Lys Leu Glu Lys Met Leu Ile Glu Lys Leu Asn Tyr Leu
 1025 1030 1035
 Val Phe Lys Asp Asn Glu Phe Asp Lys Thr Gly Gly Val Leu Arg
 1040 1045 1050
 Ala Tyr Gln Leu Thr Ala Pro Phe Glu Thr Phe Lys Lys Met Gly
 1055 1060 1065
 Lys Gln Thr Gly Ile Ile Tyr Tyr Val Pro Ala Gly Phe Thr Ser
 1070 1075 1080
 Lys Ile Cys Pro Val Thr Gly Phe Val Asn Gln Leu Tyr Pro Lys
 1085 1090 1095
 Tyr Glu Ser Val Ser Lys Ser Gln Glu Phe Phe Ser Lys Phe Asp
 1100 1105 1110
 Lys Ile Cys Tyr Asn Leu Asp Lys Gly Tyr Phe Glu Phe Ser Phe
 1115 1120 1125
 Asp Tyr Lys Asn Phe Gly Asp Lys Ala Ala Lys Gly Lys Trp Thr
 1130 1135 1140
 Ile Ala Ser Phe Gly Ser Arg Leu Ile Asn Phe Arg Asn Ser Asp
 1145 1150 1155
 Lys Asn His Asn Trp Asp Thr Arg Glu Val Tyr Pro Thr Lys Glu
 1160 1165 1170
 Leu Glu Lys Leu Leu Lys Asp Tyr Ser Ile Glu Tyr Gly His Gly
 1175 1180 1185

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Glu Cys Ile Lys Ala Ala Ile	Cys Gly Glu Ser Asp	Lys Lys Phe
1190	1195	1200
Phe Ala Lys Leu Thr Ser Val	Leu Asn Thr Ile Leu	Gln Met Arg
1205	1210	1215
Asn Ser Lys Thr Gly Thr Glu	Leu Asp Tyr Leu Ile	Ser Pro Val
1220	1225	1230
Ala Asp Val Asn Gly Asn Phe	Phe Asp Ser Arg Gln	Ala Pro Lys
1235	1240	1245
Asn Met Pro Gln Asp Ala Asp	Ala Asn Gly Ala Tyr	His Ile Gly
1250	1255	1260
Leu Lys Gly Leu Met Leu Leu	Gly Arg Ile Lys Asn	Asn Gln Glu
1265	1270	1275
Gly Lys Lys Leu Asn Leu Val	Ile Lys Asn Glu Glu	Tyr Phe Glu
1280	1285	1290
Phe Val Gln Asn Arg Asn Asn		
1295	1300	

<210> SEQ ID NO 7

<211> LENGTH: 1206

<212> TYPE: PRT

<213> ORGANISM: Lachnospiraceae sp.

<400> SEQUENCE: 7

Met Tyr Tyr Glu Ser Leu Thr Lys Gln Tyr Pro Val Ser Lys Thr Ile		
1	5	10 15
Arg Asn Glu Leu Ile Pro Ile Gly Lys Thr Leu Asp Asn Ile Arg Gln		
20	25	30
Asn Asn Ile Leu Glu Ser Asp Val Lys Arg Lys Gln Asn Tyr Glu His		
35	40	45
Val Lys Gly Ile Leu Asp Glu Tyr His Lys Gln Leu Ile Asn Glu Ala		
50	55	60
Leu Asp Asn Cys Thr Leu Pro Ser Leu Lys Ile Ala Ala Glu Ile Tyr		
65	70	75 80
Leu Lys Asn Gln Lys Glu Val Ser Asp Arg Glu Asp Phe Asn Lys Thr		
85	90	95
Gln Asp Leu Leu Arg Lys Glu Val Val Glu Lys Leu Lys Ala His Glu		
100	105	110
Asn Phe Thr Lys Ile Gly Lys Lys Asp Ile Leu Asp Leu Leu Glu Lys		
115	120	125
Leu Pro Ser Ile Ser Glu Asp Asp Tyr Asn Ala Leu Glu Ser Phe Arg		
130	135	140
Asn Phe Tyr Thr Tyr Phe Thr Ser Tyr Asn Lys Val Arg Glu Asn Leu		
145	150	155 160
Tyr Ser Asp Lys Glu Lys Ser Ser Thr Val Ala Tyr Arg Leu Ile Asn		
165	170	175
Glu Asn Phe Pro Lys Phe Leu Asp Asn Val Lys Ser Tyr Arg Phe Val		
180	185	190
Lys Thr Ala Gly Ile Leu Ala Asp Gly Leu Gly Glu Glu Glu Gln Asp		
195	200	205
Ser Leu Phe Ile Val Glu Thr Phe Asn Lys Thr Leu Thr Gln Asp Gly		
210	215	220
Ile Asp Thr Tyr Asn Ser Gln Val Gly Lys Ile Asn Ser Ser Ile Asn		
225	230	235 240
Leu Tyr Asn Gln Lys Asn Gln Lys Ala Asn Gly Phe Arg Lys Ile Pro		
245	250	255

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Lys Met Lys Met Leu Tyr Lys Gln Ile Leu Ser Asp Arg Glu Glu Ser
 260 265 270

Phe Ile Asp Glu Phe Gln Ser Asp Glu Val Leu Ile Asp Asn Val Glu
 275 280 285

Ser Tyr Gly Ser Val Leu Ile Glu Ser Leu Lys Ser Ser Lys Val Ser
 290 295 300

Ala Phe Phe Asp Ala Leu Arg Glu Ser Lys Gly Lys Asn Val Tyr Val
 305 310 315 320

Lys Asn Asp Leu Ala Lys Thr Ala Met Ser Val Ile Val Phe Glu Asn
 325 330 335

Trp Arg Thr Phe Asp Asp Leu Leu Asn Gln Glu Tyr Asp Leu Ala Asn
 340 345 350

Glu Asn Lys Lys Lys Asp Asp Lys Tyr Phe Glu Lys Arg Gln Lys Glu
 355 360 365

Leu Lys Lys Asn Lys Ser Tyr Ser Leu Glu His Leu Cys Asn Leu Ser
 370 375 380

Glu Asp Ser Cys Asn Leu Ile Glu Asn Tyr Ile His Gln Ile Ser Asp
 385 390 395 400

Asp Ile Glu Asn Ile Ile Ile Asn Asn Glu Thr Phe Leu Arg Ile Val
 405 410 415

Ile Asn Glu His Asp Arg Ser Arg Lys Leu Ala Lys Asn Arg Lys Ala
 420 425 430

Val Lys Ala Ile Lys Asp Phe Leu Asp Ser Ile Lys Val Leu Glu Arg
 435 440 445

Glu Leu Lys Leu Ile Asn Ser Ser Gly Gln Glu Leu Glu Lys Asp Leu
 450 455 460

Ile Val Tyr Ser Ala His Glu Glu Leu Leu Val Glu Leu Lys Gln Val
 465 470 475 480

Asp Ser Leu Tyr Asn Met Thr Arg Asn Tyr Leu Thr Lys Lys Pro Phe
 485 490 495

Ser Thr Glu Lys Val Lys Leu Asn Phe Asn Arg Ser Thr Leu Leu Asn
 500 505 510

Gly Trp Asp Arg Asn Lys Glu Thr Asp Asn Leu Gly Val Leu Leu Leu
 515 520 525

Lys Asp Gly Lys Tyr Tyr Leu Gly Ile Met Asn Thr Ser Ala Asn Lys
 530 535 540

Ala Phe Val Asn Pro Pro Val Ala Lys Thr Glu Lys Val Phe Lys Lys
 545 550 555 560

Val Asp Tyr Lys Leu Leu Pro Val Pro Asn Gln Met Leu Pro Lys Val
 565 570 575

Phe Phe Ala Lys Ser Asn Ile Asp Phe Tyr Asn Pro Ser Ser Glu Ile
 580 585 590

Tyr Ser Asn Tyr Lys Lys Gly Thr His Lys Lys Gly Asn Met Phe Ser
 595 600 605

Leu Glu Asp Cys His Asn Leu Ile Asp Phe Phe Lys Glu Ser Ile Ser
 610 615 620

Lys His Glu Asp Trp Ser Lys Phe Gly Phe Lys Phe Asp Thr Gln Ala
 625 630 635 640

Ser Tyr Asn Asp Ile Ser Glu Phe Tyr Arg Glu Val Glu Lys Gln Gly
 645 650 655

Tyr Lys Leu Thr Tyr Thr Asp Ile Asp Glu Thr Tyr Ile Asn Asp Leu
 660 665 670

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Ile	Glu	Arg	Asn	Glu	Leu	Tyr	Leu	Phe	Gln	Ile	Tyr	Asn	Lys	Asp	Phe	675	680	685
Ser	Met	Tyr	Ser	Lys	Gly	Lys	Leu	Asn	Leu	His	Thr	Leu	Tyr	Phe	Met	690	695	700
Met	Leu	Phe	Asp	Gln	Arg	Asn	Ile	Asp	Asp	Val	Val	Tyr	Lys	Leu	Asn	705	710	715
Gly	Glu	Ala	Glu	Val	Phe	Tyr	Arg	Pro	Ala	Ser	Ile	Ser	Glu	Asp	Glu	725	730	735
Leu	Ile	Ile	His	Lys	Ala	Gly	Glu	Glu	Ile	Lys	Asn	Lys	Asn	Pro	Asn	740	745	750
Arg	Ala	Arg	Thr	Lys	Glu	Thr	Ser	Thr	Phe	Ser	Tyr	Asp	Ile	Val	Lys	755	760	765
Asp	Lys	Arg	Tyr	Ser	Lys	Asp	Lys	Phe	Thr	Leu	His	Ile	Pro	Ile	Thr	770	775	780
Met	Asn	Phe	Gly	Val	Asp	Glu	Val	Lys	Arg	Phe	Asn	Asp	Ala	Val	Asn	785	790	795
Ser	Ala	Ile	Arg	Ile	Asp	Glu	Asn	Val	Asn	Val	Ile	Gly	Ile	Asp	Arg	805	810	815
Gly	Glu	Arg	Asn	Leu	Leu	Tyr	Val	Val	Val	Ile	Asp	Ser	Lys	Gly	Asn	820	825	830
Ile	Leu	Glu	Gln	Ile	Ser	Leu	Asn	Ser	Ile	Ile	Asn	Lys	Glu	Tyr	Asp	835	840	845
Ile	Glu	Thr	Asp	Tyr	His	Ala	Leu	Leu	Asp	Glu	Arg	Glu	Gly	Gly	Arg	850	855	860
Asp	Lys	Ala	Arg	Lys	Asp	Trp	Asn	Thr	Val	Glu	Asn	Ile	Arg	Asp	Leu	865	870	875
Lys	Ala	Gly	Leu	Tyr	Leu	Gln	Val	Val	Asn	Val	Val	Ala	Lys	Leu	Val	885	890	895
Leu	Lys	Tyr	Asn	Ala	Ile	Ile	Cys	Leu	Glu	Asp	Leu	Asn	Phe	Gly	Phe	900	905	910
Lys	Arg	Gly	Arg	Gln	Lys	Val	Glu	Lys	Gln	Val	Tyr	Gln	Lys	Phe	Glu	915	920	925
Lys	Met	Leu	Ile	Asp	Lys	Leu	Asn	Tyr	Leu	Val	Ile	Asp	Lys	Ser	Arg	930	935	940
Glu	Gln	Thr	Ser	Pro	Lys	Glu	Leu	Gly	Gly	Ala	Leu	Asn	Ala	Leu	Gln	945	950	955
Leu	Thr	Ser	Lys	Phe	Lys	Ser	Phe	Lys	Glu	Leu	Gly	Lys	Gln	Ser	Gly	965	970	975
Val	Ile	Tyr	Tyr	Val	Pro	Ala	Tyr	Leu	Thr	Ser	Lys	Ile	Asp	Pro	Thr	980	985	990
Thr	Gly	Phe	Ala	Asn	Leu	Phe	Tyr	Met	Lys	Cys	Glu	Asn	Val	Glu	Lys	995	1000	1005
Ser	Lys	Arg	Phe	Phe	Asp	Gly	Phe	Asp	Phe	Ile	Arg	Phe	Asn	Ala		1010	1015	1020
Leu	Glu	Asn	Val	Phe	Glu	Phe	Gly	Phe	Asp	Tyr	Arg	Ser	Phe	Thr		1025	1030	1035
Gln	Arg	Ala	Cys	Gly	Ile	Asn	Ser	Lys	Trp	Thr	Val	Cys	Thr	Asn		1040	1045	1050
Gly	Glu	Arg	Ile	Ile	Lys	Tyr	Arg	Asn	Pro	Asp	Lys	Asn	Asn	Met		1055	1060	1065
Phe	Asp	Glu	Lys	Val	Val	Val	Val	Thr	Asp	Glu	Met	Lys	Asn	Leu		1070	1075	1080
Phe	Glu	Gln	Tyr	Lys	Ile	Pro	Tyr	Glu	Asp	Gly	Arg	Asn	Val	Lys				

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1085	1090	1095
Asp Met Ile Ile Ser Asn Glu Glu Ala Glu Phe Tyr Arg Arg Leu		
1100	1105	1110
Tyr Arg Leu Leu Gln Gln Thr Leu Gln Met Arg Asn Ser Thr Ser		
1115	1120	1125
Asp Gly Thr Arg Asp Tyr Ile Ile Ser Pro Val Lys Asn Lys Arg		
1130	1135	1140
Glu Ala Tyr Phe Asn Ser Glu Leu Ser Asp Gly Ser Val Pro Lys		
1145	1150	1155
Asp Ala Asp Ala Asn Gly Ala Tyr Asn Ile Ala Arg Lys Gly Leu		
1160	1165	1170
Trp Val Leu Glu Gln Ile Arg Gln Lys Ser Glu Gly Glu Lys Ile		
1175	1180	1185
Asn Leu Ala Met Thr Asn Ala Glu Trp Leu Glu Tyr Ala Gln Thr		
1190	1195	1200
His Leu Leu		
1205		

<210> SEQ ID NO 8

<211> LENGTH: 1233

<212> TYPE: PRT

<213> ORGANISM: Lachnospiraceae sp.

<400> SEQUENCE: 8

Met Asp Tyr Gly Asn Gly Gln Phe Glu Arg Arg Ala Pro Leu Thr Lys		
1	5	10 15
Thr Ile Thr Leu Arg Leu Lys Pro Ile Gly Glu Thr Arg Glu Thr Ile		
	20	25 30
Arg Glu Gln Lys Leu Leu Glu Gln Asp Ala Ala Phe Arg Lys Leu Val		
	35	40 45
Glu Thr Val Thr Pro Ile Val Asp Asp Cys Ile Arg Lys Ile Ala Asp		
	50	55 60
Asn Ala Leu Cys His Phe Gly Thr Glu Tyr Asp Phe Ser Cys Leu Gly		
	65	70 75 80
Asn Ala Ile Ser Lys Asn Asp Ser Lys Ala Ile Lys Lys Glu Thr Glu		
	85	90 95
Lys Val Glu Lys Leu Leu Ala Lys Val Leu Thr Glu Asn Leu Pro Asp		
	100	105 110
Gly Leu Arg Lys Val Asn Asp Ile Asn Ser Ala Ala Phe Ile Gln Asp		
	115	120 125
Thr Leu Thr Ser Phe Val Gln Asp Asp Ala Asp Lys Arg Val Leu Ile		
	130	135 140
Gln Glu Leu Lys Gly Lys Thr Val Leu Met Gln Arg Phe Leu Thr Thr		
	145	150 155 160
Arg Ile Thr Ala Leu Thr Val Trp Leu Pro Asp Arg Val Phe Glu Asn		
	165	170 175
Phe Asn Ile Phe Ile Glu Asn Ala Glu Lys Met Arg Ile Leu Leu Asp		
	180	185 190
Ser Pro Leu Asn Glu Lys Ile Met Lys Phe Asp Pro Asp Ala Glu Gln		
	195	200 205
Tyr Ala Ser Leu Glu Phe Tyr Gly Gln Cys Leu Ser Gln Lys Asp Ile		
	210	215 220
Asp Ser Tyr Asn Leu Ile Ile Ser Gly Ile Tyr Ala Asp Asp Glu Val		
	225	230 235 240

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Lys	Asn	Pro	Gly	Ile	Asn	Glu	Ile	Val	Lys	Glu	Tyr	Asn	Gln	Gln	Ile	
				245					250					255		
Arg	Gly	Asp	Lys	Asp	Glu	Ser	Pro	Leu	Pro	Lys	Leu	Lys	Lys	Leu	His	
			260					265					270			
Lys	Gln	Ile	Leu	Met	Pro	Val	Glu	Lys	Ala	Phe	Phe	Val	Arg	Val	Leu	
		275					280					285				
Ser	Asn	Asp	Ser	Asp	Ala	Arg	Ser	Ile	Leu	Glu	Lys	Ile	Leu	Lys	Asp	
	290					295					300					
Thr	Glu	Met	Leu	Pro	Ser	Lys	Ile	Ile	Glu	Ala	Met	Lys	Glu	Ala	Asp	
305					310					315					320	
Ala	Gly	Asp	Ile	Ala	Val	Tyr	Gly	Ser	Arg	Leu	His	Glu	Leu	Ser	His	
				325					330					335		
Val	Ile	Tyr	Gly	Asp	His	Gly	Lys	Leu	Ser	Gln	Ile	Ile	Tyr	Asp	Lys	
			340					345					350			
Glu	Ser	Lys	Arg	Ile	Ser	Glu	Leu	Met	Glu	Thr	Leu	Ser	Pro	Lys	Glu	
		355					360					365				
Arg	Lys	Glu	Ser	Lys	Lys	Arg	Leu	Glu	Gly	Leu	Glu	Glu	His	Ile	Arg	
	370					375					380					
Lys	Ser	Thr	Tyr	Thr	Phe	Asp	Glu	Leu	Asn	Arg	Tyr	Ala	Glu	Lys	Asn	
385					390					395					400	
Val	Met	Ala	Ala	Tyr	Ile	Ala	Ala	Val	Glu	Glu	Ser	Cys	Ala	Glu	Ile	
				405					410					415		
Met	Arg	Lys	Glu	Lys	Asp	Leu	Arg	Thr	Leu	Leu	Ser	Lys	Glu	Asp	Val	
			420					425					430			
Lys	Ile	Arg	Gly	Asn	Arg	His	Asn	Thr	Leu	Ile	Val	Lys	Asn	Tyr	Phe	
	435						440					445				
Asn	Ala	Trp	Thr	Val	Phe	Arg	Asn	Leu	Ile	Arg	Ile	Leu	Arg	Arg	Lys	
	450					455					460					
Ser	Glu	Ala	Glu	Ile	Asp	Ser	Asp	Phe	Tyr	Asp	Val	Leu	Asp	Asp	Ser	
465					470					475					480	
Val	Glu	Val	Leu	Ser	Leu	Thr	Tyr	Lys	Gly	Glu	Asn	Leu	Cys	Arg	Ser	
				485					490					495		
Tyr	Ile	Thr	Lys	Lys	Ile	Gly	Ser	Asp	Leu	Lys	Pro	Glu	Ile	Ala	Thr	
			500					505					510			
Tyr	Gly	Ser	Ala	Leu	Arg	Pro	Asn	Ser	Arg	Trp	Trp	Ser	Pro	Gly	Glu	
		515					520					525				
Lys	Phe	Asn	Val	Lys	Phe	His	Thr	Ile	Val	Arg	Arg	Asp	Gly	Arg	Leu	
	530					535					540					
Tyr	Tyr	Phe	Ile	Leu	Pro	Lys	Gly	Ala	Lys	Pro	Val	Glu	Leu	Glu	Asp	
545					550					555					560	
Met	Asp	Gly	Asp	Ile	Glu	Cys	Leu	Gln	Met	Arg	Lys	Ile	Pro	Asn	Pro	
				565					570					575		
Thr	Ile	Phe	Leu	Pro	Lys	Leu	Val	Phe	Lys	Asp	Pro	Glu	Ala	Phe	Phe	
			580					585					590			
Arg	Asp	Asn	Pro	Glu	Ala	Asp	Glu	Phe	Val	Phe	Leu	Ser	Gly	Met	Lys	
		595					600					605				
Ala	Pro	Val	Thr	Ile	Thr	Arg	Glu	Thr	Tyr	Glu	Ala	Tyr	Arg	Tyr	Lys	
	610					615					620					
Leu	Tyr	Thr	Val	Gly	Lys	Leu	Arg	Asp	Gly	Glu	Val	Ser	Glu	Glu	Glu	
625					630					635					640	
Tyr	Lys	Arg	Ala	Leu	Leu	Gln	Val	Leu	Thr	Ala	Tyr	Lys	Glu	Phe	Leu	
				645					650					655		
Glu	Asn	Arg	Met	Ile	Tyr	Ala	Asp	Leu	Asn	Phe	Gly	Phe	Lys	Asp	Leu	

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660						665						670					
Glu	Glu	Tyr	Lys	Asp	Ser	Ser	Glu	Phe	Ile	Lys	Gln	Val	Glu	Thr	His		
675						680						685					
Asn	Thr	Phe	Met	Cys	Trp	Ala	Lys	Val	Ser	Ser	Ser	Gln	Leu	Asp	Asp		
690						695						700					
Leu	Val	Lys	Ser	Gly	Asn	Gly	Leu	Leu	Phe	Glu	Ile	Trp	Ser	Glu	Arg		
705						710						715					
Leu	Glu	Ser	Tyr	Tyr	Lys	Tyr	Gly	Asn	Glu	Lys	Val	Leu	Arg	Gly	Tyr		
725						730						735					
Glu	Gly	Val	Leu	Leu	Ser	Ile	Leu	Lys	Asp	Glu	Asn	Leu	Val	Ser	Met		
740						745						750					
Arg	Thr	Leu	Leu	Asn	Ser	Arg	Pro	Met	Leu	Val	Tyr	Arg	Pro	Lys	Glu		
755						760						765					
Ser	Ser	Lys	Pro	Met	Val	Val	His	Arg	Asp	Gly	Ser	Arg	Val	Val	Asp		
770						775						780					
Arg	Phe	Asp	Lys	Asp	Gly	Lys	Tyr	Ile	Pro	Pro	Glu	Val	His	Asp	Glu		
785						790						795					
Leu	Tyr	Arg	Phe	Phe	Asn	Asn	Leu	Leu	Ile	Lys	Glu	Lys	Leu	Gly	Glu		
805						810						815					
Lys	Ala	Arg	Lys	Ile	Leu	Asp	Asn	Lys	Lys	Val	Lys	Val	Lys	Val	Leu		
820						825						830					
Glu	Ser	Glu	Arg	Val	Lys	Trp	Ser	Lys	Phe	Tyr	Asp	Glu	Gln	Phe	Ala		
835						840						845					
Val	Thr	Phe	Ser	Val	Lys	Lys	Asn	Ala	Asp	Cys	Leu	Asp	Thr	Thr	Lys		
850						855						860					
Asp	Leu	Asn	Ala	Glu	Val	Met	Glu	Gln	Tyr	Ser	Glu	Ser	Asn	Arg	Leu		
865						870						875					
Ile	Leu	Ile	Arg	Asn	Thr	Thr	Asp	Ile	Leu	Tyr	Tyr	Leu	Val	Leu	Asp		
885						890						895					
Lys	Asn	Gly	Lys	Val	Leu	Lys	Gln	Arg	Ser	Leu	Asn	Ile	Ile	Asn	Asp		
900						905						910					
Gly	Ala	Arg	Asp	Val	Asp	Trp	Lys	Glu	Arg	Phe	Arg	Gln	Val	Thr	Lys		
915						920						925					
Asp	Arg	Asn	Glu	Gly	Tyr	Asn	Glu	Trp	Asp	Tyr	Ser	Arg	Thr	Ser	Asn		
930						935						940					
Asp	Leu	Lys	Glu	Val	Tyr	Leu	Asn	Tyr	Ala	Leu	Lys	Glu	Ile	Ala	Glu		
945						950						955					
Ala	Val	Ile	Glu	Tyr	Asn	Ala	Ile	Leu	Ile	Ile	Glu	Lys	Met	Ser	Asn		
965						970						975					
Ala	Phe	Lys	Asp	Lys	Tyr	Ser	Phe	Leu	Asp	Asp	Val	Thr	Phe	Lys	Gly		
980						985						990					
Phe	Glu	Thr	Lys	Lys	Leu	Ala	Lys	Leu	Ser	Asp	Leu	His	Phe	Arg	Gly		
995						1000						1005					
Ile	Lys	Asp	Gly	Glu	Pro	Cys	Ser	Phe	Thr	Asn	Pro	Leu	Gln	Leu			
1010						1015						1020					
Cys	Gln	Asn	Asp	Ser	Asn	Lys	Ile	Leu	Gln	Asp	Gly	Val	Ile	Phe			
1025						1030						1035					
Met	Val	Pro	Asn	Ser	Met	Thr	Arg	Ser	Leu	Asp	Pro	Asp	Thr	Gly			
1040						1045						1050					
Phe	Ile	Phe	Ala	Ile	Asn	Asp	His	Asn	Ile	Arg	Thr	Lys	Lys	Ala			
1055						1060						1065					
Lys	Leu	Asn	Phe	Leu	Ser	Lys	Phe	Asp	Gln	Leu	Lys	Val	Ser	Ser			
1070						1075						1080					

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Glu Gly Cys Leu Ile Met Lys Tyr Ser Gly Asp Ser Leu Pro Thr
 1085 1090 1095
 His Asn Thr Asp Asn Arg Val Trp Asn Cys Cys Cys Asn His Pro
 1100 1105 1110
 Ile Thr Asn Tyr Asp Arg Glu Thr Lys Lys Val Glu Phe Ile Glu
 1115 1120 1125
 Glu Pro Val Glu Glu Leu Ser Arg Val Leu Glu Glu Asn Gly Ile
 1130 1135 1140
 Glu Thr Asp Thr Glu Leu Asn Lys Leu Asn Glu Arg Glu Asn Val
 1145 1150 1155
 Pro Gly Lys Val Val Asp Ala Ile Tyr Ser Leu Val Leu Asn Tyr
 1160 1165 1170
 Leu Arg Gly Thr Val Ser Gly Val Ala Gly Gln Arg Ala Val Tyr
 1175 1180 1185
 Tyr Ser Pro Val Thr Gly Lys Lys Tyr Asp Ile Ser Phe Ile Gln
 1190 1195 1200
 Ala Met Asn Leu Asn Arg Lys Cys Asp Tyr Tyr Arg Ile Gly Ser
 1205 1210 1215
 Lys Glu Arg Gly Glu Trp Thr Asp Phe Val Ala Gln Leu Ile Asn
 1220 1225 1230

<210> SEQ ID NO 9

<211> LENGTH: 1227

<212> TYPE: PRT

<213> ORGANISM: Lachnospiraceae sp.

<400> SEQUENCE: 9

Met Ser Lys Leu Glu Lys Phe Thr Asn Cys Tyr Ser Leu Ser Lys Thr
 1 5 10 15
 Leu Arg Phe Lys Ala Ile Pro Val Gly Lys Thr Gln Glu Asn Ile Asp
 20 25 30
 Asn Lys Arg Leu Leu Val Glu Asp Glu Lys Arg Ala Glu Asp Tyr Lys
 35 40 45
 Gly Val Lys Lys Leu Leu Asp Arg Tyr Tyr Leu Ser Phe Ile Asn Asp
 50 55 60
 Val Leu His Ser Ile Lys Leu Lys Asn Leu Asn Asn Tyr Ile Ser Leu
 65 70 75 80
 Phe Arg Lys Lys Thr Arg Thr Glu Lys Glu Asn Lys Glu Leu Glu Asn
 85 90 95
 Leu Glu Ile Asn Leu Arg Lys Glu Ile Ala Lys Ala Phe Lys Gly Asn
 100 105 110
 Glu Gly Tyr Lys Ser Leu Phe Lys Lys Asp Ile Ile Glu Thr Ile Leu
 115 120 125
 Pro Glu Phe Leu Asp Asp Lys Asp Glu Ile Ala Leu Val Asn Ser Phe
 130 135 140
 Asn Gly Phe Thr Thr Ala Phe Thr Gly Phe Phe Asp Asn Arg Glu Asn
 145 150 155 160
 Met Phe Ser Glu Glu Ala Lys Ser Thr Ser Ile Ala Phe Arg Cys Ile
 165 170 175
 Asn Glu Asn Leu Thr Arg Tyr Ile Ser Asn Met Asp Ile Phe Glu Lys
 180 185 190
 Val Asp Ala Ile Phe Asp Lys His Glu Val Gln Glu Ile Lys Glu Lys
 195 200 205
 Ile Leu Asn Ser Asp Tyr Asp Val Glu Asp Phe Phe Glu Gly Glu Phe

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210					215					220					
Phe 225	Asn	Phe	Val	Leu	Thr 230	Gln	Glu	Gly	Ile	Asp 235	Val	Tyr	Asn	Ala	Ile 240
Ile	Gly	Gly	Phe	Val 245	Thr	Glu	Ser	Gly	Glu 250	Lys	Ile	Lys	Gly	Leu	Asn 255
Glu	Tyr	Ile	Asn	Leu 260	Tyr	Asn	Gln	Lys	Thr 265	Lys	Gln	Lys	Leu	Pro	Lys 270
Phe	Lys	Pro	Leu	Tyr	Lys	Gln	Val	Leu	Ser	Asp	Arg	Glu 285	Ser	Leu	Ser
Phe	Tyr	Gly	Glu	Gly	Tyr	Thr 295	Ser	Asp	Glu	Glu	Val 300	Leu	Glu	Val	Phe
Arg 305	Asn	Thr	Leu	Asn	Lys 310	Asn	Ser	Glu	Ile	Phe 315	Ser	Ser	Ile	Lys	Lys 320
Leu	Glu	Lys	Leu	Phe 325	Lys	Asn	Phe	Asp	Glu 330	Tyr	Ser	Ser	Ala	Gly	Ile 335
Phe	Val	Lys	Asn	Gly 340	Pro	Ala	Ile	Ser	Thr 345	Ile	Ser	Lys	Asp	Ile	Phe 350
Gly	Glu	Trp	Asn	Val	Ile	Arg	Asp 360	Lys	Trp	Asn	Ala	Glu 365	Tyr	Asp	Asp
Ile	His 370	Leu	Lys	Lys	Lys	Ala 375	Val	Val	Thr	Glu	Lys 380	Tyr	Glu	Asp	Asp
Arg 385	Arg	Lys	Ser	Phe	Lys 390	Lys	Ile	Gly	Ser	Phe 395	Ser	Leu	Glu	Gln	Leu 400
Gln	Glu	Tyr	Ala	Asp 405	Ala	Asp	Leu	Ser	Val 410	Val	Glu	Lys	Leu	Lys	Glu 415
Ile	Ile	Ile	Gln	Lys 420	Val	Asp	Glu	Ile	Tyr 425	Lys	Val	Tyr	Gly	Ser	Ser 430
Glu	Lys	Leu	Phe	Asp	Ala	Asp 440	Phe	Val	Leu	Glu	Lys	Ser 445	Leu	Lys	Lys 450
Asn	Asp 450	Ala	Val	Val	Ala	Ile 455	Met	Lys	Asp	Leu	Leu 460	Asp	Ser	Val	Lys 465
Ser 465	Phe	Glu	Asn	Tyr	Ile 470	Lys	Ala	Phe	Phe	Gly 475	Glu	Gly	Lys	Glu	Thr 480
Asn	Arg	Asp	Glu	Ser	Phe	Tyr	Gly	Asp	Phe 490	Val	Leu	Ala	Tyr	Asp	Ile 495
Leu	Leu	Lys	Val	Asp 500	His	Ile	Tyr	Asp 505	Ala	Ile	Arg	Asn	Tyr	Val	Thr 510
Gln	Lys	Pro	Tyr	Ser	Lys	Asp	Lys 520	Phe	Lys	Leu	Tyr	Phe 525	Gln	Asn	Pro 530
Gln 530	Phe	Met	Gly	Gly	Trp	Asp 535	Lys	Asp	Lys	Glu	Thr 540	Asp	Tyr	Arg	Ala 545
Thr 545	Ile	Leu	Arg	Tyr	Gly 550	Ser	Lys	Tyr	Tyr	Leu	Ala 555	Ile	Met	Asp	Lys 560
Lys	Tyr	Ala	Lys	Cys 565	Leu	Gln	Lys	Ile	Asp 570	Lys	Asp	Asp	Val	Asn	Gly 575
Asn	Tyr	Glu	Lys	Ile 580	Asn	Tyr	Lys	Leu 585	Leu	Pro	Gly	Pro	Asn	Lys	Met 590
Leu	Pro	Lys	Val	Phe	Phe	Ser	Lys 600	Lys	Trp	Met	Ala	Tyr 605	Tyr	Asn	Pro 610
Ser 610	Glu	Asp	Ile	Gln	Lys	Ile 615	Tyr	Lys	Asn	Gly	Thr 620	Phe	Lys	Lys	Gly 625
Asp 625	Met	Phe	Asn	Leu	Asn 630	Asp	Cys	His	Lys	Leu 635	Ile	Asp	Phe	Phe	Lys 640

Asp	Ser	Ile	Ser	Arg	Tyr	Pro	Lys	Trp	Ser	Asn	Ala	Tyr	Asp	Phe	Asn
				645						650		655			
Phe	Ser	Glu	Thr	Glu	Lys	Tyr	Lys	Asp	Ile	Ala	Gly	Phe	Tyr	Arg	Glu
				660						665		670			
Val	Glu	Glu	Gln	Gly	Tyr	Lys	Val	Ser	Phe	Glu	Ser	Ala	Ser	Lys	Lys
				675						680		685			
Glu	Val	Asp	Lys	Leu	Val	Glu	Glu	Gly	Lys	Leu	Tyr	Met	Phe	Gln	Ile
				690						695		700			
Tyr	Asn	Lys	Asp	Phe	Ser	Asp	Lys	Ser	His	Gly	Thr	Pro	Asn	Leu	His
705				710				715				720			
Thr	Met	Tyr	Phe	Lys	Leu	Leu	Phe	Asp	Glu	Asn	Asn	His	Gly	Gln	Ile
				725						730		735			
Arg	Leu	Ser	Gly	Gly	Ala	Glu	Leu	Phe	Met	Arg	Arg	Ala	Ser	Leu	Lys
				740						745		750			
Lys	Glu	Glu	Leu	Val	Val	His	Pro	Ala	Asn	Ser	Pro	Ile	Ala	Asn	Lys
				755						760		765			
Asn	Pro	Asp	Asn	Pro	Lys	Lys	Thr	Thr	Thr	Leu	Ser	Tyr	Asp	Val	Tyr
770				775				780							
Lys	Asp	Lys	Arg	Phe	Ser	Glu	Asp	Gln	Tyr	Glu	Leu	His	Ile	Pro	Ile
785				790				795				800			
Ala	Asn	Ile	Asn	Lys	Cys	Pro	Lys	Asn	Ile	Phe	Lys	Ile	Asn	Thr	Glu
				805						810		815			
Val	Arg	Val	Leu	Leu	Lys	His	Asp	Asp	Asn	Pro	Tyr	Val	Ile	Gly	Ile
				820						825		830			
Asp	Arg	Gly	Glu	Arg	Asn	Leu	Leu	Tyr	Ile	Val	Val	Val	Asp	Gly	Lys
				835						840		845			
Gly	Asn	Ile	Val	Glu	Gln	Tyr	Ser	Leu	Asn	Glu	Ile	Ile	Asn	Asn	Phe
850				855				860							
Asn	Gly	Ile	Arg	Ile	Lys	Thr	Asp	Tyr	His	Ser	Leu	Leu	Asp	Lys	Lys
865				870				875				880			
Glu	Lys	Glu	Arg	Phe	Glu	Ala	Arg	Gln	Asn	Trp	Thr	Ser	Ile	Glu	Asn
				885						890		895			
Ile	Lys	Glu	Leu	Lys	Ala	Gly	Tyr	Ile	Ser	Gln	Val	Val	His	Lys	Ile
				900						905		910			
Cys	Glu	Leu	Val	Glu	Lys	Tyr	Asp	Ala	Val	Ile	Ala	Leu	Glu	Asp	Leu
				915						920		925			
Asn	Ser	Gly	Phe	Lys	Asn	Ser	Arg	Val	Lys	Val	Glu	Lys	Gln	Val	Tyr
930				935				940							
Gln	Lys	Phe	Glu	Lys	Met	Leu	Ile	Asp	Lys	Leu	Asn	Tyr	Met	Val	Asp
945				950				955				960			
Lys	Lys	Ser	Asn	Pro	Cys	Ala	Thr	Gly	Gly	Ala	Leu	Lys	Gly	Tyr	Gln
				965						970		975			
Ile	Thr	Asn	Lys	Phe	Glu	Ser	Phe	Lys	Ser	Met	Ser	Thr	Gln	Asn	Gly
				980						985		990			
Phe	Ile	Phe	Tyr	Ile	Pro	Ala	Trp	Leu	Thr	Ser	Lys	Ile	Asp	Pro	Ser
995				1000				1005							
Thr	Gly	Phe	Val	Asn	Leu	Leu	Lys	Thr	Lys	Tyr	Thr	Ser	Ile	Ala	
1010				1015				1020							
Asp	Lys	Lys	Phe	Ile	Ser	Ser	Phe	Asp	Arg	Ile	Met	Tyr	Val	Pro	
1025				1030				1035							
Glu	Glu	Asp	Leu	Phe	Glu	Phe	Ala	Leu	Asp	Tyr	Lys	Asn	Phe	Ser	
1040				1045				1050							

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Arg Thr	Asp Ala	Asp Tyr	Ile	Lys Lys	Trp Lys	Leu	Tyr Ser	Tyr	
1055			1060			1065			
Gly Asn	Arg Ile	Arg Ile	Phe	Arg Asn	Pro Lys	Lys	Asn Asn	Val	
1070			1075			1080			
Phe Asp	Trp Glu	Glu Val	Cys	Leu Thr	Ser Ala	Tyr	Lys Glu	Leu	
1085			1090			1095			
Phe Asn	Lys Tyr	Gly Ile	Asn	Tyr Gln	Gln Gly	Asp	Ile Arg	Ala	
1100			1105			1110			
Leu Leu	Cys Glu	Gln Ser	Asp	Lys Ala	Phe Tyr	Ser	Ser Phe	Met	
1115			1120			1125			
Ala Leu	Met Ser	Leu Met	Leu	Gln Met	Arg Asn	Ser	Ile Thr	Gly	
1130			1135			1140			
Arg Thr	Asp Val	Asp Phe	Leu	Ile Ser	Pro Val	Lys	Asn Ser	Asp	
1145			1150			1155			
Gly Ile	Phe Tyr	Asp Ser	Arg	Asn Tyr	Glu Ala	Gln	Glu Asn	Ala	
1160			1165			1170			
Ile Leu	Pro Lys	Asn Ala	Asp	Ala Asn	Gly Ala	Tyr	Asn Ile	Ala	
1175			1180			1185			
Arg Lys	Val Leu	Trp Ala	Ile	Gly Gln	Phe Lys	Lys	Ala Glu	Asp	
1190			1195			1200			
Glu Lys	Leu Asp	Lys Val	Lys	Ile Ala	Ser Asn	Lys	Glu Trp	Leu	
1205			1210			1215			
Glu Tyr	Ala Gln	Thr Ser	Val	Lys His					
1220			1225						

<210> SEQ ID NO 10

<211> LENGTH: 1264

<212> TYPE: PRT

<213> ORGANISM: *Leptospira inadai*

<400> SEQUENCE: 10

Met Glu	Asp Tyr	Ser Gly	Phe Val	Asn Ile	Tyr Ser	Ile Gln	Lys Thr		
1		5		10		15			
Leu Arg	Phe Glu	Leu Lys	Pro Val	Gly Lys	Thr Leu	Glu His	Ile Glu		
	20		25		30				
Lys Lys	Gly Phe	Leu Lys	Lys Asp	Lys Ile	Arg Ala	Glu Asp	Tyr Lys		
	35		40		45				
Ala Val	Lys Lys	Ile Ile	Asp Lys	Tyr His	Arg Ala	Tyr Ile	Glu Glu		
	50		55		60				
Val Phe	Asp Ser	Val Leu	His Gln	Lys Lys	Lys Lys	Asp Lys	Thr Arg		
65		70		75		80			
Phe Ser	Thr Gln	Phe Ile	Lys Glu	Ile Lys	Glu Phe	Ser Glu	Leu Tyr		
	85		90		95				
Tyr Lys	Thr Glu	Lys Asn	Ile Pro	Asp Lys	Glu Arg	Leu Glu	Ala Leu		
	100		105		110				
Ser Glu	Lys Leu	Arg Lys	Met Leu	Val Gly	Ala Phe	Lys Gly	Glu Phe		
	115		120		125				
Ser Glu	Glu Val	Ala Glu	Lys Tyr	Asn Lys	Asn Leu	Phe Ser	Lys Glu		
	130		135		140				
Leu Ile	Arg Asn	Glu Ile	Glu Lys	Phe Cys	Glu Thr	Asp Glu	Glu Arg		
145		150		155		160			
Lys Gln	Val Ser	Asn Phe	Lys Ser	Phe Thr	Thr Tyr	Phe Thr	Gly Phe		
	165		170		175				
His Ser	Asn Arg	Gln Asn	Ile Tyr	Ser Asp	Glu Lys	Lys Ser	Thr Ala		
	180		185		190				

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Ile Gly Tyr Arg Ile Ile His Gln Asn Leu Pro Lys Phe Leu Asp Asn	195	200	205
Leu Lys Ile Ile Glu Ser Ile Gln Arg Arg Phe Lys Asp Phe Pro Trp	210	215	220
Ser Asp Leu Lys Lys Asn Leu Lys Lys Ile Asp Lys Asn Ile Lys Leu	225	230	235
Thr Glu Tyr Phe Ser Ile Asp Gly Phe Val Asn Val Leu Asn Gln Lys	245	250	255
Gly Ile Asp Ala Tyr Asn Thr Ile Leu Gly Gly Lys Ser Glu Glu Ser	260	265	270
Gly Glu Lys Ile Gln Gly Leu Asn Glu Tyr Ile Asn Leu Tyr Arg Gln	275	280	285
Lys Asn Asn Ile Asp Arg Lys Asn Pro Leu Asn Val Lys Ile Leu Phe	290	295	300
Lys Gln Ile Leu Gly Asp Arg Glu Thr Lys Ser Phe Ile Pro Glu Ala	305	310	315
Phe Pro Asp Asp Gln Ser Val Leu Asn Ser Ile Thr Glu Phe Ala Lys	325	330	335
Tyr Leu Lys Leu Asp Lys Lys Lys Lys Ser Ile Ile Ala Glu Leu Lys	340	345	350
Lys Phe Leu Ser Ser Phe Asn Arg Tyr Glu Leu Asp Gly Ile Tyr Leu	355	360	365
Ala Asn Asp Asn Ser Leu Ala Ser Ile Ser Thr Phe Leu Phe Asp Asp	370	375	380
Trp Ser Phe Ile Lys Lys Ser Val Ser Phe Lys Tyr Asp Glu Ser Val	385	390	395
Gly Asp Pro Lys Lys Lys Ile Lys Ser Pro Leu Lys Tyr Glu Lys Glu	405	410	415
Lys Glu Lys Trp Leu Lys Gln Lys Tyr Tyr Thr Ile Ser Phe Leu Asn	420	425	430
Asp Ala Ile Glu Ser Tyr Ser Lys Ser Gln Asp Glu Lys Arg Val Lys	435	440	445
Ile Arg Leu Glu Ala Tyr Phe Ala Glu Phe Lys Ser Lys Asp Asp Ala	450	455	460
Lys Lys Gln Phe Asp Leu Leu Glu Arg Ile Glu Glu Ala Tyr Ala Ile	465	470	475
Val Glu Pro Leu Leu Gly Ala Glu Tyr Pro Arg Asp Arg Asn Leu Lys	485	490	495
Ala Asp Lys Lys Glu Val Gly Lys Ile Lys Asp Phe Leu Asp Ser Ile	500	505	510
Lys Ser Leu Gln Phe Phe Leu Lys Pro Leu Leu Ser Ala Glu Ile Phe	515	520	525
Asp Glu Lys Asp Leu Gly Phe Tyr Asn Gln Leu Glu Gly Tyr Tyr Glu	530	535	540
Glu Ile Asp Ile Ser Gly His Leu Tyr Asn Lys Val Arg Asn Tyr Leu	545	550	555
Thr Gly Lys Ile Tyr Ser Lys Glu Lys Phe Lys Leu Asn Phe Glu Asn	565	570	575
Ser Thr Leu Leu Lys Gly Trp Asp Glu Asn Arg Glu Val Ala Asn Leu	580	585	590
Cys Val Ile Phe Arg Glu Asp Gln Lys Tyr Tyr Leu Gly Val Met Asp	595	600	605

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Lys 610	Glu	Asn	Asn	Thr	Ile	Leu 615	Ser	Asp	Ile	Pro	Lys 620	Val	Lys	Pro	Asn
Glu 625	Leu	Phe	Tyr	Glu	Lys 630	Met	Val	Tyr	Lys	Leu 635	Ile	Pro	Thr	Pro	His 640
Met	Gln	Leu	Pro	Arg 645	Ile	Ile	Phe	Ser	Ser	Asp 650	Asn	Leu	Ser	Ile	Tyr 655
Asn	Pro	Ser	Lys 660	Ser	Ile	Leu	Lys	Ile	Arg 665	Glu	Ala	Lys	Ser	Phe	Lys
Glu	Gly	Lys 675	Asn	Phe	Lys	Leu 680	Lys	Asp	Cys	His	Lys 685	Phe	Ile	Asp	Phe
Tyr 690	Lys	Glu	Ser	Ile	Ser	Lys 695	Asn	Glu	Asp	Trp	Ser 700	Arg	Phe	Asp	Phe
Lys 705	Phe	Ser	Lys	Thr	Ser 710	Ser	Tyr	Glu	Asn	Ile 715	Ser	Glu	Phe	Tyr	Arg 720
Glu	Val	Glu	Arg	Gln 725	Gly	Tyr	Asn	Leu	Asp 730	Phe	Lys	Lys	Val	Ser	Lys 735
Phe	Tyr	Ile	Asp 740	Ser	Leu	Val	Glu	Asp 745	Gly	Lys	Leu	Tyr	Leu	Phe	Gln
Ile	Tyr 755	Asn	Lys	Asp	Phe	Ser 760	Ile	Phe	Ser	Lys	Gly 765	Lys	Pro	Asn	Leu
His 770	Thr	Ile	Tyr	Phe	Arg 775	Ser	Leu	Phe	Ser	Lys 780	Glu	Asn	Leu	Lys	Asp
Val 785	Cys	Leu	Lys	Leu	Asn 790	Gly	Glu	Ala	Glu	Met 795	Phe	Phe	Arg	Lys	Lys 800
Ser	Ile	Asn	Tyr	Asp 805	Glu	Lys	Lys	Lys	Arg 810	Glu	Gly	His	His	Pro	Glu 815
Leu	Phe	Glu	Lys 820	Leu	Lys	Tyr	Pro	Ile	Leu 825	Lys	Asp	Lys	Arg	Tyr	Ser
Glu	Asp 835	Lys	Phe	Gln	Phe	His 840	Leu	Pro	Ile	Ser	Leu 845	Asn	Phe	Lys	Ser
Lys 850	Glu	Arg	Leu	Asn	Phe 855	Asn	Leu	Lys	Val	Asn 860	Glu	Phe	Leu	Lys	Arg
Asn 865	Lys	Asp	Ile	Asn	Ile 870	Ile	Gly	Ile	Asp	Arg 875	Gly	Glu	Arg	Asn	Leu 880
Leu	Tyr	Leu	Val	Met 885	Ile	Asn	Gln	Lys	Gly 890	Glu	Ile	Leu	Lys	Gln	Thr 895
Leu	Leu	Asp 900	Ser	Met	Gln	Ser	Gly	Lys 905	Gly	Arg	Pro	Glu	Ile	Asn	Tyr 910
Lys 915	Glu	Lys	Leu	Gln	Glu	Lys 920	Glu	Ile	Glu	Arg	Asp 925	Lys	Ala	Arg	Lys
Ser 930	Trp	Gly	Thr	Val	Glu 935	Asn	Ile	Lys	Glu	Leu 940	Lys	Glu	Gly	Tyr	Leu
Ser 945	Ile	Val	Ile	His 950	Gln	Ile	Ser	Lys	Leu 955	Met	Val	Glu	Asn	Asn	Ala 960
Ile	Val	Val	Leu	Glu 965	Asp	Leu	Asn	Ile	Gly 970	Phe	Lys	Arg	Gly	Arg	Gln 975
Lys	Val	Glu	Arg	Gln 980	Val	Tyr	Gln	Lys 985	Phe	Glu	Lys	Met	Leu	Ile	Asp 990
Lys 995	Leu	Asn	Phe	Leu	Val	Phe 1000	Lys	Glu	Asn	Lys	Pro	Thr	Glu	Pro	Gly 1005
Gly 1010	Val	Leu	Lys	Ala	Tyr	Gln 1015	Leu	Thr	Asp	Glu	Phe 1020	Gln	Ser	Phe	
Glu	Lys	Leu	Ser	Lys	Gln	Thr	Gly	Phe	Leu	Phe	Tyr	Val	Pro	Ser	

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1025	1030	1035
Trp Asn Thr Ser Lys Ile Asp Pro Arg Thr Gly Phe Ile Asp Phe		
1040	1045	1050
Leu His Pro Ala Tyr Glu Asn Ile Glu Lys Ala Lys Gln Trp Ile		
1055	1060	1065
Asn Lys Phe Asp Ser Ile Arg Phe Asn Ser Lys Met Asp Trp Phe		
1070	1075	1080
Glu Phe Thr Ala Asp Thr Arg Lys Phe Ser Glu Asn Leu Met Leu		
1085	1090	1095
Gly Lys Asn Arg Val Trp Val Ile Cys Thr Thr Asn Val Glu Arg		
1100	1105	1110
Tyr Phe Thr Ser Lys Thr Ala Asn Ser Ser Ile Gln Tyr Asn Ser		
1115	1120	1125
Ile Gln Ile Thr Glu Lys Leu Lys Glu Leu Phe Val Asp Ile Pro		
1130	1135	1140
Phe Ser Asn Gly Gln Asp Leu Lys Pro Glu Ile Leu Arg Lys Asn		
1145	1150	1155
Asp Ala Val Phe Phe Lys Ser Leu Leu Phe Tyr Ile Lys Thr Thr		
1160	1165	1170
Leu Ser Leu Arg Gln Asn Asn Gly Lys Lys Gly Glu Glu Glu Lys		
1175	1180	1185
Asp Phe Ile Leu Ser Pro Val Val Asp Ser Lys Gly Arg Phe Phe		
1190	1195	1200
Asn Ser Leu Glu Ala Ser Asp Asp Glu Pro Lys Asp Ala Asp Ala		
1205	1210	1215
Asn Gly Ala Tyr His Ile Ala Leu Lys Gly Leu Met Asn Leu Leu		
1220	1225	1230
Val Leu Asn Glu Thr Lys Glu Glu Asn Leu Ser Arg Pro Lys Trp		
1235	1240	1245
Lys Ile Lys Asn Lys Asp Trp Leu Glu Phe Val Trp Glu Arg Asn		
1250	1255	1260

Arg

<210> SEQ ID NO 11
 <211> LENGTH: 1373
 <212> TYPE: PRT
 <213> ORGANISM: Moraxella bovoculi

<400> SEQUENCE: 11

Met Leu Phe Gln Asp Phe Thr His Leu Tyr Pro Leu Ser Lys Thr Val		
1	5	10 15
Arg Phe Glu Leu Phe Ile Asp Arg Thr Leu Glu His Ile His Ala Lys		
20	25	30
Asn Phe Leu Ser Gln Asp Glu Thr Met Ala Asp Met His Gln Lys Val		
35	40	45
Lys Val Ile Leu Asp Asp Tyr His Arg Asp Phe Ile Ala Asp Met Met		
50	55	60
Gly Glu Val Lys Leu Thr Lys Leu Ala Glu Phe Tyr Asp Val Tyr Leu		
65	70	75 80
Lys Phe Arg Lys Asn Pro Lys Asp Asp Glu Leu Gln Lys Ala Gln Leu		
85	90	95
Lys Asp Leu Gln Ala Val Leu Arg Lys Glu Ile Val Lys Pro Ile Gly		
100	105	110
Asn Gly Gly Lys Tyr Lys Ala Gly Tyr Asp Arg Leu Phe Gly Ala Lys		

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115	120	125
Leu Phe Lys Asp Gly Lys Glu	Leu Gly Asp Leu Ala Lys Phe Val Ile	
130	135	140
Ala Gln Glu Gly Glu Ser Ser Pro Lys Leu Ala His Leu Ala His Phe		
145	150	155
Glu Lys Phe Ser Thr Tyr Phe Thr Gly Phe His Asp Asn Arg Lys Asn		
165	170	175
Met Tyr Ser Asp Glu Asp Lys His Thr Ala Ile Ala Tyr Arg Leu Ile		
180	185	190
His Glu Asn Leu Pro Arg Phe Ile Asp Asn Leu Gln Ile Leu Thr Thr		
195	200	205
Ile Lys Gln Lys His Ser Ala Leu Tyr Asp Gln Ile Ile Asn Glu Leu		
210	215	220
Thr Ala Ser Gly Leu Asp Val Ser Leu Ala Ser His Leu Asp Gly Tyr		
225	230	235
His Lys Leu Leu Thr Gln Glu Gly Ile Thr Ala Tyr Asn Thr Leu Leu		
245	250	255
Gly Gly Ile Ser Gly Glu Ala Gly Ser Pro Lys Ile Gln Gly Ile Asn		
260	265	270
Glu Leu Ile Asn Ser His His Asn Gln His Cys His Lys Ser Glu Arg		
275	280	285
Ile Ala Lys Leu Arg Pro Leu His Lys Gln Ile Leu Ser Asp Gly Met		
290	295	300
Ser Val Ser Phe Leu Pro Ser Lys Phe Ala Asp Asp Ser Glu Met Cys		
305	310	315
Gln Ala Val Asn Glu Phe Tyr Arg His Tyr Ala Asp Val Phe Ala Lys		
325	330	335
Val Gln Ser Leu Phe Asp Gly Phe Asp Asp His Gln Lys Asp Gly Ile		
340	345	350
Tyr Val Glu His Lys Asn Leu Asn Glu Leu Ser Lys Gln Ala Phe Gly		
355	360	365
Asp Phe Ala Leu Leu Gly Arg Val Leu Asp Gly Tyr Tyr Val Asp Val		
370	375	380
Val Asn Pro Glu Phe Asn Glu Arg Phe Ala Lys Ala Lys Thr Asp Asn		
385	390	395
Ala Lys Ala Lys Leu Thr Lys Glu Lys Asp Lys Phe Ile Lys Gly Val		
405	410	415
His Ser Leu Ala Ser Leu Glu Gln Ala Ile Glu His Tyr Thr Ala Arg		
420	425	430
His Asp Asp Glu Ser Val Gln Ala Gly Lys Leu Gly Gln Tyr Phe Lys		
435	440	445
His Gly Leu Ala Gly Val Asp Asn Pro Ile Gln Lys Ile His Asn Asn		
450	455	460
His Ser Thr Ile Lys Gly Phe Leu Glu Arg Glu Arg Pro Ala Gly Glu		
465	470	475
Arg Ala Leu Pro Lys Ile Lys Ser Gly Lys Asn Pro Glu Met Thr Gln		
485	490	495
Leu Arg Gln Leu Lys Glu Leu Leu Asp Asn Ala Leu Asn Val Ala His		
500	505	510
Phe Ala Lys Leu Leu Thr Thr Lys Thr Thr Leu Asp Asn Gln Asp Gly		
515	520	525
Asn Phe Tyr Gly Glu Phe Gly Val Leu Tyr Asp Glu Leu Ala Lys Ile		
530	535	540

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Pro Thr Leu Tyr Asn Lys Val Arg Asp Tyr Leu Ser Gln Lys Pro Phe																			
545				550				555								560			
Ser Thr Glu Lys Tyr Lys Leu Asn Phe Gly Asn Pro Thr Leu Leu Asn																			
			565					570								575			
Gly Trp Asp Leu Asn Lys Glu Lys Asp Asn Phe Gly Val Ile Leu Gln																			
			580					585								590			
Lys Asp Gly Cys Tyr Tyr Leu Ala Leu Leu Asp Lys Ala His Lys Lys																			
			595					600								605			
Val Phe Asp Asn Ala Pro Asn Thr Gly Lys Ser Ile Tyr Gln Lys Met																			
			610					615								620			
Ile Tyr Lys Tyr Leu Glu Val Arg Lys Gln Phe Pro Lys Val Phe Phe																			
								630								635			
Ser Lys Glu Ala Ile Ala Ile Asn Tyr His Pro Ser Lys Glu Leu Val																			
			645																
Glu Ile Lys Asp Lys Gly Arg Gln Arg Ser Asp Asp Glu Arg Leu Lys																			
			660																
Leu Tyr Arg Phe Ile Leu Glu Cys Leu Lys Ile His Pro Lys Tyr Asp																			
			675					680								685			
Lys Lys Phe Glu Gly Ala Ile Gly Asp Ile Gln Leu Phe Lys Lys Asp																			
			690					695								700			
Lys Lys Gly Arg Glu Val Pro Ile Ser Glu Lys Asp Leu Phe Lys Asp																			
Ile Asn Gly Ile Phe Ser Ser Lys Pro Lys Leu Glu Met Glu Asp Phe																			
Phe Ile Gly Glu Phe Lys Arg Tyr Asn Pro Ser Gln Asp Leu Val Asp																			
			740																
Gln Tyr Asn Ile Tyr Lys Lys Ile Asp Ser Asn Asp Asn Arg Lys Lys																			
Glu Asn Phe Tyr Asn Asn His Pro Lys Phe Lys Lys Asp Leu Val Arg																			
Tyr Tyr Tyr Glu Ser Met Cys Lys His Glu Glu Trp Glu Glu Ser Phe																			
Glu Phe Ser Lys Lys Leu Gln Asp Ile Gly Cys Tyr Val Asp Val Asn																			
Glu Leu Phe Thr Glu Ile Glu Thr Arg Arg Leu Asn Tyr Lys Ile Ser																			
Phe Cys Asn Ile Asn Ala Asp Tyr Ile Asp Glu Leu Val Glu Gln Gly																			
Gln Leu Tyr Leu Phe Gln Ile Tyr Asn Lys Asp Phe Ser Pro Lys Ala																			
His Gly Lys Pro Asn Leu His Thr Leu Tyr Phe Lys Ala Leu Phe Ser																			
Glu Asp Asn Leu Ala Asp Pro Ile Tyr Lys Leu Asn Gly Glu Ala Gln																			
Ile Phe Tyr Arg Lys Ala Ser Leu Asp Met Asn Glu Thr Thr Ile His																			
Arg Ala Gly Glu Val Leu Glu Asn Lys Asn Pro Asp Asn Pro Lys Lys																			
Arg Gln Phe Val Tyr Asp Ile Ile Lys Asp Lys Arg Tyr Thr Gln Lys																			
Asp Phe Met Leu His Val Pro Ile Thr Met Asn Phe Gly Val Gln Gly																			
945				950												955			960

Met	Thr	Ile	Lys	Glu	Phe	Asn	Lys	Lys	Val	Asn	Gln	Ser	Ile	Gln	Gln
			965						970			975			
Tyr	Asp	Glu	Val	Asn	Val	Ile	Gly	Ile	Asp	Arg	Gly	Glu	Arg	His	Leu
			980						985			990			
Leu	Tyr	Leu	Thr	Val	Ile	Asn	Ser	Lys	Gly	Glu	Ile	Leu	Glu	Gln	Cys
			995						1000			1005			
Ser	Leu	Asn	Asp	Ile	Thr	Thr	Ala	Ser	Ala	Asn	Gly	Thr	Gln	Met	
						1015						1020			
Thr	Thr	Pro	Tyr	His	Lys	Ile	Leu	Asp	Lys	Arg	Glu	Ile	Glu	Arg	
						1030						1035			
Leu	Asn	Ala	Arg	Val	Gly	Trp	Gly	Glu	Ile	Glu	Thr	Ile	Lys	Glu	
						1045						1050			
Leu	Lys	Ser	Gly	Tyr	Leu	Ser	His	Val	Val	His	Gln	Ile	Ser	Gln	
						1060						1065			
Leu	Met	Leu	Lys	Tyr	Asn	Ala	Ile	Val	Val	Leu	Glu	Asp	Leu	Asn	
						1075						1080			
Phe	Gly	Phe	Lys	Arg	Gly	Arg	Phe	Lys	Val	Glu	Lys	Gln	Ile	Tyr	
						1090						1095			
Gln	Asn	Phe	Glu	Asn	Ala	Leu	Ile	Lys	Lys	Leu	Asn	His	Leu	Val	
						1105						1110			
Leu	Lys	Asp	Lys	Ala	Asp	Asp	Glu	Ile	Gly	Ser	Tyr	Lys	Asn	Ala	
						1120						1125			
Leu	Gln	Leu	Thr	Asn	Asn	Phe	Thr	Asp	Leu	Lys	Ser	Ile	Gly	Lys	
						1135						1140			
Gln	Thr	Gly	Phe	Leu	Phe	Tyr	Val	Pro	Ala	Trp	Asn	Thr	Ser	Lys	
						1150						1155			
Ile	Asp	Pro	Glu	Thr	Gly	Phe	Val	Asp	Leu	Leu	Lys	Pro	Arg	Tyr	
						1165						1170			
Glu	Asn	Ile	Gln	Ala	Ser	Gln	Ala	Phe	Phe	Gly	Lys	Phe	Asp	Lys	
						1180						1185			
Ile	Cys	Tyr	Asn	Ala	Asp	Lys	Asp	Tyr	Phe	Glu	Phe	His	Ile	Asp	
						1195						1200			
Tyr	Ala	Lys	Phe	Thr	Asp	Lys	Ala	Lys	Asn	Ser	Arg	Gln	Ile	Trp	
						1210						1215			
Thr	Ile	Cys	Ser	His	Gly	Asp	Lys	Arg	Tyr	Val	Tyr	Asp	Lys	Thr	
						1225						1230			
Ala	Asn	Gln	Asn	Lys	Gly	Ala	Ala	Lys	Gly	Ile	Asn	Val	Asn	Asp	
						1240						1245			
Ile	Leu	Lys	Ser	Leu	Phe	Ala	Arg	His	His	Ile	Asn	Glu	Lys	Gln	
						1255						1260			
Pro	Asn	Leu	Val	Met	Asp	Ile	Cys	Gln	Asn	Asn	Asp	Lys	Glu	Phe	
						1270						1275			
His	Lys	Ser	Leu	Met	Tyr	Leu	Leu	Lys	Thr	Leu	Leu	Ala	Leu	Arg	
						1285						1290			
Tyr	Ser	Asn	Ala	Ser	Ser	Asp	Glu	Asp	Phe	Ile	Leu	Ser	Pro	Val	
						1300						1305			
Ala	Asn	Asp	Glu	Gly	Val	Phe	Phe	Asn	Ser	Ala	Leu	Ala	Asp	Asp	
						1315						1320			
Thr	Gln	Pro	Gln	Asn	Ala	Asp	Ala	Asn	Gly	Ala	Tyr	His	Ile	Ala	
						1330						1335			
Leu	Lys	Gly	Leu	Trp	Leu	Leu	Asn	Glu	Leu	Lys	Asn	Ser	Asp	Asp	
						1345						1350			
Leu	Asn	Lys	Val	Lys	Leu	Ala	Ile	Asp	Asn	Gln	Thr	Trp	Leu	Asn	

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1355	1360	1365
Phe Ala Gln Asn Arg		
1370		
 <210> SEQ ID NO 12		
<211> LENGTH: 1352		
<212> TYPE: PRT		
<213> ORGANISM: Paracobacteria bacterium		
 <400> SEQUENCE: 12		
Met Glu Asn Ile Phe Asp Gln Phe Ile Gly Lys Tyr Ser Leu Ser Lys		
1 5 10 15		
Thr Leu Arg Phe Glu Leu Lys Pro Val Gly Lys Thr Glu Asp Phe Leu		
20 25 30		
Lys Ile Asn Lys Val Phe Glu Lys Asp Gln Thr Ile Asp Asp Ser Tyr		
35 40 45		
Asn Gln Ala Lys Phe Tyr Phe Asp Ser Leu His Gln Lys Phe Ile Asp		
50 55 60		
Ala Ala Leu Ala Ser Asp Lys Thr Ser Glu Leu Ser Phe Gln Asn Phe		
65 70 75 80		
Ala Asp Val Leu Glu Lys Gln Asn Lys Ile Ile Leu Asp Lys Lys Arg		
85 90 95		
Glu Met Gly Ala Leu Arg Lys Arg Asp Lys Asn Ala Val Gly Ile Asp		
100 105 110		
Arg Leu Gln Lys Glu Ile Asn Asp Ala Glu Asp Ile Ile Gln Lys Glu		
115 120 125		
Lys Glu Lys Ile Tyr Lys Asp Val Arg Thr Leu Phe Asp Asn Glu Ala		
130 135 140		
Glu Ser Trp Lys Thr Tyr Tyr Gln Glu Arg Glu Val Asp Gly Lys Lys		
145 150 155 160		
Ile Thr Glu Ser Lys Ala Asp Leu Lys Gln Lys Gly Ala Asp Phe Leu		
165 170 175		
Thr Ala Ala Gly Ile Leu Lys Val Leu Lys Tyr Glu Phe Pro Glu Glu		
180 185 190		
Lys Glu Lys Glu Phe Gln Ala Lys Asn Gln Pro Ser Leu Phe Val Glu		
195 200 205		
Glu Lys Glu Asn Pro Gly Gln Lys Arg Tyr Ile Phe Asp Ser Phe Asp		
210 215 220		
Lys Phe Ala Gly Tyr Leu Thr Lys Phe Gln Gln Thr Lys Lys Asn Leu		
225 230 235 240		
Tyr Ala Ala Asp Gly Thr Ser Thr Ala Val Ala Thr Arg Ile Ala Asp		
245 250 255		
Asn Phe Ile Ile Phe His Gln Asn Thr Lys Val Phe Arg Asp Lys Tyr		
260 265 270		
Lys Asn Asn His Thr Asp Leu Gly Phe Asp Glu Glu Asn Ile Phe Glu		
275 280 285		
Ile Glu Arg Tyr Lys Asn Cys Leu Leu Gln Arg Glu Ile Glu His Ile		
290 295 300		
Lys Asn Glu Asn Ser Tyr Asn Lys Ile Ile Gly Arg Ile Asn Lys Lys		
305 310 315 320		
Ile Lys Glu Tyr Arg Asp Gln Lys Ala Lys Asp Thr Lys Leu Thr Lys		
325 330 335		
Ser Asp Phe Pro Phe Phe Lys Asn Leu Asp Lys Gln Ile Leu Gly Glu		
340 345 350		

Val	Glu	Lys	Glu	Lys	Gln	Leu	Ile	Glu	Lys	Thr	Arg	Glu	Lys	Thr	Glu
355						360						365			
Glu	Asp	Val	Leu	Ile	Glu	Arg	Phe	Lys	Glu	Phe	Ile	Glu	Asn	Asn	Glu
370						375						380			
Glu	Arg	Phe	Thr	Ala	Ala	Lys	Lys	Leu	Met	Asn	Ala	Phe	Cys	Asn	Gly
385			390						395			400			
Glu	Phe	Glu	Ser	Glu	Tyr	Glu	Gly	Ile	Tyr	Leu	Lys	Asn	Lys	Ala	Ile
			405						410			415			
Asn	Thr	Ile	Ser	Arg	Arg	Trp	Phe	Val	Ser	Asp	Arg	Asp	Phe	Glu	Leu
			420						425			430			
Lys	Leu	Pro	Gln	Gln	Lys	Ser	Lys	Asn	Lys	Ser	Glu	Lys	Asn	Glu	Pro
435						440						445			
Lys	Val	Lys	Lys	Phe	Ile	Ser	Ile	Ala	Glu	Ile	Lys	Asn	Ala	Val	Glu
450						455						460			
Glu	Leu	Asp	Gly	Asp	Ile	Phe	Lys	Ala	Val	Phe	Tyr	Asp	Lys	Lys	Ile
465			470						475			480			
Ile	Ala	Gln	Gly	Gly	Ser	Lys	Leu	Glu	Gln	Phe	Leu	Val	Ile	Trp	Lys
			485						490			495			
Tyr	Glu	Phe	Glu	Tyr	Leu	Phe	Arg	Asp	Ile	Glu	Arg	Glu	Asn	Gly	Glu
			500			505						510			
Lys	Leu	Leu	Gly	Tyr	Asp	Ser	Cys	Leu	Lys	Ile	Ala	Lys	Gln	Leu	Gly
515						520						525			
Ile	Phe	Pro	Gln	Glu	Lys	Glu	Ala	Arg	Glu	Lys	Ala	Thr	Ala	Val	Ile
530						535						540			
Lys	Asn	Tyr	Ala	Asp	Ala	Gly	Leu	Gly	Ile	Phe	Gln	Met	Met	Lys	Tyr
545			550						555			560			
Phe	Ser	Leu	Asp	Asp	Lys	Asp	Arg	Lys	Asn	Thr	Pro	Gly	Gln	Leu	Ser
			565						570			575			
Thr	Asn	Phe	Tyr	Ala	Glu	Tyr	Asp	Gly	Tyr	Tyr	Lys	Asp	Phe	Glu	Phe
			580			585						590			
Ile	Lys	Tyr	Tyr	Asn	Glu	Phe	Arg	Asn	Phe	Ile	Thr	Lys	Lys	Pro	Phe
595						600						605			
Asp	Glu	Asp	Lys	Ile	Lys	Leu	Asn	Phe	Glu	Asn	Gly	Ala	Leu	Leu	Lys
610						615						620			
Gly	Trp	Asp	Glu	Asn	Lys	Glu	Tyr	Asp	Phe	Met	Gly	Val	Ile	Leu	Lys
625			630						635			640			
Lys	Glu	Gly	Arg	Leu	Tyr	Leu	Gly	Ile	Met	His	Lys	Asn	His	Arg	Lys
			645						650			655			
Leu	Phe	Gln	Ser	Met	Gly	Asn	Ala	Lys	Gly	Asp	Asn	Ala	Asn	Arg	Tyr
			660			665						670			
Gln	Lys	Met	Ile	Tyr	Lys	Gln	Ile	Ala	Asp	Ala	Ser	Lys	Asp	Val	Pro
675						680						685			
Arg	Leu	Leu	Leu	Thr	Ser	Lys	Lys	Ala	Met	Glu	Lys	Phe	Lys	Pro	Ser
690						695						700			
Gln	Glu	Ile	Leu	Arg	Ile	Lys	Lys	Glu	Lys	Thr	Phe	Lys	Arg	Glu	Ser
705			710						715			720			
Lys	Asn	Phe	Ser	Leu	Arg	Asp	Leu	His	Ala	Leu	Ile	Glu	Tyr	Tyr	Arg
			725						730			735			
Asn	Cys	Ile	Pro	Gln	Tyr	Ser	Asn	Trp	Ser	Phe	Tyr	Asp	Phe	Gln	Phe
			740			745						750			
Gln	Asp	Thr	Gly	Lys	Tyr	Gln	Asn	Ile	Lys	Glu	Phe	Thr	Asp	Asp	Val
755						760						765			
Gln	Lys	Tyr	Gly	Tyr	Lys										

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770	775	780
Ile Asn Gln Ala Leu Asn Glu Gly Lys Met Tyr Leu Phe Glu Val Val 785 790 795 800		
Asn Lys Asp Ile Tyr Asn Thr Lys Asn Gly Ser Lys Asn Leu His Thr 805 810 815		
Leu Tyr Phe Glu His Ile Leu Ser Ala Glu Asn Leu Asn Asp Pro Val 820 825 830		
Phe Lys Leu Ser Gly Met Ala Glu Ile Phe Gln Arg Gln Pro Ser Val 835 840 845		
Asn Glu Arg Glu Lys Ile Thr Thr Gln Lys Asn Gln Cys Ile Leu Asp 850 855 860		
Lys Gly Asp Arg Ala Tyr Lys Tyr Arg Arg Tyr Thr Glu Lys Lys Ile 865 870 875 880		
Met Phe His Met Ser Leu Val Leu Asn Thr Gly Lys Gly Glu Ile Lys 885 890 895		
Gln Val Gln Phe Asn Lys Ile Ile Asn Gln Arg Ile Ser Ser Ser Asp 900 905 910		
Asn Glu Met Arg Val Asn Val Ile Gly Ile Asp Arg Gly Glu Lys Asn 915 920 925		
Leu Leu Tyr Tyr Ser Val Val Lys Gln Asn Gly Glu Ile Ile Glu Gln 930 935 940		
Ala Ser Leu Asn Glu Ile Asn Gly Val Asn Tyr Arg Asp Lys Leu Ile 945 950 955 960		
Glu Arg Glu Lys Glu Arg Leu Lys Asn Arg Gln Ser Trp Lys Pro Val 965 970 975		
Val Lys Ile Lys Asp Leu Lys Lys Gly Tyr Ile Ser His Val Ile His 980 985 990		
Lys Ile Cys Gln Leu Ile Glu Lys Tyr Ser Ala Ile Val Val Leu Glu 995 1000 1005		
Asp Leu Asn Met Arg Phe Lys Gln Ile Arg Gly Gly Ile Glu Arg 1010 1015 1020		
Ser Val Tyr Gln Gln Phe Glu Lys Ala Leu Ile Asp Lys Leu Gly 1025 1030 1035		
Tyr Leu Val Phe Lys Asp Asn Arg Asp Leu Arg Ala Pro Gly Gly 1040 1045 1050		
Val Leu Asn Gly Tyr Gln Leu Ser Ala Pro Phe Val Ser Phe Glu 1055 1060 1065		
Lys Met Arg Lys Gln Thr Gly Ile Leu Phe Tyr Thr Gln Ala Glu 1070 1075 1080		
Tyr Thr Ser Lys Thr Asp Pro Ile Thr Gly Phe Arg Lys Asn Val 1085 1090 1095		
Tyr Ile Ser Asn Ser Ala Ser Leu Asp Lys Ile Lys Glu Ala Val 1100 1105 1110		
Lys Lys Phe Asp Ala Ile Gly Trp Asp Gly Lys Glu Gln Ser Tyr 1115 1120 1125		
Phe Phe Lys Tyr Asn Pro Tyr Asn Leu Ala Asp Glu Lys Tyr Lys 1130 1135 1140		
Asn Ser Thr Val Ser Lys Glu Trp Ala Ile Phe Ala Ser Ala Pro 1145 1150 1155		
Arg Ile Arg Arg Gln Lys Gly Glu Asp Gly Tyr Trp Lys Tyr Asp 1160 1165 1170		
Arg Val Lys Val Asn Glu Glu Phe Glu Lys Leu Leu Lys Val Trp 1175 1180 1185		

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Asn Phe Val Asn Pro Lys Ala Thr Asp Ile Lys Gln Glu Ile Ile
1190                      1195                      1200

Lys Lys Ile Lys Ala Gly Asp Leu Gln Gly Glu Lys Glu Leu Asp
1205                      1210                      1215

Gly Arg Leu Arg Asn Phe Trp His Ser Phe Ile Tyr Leu Phe Asn
1220                      1225                      1230

Leu Val Leu Glu Leu Arg Asn Ser Phe Ser Leu Gln Ile Lys Ile
1235                      1240                      1245

Lys Ala Gly Glu Val Ile Ala Val Asp Glu Gly Val Asp Phe Ile
1250                      1255                      1260

Ala Ser Pro Val Lys Pro Phe Phe Thr Thr Pro Asn Pro Tyr Ile
1265                      1270                      1275

Pro Ser Asn Leu Cys Trp Leu Ala Val Glu Asn Ala Asp Ala Asn
1280                      1285                      1290

Gly Ala Tyr Asn Ile Ala Arg Lys Gly Val Met Ile Leu Lys Lys
1295                      1300                      1305

Ile Arg Glu His Ala Lys Lys Asp Pro Glu Phe Lys Lys Leu Pro
1310                      1315                      1320

Asn Leu Phe Ile Ser Asn Ala Glu Trp Asp Glu Ala Ala Arg Asp
1325                      1330                      1335

Trp Gly Lys Tyr Ala Gly Thr Thr Ala Leu Asn Leu Asp His
1340                      1345                      1350

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<210> SEQ ID NO 13

<211> LENGTH: 1260

<212> TYPE: PRT

<213> ORGANISM: Porphyromonas crevioricanis

<400> SEQUENCE: 13

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Met Asp Ser Leu Lys Asp Phe Thr Asn Leu Tyr Pro Val Ser Lys Thr
1      5      10      15

Leu Arg Phe Glu Leu Lys Pro Val Gly Lys Thr Leu Glu Asn Ile Glu
20     25     30

Lys Ala Gly Ile Leu Lys Glu Asp Glu His Arg Ala Glu Ser Tyr Arg
35     40     45

Arg Val Lys Lys Ile Ile Asp Thr Tyr His Lys Val Phe Ile Asp Ser
50     55     60

Ser Leu Glu Asn Met Ala Lys Met Gly Ile Glu Asn Glu Ile Lys Ala
65     70     75     80

Met Leu Gln Ser Phe Cys Glu Leu Tyr Lys Lys Asp His Arg Thr Glu
85     90     95

Gly Glu Asp Lys Ala Leu Asp Lys Ile Arg Ala Val Leu Arg Gly Leu
100    105    110

Ile Val Gly Ala Phe Thr Gly Val Cys Gly Arg Arg Glu Asn Thr Val
115    120    125

Gln Asn Glu Lys Tyr Glu Ser Leu Phe Lys Glu Lys Leu Ile Lys Glu
130    135    140

Ile Leu Pro Asp Phe Val Leu Ser Thr Glu Ala Glu Ser Leu Pro Phe
145    150    155    160

Ser Val Glu Glu Ala Thr Arg Ser Leu Lys Glu Phe Asp Ser Phe Thr
165    170    175

Ser Tyr Phe Ala Gly Phe Tyr Glu Asn Arg Lys Asn Ile Tyr Ser Thr
180    185    190

Lys Pro Gln Ser Thr Ala Ile Ala Tyr Arg Leu Ile His Glu Asn Leu

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195	200	205
Pro Lys Phe Ile Asp Asn Ile Leu Val Phe Gln Lys Ile Lys Glu Pro 210 215 220		
Ile Ala Lys Glu Leu Glu His Ile Arg Ala Asp Phe Ser Ala Gly Gly 225 230 235 240		
Tyr Ile Lys Lys Asp Glu Arg Leu Glu Asp Ile Phe Ser Leu Asn Tyr 245 250 255		
Tyr Ile His Val Leu Ser Gln Ala Gly Ile Glu Lys Tyr Asn Ala Leu 260 265 270		
Ile Gly Lys Ile Val Thr Glu Gly Asp Gly Glu Met Lys Gly Leu Asn 275 280 285		
Glu His Ile Asn Leu Tyr Asn Gln Gln Arg Gly Arg Glu Asp Arg Leu 290 295 300		
Pro Leu Phe Arg Pro Leu Tyr Lys Gln Ile Leu Ser Asp Arg Glu Gln 305 310 315 320		
Leu Ser Tyr Leu Pro Glu Ser Phe Glu Lys Asp Glu Glu Leu Leu Arg 325 330 335		
Ala Leu Lys Glu Phe Tyr Asp His Ile Ala Glu Asp Ile Leu Gly Arg 340 345 350		
Thr Gln Gln Leu Met Thr Ser Ile Ser Glu Tyr Asp Leu Ser Arg Ile 355 360 365		
Tyr Val Arg Asn Asp Ser Gln Leu Thr Asp Ile Ser Lys Lys Met Leu 370 375 380		
Gly Asp Trp Asn Ala Ile Tyr Met Ala Arg Glu Arg Ala Tyr Asp His 385 390 395 400		
Glu Gln Ala Pro Lys Arg Ile Thr Ala Lys Tyr Glu Arg Asp Arg Ile 405 410 415		
Lys Ala Leu Lys Gly Glu Glu Ser Ile Ser Leu Ala Asn Leu Asn Ser 420 425 430		
Cys Ile Ala Phe Leu Asp Asn Val Arg Asp Cys Arg Val Asp Thr Tyr 435 440 445		
Leu Ser Thr Leu Gly Gln Lys Glu Gly Pro His Gly Leu Ser Asn Leu 450 455 460		
Val Glu Asn Val Phe Ala Ser Tyr His Glu Ala Glu Gln Leu Leu Ser 465 470 475 480		
Phe Pro Tyr Pro Glu Glu Asn Asn Leu Ile Gln Asp Lys Asp Asn Val 485 490 495		
Val Leu Ile Lys Asn Leu Leu Asp Asn Ile Ser Asp Leu Gln Arg Phe 500 505 510		
Leu Lys Pro Leu Trp Gly Met Gly Asp Glu Pro Asp Lys Asp Glu Arg 515 520 525		
Phe Tyr Gly Glu Tyr Asn Tyr Ile Arg Gly Ala Leu Asp Gln Val Ile 530 535 540		
Pro Leu Tyr Asn Lys Val Arg Asn Tyr Leu Thr Arg Lys Pro Tyr Ser 545 550 555 560		
Thr Arg Lys Val Lys Leu Asn Phe Gly Asn Ser Gln Leu Leu Ser Gly 565 570 575		
Trp Asp Arg Asn Lys Glu Lys Asp Asn Ser Cys Val Ile Leu Arg Lys 580 585 590		
Gly Gln Asn Phe Tyr Leu Ala Ile Met Asn Asn Arg His Lys Arg Ser 595 600 605		
Phe Glu Asn Lys Met Leu Pro Glu Tyr Lys Glu Gly Glu Pro Tyr Phe 610 615 620		

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Glu Lys Met Asp Tyr Lys Phe Leu Pro Asp Pro Asn Lys Met Leu Pro
 625 630 635 640
 Lys Val Phe Leu Ser Lys Lys Gly Ile Glu Ile Tyr Lys Pro Ser Pro
 645 650 655
 Lys Leu Leu Glu Gln Tyr Gly His Gly Thr His Lys Lys Gly Asp Thr
 660 665 670
 Phe Ser Met Asp Asp Leu His Glu Leu Ile Asp Phe Phe Lys His Ser
 675 680 685
 Ile Glu Ala His Glu Asp Trp Lys Gln Phe Gly Phe Lys Phe Ser Asp
 690 695 700
 Thr Ala Thr Tyr Glu Asn Val Ser Ser Phe Tyr Arg Glu Val Glu Asp
 705 710 715 720
 Gln Gly Tyr Lys Leu Ser Phe Arg Lys Val Ser Glu Ser Tyr Val Tyr
 725 730 735
 Ser Leu Ile Asp Gln Gly Lys Leu Tyr Leu Phe Gln Ile Tyr Asn Lys
 740 745 750
 Asp Phe Ser Pro Cys Ser Lys Gly Thr Pro Asn Leu His Thr Leu Tyr
 755 760 765
 Trp Arg Met Leu Phe Asp Glu Arg Asn Leu Ala Asp Val Ile Tyr Lys
 770 775 780
 Leu Asp Gly Lys Ala Glu Ile Phe Phe Arg Glu Lys Ser Leu Lys Asn
 785 790 795 800
 Asp His Pro Thr His Pro Ala Gly Lys Pro Ile Lys Lys Lys Ser Arg
 805 810 815
 Gln Lys Lys Gly Glu Glu Ser Leu Phe Glu Tyr Asp Leu Val Lys Asp
 820 825 830
 Arg Arg Tyr Thr Met Asp Lys Phe Gln Phe His Val Pro Ile Thr Met
 835 840 845
 Asn Phe Lys Cys Ser Ala Gly Ser Lys Val Asn Asp Met Val Asn Ala
 850 855 860
 His Ile Arg Glu Ala Lys Asp Met His Val Ile Gly Ile Asp Arg Gly
 865 870 875 880
 Glu Arg Asn Leu Leu Tyr Ile Cys Val Ile Asp Ser Arg Gly Thr Ile
 885 890 895
 Leu Asp Gln Ile Ser Leu Asn Thr Ile Asn Asp Ile Asp Tyr His Asp
 900 905 910
 Leu Leu Glu Ser Arg Asp Lys Asp Arg Gln Gln Glu His Arg Asn Trp
 915 920 925
 Gln Thr Ile Glu Gly Ile Lys Glu Leu Lys Gln Gly Tyr Leu Ser Gln
 930 935 940
 Ala Val His Arg Ile Ala Glu Leu Met Val Ala Tyr Lys Ala Val Val
 945 950 955 960
 Ala Leu Glu Asp Leu Asn Met Gly Phe Lys Arg Gly Arg Gln Lys Val
 965 970 975
 Glu Ser Ser Val Tyr Gln Gln Phe Glu Lys Gln Leu Ile Asp Lys Leu
 980 985 990
 Asn Tyr Leu Val Asp Lys Lys Lys Arg Pro Glu Asp Ile Gly Gly Leu
 995 1000 1005
 Leu Arg Ala Tyr Gln Phe Thr Ala Pro Phe Lys Ser Phe Lys Glu
 1010 1015 1020
 Met Gly Lys Gln Asn Gly Phe Leu Phe Tyr Ile Pro Ala Trp Asn
 1025 1030 1035

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Thr	Ser	Asn	Ile	Asp	Pro	Thr	Thr	Gly	Phe	Val	Asn	Leu	Phe	His
1040						1045					1050			
Val	Gln	Tyr	Glu	Asn	Val	Asp	Lys	Ala	Lys	Ser	Phe	Phe	Gln	Lys
1055						1060					1065			
Phe	Asp	Ser	Ile	Ser	Tyr	Asn	Pro	Lys	Lys	Asp	Trp	Phe	Glu	Phe
1070						1075					1080			
Ala	Phe	Asp	Tyr	Lys	Asn	Phe	Thr	Lys	Lys	Ala	Glu	Gly	Ser	Arg
1085						1090					1095			
Ser	Met	Trp	Ile	Leu	Cys	Thr	His	Gly	Ser	Arg	Ile	Lys	Asn	Phe
1100						1105					1110			
Arg	Asn	Ser	Gln	Lys	Asn	Gly	Gln	Trp	Asp	Ser	Glu	Glu	Phe	Ala
1115						1120					1125			
Leu	Thr	Glu	Ala	Phe	Lys	Ser	Leu	Phe	Val	Arg	Tyr	Glu	Ile	Asp
1130						1135					1140			
Tyr	Thr	Ala	Asp	Leu	Lys	Thr	Ala	Ile	Val	Asp	Glu	Lys	Gln	Lys
1145						1150					1155			
Asp	Phe	Phe	Val	Asp	Leu	Leu	Lys	Leu	Phe	Lys	Leu	Thr	Val	Gln
1160						1165					1170			
Met	Arg	Asn	Ser	Trp	Lys	Glu	Lys	Asp	Leu	Asp	Tyr	Leu	Ile	Ser
1175						1180					1185			
Pro	Val	Ala	Gly	Ala	Asp	Gly	Arg	Phe	Phe	Asp	Thr	Arg	Glu	Gly
1190						1195					1200			
Asn	Lys	Ser	Leu	Pro	Lys	Asp	Ala	Asp	Ala	Asn	Gly	Ala	Tyr	Asn
1205						1210					1215			
Ile	Ala	Leu	Lys	Gly	Leu	Trp	Ala	Leu	Arg	Gln	Ile	Arg	Gln	Thr
1220						1225					1230			
Ser	Glu	Gly	Gly	Lys	Leu	Lys	Leu	Ala	Ile	Ser	Asn	Lys	Glu	Trp
1235						1240					1245			
Leu	Gln	Phe	Val	Gln	Glu	Arg	Ser	Tyr	Glu	Lys	Asp			
1250						1255					1260			

<210> SEQ ID NO 14

<211> LENGTH: 1324

<212> TYPE: PRT

<213> ORGANISM: Prevotella disiens

<400> SEQUENCE: 14

Met	Glu	Asn	Tyr	Gln	Glu	Phe	Thr	Asn	Leu	Phe	Gln	Leu	Asn	Lys	Thr
1				5				10					15		
Leu	Arg	Phe	Glu	Leu	Lys	Pro	Ile	Gly	Lys	Thr	Cys	Glu	Leu	Leu	Glu
	20						25					30			
Glu	Gly	Lys	Ile	Phe	Ala	Ser	Gly	Ser	Phe	Leu	Glu	Lys	Asp	Lys	Val
	35					40					45				
Arg	Ala	Asp	Asn	Val	Ser	Tyr	Val	Lys	Lys	Glu	Ile	Asp	Lys	Lys	His
	50				55					60					
Lys	Ile	Phe	Ile	Glu	Glu	Thr	Leu	Ser	Ser	Phe	Ser	Ile	Ser	Asn	Asp
65				70				75				80			
Leu	Leu	Lys	Gln	Tyr	Phe	Asp	Cys	Tyr	Asn	Glu	Leu	Lys	Ala	Phe	Lys
		85					90					95			
Lys	Asp	Cys	Lys	Ser	Asp	Glu	Glu	Glu	Val	Lys	Lys	Thr	Ala	Leu	Arg
	100						105					110			
Asn	Lys	Cys	Thr	Ser	Ile	Gln	Arg	Ala	Met	Arg	Glu	Ala	Ile	Ser	Gln
	115					120					125				
Ala	Phe	Leu	Lys	Ser	Pro	Gln	Lys	Lys	Leu	Leu	Ala	Ile	Lys	Asn	Leu
	130					135					140				

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Ile	Glu	Asn	Val	Phe	Lys	Ala	Asp	Glu	Asn	Val	Gln	His	Phe	Ser	Glu	
145					150					155					160	
Phe	Thr	Ser	Tyr	Phe	Ser	Gly	Phe	Glu	Thr	Asn	Arg	Glu	Asn	Phe	Tyr	
				165					170					175		
Ser	Asp	Glu	Glu	Lys	Ser	Thr	Ser	Ile	Ala	Tyr	Arg	Leu	Val	His	Asp	
		180						185					190			
Asn	Leu	Pro	Ile	Phe	Ile	Lys	Asn	Ile	Tyr	Ile	Phe	Glu	Lys	Leu	Lys	
		195					200					205				
Glu	Gln	Phe	Asp	Ala	Lys	Thr	Leu	Ser	Glu	Ile	Phe	Glu	Asn	Tyr	Lys	
	210					215					220					
Leu	Tyr	Val	Ala	Gly	Ser	Ser	Leu	Asp	Glu	Val	Phe	Ser	Leu	Glu	Tyr	
225					230					235					240	
Phe	Asn	Asn	Thr	Leu	Thr	Gln	Lys	Gly	Ile	Asp	Asn	Tyr	Asn	Ala	Val	
			245						250					255		
Ile	Gly	Lys	Ile	Val	Lys	Glu	Asp	Lys	Gln	Glu	Ile	Gln	Gly	Leu	Asn	
		260						265					270			
Glu	His	Ile	Asn	Leu	Tyr	Asn	Gln	Lys	His	Lys	Asp	Arg	Arg	Leu	Pro	
	275						280					285				
Phe	Phe	Ile	Ser	Leu	Lys	Lys	Gln	Ile	Leu	Ser	Asp	Arg	Glu	Ala	Leu	
	290					295					300					
Ser	Trp	Leu	Pro	Asp	Met	Phe	Lys	Asn	Asp	Ser	Glu	Val	Ile	Asp	Ala	
305					310					315					320	
Leu	Lys	Gly	Phe	Tyr	Ile	Glu	Asp	Gly	Phe	Glu	Asn	Asn	Val	Leu	Thr	
			325						330					335		
Pro	Leu	Ala	Thr	Leu	Leu	Ser	Ser	Leu	Asp	Lys	Tyr	Asn	Leu	Asn	Gly	
		340						345					350			
Ile	Phe	Ile	Arg	Asn	Asn	Glu	Ala	Leu	Ser	Ser	Leu	Ser	Gln	Asn	Val	
	355					360						365				
Tyr	Arg	Asn	Phe	Ser	Ile	Asp	Glu	Ala	Ile	Asp	Ala	Gln	Asn	Ala	Glu	
	370					375				380						
Leu	Gln	Thr	Phe	Asn	Asn	Tyr	Glu	Leu	Ile	Ala	Asn	Ala	Leu	Arg	Ala	
385					390					395					400	
Lys	Ile	Lys	Lys	Glu	Thr	Lys	Gln	Gly	Arg	Lys	Ser	Phe	Glu	Lys	Tyr	
			405						410					415		
Glu	Glu	Tyr	Ile	Asp	Lys	Lys	Val	Lys	Ala	Ile	Asp	Ser	Leu	Ser	Ile	
		420						425					430			
Gln	Glu	Ile	Asn	Glu	Leu	Val	Glu	Asn	Tyr	Val	Ser	Glu	Phe	Asn	Ser	
	435						440					445				
Asn	Ser	Gly	Asn	Met	Pro	Arg	Lys	Val	Glu	Asp	Tyr	Phe	Ser	Leu	Met	
	450					455					460					
Arg	Lys	Gly	Asp	Phe	Gly	Ser	Asn	Asp	Leu	Ile	Glu	Asn	Ile	Lys	Thr	
465					470					475					480	
Lys	Leu	Ser	Ala	Ala	Glu	Lys	Leu	Leu	Gly	Thr	Lys	Tyr	Gln	Glu	Thr	
			485						490					495		
Ala	Lys	Asp	Ile	Phe	Lys	Lys	Asp	Glu	Asn	Ser	Lys	Leu	Ile	Lys	Glu	
		500						505					510			
Leu	Leu	Asp	Ala	Thr	Lys	Gln	Phe	Gln	His	Phe	Ile	Lys	Pro	Leu	Leu	
		515					520					525				
Gly	Thr	Gly	Glu	Glu	Ala	Asp	Arg	Asp	Leu	Val	Phe	Tyr	Gly	Asp	Phe	
	530					535					540					
Leu	Pro	Leu	Tyr	Glu	Lys	Phe	Glu	Glu	Leu	Thr	Leu	Leu	Tyr	Asn	Lys	
545					550					555					560	

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Val	Arg	Asn	Arg	Leu	Thr	Gln	Lys	Pro	Tyr	Ser	Lys	Asp	Lys	Ile	Arg	
				565					570					575		
Leu	Cys	Phe	Asn	Lys	Pro	Lys	Leu	Met	Thr	Gly	Trp	Val	Asp	Ser	Lys	
			580					585					590			
Thr	Glu	Lys	Ser	Asp	Asn	Gly	Thr	Gln	Tyr	Gly	Gly	Tyr	Leu	Phe	Arg	
		595					600					605				
Lys	Lys	Asn	Glu	Ile	Gly	Glu	Tyr	Asp	Tyr	Phe	Leu	Gly	Ile	Ser	Ser	
	610					615					620					
Lys	Ala	Gln	Leu	Phe	Arg	Lys	Asn	Glu	Ala	Val	Ile	Gly	Asp	Tyr	Glu	
625					630					635					640	
Arg	Leu	Asp	Tyr	Tyr	Gln	Pro	Lys	Ala	Asn	Thr	Ile	Tyr	Gly	Ser	Ala	
			645						650					655		
Tyr	Glu	Gly	Glu	Asn	Ser	Tyr	Lys	Glu	Asp	Lys	Lys	Arg	Leu	Asn	Lys	
			660					665					670			
Val	Ile	Ile	Ala	Tyr	Ile	Glu	Gln	Ile	Lys	Gln	Thr	Asn	Ile	Lys	Lys	
	675						680					685				
Ser	Ile	Ile	Glu	Ser	Ile	Ser	Lys	Tyr	Pro	Asn	Ile	Ser	Asp	Asp	Asp	
	690					695					700					
Lys	Val	Thr	Pro	Ser	Ser	Leu	Leu	Glu	Lys	Ile	Lys	Lys	Val	Ser	Ile	
705					710					715					720	
Asp	Ser	Tyr	Asn	Gly	Ile	Leu	Ser	Phe	Lys	Ser	Phe	Gln	Ser	Val	Asn	
			725						730					735		
Lys	Glu	Val	Ile	Asp	Asn	Leu	Leu	Lys	Thr	Ile	Ser	Pro	Leu	Lys	Asn	
		740						745					750			
Lys	Ala	Glu	Phe	Leu	Asp	Leu	Ile	Asn	Lys	Asp	Tyr	Gln	Ile	Phe	Thr	
	755						760					765				
Glu	Val	Gln	Ala	Val	Ile	Asp	Glu	Ile	Cys	Lys	Gln	Lys	Thr	Phe	Ile	
	770					775					780					
Tyr	Phe	Pro	Ile	Ser	Asn	Val	Glu	Leu	Glu	Lys	Glu	Met	Gly	Asp	Lys	
785					790					795					800	
Asp	Lys	Pro	Leu	Cys	Leu	Phe	Gln	Ile	Ser	Asn	Lys	Asp	Leu	Ser	Phe	
			805						810					815		
Ala	Lys	Thr	Phe	Ser	Ala	Asn	Leu	Arg	Lys	Lys	Arg	Gly	Ala	Glu	Asn	
			820					825					830			
Leu	His	Thr	Met	Leu	Phe	Lys	Ala	Leu	Met	Glu	Gly	Asn	Gln	Asp	Asn	
	835						840					845				
Leu	Asp	Leu	Gly	Ser	Gly	Ala	Ile	Phe	Tyr	Arg	Ala	Lys	Ser	Leu	Asp	
	850					855					860					
Gly	Asn	Lys	Pro	Thr	His	Pro	Ala	Asn	Glu	Ala	Ile	Lys	Cys	Arg	Asn	
865					870					875					880	
Val	Ala	Asn	Lys	Asp	Lys	Val	Ser	Leu	Phe	Thr	Tyr	Asp	Ile	Tyr	Lys	
			885						890					895		
Asn	Arg	Arg	Tyr	Met	Glu	Asn	Lys	Phe	Leu	Phe	His	Leu	Ser	Ile	Val	
			900					905					910			
Gln	Asn	Tyr	Lys	Ala	Ala	Asn	Asp	Ser	Ala	Gln	Leu	Asn	Ser	Ser	Ala	
		915					920					925				
Thr	Glu	Tyr	Ile	Arg	Lys	Ala	Asp	Asp	Leu	His	Ile	Ile	Gly	Ile	Asp	
	930					935					940					
Arg	Gly	Glu	Arg	Asn	Leu	Leu	Tyr	Tyr	Ser	Val	Ile	Asp	Met	Lys	Gly	
945				950						955					960	
Asn	Ile	Val	Glu	Gln	Asp	Ser	Leu	Asn	Ile	Ile	Arg	Asn	Asn	Asp	Leu	
			965						970					975		
Glu	Thr	Asp	Tyr	His	Asp	Leu	Leu	Asp	Lys	Arg	Glu	Lys	Glu	Arg	Lys	

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980						985					990				
Ala	Asn	Arg	Gln	Asn	Trp	Glu	Ala	Val	Glu	Gly	Ile	Lys	Asp	Leu	Lys
	995						1000					1005			
Lys	Gly	Tyr	Leu	Ser	Gln	Ala	Val	His	Gln	Ile	Ala	Gln	Leu	Met	
	1010					1015					1020				
Leu	Lys	Tyr	Asn	Ala	Ile	Ile	Ala	Leu	Glu	Asp	Leu	Gly	Gln	Met	
	1025					1030					1035				
Phe	Val	Thr	Arg	Gly	Gln	Lys	Ile	Glu	Lys	Ala	Val	Tyr	Gln	Gln	
	1040					1045					1050				
Phe	Glu	Lys	Ser	Leu	Val	Asp	Lys	Leu	Ser	Tyr	Leu	Val	Asp	Lys	
	1055					1060					1065				
Lys	Arg	Pro	Tyr	Asn	Glu	Leu	Gly	Gly	Ile	Leu	Lys	Ala	Tyr	Gln	
	1070					1075					1080				
Leu	Ala	Ser	Ser	Ile	Thr	Lys	Asn	Asn	Ser	Asp	Lys	Gln	Asn	Gly	
	1085					1090					1095				
Phe	Leu	Phe	Tyr	Val	Pro	Ala	Trp	Asn	Thr	Ser	Lys	Ile	Asp	Pro	
	1100					1105					1110				
Val	Thr	Gly	Phe	Thr	Asp	Leu	Leu	Arg	Pro	Lys	Ala	Met	Thr	Ile	
	1115					1120					1125				
Lys	Glu	Ala	Gln	Asp	Phe	Phe	Gly	Ala	Phe	Asp	Asn	Ile	Ser	Tyr	
	1130					1135					1140				
Asn	Asp	Lys	Gly	Tyr	Phe	Glu	Phe	Glu	Thr	Asn	Tyr	Asp	Lys	Phe	
	1145					1150					1155				
Lys	Ile	Arg	Met	Lys	Ser	Ala	Gln	Thr	Arg	Trp	Thr	Ile	Cys	Thr	
	1160					1165					1170				
Phe	Gly	Asn	Arg	Ile	Lys	Arg	Lys	Lys	Asp	Lys	Asn	Tyr	Trp	Asn	
	1175					1180					1185				
Tyr	Glu	Glu	Val	Glu	Leu	Thr	Glu	Glu	Phe	Lys	Lys	Leu	Phe	Lys	
	1190					1195					1200				
Asp	Ser	Asn	Ile	Asp	Tyr	Glu	Asn	Cys	Asn	Leu	Lys	Glu	Glu	Ile	
	1205					1210					1215				
Gln	Asn	Lys	Asp	Asn	Arg	Lys	Phe	Phe	Asp	Asp	Leu	Ile	Lys	Leu	
	1220					1225					1230				
Leu	Gln	Leu	Thr	Leu	Gln	Met	Arg	Asn	Ser	Asp	Asp	Lys	Gly	Asn	
	1235					1240					1245				
Asp	Tyr	Ile	Ile	Ser	Pro	Val	Ala	Asn	Ala	Glu	Gly	Gln	Phe	Phe	
	1250					1255					1260				
Asp	Ser	Arg	Asn	Gly	Asp	Lys	Lys	Leu	Pro	Leu	Asp	Ala	Asp	Ala	
	1265					1270					1275				
Asn	Gly	Ala	Tyr	Asn	Ile	Ala	Arg	Lys	Gly	Leu	Trp	Asn	Ile	Arg	
	1280					1285					1290				
Gln	Ile	Lys	Gln	Thr	Lys	Asn	Lys	Asp	Asp	Leu	Asn	Leu	Ser	Ile	
	1295					1300					1305				
Ser	Ser	Thr	Glu	Trp	Leu	Asp	Phe	Val	Arg	Glu	Lys	Pro	Tyr	Leu	
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Lys

<210> SEQ ID NO 15
 <211> LENGTH: 1484
 <212> TYPE: PRT
 <213> ORGANISM: Peregrinibacteria bacterium
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1073)..(1073)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

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<400> SEQUENCE: 15

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Met Ser Asn Phe Phe Lys Asn Phe Thr Asn Leu Tyr Glu Leu Ser Lys
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Thr Leu Arg Phe Glu Leu Lys Pro Val Gly Asp Thr Leu Thr Asn Met
20      25      30

Lys Asp His Leu Glu Tyr Asp Glu Lys Leu Gln Thr Phe Leu Lys Asp
35      40      45

Gln Asn Ile Asp Asp Ala Tyr Gln Ala Leu Lys Pro Gln Phe Asp Glu
50      55      60

Ile His Glu Glu Phe Ile Thr Asp Ser Leu Glu Ser Lys Lys Ala Lys
65      70      75      80

Glu Ile Asp Phe Ser Glu Tyr Leu Asp Leu Phe Gln Glu Lys Lys Glu
85      90      95

Leu Asn Asp Ser Glu Lys Lys Leu Arg Asn Lys Ile Gly Glu Thr Phe
100     105     110

Asn Lys Ala Gly Glu Lys Trp Lys Lys Glu Lys Tyr Pro Gln Tyr Glu
115     120     125

Trp Lys Lys Gly Ser Lys Ile Ala Asn Gly Ala Asp Ile Leu Ser Cys
130     135     140

Gln Asp Met Leu Gln Phe Ile Lys Tyr Lys Asn Pro Glu Asp Glu Lys
145     150     155     160

Ile Lys Asn Tyr Ile Asp Asp Thr Leu Lys Gly Phe Phe Thr Tyr Phe
165     170     175

Gly Gly Phe Asn Gln Asn Arg Ala Asn Tyr Tyr Glu Thr Lys Lys Glu
180     185     190

Ala Ser Thr Ala Val Ala Thr Arg Ile Val His Glu Asn Leu Pro Lys
195     200     205

Phe Cys Asp Asn Val Ile Gln Phe Lys His Ile Ile Lys Arg Lys Lys
210     215     220

Asp Gly Thr Val Glu Lys Thr Glu Arg Lys Thr Glu Tyr Leu Asn Ala
225     230     235     240

Tyr Gln Tyr Leu Lys Asn Asn Asn Lys Ile Thr Gln Ile Lys Asp Ala
245     250     255

Glu Thr Glu Lys Met Ile Glu Ser Thr Pro Ile Ala Glu Lys Ile Phe
260     265     270

Asp Val Tyr Tyr Phe Ser Ser Cys Leu Ser Gln Lys Gln Ile Glu Glu
275     280     285

Tyr Asn Arg Ile Ile Gly His Tyr Asn Leu Leu Ile Asn Leu Tyr Asn
290     295     300

Gln Ala Lys Arg Ser Glu Gly Lys His Leu Ser Ala Asn Glu Lys Lys
305     310     315     320

Tyr Lys Asp Leu Pro Lys Phe Lys Thr Leu Tyr Lys Gln Ile Gly Cys
325     330     335

Gly Lys Lys Lys Asp Leu Phe Tyr Thr Ile Lys Cys Asp Thr Glu Glu
340     345     350

Glu Ala Asn Lys Ser Arg Asn Glu Gly Lys Glu Ser His Ser Val Glu
355     360     365

Glu Ile Ile Asn Lys Ala Gln Glu Ala Ile Asn Lys Tyr Phe Lys Ser
370     375     380

Asn Asn Asp Cys Glu Asn Ile Asn Thr Val Pro Asp Phe Ile Asn Tyr
385     390     395     400

Ile Leu Thr Lys Glu Asn Tyr Glu Gly Val Tyr Trp Ser Lys Ala Ala

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405								410				415			
Met	Asn	Thr	Ile	Ser	Asp	Lys	Tyr	Phe	Ala	Asn	Tyr	His	Asp	Leu	Gln
			420				425						430		
Asp	Arg	Leu	Lys	Glu	Ala	Lys	Val	Phe	Gln	Lys	Ala	Asp	Lys	Lys	Ser
			435				440				445				
Glu	Asp	Asp	Ile	Lys	Ile	Pro	Glu	Ala	Ile	Glu	Leu	Ser	Gly	Leu	Phe
			450				455				460				
Gly	Val	Leu	Asp	Ser	Leu	Ala	Asp	Trp	Gln	Thr	Thr	Leu	Phe	Lys	Ser
			465				470				475			480	
Ser	Ile	Leu	Ser	Asn	Glu	Lys	Leu	Lys	Ile	Ile	Thr	Asp	Ser	Gln	Thr
			485				490						495		
Pro	Ser	Glu	Ala	Leu	Leu	Lys	Met	Ile	Phe	Asn	Asp	Ile	Glu	Lys	Asn
			500				505						510		
Met	Glu	Ser	Phe	Leu	Lys	Glu	Thr	Asn	Asp	Ile	Ile	Thr	Leu	Lys	Lys
			515				520				525				
Tyr	Lys	Gly	Asn	Lys	Glu	Gly	Thr	Glu	Lys	Ile	Lys	Gln	Trp	Phe	Asp
			530				535				540				
Tyr	Thr	Leu	Ala	Ile	Asn	Arg	Met	Leu	Lys	Tyr	Phe	Leu	Val	Lys	Glu
			545				550				555			560	
Asn	Lys	Ile	Lys	Gly	Asn	Ser	Leu	Asp	Thr	Asn	Ile	Ser	Glu	Ala	Leu
			565				570						575		
Lys	Thr	Leu	Ile	Tyr	Ser	Asp	Asp	Ala	Glu	Trp	Phe	Lys	Trp	Tyr	Asp
			580				585						590		
Ala	Leu	Arg	Asn	Tyr	Leu	Thr	Gln	Lys	Pro	Gln	Asp	Glu	Ala	Lys	Glu
			595				600				605				
Asn	Lys	Leu	Lys	Leu	Asn	Phe	Asp	Asn	Pro	Ser	Leu	Ala	Gly	Gly	Trp
			610				615				620				
Asp	Val	Asn	Lys	Glu	Cys	Ser	Asn	Phe	Cys	Val	Ile	Leu	Lys	Asp	Lys
			625				630				635			640	
Asn	Glu	Lys	Lys	Tyr	Leu	Ala	Met	Ile	Lys	Lys	Gly	Glu	Asn	Thr	Leu
			645				650						655		
Phe	Gln	Lys	Glu	Trp	Thr	Glu	Gly	Arg	Gly	Lys	Asn	Leu	Thr	Lys	Lys
			660				665						670		
Ser	Asn	Pro	Leu	Phe	Glu	Ile	Asn	Asn	Cys	Glu	Ile	Leu	Ser	Lys	Met
			675				680						685		
Glu	Tyr	Asp	Phe	Trp	Ala	Asp	Val	Ser	Lys	Met	Ile	Pro	Lys	Cys	Ser
			690				695				700				
Thr	Gln	Leu	Lys	Ala	Val	Val	Asn	His	Phe	Lys	Gln	Ser	Asp	Asn	Glu
			705				710				715			720	
Phe	Ile	Phe	Pro	Ile	Gly	Tyr	Lys	Val	Thr	Ser	Gly	Glu	Lys	Phe	Arg
			725				730						735		
Glu	Glu	Cys	Lys	Ile	Ser	Lys	Gln	Asp	Phe	Glu	Leu	Asn	Asn	Lys	Val
			740				745						750		
Phe	Asn	Lys	Asn	Glu	Leu	Ser	Val	Thr	Ala	Met	Arg	Tyr	Asp	Leu	Ser
			755				760						765		
Ser	Thr	Gln	Glu	Lys	Gln	Tyr	Ile	Lys	Ala	Phe	Gln	Lys	Glu	Tyr	Trp
			770				775				780				
Glu	Leu	Leu	Phe	Lys	Gln	Glu	Lys	Arg	Asp	Thr	Lys	Leu	Thr	Asn	Asn
			785				790				795			800	
Glu	Ile	Phe	Asn	Glu	Trp	Ile	Asn	Phe	Cys	Asn	Lys	Lys	Tyr	Ser	Glu
			805				810						815		
Leu	Leu	Ser	Trp	Glu	Arg	Lys	Tyr	Lys	Asp	Ala	Leu	Thr	Asn	Trp	Ile
			820				825						830		

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Asn Phe Cys Lys Tyr Phe Leu Ser Lys Tyr Pro Lys Thr Thr Leu Phe	835	840	845
Asn Tyr Ser Phe Lys Glu Ser Glu Asn Tyr Asn Ser Leu Asp Glu Phe	850	855	860
Tyr Arg Asp Val Asp Ile Cys Ser Tyr Lys Leu Asn Ile Asn Thr Thr	865	870	875 880
Ile Asn Lys Ser Ile Leu Asp Arg Leu Val Glu Glu Gly Lys Leu Tyr	885	890	895
Leu Phe Glu Ile Lys Asn Gln Asp Ser Asn Asp Gly Lys Ser Ile Gly	900	905	910
His Lys Asn Asn Leu His Thr Ile Tyr Trp Asn Ala Ile Phe Glu Asn	915	920	925
Phe Asp Asn Arg Pro Lys Leu Asn Gly Glu Ala Glu Ile Phe Tyr Arg	930	935	940
Lys Ala Ile Ser Lys Asp Lys Leu Gly Ile Val Lys Gly Lys Lys Thr	945	950	955 960
Lys Asn Gly Thr Trp Ile Ile Lys Asn Tyr Arg Phe Ser Lys Glu Lys	965	970	975
Phe Ile Leu His Val Pro Ile Thr Leu Asn Phe Cys Ser Asn Asn Glu	980	985	990
Tyr Val Asn Asp Ile Val Asn Thr Lys Phe Tyr Asn Phe Ser Asn Leu	995	1000	1005
His Phe Leu Gly Ile Asp Arg Gly Glu Lys His Leu Ala Tyr Tyr	1010	1015	1020
Ser Leu Val Asn Lys Asn Gly Glu Ile Val Asp Gln Gly Thr Leu	1025	1030	1035
Asn Leu Pro Phe Thr Asp Lys Asp Gly Asn Gln Arg Ser Ile Lys	1040	1045	1050
Lys Glu Lys Tyr Phe Tyr Asn Lys Gln Glu Asp Lys Trp Glu Ala	1055	1060	1065
Lys Glu Val Asp Xaa Trp Asn Tyr Asn Asp Leu Leu Asp Ala Met	1070	1075	1080
Ala Ser Asn Arg Asp Met Ala Arg Lys Asn Trp Gln Arg Ile Gly	1085	1090	1095
Thr Ile Lys Glu Ala Lys Asn Gly Tyr Val Ser Leu Val Ile Arg	1100	1105	1110
Lys Ile Ala Asp Leu Ala Val Asn Asn Glu Arg Pro Ala Phe Ile	1115	1120	1125
Val Leu Glu Asp Leu Asn Thr Gly Phe Lys Arg Ser Arg Gln Lys	1130	1135	1140
Ile Asp Lys Ser Val Tyr Gln Lys Phe Glu Leu Ala Leu Ala Lys	1145	1150	1155
Lys Leu Asn Phe Leu Val Asp Lys Asn Ala Lys Arg Asp Glu Ile	1160	1165	1170
Gly Ser Pro Thr Lys Ala Leu Gln Leu Thr Pro Pro Val Asn Asn	1175	1180	1185
Tyr Gly Asp Ile Glu Asn Lys Lys Gln Ala Gly Ile Met Leu Tyr	1190	1195	1200
Thr Arg Ala Asn Tyr Thr Ser Gln Thr Asp Pro Ala Thr Gly Trp	1205	1210	1215
Arg Lys Thr Ile Tyr Leu Lys Ala Gly Pro Glu Glu Thr Thr Tyr	1220	1225	1230

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Lys	Lys	Asp	Gly	Lys	Ile	Lys	Asn	Lys	Ser	Val	Lys	Asp	Gln	Ile
1235						1240					1245			
Ile	Glu	Thr	Phe	Thr	Asp	Ile	Gly	Phe	Asp	Gly	Lys	Asp	Tyr	Tyr
1250						1255					1260			
Phe	Glu	Tyr	Asp	Lys	Gly	Glu	Phe	Val	Asp	Glu	Lys	Thr	Gly	Glu
1265						1270					1275			
Ile	Lys	Pro	Lys	Lys	Trp	Arg	Leu	Tyr	Ser	Gly	Glu	Asn	Gly	Lys
1280						1285					1290			
Ser	Leu	Asp	Arg	Phe	Arg	Gly	Glu	Arg	Glu	Lys	Asp	Lys	Tyr	Glu
1295						1300					1305			
Trp	Lys	Ile	Asp	Lys	Ile	Asp	Ile	Val	Lys	Ile	Leu	Asp	Asp	Leu
1310						1315					1320			
Phe	Val	Asn	Phe	Asp	Lys	Asn	Ile	Ser	Leu	Leu	Lys	Gln	Leu	Lys
1325						1330					1335			
Glu	Gly	Val	Glu	Leu	Thr	Arg	Asn	Asn	Glu	His	Gly	Thr	Gly	Glu
1340						1345					1350			
Ser	Leu	Arg	Phe	Ala	Ile	Asn	Leu	Ile	Gln	Gln	Ile	Arg	Asn	Thr
1355						1360					1365			
Gly	Asn	Asn	Glu	Arg	Asp	Asn	Asp	Phe	Ile	Leu	Ser	Pro	Val	Arg
1370						1375					1380			
Asp	Glu	Asn	Gly	Lys	His	Phe	Asp	Ser	Arg	Glu	Tyr	Trp	Asp	Lys
1385						1390					1395			
Glu	Thr	Lys	Gly	Glu	Lys	Ile	Ser	Met	Pro	Ser	Ser	Gly	Asp	Ala
1400						1405					1410			
Asn	Gly	Ala	Phe	Asn	Ile	Ala	Arg	Lys	Gly	Ile	Ile	Met	Asn	Ala
1415						1420					1425			
His	Ile	Leu	Ala	Asn	Ser	Asp	Ser	Lys	Asp	Leu	Ser	Leu	Phe	Val
1430						1435					1440			
Ser	Asp	Glu	Glu	Trp	Asp	Leu	His	Leu	Asn	Asn	Lys	Thr	Glu	Trp
1445						1450					1455			
Lys	Lys	Gln	Leu	Asn	Ile	Phe	Ser	Ser	Arg	Lys	Ala	Met	Ala	Lys
1460						1465					1470			
Arg	Lys	Lys	Lys	Arg	Pro	Ala	Ala	Thr	Lys	Lys				
1475						1480								

<210> SEQ ID NO 16

<211> LENGTH: 1245

<212> TYPE: PRT

<213> ORGANISM: Porphyromonas macacae

<400> SEQUENCE: 16

Met	Lys	Thr	Gln	His	Phe	Phe	Glu	Asp	Phe	Thr	Ser	Leu	Tyr	Ser	Leu
1					5					10				15	
Ser	Lys	Thr	Ile	Arg	Phe	Glu	Leu	Lys	Pro	Ile	Gly	Lys	Thr	Leu	Glu
			20					25					30		
Asn	Ile	Lys	Lys	Asn	Gly	Leu	Ile	Arg	Arg	Asp	Glu	Gln	Arg	Leu	Asp
			35					40					45		
Asp	Tyr	Glu	Lys	Leu	Lys	Lys	Val	Ile	Asp	Glu	Tyr	His	Glu	Asp	Phe
		50				55					60				
Ile	Ala	Asn	Ile	Leu	Ser	Ser	Phe	Ser	Phe	Ser	Glu	Glu	Ile	Leu	Gln
65					70					75				80	
Ser	Tyr	Ile	Gln	Asn	Leu	Ser	Ile	Ser	Glu	Ala	Arg	Ala	Lys	Ile	Glu
				85					90					95	
Lys	Thr	Met	Arg	Asp	Thr	Leu	Ala	Lys	Ala	Phe	Ser	Glu	Asp	Glu	Arg
			100					105						110	

Tyr	Lys	Ser	Ile	Phe	Lys	Lys	Glu	Leu	Val	Lys	Lys	Asp	Ile	Pro	Val
115															
Trp	Cys	Pro	Ala	Tyr	Lys	Ser	Leu	Cys	Lys	Lys	Phe	Asp	Asn	Phe	Thr
130															
Thr	Ser	Leu	Val	Pro	Phe	His	Glu	Asn	Arg	Lys	Asn	Leu	Tyr	Thr	Ser
145															
Asn	Glu	Ile	Thr	Ala	Ser	Ile	Pro	Tyr	Arg	Ile	Val	His	Val	Asn	Leu
165															
Pro	Lys	Phe	Ile	Gln	Asn	Ile	Glu	Ala	Leu	Cys	Glu	Leu	Gln	Lys	Lys
180															
Met	Gly	Ala	Asp	Leu	Tyr	Leu	Glu	Met	Met	Glu	Asn	Leu	Arg	Asn	Val
195															
Trp	Pro	Ser	Phe	Val	Lys	Thr	Pro	Asp	Asp	Leu	Cys	Asn	Leu	Lys	Thr
210															
Tyr	Asn	His	Leu	Met	Val	Gln	Ser	Ser	Ile	Ser	Glu	Tyr	Asn	Arg	Phe
225															
Val	Gly	Gly	Tyr	Ser	Thr	Glu	Asp	Gly	Thr	Lys	His	Gln	Gly	Ile	Asn
245															
Glu	Trp	Ile	Asn	Ile	Tyr	Arg	Gln	Arg	Asn	Lys	Glu	Met	Arg	Leu	Pro
260															
Gly	Leu	Val	Phe	Leu	His	Lys	Gln	Ile	Leu	Ala	Lys	Val	Asp	Ser	Ser
275															
Ser	Phe	Ile	Ser	Asp	Thr	Leu	Glu	Asn	Asp	Asp	Gln	Val	Phe	Cys	Val
290															
Leu	Arg	Gln	Phe	Arg	Lys	Leu	Phe	Trp	Asn	Thr	Val	Ser	Ser	Lys	Glu
305															
Asp	Asp	Ala	Ala	Ser	Leu	Lys	Asp	Leu	Phe	Cys	Gly	Leu	Ser	Gly	Tyr
325															
Asp	Pro	Glu	Ala	Ile	Tyr	Val	Ser	Asp	Ala	His	Leu	Ala	Thr	Ile	Ser
340															
Lys	Asn	Ile	Phe	Asp	Arg	Trp	Asn	Tyr	Ile	Ser	Asp	Ala	Ile	Arg	Arg
355															
Lys	Thr	Glu	Val	Leu	Met	Pro	Arg	Lys	Lys	Glu	Ser	Val	Glu	Arg	Tyr
370															
Ala	Glu	Lys	Ile	Ser	Lys	Gln	Ile	Lys	Lys	Arg	Gln	Ser	Tyr	Ser	Leu
385															
Ala	Glu	Leu	Asp	Asp	Leu	Leu	Ala	His	Tyr	Ser	Glu	Glu	Ser	Leu	Pro
405															
Ala	Gly	Phe	Ser	Leu	Leu	Ser	Tyr	Phe	Thr	Ser	Leu	Gly	Gly	Gln	Lys
420															
Tyr	Leu	Val	Ser	Asp	Gly	Glu	Val	Ile	Leu	Tyr	Glu	Glu	Gly	Ser	Asn
435															
Ile	Trp	Asp	Glu	Val	Leu	Ile	Ala	Phe	Arg	Asp	Leu	Gln	Val	Ile	Leu
450															
Asp	Lys	Asp	Phe	Thr	Glu	Lys	Lys	Leu	Gly	Lys	Asp	Glu	Glu	Ala	Val
465															
Ser	Val	Ile	Lys	Lys	Ala	Leu	Asp	Ser	Ala	Leu	Arg	Leu	Arg	Lys	Phe
485															
Phe	Asp	Leu	Leu	Ser	Gly	Thr	Gly	Ala	Glu	Ile	Arg	Arg	Asp	Ser	Ser
500															
Phe	Tyr	Ala	Leu	Tyr	Thr	Asp	Arg	Met	Asp	Lys	Leu	Lys	Gly	Leu	Leu
515															

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Lys 530	Met	Tyr	Asp	Lys	Val	Arg	Asn	Tyr	Leu	Thr	Lys 540	Lys	Pro	Tyr	Ser
Ile 545	Glu	Lys	Phe	Lys	Leu 550	His	Phe	Asp	Asn	Pro	Ser 555	Leu	Leu	Ser	Gly 560
Trp	Asp	Lys	Asn	Lys 565	Glu	Leu	Asn	Asn	Leu	Ser 570	Val	Ile	Phe	Arg 575	Gln
Asn	Gly	Tyr	Tyr 580	Tyr	Leu	Gly	Ile	Met 585	Thr	Pro	Lys	Gly	Lys 590	Asn	Leu
Phe	Lys	Thr 595	Leu	Pro	Lys	Leu	Gly 600	Ala	Glu	Glu	Met	Phe 605	Tyr	Glu	Lys
Met 610	Glu	Tyr	Lys	Gln	Ile	Ala 615	Glu	Pro	Met	Leu	Met 620	Leu	Pro	Lys	Val
Phe 625	Phe	Pro	Lys	Lys	Thr 630	Lys	Pro	Ala	Phe	Ala 635	Pro	Asp	Gln	Ser	Val 640
Val	Asp	Ile	Tyr	Asn 645	Lys	Lys	Thr	Phe	Lys 650	Thr	Gly	Gln	Lys	Gly 655	Phe
Asn	Lys	Lys	Asp 660	Leu	Tyr	Arg	Leu	Ile 665	Asp	Phe	Tyr	Lys	Glu 670	Ala	Leu
Thr	Val	His 675	Glu	Trp	Lys	Leu	Phe 680	Asn	Phe	Ser	Phe 685	Ser	Pro	Thr	Glu
Gln	Tyr	Arg	Asn	Ile	Gly 690	Glu	Phe 695	Phe	Asp	Glu	Val 700	Arg	Glu	Gln	Ala
Tyr 705	Lys	Val	Ser	Met	Val 710	Asn	Val	Pro	Ala	Ser 715	Tyr	Ile	Asp	Glu	Ala 720
Val	Glu	Asn	Gly	Lys 725	Leu	Tyr	Leu	Phe	Gln 730	Ile	Tyr	Asn	Lys	Asp 735	Phe
Ser	Pro	Tyr	Ser 740	Lys	Gly	Ile	Pro	Asn 745	Leu	His	Thr	Leu	Tyr 750	Trp	Lys
Ala	Leu	Phe 755	Ser	Glu	Gln	Asn	Gln 760	Ser	Arg	Val	Tyr	Lys 765	Leu	Cys	Gly
Gly 770	Gly	Glu	Leu	Phe	Tyr	Arg 775	Lys	Ala	Ser	Leu	His 780	Met	Gln	Asp	Thr
Thr 785	Val	His	Pro	Lys	Gly 790	Ile	Ser	Ile	His 795	Lys	Lys	Asn	Leu	Asn 800	Lys
Lys	Gly	Glu	Thr 805	Ser	Leu	Phe	Asn	Tyr	Asp 810	Leu	Val	Lys	Asp	Lys 815	Arg
Phe	Thr	Glu	Asp 820	Lys	Phe	Phe	Phe	His 825	Val	Pro	Ile	Ser	Ile 830	Asn	Tyr
Lys	Asn	Lys 835	Lys	Ile	Thr	Asn	Val 840	Asn	Gln	Met	Val	Arg 845	Asp	Tyr	Ile
Ala 850	Gln	Asn	Asp	Asp	Leu	Gln 855	His	Gly	Ile	Asp	Arg 860	Gly	Glu	Arg	Asn
Leu 865	Leu	Tyr	Ile	Ser	Arg 870	Ile	Asp	Thr	Arg	Gly 875	Asn	Leu	Leu	Glu 880	Gln
Phe	Ser	Leu	Asn 885	Val	Ile	Glu	Ser	Asp	Lys 890	Gly	Asp	Leu	Arg	Thr 895	Asp
Tyr	Gln	Lys 900	Ile	Leu	Gly	Asp	Arg 905	Glu	Gln	Glu	Arg	Leu	Arg 910	Arg	Arg
Gln 915	Glu	Trp	Lys	Ser	Ile	Glu	Ser 920	Ile	Lys	Asp	Leu	Lys 925	Asp	Gly	Tyr
Met 930	Ser	Gln	Val	Val	His 935	Lys	Ile	Cys	Asn	Met	Val 940	Val	Glu	His	Lys
Ala	Ile	Val	Val	Leu	Glu	Asn	Leu	Asn	Leu	Ser	Phe	Met	Lys	Gly	Arg

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945	950	955	960
Lys Lys Val Glu Lys Ser Val Tyr Glu Lys Phe Glu Arg Met Leu Val			
	965	970	975
Asp Lys Leu Asn Tyr Leu Val Val Asp Lys Lys Asn Leu Ser Asn Glu			
	980	985	990
Pro Gly Gly Leu Tyr Ala Ala Tyr Gln Leu Thr Asn Pro Leu Phe Ser			
	995	1000	1005
Phe Glu Glu Leu His Arg Tyr Pro Gln Ser Gly Ile Leu Phe Phe			
	1010	1015	1020
Val Asp Pro Trp Asn Thr Ser Leu Thr Asp Pro Ser Thr Gly Phe			
	1025	1030	1035
Val Asn Leu Leu Gly Arg Ile Asn Tyr Thr Asn Val Gly Asp Ala			
	1040	1045	1050
Arg Lys Phe Phe Asp Arg Phe Asn Ala Ile Arg Tyr Asp Gly Lys			
	1055	1060	1065
Gly Asn Ile Leu Phe Asp Leu Asp Leu Ser Arg Phe Asp Val Arg			
	1070	1075	1080
Val Glu Thr Gln Arg Lys Leu Trp Thr Leu Thr Thr Phe Gly Ser			
	1085	1090	1095
Arg Ile Ala Lys Ser Lys Lys Ser Gly Lys Trp Met Val Glu Arg			
	1100	1105	1110
Ile Glu Asn Leu Ser Leu Cys Phe Leu Glu Leu Phe Glu Gln Phe			
	1115	1120	1125
Asn Ile Gly Tyr Arg Val Glu Lys Asp Leu Lys Lys Ala Ile Leu			
	1130	1135	1140
Ser Gln Asp Arg Lys Glu Phe Tyr Val Arg Leu Ile Tyr Leu Phe			
	1145	1150	1155
Asn Leu Met Met Gln Ile Arg Asn Ser Asp Gly Glu Glu Asp Tyr			
	1160	1165	1170
Ile Leu Ser Pro Ala Leu Asn Glu Lys Asn Leu Gln Phe Asp Ser			
	1175	1180	1185
Arg Leu Ile Glu Ala Lys Asp Leu Pro Val Asp Ala Asp Ala Asn			
	1190	1195	1200
Gly Ala Tyr Asn Val Ala Arg Lys Gly Leu Met Val Val Gln Arg			
	1205	1210	1215
Ile Lys Arg Gly Asp His Glu Ser Ile His Arg Ile Gly Arg Ala			
	1220	1225	1230
Gln Trp Leu Arg Tyr Val Gln Glu Gly Ile Val Glu			
	1235	1240	1245

<210> SEQ ID NO 17

<211> LENGTH: 1250

<212> TYPE: PRT

<213> ORGANISM: Smithella sp.

<400> SEQUENCE: 17

Met Gln Thr Leu Phe Glu Asn Phe Thr Asn Gln Tyr Pro Val Ser Lys			
1	5	10	15
Thr Leu Arg Phe Glu Leu Ile Pro Gln Gly Lys Thr Lys Asp Phe Ile			
	20	25	30
Glu Gln Lys Gly Leu Leu Lys Lys Asp Glu Asp Arg Ala Glu Lys Tyr			
	35	40	45
Lys Lys Val Lys Asn Ile Ile Asp Glu Tyr His Lys Asp Phe Ile Glu			
	50	55	60

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Lys 65	Ser	Leu	Asn	Gly	Leu 70	Lys	Leu	Asp	Gly	Leu 75	Glu	Lys	Tyr	Lys	Thr 80
Leu	Tyr	Leu	Lys 85	Gln	Glu	Lys	Asp	Asp	Lys 90	Asp	Lys	Lys	Ala	Phe	Asp 95
Lys	Glu	Lys	Glu 100	Asn	Leu	Arg	Lys	Gln	Ile 105	Ala	Asn	Ala	Phe	Arg	Asn 110
Asn	Glu	Lys 115	Phe	Lys	Thr	Leu	Phe	Ala	Lys 120	Glu	Leu	Ile	Lys	Asn	Asp 125
Leu	Met	Ser 130	Phe	Ala	Cys	Glu 135	Glu	Asp	Lys	Lys	Asn 140	Val	Lys	Glu	Phe
Glu	Ala	Phe	Thr	Thr	Tyr 150	Phe	Thr	Gly	Phe	His 155	Gln	Asn	Arg	Ala	Asn 160
Met	Tyr	Val	Ala 165	Asp	Glu	Lys	Arg	Thr	Ala 170	Ile	Ala	Ser	Arg	Leu	Ile 175
His	Glu	Asn	Leu 180	Pro	Lys	Phe	Ile	Asp 185	Asn	Ile	Lys	Ile	Phe	Glu	Lys 190
Met	Lys	Lys 195	Glu	Ala	Pro	Glu	Leu	Leu	Ser 200	Pro	Phe	Asn	Gln	Thr	Leu 205
Lys	Asp	Met 210	Lys	Asp	Val	Ile 215	Lys	Gly	Thr	Thr	Leu 220	Glu	Glu	Ile	Phe
Ser	Leu	Asp	Tyr	Phe	Asn 230	Lys	Thr	Leu	Thr	Gln 235	Ser	Gly	Ile	Asp	Ile 240
Tyr	Asn	Ser	Val 245	Ile	Gly	Gly	Arg	Thr	Pro 250	Glu	Glu	Gly	Lys	Thr	Lys 255
Ile	Lys	Gly	Leu 260	Asn	Glu	Tyr	Ile	Asn 265	Thr	Asp	Phe	Asn	Gln	Lys	Gln 270
Thr	Asp	Lys 275	Lys	Lys	Arg	Gln	Pro 280	Lys	Phe	Lys	Gln	Leu	Tyr	Lys	Gln 285
Ile	Leu	Ser 290	Asp	Arg	Gln	Ser 295	Leu	Ser	Phe	Ile 300	Ala	Glu	Ala	Phe	Lys
Asn	Asp	Thr	Glu	Ile	Leu 310	Glu	Ala	Ile	Glu	Lys 315	Phe	Tyr	Val	Asn	Glu 320
Leu	Leu	His	Phe 325	Ser	Asn	Glu	Gly	Lys	Ser 330	Thr	Asn	Val	Leu	Asp	Ala 335
Ile	Lys	Asn	Ala 340	Val	Ser	Asn	Leu	Glu 345	Ser	Phe	Asn	Leu	Thr	Lys	Met 350
Tyr	Phe	Arg 355	Ser	Gly	Ala	Ser	Leu 360	Thr	Asp	Val	Ser	Arg	Lys	Val	Phe 365
Gly	Glu	Trp 370	Ser	Ile	Ile	Asn 375	Arg	Ala	Leu	Asp	Asn 380	Tyr	Tyr	Ala	Thr
Thr	Tyr	Pro	Ile	Lys	Pro 390	Arg	Glu	Lys	Ser	Glu 395	Lys	Tyr	Glu	Glu	Arg 400
Lys	Glu	Lys	Trp 405	Leu	Lys	Gln	Asp	Phe	Asn 410	Val	Ser	Leu	Ile	Gln	Thr 415
Ala	Ile	Asp	Glu 420	Tyr	Asp	Asn	Glu	Thr	Val 425	Lys	Gly	Lys	Asn	Ser	Gly 430
Lys	Val	Ile	Ala 435	Asp	Tyr	Phe	Ala 440	Lys	Phe	Cys	Asp	Asp	Lys	Glu	Thr 445
Asp	Leu	Ile	Gln	Lys	Val 455	Asn	Glu	Gly	Tyr	Ile 460	Ala	Val	Lys	Asp	Leu 465
Leu	Asn	Thr	Pro	Cys	Pro 470	Glu	Asn	Glu	Lys	Leu 475	Gly	Ser	Asn	Lys	Asp 480
Gln	Val	Lys	Gln	Ile	Lys	Ala	Phe	Met	Asp	Ser	Ile	Met	Asp	Ile	Met

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485								490					495				
His	Phe	Val	Arg	Pro	Leu	Ser	Leu	Lys	Asp	Thr	Asp	Lys	Glu	Lys	Asp		
			500					505					510				
Glu	Thr	Phe	Tyr	Ser	Leu	Phe	Thr	Pro	Leu	Tyr	Asp	His	Leu	Thr	Gln		
		515					520					525					
Thr	Ile	Ala	Leu	Tyr	Asn	Lys	Val	Arg	Asn	Tyr	Leu	Thr	Gln	Lys	Pro		
	530					535					540						
Tyr	Ser	Thr	Glu	Lys	Ile	Lys	Leu	Asn	Phe	Glu	Asn	Ser	Thr	Leu	Leu		
545					550					555					560		
Gly	Gly	Trp	Asp	Leu	Asn	Lys	Glu	Thr	Asp	Asn	Thr	Ala	Ile	Ile	Leu		
			565						570					575			
Arg	Lys	Asp	Asn	Leu	Tyr	Tyr	Leu	Gly	Ile	Met	Asp	Lys	Arg	His	Asn		
			580					585					590				
Arg	Ile	Phe	Arg	Asn	Val	Pro	Lys	Ala	Asp	Lys	Lys	Asp	Phe	Cys	Tyr		
		595					600					605					
Glu	Lys	Met	Val	Tyr	Lys	Leu	Leu	Pro	Gly	Ala	Asn	Lys	Met	Leu	Pro		
	610					615					620						
Lys	Val	Phe	Phe	Ser	Gln	Ser	Arg	Ile	Gln	Glu	Phe	Thr	Pro	Ser	Ala		
625					630					635					640		
Lys	Leu	Leu	Glu	Asn	Tyr	Ala	Asn	Glu	Thr	His	Lys	Lys	Gly	Asp	Asn		
			645						650					655			
Phe	Asn	Leu	Asn	His	Cys	His	Lys	Leu	Ile	Asp	Phe	Phe	Lys	Asp	Ser		
		660						665					670				
Ile	Asn	Lys	His	Glu	Asp	Trp	Lys	Asn	Phe	Asp	Phe	Arg	Phe	Ser	Ala		
	675						680					685					
Thr	Ser	Thr	Tyr	Ala	Asp	Leu	Ser	Gly	Phe	Tyr	His	Glu	Val	Glu	His		
	690					695					700						
Gln	Gly	Tyr	Lys	Ile	Ser	Phe	Gln	Ser	Val	Ala	Asp	Ser	Phe	Ile	Asp		
705					710					715					720		
Asp	Leu	Val	Asn	Glu	Gly	Lys	Leu	Tyr	Leu	Phe	Gln	Ile	Tyr	Asn	Lys		
			725						730					735			
Asp	Phe	Ser	Pro	Phe	Ser	Lys	Gly	Lys	Pro	Asn	Leu	His	Thr	Leu	Tyr		
		740						745					750				
Trp	Lys	Met	Leu	Phe	Asp	Glu	Asn	Asn	Leu	Lys	Asp	Val	Val	Tyr	Lys		
		755					760					765					
Leu	Asn	Gly	Glu	Ala	Glu	Val	Phe	Tyr	Arg	Lys	Lys	Ser	Ile	Ala	Glu		
	770					775					780						
Lys	Asn	Thr	Thr	Ile	His	Lys	Ala	Asn	Glu	Ser	Ile	Ile	Asn	Lys	Asn		
785					790					795					800		
Pro	Asp	Asn	Pro	Lys	Ala	Thr	Ser	Thr	Phe	Asn	Tyr	Asp	Ile	Val	Lys		
			805						810					815			
Asp	Lys	Arg	Tyr	Thr	Ile	Asp	Lys	Phe	Gln	Phe	His	Ile	Pro	Ile	Thr		
		820						825					830				
Met	Asn	Phe	Lys	Ala	Glu	Gly	Ile	Phe	Asn	Met	Asn	Gln	Arg	Val	Asn		
		835					840					845					
Gln	Phe	Leu	Lys	Ala	Asn	Pro	Asp	Ile	Asn	Ile	Ile	Gly	Ile	Asp	Arg		
	850					855					860						
Gly	Glu	Arg	His	Leu	Leu	Tyr	Tyr	Ala	Leu	Ile	Asn	Gln	Lys	Gly	Lys		
865					870					875					880		
Ile	Leu	Lys	Gln	Asp	Thr	Leu	Asn	Val	Ile	Ala	Asn	Glu	Lys	Gln	Lys		
			885						890					895			
Val	Asp	Tyr	His	Asn	Leu	Leu	Asp	Lys	Lys	Glu	Gly	Asp	Arg	Ala	Thr		
			900					905					910				

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Ala Arg Gln Glu Trp Gly Val Ile Glu Thr Ile Lys Glu Leu Lys Glu
 915 920 925

Gly Tyr Leu Ser Gln Val Ile His Lys Leu Thr Asp Leu Met Ile Glu
 930 935 940

Asn Asn Ala Ile Ile Val Met Glu Asp Leu Asn Phe Gly Phe Lys Arg
 945 950 955 960

Gly Arg Gln Lys Val Glu Lys Gln Val Tyr Gln Lys Phe Glu Lys Met
 965 970 975

Leu Ile Asp Lys Leu Asn Tyr Leu Val Asp Lys Asn Lys Lys Ala Asn
 980 985 990

Glu Leu Gly Gly Leu Leu Asn Ala Phe Gln Leu Ala Asn Lys Phe Glu
 995 1000 1005

Ser Phe Gln Lys Met Gly Lys Gln Asn Gly Phe Ile Phe Tyr Val
 1010 1015 1020

Pro Ala Trp Asn Thr Ser Lys Thr Asp Pro Ala Thr Gly Phe Ile
 1025 1030 1035

Asp Phe Leu Lys Pro Arg Tyr Glu Asn Leu Asn Gln Ala Lys Asp
 1040 1045 1050

Phe Phe Glu Lys Phe Asp Ser Ile Arg Leu Asn Ser Lys Ala Asp
 1055 1060 1065

Tyr Phe Glu Phe Ala Phe Asp Phe Lys Asn Phe Thr Glu Lys Ala
 1070 1075 1080

Asp Gly Gly Arg Thr Lys Trp Thr Val Cys Thr Thr Asn Glu Asp
 1085 1090 1095

Arg Tyr Gln Trp Asn Arg Ala Leu Asn Asn Asn Arg Gly Ser Gln
 1100 1105 1110

Glu Lys Tyr Asp Ile Thr Ala Glu Leu Lys Ser Leu Phe Asp Gly
 1115 1120 1125

Lys Val Asp Tyr Lys Ser Gly Lys Asp Leu Lys Gln Gln Ile Ala
 1130 1135 1140

Ser Gln Glu Ser Ala Asp Phe Phe Lys Ala Leu Met Lys Asn Leu
 1145 1150 1155

Ser Ile Thr Leu Ser Leu Arg His Asn Asn Gly Glu Lys Gly Asp
 1160 1165 1170

Asn Glu Gln Asp Tyr Ile Leu Ser Pro Val Ala Asp Ser Lys Gly
 1175 1180 1185

Arg Phe Phe Asp Ser Arg Lys Ala Asp Asp Asp Met Pro Lys Asn
 1190 1195 1200

Ala Asp Ala Asn Gly Ala Tyr His Ile Ala Leu Lys Gly Leu Trp
 1205 1210 1215

Cys Leu Glu Gln Ile Ser Lys Thr Asp Asp Leu Lys Lys Val Lys
 1220 1225 1230

Leu Ala Ile Ser Asn Lys Glu Trp Leu Glu Phe Val Gln Thr Leu
 1235 1240 1245

Lys Gly
 1250

<210> SEQ ID NO 18

<211> LENGTH: 3987

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 18

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atggccggga gcaagaagcg ccggataaag caggacacgc agttcgaggg cttaccaaac	60
ctgtaccaag tctccaagac gctccgggtc gagcttatcc cgcaaggga gaccctgaaa	120
cacatccagg aacaaggttt catcgaggag gacaaggccc gcaacgacca ctacaaggag	180
ctcaagccca taatcgatcg gatctacaag acgtacgccc accagtgcct ccaactggtg	240
cagctcgact gggagaacct gagcgccgcc attgacagct accgcaagga aaagacggag	300
gagacgcgca acgcccctat tgaggagcaa gccacctacc gcaacgccat ccacgactac	360
ttcatcgggc gcaccgacaa cctgacggac gcgatcaaca agcggccagc ggaaatctac	420
aaaggccttt tcaaggccga gctcttcaac ggaagggtcc taaaacagct cgggactgtc	480
acgacaaccg agcatgagaa cgcccctcct cgcagcttcg acaagttcac cacatacttc	540
tcgggctttc accggaaccg caagaacgtt ttcagcgccc aggcacatctc caccgccatc	600
ccgcacagga tcgtccagga caacttcccc aagttcaagg agaactgcc catcttcacg	660
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<210> SEQ ID NO 19

<211> LENGTH: 3987

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 19

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<210> SEQ ID NO 20

<211> LENGTH: 3987

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 20

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<210> SEQ ID NO 21

<211> LENGTH: 1592

<212> TYPE: DNA

<213> ORGANISM: Medicago truncatula

<400> SEQUENCE: 21

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<210> SEQ ID NO 22
<211> LENGTH: 2000
<212> TYPE: DNA
<213> ORGANISM: Zea mays

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<400> SEQUENCE: 22

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gtcgtgcccc tctctagaga taaagagcat tgcgtgtcta aagtataaaa aattaccaca 60
tatttttttg tcacacttat ttgaagtgtg gtttatctat ctctatacat atatttaaac 120
ttcactctac aaataatata gtctataata ctaaaaaat attagtgttt tagaggatca 180
tataaataaa ctgctagaca tgggtctaaag gataattgaa tattttgaca atctacagtt 240
ttatcttttt agtgtgcatg tgatctctct gttttttttg caaatagctt gacctatata 300
atacttcato cattttatta gtacatccat ttaggattta gggttgatgg tttctataga 360
ctaattttta gtacatccat tttattcttt ttagtctcta aattttttaa aactaaaact 420
ctattttagt tttttattta ataatttaga tataaaatga aataaaataa attgactaca 480
aataaaacaa atacccttta agaataaaaa aaactaagca aacatttttc ttgtttcgag 540
tagataatga caggctgttc aacgcgctcg acgagtctaa cggacaccaa ccagcgaacc 600
agcagcgtcg cgtcgggcca agcgaagcag acggcacggc atctctgtag ctgcctctgg 660
acccctctcg agagtccgcg tccaccgttg gacttgctcc gctgtcggca tccagaaatt 720
gCGTggcgga gCGgcagacg tgaggcggca cggcaggcgg cctcttcctc ctctcacggc 780
acCGgcagct acgggggatt cctttcccaC cGctccttcg ctttccttc ctgcCCgc 840
gtaataaata gacacccctt ccacaccctt tttcccaac ctCGtggtcg ttCGgagcgc 900
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cCGctcatcc tccccccccc cctctctcta ccttctctag atCGcgatc cGgtccatgg 1020
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atgtttgtga tgatgtggtc tggttgggCG gtcgttctag atCGgagtag gatactgttt 1140
caagctacct ggtggattta ttaattttgt atctgtatgt gtgtgccata catcttcata 1200
gttacgagtt taagatgatg gatggaaata tcgatctagg atagggtatac atgttgatgc 1260
gggttttact gatgcataa cagagatgct ttttttctcg cttgggtgtg atgatatggt 1320
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attaaggat aaagggtcgt tctagatcgg agtagaatac tgtttcaaac tacctggtgg 1440
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ggatggaaat atcgatctag gataggata catgttgatg cgggttttac tgatgcatac 1560
acagagatgc tttttttcgc ttggttgtga tgatgtggtc tggttgggCG gtcgttctag 1620
atcggagtag aatactgttt caaactacct ggtggattta ttaattttgt atctttatgt 1680
gtgtgccata catcttcata gttacgagtt taagatgatg gatggaaata ttgatctagg 1740
atagggtatac atgttgatgt ggggttttact gatgcataa catgatggca tatgcggcat 1800

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ctattcatat gctctaacct tgagtaccta tctattataa taaacaagta tgttttataa 1860
ttattttgat cttgatatac ttggatgatg gcatatgcag cagctatatg tggatttttt 1920
agccctgcct tcatacgeta tttatttgct tggtagtggt tcttttgtcc gatgctcacc 1980
ctggtggttg gtgatacttc 2000

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<210> SEQ ID NO 23
<211> LENGTH: 228
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus

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<400> SEQUENCE: 23

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Ser Ser Glu Thr Gly Pro Val Ala Val Asp Pro Thr Leu Arg Arg Arg
1      5      10      15
Ile Glu Pro His Glu Phe Glu Val Phe Phe Asp Pro Arg Glu Leu Arg
20     25     30
Lys Glu Thr Cys Leu Leu Tyr Glu Ile Asn Trp Gly Gly Arg His Ser
35     40     45
Ile Trp Arg His Thr Ser Gln Asn Thr Asn Lys His Val Glu Val Asn
50     55     60
Phe Ile Glu Lys Phe Thr Thr Glu Arg Tyr Phe Cys Pro Asn Thr Arg
65     70     75     80
Cys Ser Ile Thr Trp Phe Leu Ser Trp Ser Pro Cys Gly Glu Cys Ser
85     90     95
Arg Ala Ile Thr Glu Phe Leu Ser Arg Tyr Pro His Val Thr Leu Phe
100    105    110
Ile Tyr Ile Ala Arg Leu Tyr His His Ala Asp Pro Arg Asn Arg Gln
115    120    125
Gly Leu Arg Asp Leu Ile Ser Ser Gly Val Thr Ile Gln Ile Met Thr
130    135    140
Glu Gln Glu Ser Gly Tyr Cys Trp Arg Asn Phe Val Asn Tyr Ser Pro
145    150    155    160
Ser Asn Glu Ala His Trp Pro Arg Tyr Pro His Leu Trp Val Arg Leu
165    170    175
Tyr Val Leu Glu Leu Tyr Cys Ile Ile Leu Gly Leu Pro Pro Cys Leu
180    185    190
Asn Ile Leu Arg Arg Lys Gln Pro Gln Leu Thr Phe Phe Thr Ile Ala
195    200    205
Leu Gln Ser Cys His Tyr Gln Arg Leu Pro Pro His Ile Leu Trp Ala
210    215    220
Thr Gly Leu Lys
225

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<210> SEQ ID NO 24
<211> LENGTH: 199
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 24

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Met Glu Ala Ser Pro Ala Ser Gly Pro Arg His Leu Met Asp Pro His
1      5      10      15
Ile Phe Thr Ser Asn Phe Asn Asn Gly Ile Gly Arg His Lys Thr Tyr
20     25     30
Leu Cys Tyr Glu Val Glu Arg Leu Asp Asn Gly Thr Ser Val Lys Met
35     40     45
Asp Gln His Arg Gly Phe Leu His Asn Gln Ala Lys Asn Leu Leu Cys

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50	55	60
Gly Phe Tyr Gly Arg His Ala Glu Leu Arg Phe Leu Asp Leu Val Pro		
65	70	75 80
Ser Leu Gln Leu Asp Pro Ala Gln Ile Tyr Arg Val Thr Trp Phe Ile		
	85	90 95
Ser Trp Ser Pro Cys Phe Ser Trp Gly Cys Ala Gly Glu Val Arg Ala		
	100	105 110
Phe Leu Gln Glu Asn Thr His Val Arg Leu Arg Ile Phe Ala Ala Arg		
	115	120 125
Ile Tyr Asp Tyr Asp Pro Leu Tyr Lys Glu Ala Leu Gln Met Leu Arg		
	130	135 140
Asp Ala Gly Ala Gln Val Ser Ile Met Thr Tyr Asp Glu Phe Lys His		
	145	150 155 160
Cys Trp Asp Thr Phe Val Asp His Gln Gly Cys Pro Phe Gln Pro Trp		
	165	170 175
Asp Gly Leu Asp Glu His Ser Gln Ala Leu Ser Gly Arg Leu Arg Ala		
	180	185 190
Ile Leu Gln Asn Gln Gly Asn		
	195	

<210> SEQ ID NO 25

<211> LENGTH: 621

<212> TYPE: DNA

<213> ORGANISM: Petromyzon marinus

<400> SEQUENCE: 25

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acagatgcag agtatgtgag aattcacgaa aagctggaca tctatacctt caagaagcag      60
ttctttaaca ataagaagtc tgtgagccat aggtgctacg tgctgttcga gctgaagaga      120
aggggtgaaa gaagggcatg tttttggggg tatgctgtga acaagcccca gtctggaact      180
gagagaggca ttcacgccga aattttcagc atcagaaaagg tggaggaata cctgagggat      240
aaccctggac agtttacaat taattggtat tctagctggt ctccatgcgc tgactgtgcc      300
gagaagatcc tggaatggta caaccaggag ctgagaggaa atggccatac cctgaagatt      360
tgggcctgca agctgtacta tgaagaagaac gcaagaaatc agatcggaact gtggaacctg      420
agggataatg gtgtggggct gaacgtgatg gtgtccgagc actatcagtg ctgtagaaag      480
attttcattc agtcctcaca taatcagctg aacgagaata gatggctgga aaagactctg      540
aagagggctg agaagagaag gtccgaactg tcaattatga tccaggtgaa gatcctgcac      600
accactaagt cacctgccgt g                                     621

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<210> SEQ ID NO 26

<211> LENGTH: 160

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 26

Phe Glu Arg Asn Tyr Asp Pro Arg Glu Leu Arg Lys Glu Thr Tyr Leu		
1	5	10 15
Leu Tyr Glu Ile Lys Trp Gly Lys Ser Gly Lys Leu Trp Arg His Trp		
	20	25 30
Cys Gln Asn Asn Arg Thr Gln His Ala Glu Val Tyr Phe Leu Glu Asn		
	35	40 45
Ile Phe Asn Ala Arg Arg Phe Asn Pro Ser Thr His Cys Ser Ile Thr		

-continued

50	55	60	
Trp Tyr Leu Ser Trp Ser Pro Cys Ala Glu Cys Ser Gln Lys Ile Val			
65	70	75	80
Asp Phe Leu Lys Glu His Pro Asn Val Leu Glu Ile Tyr Val Ala Arg			
	85	90	95
Leu Tyr Tyr His Glu Asp Glu Arg Asn Arg Gln Gly Leu Arg Asp Leu			
	100	105	110
Val Asn Ser Gly Val Thr Ile Arg Ile Met Asp Leu Pro Asp Tyr Asn			
	115	120	125
Tyr Cys Trp Lys Thr Phe Val Ser Asp Gln Gly Gly Asp Glu Asp Tyr			
	130	135	140
Trp Pro Gly His Phe Ala Pro Trp Ile Lys Gln Tyr Ser Leu Lys Leu			
145	150	155	160

<210> SEQ ID NO 27
 <211> LENGTH: 207
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 27

Thr Asp Ala Glu Tyr Val Arg Ile His Glu Lys Leu Asp Ile Tyr Thr			
1	5	10	15
Phe Lys Lys Gln Phe Ser Asn Asn Lys Lys Ser Val Ser His Arg Cys			
	20	25	30
Tyr Val Leu Phe Glu Leu Lys Arg Arg Gly Glu Arg Arg Ala Cys Phe			
	35	40	45
Trp Gly Tyr Ala Val Asn Lys Pro Gln Ser Gly Thr Glu Arg Gly Ile			
	50	55	60
His Ala Glu Ile Phe Ser Ile Arg Lys Val Glu Glu Tyr Leu Arg Asp			
65	70	75	80
Asn Pro Gly Gln Phe Thr Ile Asn Trp Tyr Ser Ser Trp Ser Pro Cys			
	85	90	95
Ala Asp Cys Ala Glu Lys Ile Leu Glu Trp Tyr Asn Gln Glu Leu Arg			
	100	105	110
Gly Asn Gly His Thr Leu Lys Ile Trp Val Cys Lys Leu Tyr Tyr Glu			
	115	120	125
Lys Asn Ala Arg Asn Gln Ile Gly Leu Trp Asn Leu Arg Asp Asn Gly			
	130	135	140
Val Gly Leu Asn Val Met Val Ser Glu His Tyr Gln Cys Cys Arg Lys			
145	150	155	160
Ile Phe Ile Gln Ser Ser His Asn Gln Leu Asn Glu Asn Arg Trp Leu			
	165	170	175
Glu Lys Thr Leu Lys Arg Ala Glu Lys Arg Arg Ser Glu Leu Ser Ile			
	180	185	190
Met Phe Gln Val Lys Ile Leu His Thr Thr Lys Ser Pro Ala Val			
	195	200	205

<210> SEQ ID NO 28
 <211> LENGTH: 228
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 28

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Ser Ser Lys Thr Gly Pro Val Ala Val Asp Pro Thr Leu Arg Arg Arg
1      5      10      15

Ile Glu Pro His Glu Phe Glu Val Phe Phe Asp Pro Arg Glu Leu Arg
      20      25      30

Lys Glu Thr Cys Leu Leu Tyr Glu Ile Asn Trp Gly Gly Arg His Ser
      35      40      45

Ile Trp Arg His Thr Ser Gln Asn Thr Asn Lys His Val Glu Val Asn
50      55      60

Phe Ile Glu Lys Phe Thr Thr Glu Arg Tyr Phe Cys Pro Asn Thr Arg
65      70      75      80

Cys Ser Ile Thr Trp Phe Leu Ser Trp Ser Pro Cys Gly Glu Cys Ser
      85      90      95

Arg Ala Ile Thr Glu Phe Leu Ser Arg Tyr Pro Asn Val Thr Leu Phe
100      105      110

Ile Tyr Ile Ala Arg Leu Tyr His Leu Ala Asn Pro Arg Asn Arg Gln
115      120      125

Gly Leu Arg Asp Leu Ile Ser Ser Gly Val Thr Ile Gln Ile Met Thr
130      135      140

Glu Gln Glu Ser Gly Tyr Cys Trp His Asn Phe Val Asn Tyr Ser Pro
145      150      155      160

Ser Asn Glu Ser His Trp Pro Arg Tyr Pro His Leu Trp Val Arg Leu
165      170      175

Tyr Val Leu Glu Leu Tyr Cys Ile Ile Leu Gly Leu Pro Pro Cys Leu
180      185      190

Asn Ile Leu Arg Arg Lys Gln Ser Gln Leu Thr Ser Phe Thr Ile Ala
195      200      205

Leu Gln Ser Cys His Tyr Gln Arg Leu Pro Pro His Ile Leu Trp Ala
210      215      220

Thr Gly Leu Lys
225

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<210> SEQ ID NO 29
<211> LENGTH: 162
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 29

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Ser Phe Glu Arg Asn Tyr Asp Pro Arg Glu Leu Arg Lys Glu Thr Tyr
1      5      10      15

Leu Leu Tyr Glu Ile Lys Trp Gly Lys Ser Gly Lys Leu Trp Arg His
      20      25      30

Trp Cys Gln Asn Asn Arg Thr Gln His Ala Glu Val Tyr Phe Leu Glu
      35      40      45

Asn Ile Phe Asn Ala Arg Arg Phe Asn Pro Ser Thr His Cys Ser Ile
50      55      60

Thr Trp Tyr Leu Ser Trp Ser Pro Cys Ala Glu Cys Ser Gln Lys Ile
65      70      75      80

Val Asp Phe Leu Lys Glu His Pro Asn Val Asn Leu Glu Ile Tyr Val
      85      90      95

Ala Arg Leu Tyr Tyr Pro Glu Asn Glu Arg Asn Arg Gln Gly Leu Arg
100      105      110

Asp Leu Val Asn Ser Gly Val Thr Ile Arg Ile Met Asp Leu Pro Asp
115      120      125

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Tyr Asn Tyr Cys Trp Lys Thr Phe Val Ser Asp Gln Gly Gly Asp Glu
130 135 140

Asp Tyr Trp Pro Gly His Phe Ala Pro Trp Ile Lys Gln Tyr Ser Leu
145 150 155 160

Lys Leu

<210> SEQ ID NO 30

<211> LENGTH: 166

<212> TYPE: PRT

<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 30

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr
1 5 10 15

Leu Ala Lys Arg Ala Trp Asp Glu Arg Glu Val Pro Val Gly Ala Val
20 25 30

Leu Val His Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Pro Ile
35 40 45

Gly Arg His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln
50 55 60

Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr
65 70 75 80

Val Thr Leu Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser
85 90 95

Arg Ile Gly Arg Val Val Phe Gly Ala Arg Asp Ala Lys Thr Gly Ala
100 105 110

Ala Gly Ser Leu Met Asp Val Leu His His Pro Gly Met Asn His Arg
115 120 125

Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu Leu
130 135 140

Ser Asp Phe Phe Arg Met Arg Arg Gln Glu Ile Lys Ala Gln Lys Lys
145 150 155 160

Ala Gln Ser Ser Thr Asp
165

<210> SEQ ID NO 31

<211> LENGTH: 166

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 31

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr
1 5 10 15

Leu Ala Lys Arg Ala Arg Asp Glu Arg Glu Val Pro Val Gly Ala Val
20 25 30

Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala Ile
35 40 45

Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln
50 55 60

Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr
65 70 75 80

Val Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser
85 90 95

Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ala Lys Thr Gly Ala
100 105 110

-continued

Ala Gly Ser Leu Met Asp Val Leu His Tyr Pro Gly Met Asn His Arg
 115 120 125

Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu Leu
 130 135 140

Cys Tyr Phe Phe Arg Met Pro Arg Gln Val Phe Asn Ala Gln Lys Lys
 145 150 155 160

Ala Gln Ser Ser Thr Asp
 165

<210> SEQ ID NO 32
 <211> LENGTH: 166
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 32

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr
 1 5 10 15

Leu Ala Lys Arg Ala Trp Asp Glu Arg Glu Val Pro Val Gly Ala Val
 20 25 30

Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ser Ile
 35 40 45

Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln
 50 55 60

Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr
 65 70 75 80

Val Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser
 85 90 95

Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ala Lys Thr Gly Ala
 100 105 110

Ala Gly Ser Leu Met Asp Val Leu His Tyr Pro Gly Met Asn His Arg
 115 120 125

Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu Leu
 130 135 140

Cys Tyr Phe Phe Arg Met Arg Arg Gln Val Phe Asn Ala Gln Lys Lys
 145 150 155 160

Ala Gln Ser Ser Thr Asp
 165

<210> SEQ ID NO 33
 <211> LENGTH: 166
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 33

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr
 1 5 10 15

Leu Ala Lys Arg Ala Leu Asp Glu Arg Glu Val Pro Val Gly Ala Val
 20 25 30

Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala Ile
 35 40 45

Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln
 50 55 60

Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr

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65	70	75	80
Val Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser			
	85	90	95
Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ala Lys Thr Gly Ala			
	100	105	110
Ala Gly Ser Leu Met Asp Val Leu His Tyr Pro Gly Met Asn His Arg			
	115	120	125
Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Asn Ala Leu Leu			
	130	135	140
Cys Tyr Phe Phe Arg Met Arg Arg Gln Val Phe Asn Ala Gln Lys Lys			
145	150	155	160
Ala Gln Ser Ser Thr Asp			
	165		

<210> SEQ ID NO 34
 <211> LENGTH: 166
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 34

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr			
1	5	10	15
Leu Ala Lys Arg Ala Leu Asp Glu Arg Glu Val Pro Val Gly Ala Val			
	20	25	30
Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala Ile			
	35	40	45
Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln			
	50	55	60
Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr			
65	70	75	80
Val Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser			
	85	90	95
Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ala Lys Thr Gly Ala			
	100	105	110
Ala Gly Ser Leu Met Asp Val Leu His Tyr Pro Gly Met Asn His Arg			
	115	120	125
Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Asn Ala Leu Leu			
	130	135	140
Cys Tyr Phe Phe Arg Met Pro Arg Gln Val Phe Asn Ala Gln Lys Lys			
145	150	155	160
Ala Gln Ser Ser Thr Asp			
	165		

<210> SEQ ID NO 35
 <211> LENGTH: 1763
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 35

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr			
1	5	10	15
Leu Ala Lys Arg Ala Trp Asp Glu Arg Glu Val Pro Val Gly Ala Val			
	20	25	30

Leu	Val	His	Asn	Asn	Arg	Val	Ile	Gly	Glu	Gly	Trp	Asn	Arg	Pro	Ile
35						40						45			
Gly	Arg	His	Asp	Pro	Thr	Ala	His	Ala	Glu	Ile	Met	Ala	Leu	Arg	Gln
50						55			60						
Gly	Gly	Leu	Val	Met	Gln	Asn	Tyr	Arg	Leu	Ile	Asp	Ala	Thr	Leu	Tyr
65			70						75			80			
Val	Thr	Leu	Glu	Pro	Cys	Val	Met	Cys	Ala	Gly	Ala	Met	Ile	His	Ser
			85						90			95			
Arg	Ile	Gly	Arg	Val	Val	Phe	Gly	Ala	Arg	Asp	Ala	Lys	Thr	Gly	Ala
			100			105						110			
Ala	Gly	Ser	Leu	Met	Asp	Val	Leu	His	His	Pro	Gly	Met	Asn	His	Arg
115						120						125			
Val	Glu	Ile	Thr	Glu	Gly	Ile	Leu	Ala	Asp	Glu	Cys	Ala	Ala	Leu	Leu
130						135			140						
Ser	Asp	Phe	Phe	Arg	Met	Arg	Arg	Gln	Glu	Ile	Lys	Ala	Gln	Lys	Lys
145			150						155			160			
Ala	Gln	Ser	Ser	Thr	Asp	Ser	Gly	Gly	Ser	Ser	Gly	Gly	Ser	Ser	Gly
			165						170			175			
Ser	Glu	Thr	Pro	Gly	Thr	Ser	Glu	Ser	Ala	Thr	Pro	Glu	Ser	Ser	Gly
			180			185						190			
Gly	Ser	Ser	Gly	Gly	Ser	Ser	Glu	Val	Glu	Phe	Ser	His	Glu	Tyr	Trp
195						200						205			
Met	Arg	His	Ala	Leu	Thr	Leu	Ala	Lys	Arg	Ala	Arg	Asp	Glu	Arg	Glu
210						215			220						
Val	Pro	Val	Gly	Ala	Val	Leu	Val	Leu	Asn	Asn	Arg	Val	Ile	Gly	Glu
225			230						235			240			
Gly	Trp	Asn	Arg	Ala	Ile	Gly	Leu	His	Asp	Pro	Thr	Ala	His	Ala	Glu
			245						250			255			
Ile	Met	Ala	Leu	Arg	Gln	Gly	Gly	Leu	Val	Met	Gln	Asn	Tyr	Arg	Leu
			260			265						270			
Ile	Asp	Ala	Thr	Leu	Tyr	Val	Thr	Phe	Glu	Pro	Cys	Val	Met	Cys	Ala
275						280						285			
Gly	Ala	Met	Ile	His	Ser	Arg	Ile	Gly	Arg	Val	Val	Phe	Gly	Val	Arg
290						295			300						
Asn	Ala	Lys	Thr	Gly	Ala	Ala	Gly	Ser	Leu	Met	Asp	Val	Leu	His	Tyr
305			310						315			320			
Pro	Gly	Met	Asn	His	Arg	Val	Glu	Ile	Thr	Glu	Gly	Ile	Leu	Ala	Asp
			325			330						335			
Glu	Cys	Ala	Ala	Leu	Leu	Cys	Tyr	Phe	Phe	Arg	Met	Pro	Arg	Gln	Val
			340			345						350			
Phe	Asn	Ala	Gln	Lys	Lys	Ala	Gln	Ser	Ser	Thr	Asp	Ser	Gly	Gly	Ser
355			360						365						
Ser	Gly	Gly	Ser	Ser	Gly	Ser	Glu	Thr	Pro	Gly	Thr	Ser	Glu	Ser	Ala
370			375						380						
Thr	Pro	Glu	Ser	Ser	Gly	Gly	Ser	Ser	Gly	Gly	Ser	Asp	Lys	Lys	Tyr
385			390						395			400			
Ser	Ile	Gly	Leu	Ala	Ile	Gly	Thr	Asn	Ser	Val	Gly	Trp	Ala	Val	Ile
			405			410						415			
Thr	Asp	Glu	Tyr	Lys	Val	Pro	Ser	Lys	Lys	Phe	Lys	Val	Leu	Gly	Asn
			420			425						430			
Thr	Asp	Arg	His	Ser	Ile	Lys	Lys	Asn	Leu	Ile	Gly	Ala	Leu	Leu	Phe
435			440			445									
Asp	Ser	Gly	Glu	Thr	Ala	Glu	Ala	Thr</							

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450				455				460							
Arg 465	Arg	Tyr	Thr	Arg	Arg 470	Lys	Asn	Arg	Ile	Cys 475	Tyr	Leu	Gln	Glu	Ile 480
Phe	Ser	Asn	Glu	Met 485	Ala	Lys	Val	Asp 490	Asp	Ser	Phe	Phe	His	Arg 495	Leu
Glu	Glu	Ser	Phe	Leu 500	Val	Glu	Glu	Asp 505	Lys	Lys	His	Glu	Arg 510	His	Pro
Ile	Phe	Gly 515	Asn	Ile	Val	Asp 520	Glu	Val	Ala	Tyr	His	Glu 525	Lys	Tyr	Pro
Thr	Ile 530	Tyr	His	Leu	Arg	Lys 535	Lys	Leu	Val	Asp 540	Ser	Thr	Asp	Lys	Ala
Asp 545	Leu	Arg	Leu	Ile	Tyr 550	Leu	Ala	Leu	Ala	His 555	Met	Ile	Lys	Phe	Arg 560
Gly	His	Phe	Leu	Ile 565	Glu	Gly	Asp	Leu	Asn 570	Pro	Asp	Asn	Ser	Asp 575	Val
Asp	Lys	Leu	Phe	Ile 580	Gln	Leu	Val	Gln 585	Thr	Tyr	Asn	Gln	Leu 590	Phe	Glu
Glu	Asn	Pro 595	Ile	Asn	Ala	Ser	Gly 600	Val	Asp	Ala	Lys	Ala 605	Ile	Leu	Ser
Ala	Arg 610	Leu	Ser	Lys	Ser	Arg 615	Arg	Leu	Glu	Asn	Leu 620	Ile	Ala	Gln	Leu
Pro 625	Gly	Glu	Lys	Lys	Asn 630	Gly	Leu	Phe	Gly	Asn 635	Leu	Ile	Ala	Leu	Ser 640
Leu	Gly	Leu	Thr	Pro 645	Asn	Phe	Lys	Ser	Asn 650	Phe	Asp	Leu	Ala	Glu 655	Asp
Ala	Lys	Leu	Gln 660	Leu	Ser	Lys	Asp	Thr 665	Tyr	Asp	Asp	Asp	Leu 670	Asp	Asn
Leu	Leu	Ala 675	Gln	Ile	Gly	Asp	Gln 680	Tyr	Ala	Asp	Leu	Phe 685	Leu	Ala	Ala
Lys 690	Asn	Leu	Ser	Asp	Ala	Ile 695	Leu	Leu	Ser	Asp	Ile 700	Leu	Arg	Val	Asn
Thr 705	Glu	Ile	Thr	Lys	Ala 710	Pro	Leu	Ser	Ala	Ser 715	Met	Ile	Lys	Arg	Tyr 720
Asp	Glu	His	His	Gln 725	Asp	Leu	Thr	Leu	Leu 730	Lys	Ala	Leu	Val	Arg 735	Gln
Gln	Leu	Pro	Glu	Lys	Tyr	Lys	Glu	Ile 745	Phe	Phe	Asp	Gln	Ser 750	Lys	Asn
Gly	Tyr	Ala 755	Gly	Tyr	Ile	Asp	Gly 760	Gly	Ala	Ser	Gln	Glu 765	Glu	Phe	Tyr
Lys 770	Phe	Ile	Lys	Pro	Ile	Leu 775	Glu	Lys	Met	Asp	Gly 780	Thr	Glu	Glu	Leu
Leu 785	Val	Lys	Leu	Asn	Arg 790	Glu	Asp	Leu	Leu	Arg 795	Lys	Gln	Arg	Thr	Phe 800
Asp	Asn	Gly	Ser	Ile 805	Pro	His	Gln	Ile	His 810	Leu	Gly	Glu	Leu	His 815	Ala
Ile	Leu	Arg	Arg	Gln 820	Glu	Asp	Phe	Tyr	Pro 825	Phe	Leu	Lys	Asp 830	Asn	Arg
Glu	Lys	Ile 835	Glu	Lys	Ile	Leu	Thr 840	Phe	Arg	Ile	Pro	Tyr 845	Tyr	Val	Gly
Pro 850	Leu	Ala	Arg	Gly	Asn 855	Ser	Arg	Phe	Ala	Trp	Met 860	Thr	Arg	Lys	Ser
Glu 865	Glu	Thr	Ile	Thr	Pro 870	Trp	Asn	Phe	Glu	Glu	Val	Val	Asp	Lys	Gly 880

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Ala Ser Ala Gln Ser Phe Ile Glu Arg Met Thr Asn Phe Asp Lys Asn	885	890	895
Leu Pro Asn Glu Lys Val Leu Pro Lys His Ser Leu Leu Tyr Glu Tyr	900	905	910
Phe Thr Val Tyr Asn Glu Leu Thr Lys Val Lys Tyr Val Thr Glu Gly	915	920	925
Met Arg Lys Pro Ala Phe Leu Ser Gly Glu Gln Lys Lys Ala Ile Val	930	935	940
Asp Leu Leu Phe Lys Thr Asn Arg Lys Val Thr Val Lys Gln Leu Lys	945	950	955
Glu Asp Tyr Phe Lys Lys Ile Glu Cys Phe Asp Ser Val Glu Ile Ser	965	970	975
Gly Val Glu Asp Arg Phe Asn Ala Ser Leu Gly Thr Tyr His Asp Leu	980	985	990
Leu Lys Ile Ile Lys Asp Lys Asp Phe Leu Asp Asn Glu Glu Asn Glu	995	1000	1005
Asp Ile Leu Glu Asp Ile Val Leu Thr Leu Thr Leu Phe Glu Asp	1010	1015	1020
Arg Glu Met Ile Glu Glu Arg Leu Lys Thr Tyr Ala His Leu Phe	1025	1030	1035
Asp Asp Lys Val Met Lys Gln Leu Lys Arg Arg Arg Tyr Thr Gly	1040	1045	1050
Trp Gly Arg Leu Ser Arg Lys Leu Ile Asn Gly Ile Arg Asp Lys	1055	1060	1065
Gln Ser Gly Lys Thr Ile Leu Asp Phe Leu Lys Ser Asp Gly Phe	1070	1075	1080
Ala Asn Arg Asn Phe Met Gln Leu Ile His Asp Asp Ser Leu Thr	1085	1090	1095
Phe Lys Glu Asp Ile Gln Lys Ala Gln Val Ser Gly Gln Gly Asp	1100	1105	1110
Ser Leu His Glu His Ile Ala Asn Leu Ala Gly Ser Pro Ala Ile	1115	1120	1125
Lys Lys Gly Ile Leu Gln Thr Val Lys Val Val Asp Glu Leu Val	1130	1135	1140
Lys Val Met Gly Arg His Lys Pro Glu Asn Ile Val Ile Glu Met	1145	1150	1155
Ala Arg Glu Asn Gln Thr Thr Gln Lys Gly Gln Lys Asn Ser Arg	1160	1165	1170
Glu Arg Met Lys Arg Ile Glu Glu Gly Ile Lys Glu Leu Gly Ser	1175	1180	1185
Gln Ile Leu Lys Glu His Pro Val Glu Asn Thr Gln Leu Gln Asn	1190	1195	1200
Glu Lys Leu Tyr Leu Tyr Tyr Leu Gln Asn Gly Arg Asp Met Tyr	1205	1210	1215
Val Asp Gln Glu Leu Asp Ile Asn Arg Leu Ser Asp Tyr Asp Val	1220	1225	1230
Asp His Ile Val Pro Gln Ser Phe Leu Lys Asp Asp Ser Ile Asp	1235	1240	1245
Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg Gly Lys Ser Asp	1250	1255	1260
Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys Asn Tyr Trp	1265	1270	1275

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Arg Gln Leu Leu Asn Ala Lys	Leu Ile Thr Gln Arg	Lys Phe Asp
1280	1285	1290
Asn Leu Thr Lys Ala Glu Arg	Gly Gly Leu Ser Glu	Leu Asp Lys
1295	1300	1305
Ala Gly Phe Ile Lys Arg Gln	Leu Val Glu Thr Arg	Gln Ile Thr
1310	1315	1320
Lys His Val Ala Gln Ile Leu	Asp Ser Arg Met Asn	Thr Lys Tyr
1325	1330	1335
Asp Glu Asn Asp Lys Leu Ile	Arg Glu Val Lys Val	Ile Thr Leu
1340	1345	1350
Lys Ser Lys Leu Val Ser Asp	Phe Arg Lys Asp Phe	Gln Phe Tyr
1355	1360	1365
Lys Val Arg Glu Ile Asn Asn	Tyr His His Ala His	Asp Ala Tyr
1370	1375	1380
Leu Asn Ala Val Val Gly Thr	Ala Leu Ile Lys Lys	Tyr Pro Lys
1385	1390	1395
Leu Glu Ser Glu Phe Val Tyr	Gly Asp Tyr Lys Val	Tyr Asp Val
1400	1405	1410
Arg Lys Met Ile Ala Lys Ser	Glu Gln Glu Ile Gly	Lys Ala Thr
1415	1420	1425
Ala Lys Tyr Phe Phe Tyr Ser	Asn Ile Met Asn Phe	Phe Lys Thr
1430	1435	1440
Glu Ile Thr Leu Ala Asn Gly	Glu Ile Arg Lys Arg	Pro Leu Ile
1445	1450	1455
Glu Thr Asn Gly Glu Thr Gly	Glu Ile Val Trp Asp	Lys Gly Arg
1460	1465	1470
Asp Phe Ala Thr Val Arg Lys	Val Leu Ser Met Pro	Gln Val Asn
1475	1480	1485
Ile Val Lys Lys Thr Glu Val	Gln Thr Gly Gly Phe	Ser Lys Glu
1490	1495	1500
Ser Ile Leu Pro Lys Arg Asn	Ser Asp Lys Leu Ile	Ala Arg Lys
1505	1510	1515
Lys Asp Trp Asp Pro Lys Lys	Tyr Gly Gly Phe Asp	Ser Pro Thr
1520	1525	1530
Val Ala Tyr Ser Val Leu Val	Val Ala Lys Val Glu	Lys Gly Lys
1535	1540	1545
Ser Lys Lys Leu Lys Ser Val	Lys Glu Leu Leu Gly	Ile Thr Ile
1550	1555	1560
Met Glu Arg Ser Ser Phe Glu	Lys Asn Pro Ile Asp	Phe Leu Glu
1565	1570	1575
Ala Lys Gly Tyr Lys Glu Val	Lys Lys Asp Leu Ile	Ile Lys Leu
1580	1585	1590
Pro Lys Tyr Ser Leu Phe Glu	Leu Glu Asn Gly Arg	Lys Arg Met
1595	1600	1605
Leu Ala Ser Ala Gly Glu Leu	Gln Lys Gly Asn Glu	Leu Ala Leu
1610	1615	1620
Pro Ser Lys Tyr Val Asn Phe	Leu Tyr Leu Ala Ser	His Tyr Glu
1625	1630	1635
Lys Leu Lys Gly Ser Pro Glu	Asp Asn Glu Gln Lys	Gln Leu Phe
1640	1645	1650
Val Glu Gln His Lys His Tyr	Leu Asp Glu Ile Ile	Glu Gln Ile
1655	1660	1665
Ser Glu Phe Ser Lys Arg Val	Ile Leu Ala Asp Ala	Asn Leu Asp

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1670	1675	1680
Lys Val Leu Ser Ala Tyr Asn Lys His Arg Asp Lys Pro Ile Arg		
1685	1690	1695
Glu Gln Ala Glu Asn Ile Ile His Leu Phe Thr Leu Thr Asn Leu		
1700	1705	1710
Gly Ala Pro Ala Ala Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg		
1715	1720	1725
Lys Arg Tyr Thr Ser Thr Lys Glu Val Leu Asp Ala Thr Leu Ile		
1730	1735	1740
His Gln Ser Ile Thr Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser		
1745	1750	1755
Gln Leu Gly Gly Asp		
1760		

<210> SEQ ID NO 36
 <211> LENGTH: 1565
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 36

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr		
1	5	10 15
Leu Ala Lys Arg Ala Arg Asp Glu Arg Glu Val Pro Val Gly Ala Val		
20	25	30
Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala Ile		
35	40	45
Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln		
50	55	60
Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr		
65	70	75 80
Val Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser		
85	90	95
Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ser Lys Arg Gly Ala		
100	105	110
Ala Gly Ser Leu Met Asn Val Leu Asn Tyr Pro Gly Met Asn His Arg		
115	120	125
Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu Leu		
130	135	140
Cys Asp Phe Tyr Arg Met Pro Arg Gln Val Phe Asn Ala Gln Lys Lys		
145	150	155 160
Ala Gln Ser Ser Ile Asn Ser Gly Gly Ser Ser Gly Gly Ser Ser Gly		
165	170	175
Ser Glu Thr Pro Gly Thr Ser Glu Ser Ala Thr Pro Glu Ser Ser Gly		
180	185	190
Gly Ser Ser Gly Gly Ser Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile		
195	200	205
Gly Thr Asn Ser Val Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val		
210	215	220
Pro Ser Lys Lys Phe Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile		
225	230	235 240
Lys Lys Asn Leu Ile Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala		
245	250	255
Glu Ala Thr Arg Leu Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg		

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260					265					270					
Lys	Asn	Arg	Ile	Cys	Tyr	Leu	Gln	Glu	Ile	Phe	Ser	Asn	Glu	Met	Ala
		275					280					285			
Lys	Val	Asp	Asp	Ser	Phe	Phe	His	Arg	Leu	Glu	Glu	Ser	Phe	Leu	Val
	290					295					300				
Glu	Glu	Asp	Lys	Lys	His	Glu	Arg	His	Pro	Ile	Phe	Gly	Asn	Ile	Val
305					310					315				320	
Asp	Glu	Val	Ala	Tyr	His	Glu	Lys	Tyr	Pro	Thr	Ile	Tyr	His	Leu	Arg
			325						330					335	
Lys	Lys	Leu	Val	Asp	Ser	Thr	Asp	Lys	Ala	Asp	Leu	Arg	Leu	Ile	Tyr
		340						345					350		
Leu	Ala	Leu	Ala	His	Met	Ile	Lys	Phe	Arg	Gly	His	Phe	Leu	Ile	Glu
		355					360					365			
Gly	Asp	Leu	Asn	Pro	Asp	Asn	Ser	Asp	Val	Asp	Lys	Leu	Phe	Ile	Gln
370						375					380				
Leu	Val	Gln	Thr	Tyr	Asn	Gln	Leu	Phe	Glu	Glu	Asn	Pro	Ile	Asn	Ala
385					390					395					400
Ser	Gly	Val	Asp	Ala	Lys	Ala	Ile	Leu	Ser	Ala	Arg	Leu	Ser	Lys	Ser
				405					410					415	
Arg	Arg	Leu	Glu	Asn	Leu	Ile	Ala	Gln	Leu	Pro	Gly	Glu	Lys	Lys	Asn
			420					425					430		
Gly	Leu	Phe	Gly	Asn	Leu	Ile	Ala	Leu	Ser	Leu	Gly	Leu	Thr	Pro	Asn
	435					440						445			
Phe	Lys	Ser	Asn	Phe	Asp	Leu	Ala	Glu	Asp	Ala	Lys	Leu	Gln	Leu	Ser
450						455					460				
Lys	Asp	Thr	Tyr	Asp	Asp	Asp	Leu	Asp	Asn	Leu	Leu	Ala	Gln	Ile	Gly
465				470					475						480
Asp	Gln	Tyr	Ala	Asp	Leu	Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala
			485						490					495	
Ile	Leu	Leu	Ser	Asp	Ile	Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala
			500					505						510	
Pro	Leu	Ser	Ala	Ser	Met	Ile	Lys	Arg	Tyr	Asp	Glu	His	His	Gln	Asp
	515						520						525		
Leu	Thr	Leu	Leu	Lys	Ala	Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr
	530					535						540			
Lys	Glu	Ile	Phe	Phe	Asp	Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Ile
545					550					555					560
Asp	Gly	Gly	Ala	Ser	Gln	Glu	Glu	Phe	Tyr	Lys	Phe	Ile	Lys	Pro	Ile
			565						570					575	
Leu	Glu	Lys	Met	Asp	Gly	Thr	Glu	Glu	Leu	Leu	Val	Lys	Leu	Asn	Arg
			580					585					590		
Glu	Asp	Leu	Leu	Arg	Lys	Gln	Arg	Thr	Phe	Asp	Asn	Gly	Ser	Ile	Pro
	595					600						605			
His	Gln	Ile	His	Leu	Gly	Glu	Leu	His	Ala	Ile	Leu	Arg	Arg	Gln	Glu
	610					615						620			
Asp	Phe	Tyr	Pro	Phe	Leu	Lys	Asp	Asn	Arg	Glu	Lys	Ile	Glu	Lys	Ile
625					630					635					640
Leu	Thr	Phe	Arg	Ile	Pro	Tyr	Tyr	Val	Gly	Pro	Leu	Ala	Arg	Gly	Asn
			645						650					655	
Ser	Arg	Phe	Ala	Trp	Met	Thr	Arg	Lys	Ser	Glu	Glu	Thr	Ile	Thr	Pro
		660						665					670		
Trp	Asn	Phe	Glu	Glu	Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe
	675						680					685			

Ile	Glu	Arg	Met	Thr	Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val
690					695					700					
Leu	Pro	Lys	His	Ser	Leu	Leu	Tyr	Glu	Tyr	Phe	Thr	Val	Tyr	Asn	Glu
705					710					715					720
Leu	Thr	Lys	Val	Lys	Tyr	Val	Thr	Glu	Gly	Met	Arg	Lys	Pro	Ala	Phe
				725					730					735	
Leu	Ser	Gly	Glu	Gln	Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr
			740					745					750		
Asn	Arg	Lys	Val	Thr	Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys
		755					760					765			
Ile	Glu	Cys	Phe	Asp	Ser	Val	Glu	Ile	Ser	Gly	Val	Glu	Asp	Arg	Phe
770						775					780				
Asn	Ala	Ser	Leu	Gly	Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp
785					790					795					800
Lys	Asp	Phe	Leu	Asp	Asn	Glu	Glu	Asn	Glu	Asp	Ile	Leu	Glu	Asp	Ile
			805						810					815	
Val	Leu	Thr	Leu	Thr	Leu	Phe	Glu	Asp	Arg	Glu	Met	Ile	Glu	Glu	Arg
			820					825					830		
Leu	Lys	Thr	Tyr	Ala	His	Leu	Phe	Asp	Asp	Lys	Val	Met	Lys	Gln	Leu
		835					840					845			
Lys	Arg	Arg	Arg	Tyr	Thr	Gly	Trp	Gly	Arg	Leu	Ser	Arg	Lys	Leu	Ile
850						855					860				
Asn	Gly	Ile	Arg	Asp	Lys	Gln	Ser	Gly	Lys	Thr	Ile	Leu	Asp	Phe	Leu
865					870					875					880
Lys	Ser	Asp	Gly	Phe	Ala	Asn	Arg	Asn	Phe	Met	Gln	Leu	Ile	His	Asp
			885						890					895	
Asp	Ser	Leu	Thr	Phe	Lys	Glu	Asp	Ile	Gln	Lys	Ala	Gln	Val	Ser	Gly
			900					905					910		
Gln	Gly	Asp	Ser	Leu	His	Glu	His	Ile	Ala	Asn	Leu	Ala	Gly	Ser	Pro
		915					920					925			
Ala	Ile	Lys	Lys	Gly	Ile	Leu	Gln	Thr	Val	Lys	Val	Val	Asp	Glu	Leu
930						935					940				
Val	Lys	Val	Met	Gly	Arg	His	Lys	Pro	Glu	Asn	Ile	Val	Ile	Glu	Met
945					950					955					960
Ala	Arg	Glu	Asn	Gln	Thr	Thr	Gln	Lys	Gly	Gln	Lys	Asn	Ser	Arg	Glu
			965						970					975	
Arg	Met	Lys	Arg	Ile	Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	Ser	Gln	Ile
			980					985					990		
Leu	Lys	Glu	His	Pro	Val	Glu	Asn	Thr	Gln	Leu	Gln	Asn	Glu	Lys	Leu
		995					1000					1005			
Tyr	Leu	Tyr	Tyr	Leu	Gln	Asn	Gly	Arg	Asp	Met	Tyr	Val	Asp	Gln	
1010						1015					1020				
Glu	Leu	Asp	Ile	Asn	Arg	Leu	Ser	Asp	Tyr	Asp	Val	Asp	His	Ile	
1025						1030					1035				
Val	Pro	Gln</													

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Lys 1100	Ala	Glu	Arg	Gly	Gly	Leu 1105	Ser	Glu	Leu	Asp	Lys 1110	Ala	Gly	Phe
Ile 1115	Lys	Arg	Gln	Leu	Val	Glu 1120	Thr	Arg	Gln	Ile	Thr 1125	Lys	His	Val
Ala 1130	Gln	Ile	Leu	Asp	Ser	Arg 1135	Met	Asn	Thr	Lys	Tyr 1140	Asp	Glu	Asn
Asp 1145	Lys	Leu	Ile	Arg	Glu	Val 1150	Lys	Val	Ile	Thr	Leu 1155	Lys	Ser	Lys
Leu 1160	Val	Ser	Asp	Phe	Arg	Lys 1165	Asp	Phe	Gln	Phe	Tyr 1170	Lys	Val	Arg
Glu 1175	Ile	Asn	Asn	Tyr	His	His 1180	Ala	His	Asp	Ala	Tyr 1185	Leu	Asn	Ala
Val 1190	Val	Gly	Thr	Ala	Leu	Ile 1195	Lys	Lys	Tyr	Pro	Lys 1200	Leu	Glu	Ser
Glu 1205	Phe	Val	Tyr	Gly	Asp	Tyr 1210	Lys	Val	Tyr	Asp	Val 1215	Arg	Lys	Met
Ile 1220	Ala	Lys	Ser	Glu	Gln	Glu 1225	Ile	Gly	Lys	Ala	Thr 1230	Ala	Lys	Tyr
Phe 1235	Phe	Tyr	Ser	Asn	Ile	Met 1240	Asn	Phe	Phe	Lys	Thr 1245	Glu	Ile	Thr
Leu 1250	Ala	Asn	Gly	Glu	Ile	Arg 1255	Lys	Arg	Pro	Leu	Ile 1260	Glu	Thr	Asn
Gly 1265	Glu	Thr	Gly	Glu	Ile	Val 1270	Trp	Asp	Lys	Gly	Arg 1275	Asp	Phe	Ala
Thr 1280	Val	Arg	Lys	Val	Leu	Ser 1285	Met	Pro	Gln	Val	Asn 1290	Ile	Val	Lys
Lys 1295	Thr	Glu	Val	Gln	Thr	Gly 1300	Gly	Phe	Ser	Lys	Glu 1305	Ser	Ile	Leu
Pro 1310	Lys	Arg	Asn	Ser	Asp	Lys 1315	Leu	Ile	Ala	Arg	Lys 1320	Lys	Asp	Trp
Asp 1325	Pro	Lys	Lys	Tyr	Gly	Gly 1330	Phe	Asp	Ser	Pro	Thr 1335	Val	Ala	Tyr
Ser 1340	Val	Leu	Val	Val	Ala	Lys 1345	Val	Glu	Lys	Gly	Lys 1350	Ser	Lys	Lys
Leu 1355	Lys	Ser	Val	Lys	Glu	Leu 1360	Leu	Gly	Ile	Thr	Ile 1365	Met	Glu	Arg
Ser 1370	Ser	Phe	Glu	Lys	Asn	Pro 1375	Ile	Asp	Phe	Leu	Glu 1380	Ala	Lys	Gly
Tyr 1385	Lys	Glu	Val	Lys	Lys	Asp 1390	Leu	Ile	Ile	Lys	Leu 1395	Pro	Lys	Tyr
Ser 1400	Leu	Phe	Glu	Leu	Glu	Asn 1405	Gly	Arg	Lys	Arg	Met 1410	Leu	Ala	Ser
Ala 1415	Gly	Glu	Leu	Gln	Lys	Gly 1420	Asn	Glu	Leu	Ala	Leu 1425	Pro	Ser	Lys
Tyr 1430	Val	Asn	Phe	Leu	Tyr	Leu 1435	Ala	Ser	His	Tyr	Glu 1440	Lys	Leu	Lys
Gly 1445	Ser	Pro	Glu	Asp	Asn	Glu 1450	Gln	Lys	Gln	Leu	Phe 1455	Val	Glu	Gln
His 1460	Lys	His	Tyr	Leu	Asp	Glu 1465	Ile	Ile	Glu	Gln	Ile 1470	Ser	Glu	Phe
Ser 1475	Lys	Arg	Val	Ile	Leu	Ala 1480	Asp	Ala	Asn	Leu	Asp 1485	Lys	Val	Leu
Ser 1490	Ala	Tyr	Asn	Lys	His	Arg	Asp	Lys	Pro	Ile	Arg	Glu	Gln	Ala

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1490	1495	1500
Glu Asn Ile Ile His Leu Phe Thr Leu Thr Asn Leu Gly Ala Pro		
1505	1510	1515
Ala Ala Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Arg Tyr		
1520	1525	1530
Thr Ser Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser		
1535	1540	1545
Ile Thr Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly		
1550	1555	1560
Gly Asp		
1565		

<210> SEQ ID NO 37

<211> LENGTH: 1565

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 37

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr		
1	5	10
Leu Ala Lys Arg Ala Arg Asp Glu Arg Glu Val Pro Val Gly Ala Val		
20	25	30
Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala Ile		
35	40	45
Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln		
50	55	60
Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Tyr Asp Ala Thr Leu Tyr		
65	70	75
Ser Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser		
85	90	95
Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ala Lys Thr Gly Ala		
100	105	110
Ala Gly Ser Leu Met Asp Val Leu His His Pro Gly Met Asn His Arg		
115	120	125
Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu Leu		
130	135	140
Cys Arg Phe Phe Arg Met Pro Arg Arg Val Phe Asn Ala Gln Lys Lys		
145	150	155
Ala Gln Ser Ser Thr Asp Ser Gly Gly Ser Ser Gly Gly Ser Ser Gly		
165	170	175
Ser Glu Thr Pro Gly Thr Ser Glu Ser Ala Thr Pro Glu Ser Ser Gly		
180	185	190
Gly Ser Ser Gly Gly Ser Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile		
195	200	205
Gly Thr Asn Ser Val Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val		
210	215	220
Pro Ser Lys Lys Phe Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile		
225	230	235
Lys Lys Asn Leu Ile Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala		
245	250	255
Glu Ala Thr Arg Leu Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg		
260	265	270
Lys Asn Arg Ile Cys Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala		

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275					280					285					
Lys	Val	Asp	Asp	Ser	Phe	Phe	His	Arg	Leu	Glu	Glu	Ser	Phe	Leu	Val
290						295					300				
Glu	Glu	Asp	Lys	Lys	His	Glu	Arg	His	Pro	Ile	Phe	Gly	Asn	Ile	Val
305					310					315					320
Asp	Glu	Val	Ala	Tyr	His	Glu	Lys	Tyr	Pro	Thr	Ile	Tyr	His	Leu	Arg
				325					330					335	
Lys	Lys	Leu	Val	Asp	Ser	Thr	Asp	Lys	Ala	Asp	Leu	Arg	Leu	Ile	Tyr
		340						345					350		
Leu	Ala	Leu	Ala	His	Met	Ile	Lys	Phe	Arg	Gly	His	Phe	Leu	Ile	Glu
	355						360					365			
Gly	Asp	Leu	Asn	Pro	Asp	Asn	Ser	Asp	Val	Asp	Lys	Leu	Phe	Ile	Gln
370					375						380				
Leu	Val	Gln	Thr	Tyr	Asn	Gln	Leu	Phe	Glu	Glu	Asn	Pro	Ile	Asn	Ala
385					390					395					400
Ser	Gly	Val	Asp	Ala	Lys	Ala	Ile	Leu	Ser	Ala	Arg	Leu	Ser	Lys	Ser
				405					410					415	
Arg	Arg	Leu	Glu	Asn	Leu	Ile	Ala	Gln	Leu	Pro	Gly	Glu	Lys	Lys	Asn
		420						425					430		
Gly	Leu	Phe	Gly	Asn	Leu	Ile	Ala	Leu	Ser	Leu	Gly	Leu	Thr	Pro	Asn
	435						440						445		
Phe	Lys	Ser	Asn	Phe	Asp	Leu	Ala	Glu	Asp	Ala	Lys	Leu	Gln	Leu	Ser
450					455						460				
Lys	Asp	Thr	Tyr	Asp	Asp	Asp	Leu	Asp	Asn	Leu	Leu	Ala	Gln	Ile	Gly
465				470					475						480
Asp	Gln	Tyr	Ala	Asp	Leu	Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala
				485					490					495	
Ile	Leu	Leu	Ser	Asp	Ile	Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala
		500						505					510		
Pro	Leu	Ser	Ala	Ser	Met	Ile	Lys	Arg	Tyr	Asp	Glu	His	His	Gln	Asp
	515						520						525		
Leu	Thr	Leu	Leu	Lys	Ala	Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr
530					535							540			
Lys	Glu	Ile	Phe	Phe	Asp	Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Ile
545					550					555					560
Asp	Gly	Gly	Ala	Ser	Gln	Glu	Glu	Phe	Tyr	Lys	Phe	Ile	Lys	Pro	Ile
				565					570					575	
Leu	Glu	Lys	Met	Asp	Gly	Thr	Glu	Glu	Leu	Leu	Val	Lys	Leu	Asn	Arg
		580						585					590		
Glu	Asp	Leu	Leu	Arg	Lys	Gln	Arg	Thr	Phe	Asp	Asn	Gly	Ser	Ile	Pro
	595						600					605			
His	Gln	Ile	His	Leu	Gly	Glu	Leu	His	Ala	Ile	Leu	Arg	Arg	Gln	Glu
610					615						620				
Asp	Phe	Tyr	Pro	Phe	Leu	Lys	Asp	Asn	Arg	Glu	Lys	Ile	Glu	Lys	Ile
625					630					635					640
Leu	Thr	Phe	Arg	Ile	Pro	Tyr	Tyr	Val	Gly	Pro	Leu	Ala	Arg	Gly	Asn
				645					650					655	
Ser	Arg	Phe	Ala	Trp	Met	Thr	Arg	Lys	Ser	Glu	Glu	Thr	Ile	Thr	Pro
		660						665					670		
Trp	Asn	Phe	Glu	Glu	Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe
	675						680					685			
Ile	Glu	Arg	Met	Thr	Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val
690						695					700				

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Leu Pro Lys His Ser Leu Leu Tyr Glu Tyr Phe Thr Val Tyr Asn Glu
 705 710 715 720
 Leu Thr Lys Val Lys Tyr Val Thr Glu Gly Met Arg Lys Pro Ala Phe
 725 730 735
 Leu Ser Gly Glu Gln Lys Lys Ala Ile Val Asp Leu Leu Phe Lys Thr
 740 745 750
 Asn Arg Lys Val Thr Val Lys Gln Leu Lys Glu Asp Tyr Phe Lys Lys
 755 760 765
 Ile Glu Cys Phe Asp Ser Val Glu Ile Ser Gly Val Glu Asp Arg Phe
 770 775 780
 Asn Ala Ser Leu Gly Thr Tyr His Asp Leu Leu Lys Ile Ile Lys Asp
 785 790 795 800
 Lys Asp Phe Leu Asp Asn Glu Glu Asn Glu Asp Ile Leu Glu Asp Ile
 805 810 815
 Val Leu Thr Leu Thr Leu Phe Glu Asp Arg Glu Met Ile Glu Glu Arg
 820 825 830
 Leu Lys Thr Tyr Ala His Leu Phe Asp Asp Lys Val Met Lys Gln Leu
 835 840 845
 Lys Arg Arg Arg Tyr Thr Gly Trp Gly Arg Leu Ser Arg Lys Leu Ile
 850 855 860
 Asn Gly Ile Arg Asp Lys Gln Ser Gly Lys Thr Ile Leu Asp Phe Leu
 865 870 875 880
 Lys Ser Asp Gly Phe Ala Asn Arg Asn Phe Met Gln Leu Ile His Asp
 885 890 895
 Asp Ser Leu Thr Phe Lys Glu Asp Ile Gln Lys Ala Gln Val Ser Gly
 900 905 910
 Gln Gly Asp Ser Leu His Glu His Ile Ala Asn Leu Ala Gly Ser Pro
 915 920 925
 Ala Ile Lys Lys Gly Ile Leu Gln Thr Val Lys Val Val Asp Glu Leu
 930 935 940
 Val Lys Val Met Gly Arg His Lys Pro Glu Asn Ile Val Ile Glu Met
 945 950 955 960
 Ala Arg Glu Asn Gln Thr Thr Gln Lys Gly Gln Lys Asn Ser Arg Glu
 965 970 975
 Arg Met Lys Arg Ile Glu Glu Gly Ile Lys Glu Leu Gly Ser Gln Ile
 980 985 990
 Leu Lys Glu His Pro Val Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu
 995 1000 1005
 Tyr Leu Tyr Tyr Leu Gln Asn Gly Arg Asp Met Tyr Val Asp Gln
 1010 1015 1020
 Glu Leu Asp Ile Asn Arg Leu Ser Asp Tyr Asp Val Asp His Ile
 1025 1030 1035
 Val Pro Gln Ser Phe Leu Lys Asp Asp Ser Ile Asp Asn Lys Val
 1040 1045 1050
 Leu Thr Arg Ser Asp Lys Asn Arg Gly Lys Ser Asp Asn Val Pro
 1055 1060 1065
 Ser Glu Glu Val Val Lys Lys Met Lys Asn Tyr Trp Arg Gln Leu
 1070 1075 1080
 Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys Phe Asp Asn Leu Thr
 1085 1090 1095
 Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp Lys Ala Gly Phe
 1100 1105 1110

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Ile	Lys	Arg	Gln	Leu	Val	Glu	Thr	Arg	Gln	Ile	Thr	Lys	His	Val
1115						1120					1125			
Ala	Gln	Ile	Leu	Asp	Ser	Arg	Met	Asn	Thr	Lys	Tyr	Asp	Glu	Asn
1130						1135					1140			
Asp	Lys	Leu	Ile	Arg	Glu	Val	Lys	Val	Ile	Thr	Leu	Lys	Ser	Lys
1145						1150					1155			
Leu	Val	Ser	Asp	Phe	Arg	Lys	Asp	Phe	Gln	Phe	Tyr	Lys	Val	Arg
1160						1165					1170			
Glu	Ile	Asn	Asn	Tyr	His	His	Ala	His	Asp	Ala	Tyr	Leu	Asn	Ala
1175						1180					1185			
Val	Val	Gly	Thr	Ala	Leu	Ile	Lys	Lys	Tyr	Pro	Lys	Leu	Glu	Ser
1190						1195					1200			
Glu	Phe	Val	Tyr	Gly	Asp	Tyr	Lys	Val	Tyr	Asp	Val	Arg	Lys	Met
1205						1210					1215			
Ile	Ala	Lys	Ser	Glu	Gln	Glu	Ile	Gly	Lys	Ala	Thr	Ala	Lys	Tyr
1220						1225					1230			
Phe	Phe	Tyr	Ser	Asn	Ile	Met	Asn	Phe	Phe	Lys	Thr	Glu	Ile	Thr
1235						1240					1245			
Leu	Ala	Asn	Gly	Glu	Ile	Arg	Lys	Arg	Pro	Leu	Ile	Glu	Thr	Asn
1250						1255					1260			
Gly	Glu	Thr	Gly	Glu	Ile	Val	Trp	Asp	Lys	Gly	Arg	Asp	Phe	Ala
1265						1270					1275			
Thr	Val	Arg	Lys	Val	Leu	Ser	Met	Pro	Gln	Val	Asn	Ile	Val	Lys
1280						1285					1290			
Lys	Thr	Glu	Val	Gln	Thr	Gly	Gly	Phe	Ser	Lys	Glu	Ser	Ile	Leu
1295						1300					1305			
Pro	Lys	Arg	Asn	Ser	Asp	Lys	Leu	Ile	Ala	Arg	Lys	Lys	Asp	Trp
1310						1315					1320			
Asp	Pro	Lys	Lys	Tyr	Gly	Gly	Phe	Asp	Ser	Pro	Thr	Val	Ala	Tyr
1325						1330					1335			
Ser	Val	Leu	Val	Val	Ala	Lys	Val	Glu	Lys	Gly	Lys	Ser	Lys	Lys
1340						1345					1350			
Leu	Lys	Ser	Val	Lys	Glu	Leu	Leu	Gly	Ile	Thr	Ile	Met	Glu	Arg
1355						1360					1365			
Ser	Ser	Phe	Glu	Lys	Asn	Pro	Ile	Asp	Phe	Leu	Glu	Ala	Lys	Gly
1370						1375					1380			
Tyr	Lys	Glu	Val	Lys	Lys	Asp	Leu	Ile	Ile	Lys	Leu	Pro	Lys	Tyr
1385						1390					1395			
Ser	Leu	Phe	Glu	Leu	Glu	Asn	Gly	Arg	Lys	Arg	Met	Leu	Ala	Ser
1400						1405					1410			
Ala	Gly	Glu	Leu	Gln	Lys	Gly	Asn	Glu	Leu	Ala	Leu	Pro	Ser	Lys
1415						1420					1425			
Tyr	Val	Asn	Phe	Leu	Tyr	Leu	Ala	Ser	His	Tyr	Glu	Lys	Leu	Lys
1430						1435					1440			
Gly	Ser	Pro	Glu	Asp	Asn	Glu	Gln	Lys	Gln	Leu	Phe	Val	Glu	Gln
1445						1450					1455			
His	Lys	His	Tyr	Leu	Asp	Glu	Ile	Ile	Glu	Gln	Ile	Ser	Glu	Phe
1460						1465					1470			
Ser	Lys	Arg	Val	Ile	Leu	Ala	Asp	Ala	Asn	Leu	Asp	Lys	Val	Leu
1475						1480					1485			
Ser	Ala	Tyr	Asn	Lys	His	Arg	Asp	Lys	Pro	Ile	Arg	Glu	Gln	Ala
1490						1495					1500			
Glu	Asn	Ile	Ile	His	Leu	Phe	Thr	Leu	Thr	Asn	Leu	Gly	Ala	Pro

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1505	1510	1515
Ala Ala Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Arg Tyr		
1520	1525	1530
Thr Ser Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser		
1535	1540	1545
Ile Thr Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly		
1550	1555	1560
Gly Asp		
1565		

<210> SEQ ID NO 38
 <211> LENGTH: 364
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 38

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr		
1	5	10 15
Leu Ala Lys Arg Ala Trp Asp Glu Arg Glu Val Pro Val Gly Ala Val		
20	25	30
Leu Val His Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Pro Ile		
35	40	45
Gly Arg His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln		
50	55	60
Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr		
65	70	75 80
Val Thr Leu Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser		
85	90	95
Arg Ile Gly Arg Val Val Phe Gly Ala Arg Asp Ala Lys Thr Gly Ala		
100	105	110
Ala Gly Ser Leu Met Asp Val Leu His His Pro Gly Met Asn His Arg		
115	120	125
Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu Leu		
130	135	140
Ser Asp Phe Phe Arg Met Arg Arg Gln Glu Ile Lys Ala Gln Lys Lys		
145	150	155 160
Ala Gln Ser Ser Thr Asp Ser Gly Gly Ser Ser Gly Gly Ser Ser Gly		
165	170	175
Ser Glu Thr Pro Gly Thr Ser Glu Ser Ala Thr Pro Glu Ser Ser Gly		
180	185	190
Gly Ser Ser Gly Gly Ser Ser Glu Val Glu Phe Ser His Glu Tyr Trp		
195	200	205
Met Arg His Ala Leu Thr Leu Ala Lys Arg Ala Arg Asp Glu Arg Glu		
210	215	220
Val Pro Val Gly Ala Val Leu Val Leu Asn Asn Arg Val Ile Gly Glu		
225	230	235 240
Gly Trp Asn Arg Ala Ile Gly Leu His Asp Pro Thr Ala His Ala Glu		
245	250	255
Ile Met Ala Leu Arg Gln Gly Gly Leu Val Met Gln Asn Tyr Arg Leu		
260	265	270
Ile Asp Ala Thr Leu Tyr Val Thr Phe Glu Pro Cys Val Met Cys Ala		
275	280	285
Gly Ala Met Ile His Ser Arg Ile Gly Arg Val Val Phe Gly Val Arg		

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290	295	300
Asn Ala Lys Thr Gly Ala Ala Gly Ser Leu Met Asp Val Leu His Tyr		
305	310	315 320
Pro Gly Met Asn His Arg Val Glu Ile Thr Glu Gly Ile Leu Ala Asp		
	325	330 335
Glu Cys Ala Ala Leu Leu Cys Tyr Phe Phe Arg Met Pro Arg Gln Val		
	340	345 350
Phe Asn Ala Gln Lys Lys Ala Gln Ser Ser Thr Asp		
	355	360

<210> SEQ ID NO 39
 <211> LENGTH: 167
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 39

Met Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu		
1	5	10 15
Thr Leu Ala Lys Arg Ala Arg Asp Glu Arg Glu Val Pro Val Gly Ala		
	20	25 30
Val Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala		
	35	40 45
Ile Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg		
	50	55 60
Gln Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Tyr Asp Ala Thr Leu		
	65	70 75 80
Tyr Ser Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His		
	85	90 95
Ser Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ala Lys Thr Gly		
	100	105 110
Ala Ala Gly Ser Leu Met Asp Val Leu His His Pro Gly Met Asn His		
	115	120 125
Arg Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu		
	130	135 140
Leu Cys Arg Phe Phe Arg Met Pro Arg Arg Val Phe Asn Ala Gln Lys		
	145	150 155 160
Lys Ala Gln Ser Ser Thr Asp		
	165	

<210> SEQ ID NO 40
 <211> LENGTH: 167
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 40

Met Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu		
1	5	10 15
Thr Leu Ala Lys Arg Ala Arg Asp Glu Arg Glu Val Pro Val Gly Ala		
	20	25 30
Val Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala		
	35	40 45
Ile Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg		
	50	55 60

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Gln Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu
65 70 75 80

Tyr Val Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His
85 90 95

Ser Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ser Lys Arg Gly
100 105 110

Ala Ala Gly Ser Leu Met Asn Val Leu Asn Tyr Pro Gly Met Asn His
115 120 125

Arg Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu
130 135 140

Leu Cys Asp Phe Tyr Arg Met Pro Arg Gln Val Phe Asn Ala Gln Lys
145 150 155 160

Lys Ala Gln Ser Ser Ile Asn
165

<210> SEQ ID NO 41
 <211> LENGTH: 83
 <212> TYPE: PRT
 <213> ORGANISM: Bacillus phage AR9

<400> SEQUENCE: 41

Thr Asn Leu Ser Asp Ile Ile Glu Lys Glu Thr Gly Lys Gln Leu Val
1 5 10 15

Ile Gln Glu Ser Ile Leu Met Leu Pro Glu Glu Val Glu Glu Val Ile
20 25 30

Gly Asn Lys Pro Glu Ser Asp Ile Leu Val His Thr Ala Tyr Asp Glu
35 40 45

Ser Thr Asp Glu Asn Val Met Leu Leu Thr Ser Asp Ala Pro Glu Tyr
50 55 60

Lys Pro Trp Ala Leu Val Ile Gln Asp Ser Asn Gly Glu Asn Lys Ile
65 70 75 80

Lys Met Leu

<210> SEQ ID NO 42
 <211> LENGTH: 19
 <212> TYPE: RNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(19)
 <223> OTHER INFORMATION: n is a, c, g, or u

<400> SEQUENCE: 42

nnnnnnnnnn nnnnnnnnnn

19

<210> SEQ ID NO 43
 <211> LENGTH: 22
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (4)..(22)
 <223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 43

aaannnnnnn nnnnnnnnnn nn

22

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<210> SEQ ID NO 44
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (4)..(22)
<223> OTHER INFORMATION: n is a, c, g, or t

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<400> SEQUENCE: 44

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tttnnnnnnnn nnnnnnnnnn nn

```

22

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<210> SEQ ID NO 45
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 45

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Glu Glu Leu Leu Ser Lys Asn Tyr His Leu Glu Asn Glu Val Ala Arg
1          5          10          15
Leu Lys Lys Gly Ser Gly Ser Gly
20

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<210> SEQ ID NO 46
<211> LENGTH: 241
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 46

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Glu Glu Glu Leu Leu Ser Lys Asn Tyr His Leu Glu Asn Glu Val Ala
1          5          10          15
Arg Leu Lys Lys Gly Ser Gly Ser Gly Glu Glu Leu Leu Ser Lys Asn
20          25          30
Tyr His Leu Glu Asn Glu Val Ala Arg Leu Lys Lys Gly Ser Gly Ser
35          40          45
Gly Glu Glu Leu Leu Ser Lys Asn Tyr His Leu Glu Asn Glu Val Ala
50          55          60
Arg Leu Lys Lys Gly Ser Gly Ser Gly Glu Glu Leu Leu Ser Lys Asn
65          70          75          80
Tyr His Leu Glu Asn Glu Val Ala Arg Leu Lys Lys Gly Ser Gly Ser
85          90          95
Gly Glu Glu Leu Leu Ser Lys Asn Tyr His Leu Glu Asn Glu Val Ala
100         105         110
Arg Leu Lys Lys Gly Ser Gly Ser Gly Glu Glu Leu Leu Ser Lys Asn
115         120         125
Tyr His Leu Glu Asn Glu Val Ala Arg Leu Lys Lys Gly Ser Gly Ser
130         135         140
Gly Glu Glu Leu Leu Ser Lys Asn Tyr His Leu Glu Asn Glu Val Ala
145         150         155         160
Arg Leu Lys Lys Gly Ser Gly Ser Gly Glu Glu Leu Leu Ser Lys Asn
165         170         175
Tyr His Leu Glu Asn Glu Val Ala Arg Leu Lys Lys Gly Ser Gly Ser
180         185         190
Gly Glu Glu Leu Leu Ser Lys Asn Tyr His Leu Glu Asn Glu Val Ala
195         200         205

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Arg Leu Lys Lys Gly Ser Gly Ser Gly Glu Glu Leu Leu Ser Lys Asn
 210 215 220

Tyr His Leu Glu Asn Glu Val Ala Arg Leu Lys Lys Gly Ser Gly Ser
 225 230 235 240

Gly

<210> SEQ ID NO 47
 <211> LENGTH: 277
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 47

Met Gly Pro Asp Ile Val Met Thr Gln Ser Pro Ser Ser Leu Ser Ala
 1 5 10 15

Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ser Ser Thr Gly Ala
 20 25 30

Val Thr Thr Ser Asn Tyr Ala Ser Trp Val Gln Glu Lys Pro Gly Lys
 35 40 45

Leu Phe Lys Gly Leu Ile Gly Gly Thr Asn Asn Arg Ala Pro Gly Val
 50 55 60

Pro Ser Arg Phe Ser Gly Ser Leu Ile Gly Asp Lys Ala Thr Leu Thr
 65 70 75 80

Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Phe Cys Ala Leu
 85 90 95

Trp Tyr Ser Asn His Trp Val Phe Gly Gln Gly Thr Lys Val Glu Leu
 100 105 110

Lys Arg Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 115 120 125

Ser Ser Gly Gly Gly Ser Glu Val Lys Leu Leu Glu Ser Gly Gly Gly
 130 135 140

Leu Val Gln Pro Gly Gly Ser Leu Lys Leu Ser Cys Ala Val Ser Gly
 145 150 155 160

Phe Ser Leu Thr Asp Tyr Gly Val Asn Trp Val Arg Gln Ala Pro Gly
 165 170 175

Arg Gly Leu Glu Trp Ile Gly Val Ile Trp Gly Asp Gly Ile Thr Asp
 180 185 190

Tyr Asn Ser Ala Leu Lys Asp Arg Phe Ile Ile Ser Lys Asp Asn Gly
 195 200 205

Lys Asn Thr Val Tyr Leu Gln Met Ser Lys Val Arg Ser Asp Asp Thr
 210 215 220

Ala Leu Tyr Tyr Cys Val Thr Gly Leu Phe Asp Tyr Trp Gly Gln Gly
 225 230 235 240

Thr Leu Val Thr Val Ser Ser Tyr Pro Tyr Asp Val Pro Asp Tyr Ala
 245 250 255

Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 260 265 270

Gly Gly Gly Gly Ser
 275

<210> SEQ ID NO 48
 <211> LENGTH: 66
 <212> TYPE: DNA
 <213> ORGANISM: Saccharomyces bayanus

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<400> SEQUENCE: 48

ttcttgcgtg acttatagat cgctacgtta ttccaatttt gaaaatctga gtctctgggag 60
 tgcgga 66

<210> SEQ ID NO 49

<211> LENGTH: 605

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 49

Met Ser Gly Trp Glu Ser Tyr Tyr Lys Thr Glu Gly Asp Glu Glu Ala
 1 5 10 15
 Glu Glu Glu Gln Glu Glu Asn Leu Glu Ala Ser Gly Asp Tyr Lys Tyr
 20 25 30
 Ser Gly Arg Asp Ser Leu Ile Phe Leu Val Asp Ala Ser Lys Ala Met
 35 40 45
 Phe Glu Ser Gln Ser Glu Asp Glu Leu Thr Pro Phe Asp Met Ser Ile
 50 55 60
 Gln Cys Ile Gln Ser Val Tyr Ile Ser Lys Ile Ile Ser Ser Asp Arg
 65 70 75 80
 Asp Leu Leu Ala Trp Phe Tyr Gly Thr Glu Lys Asp Lys Asn Ser Val
 85 90 95
 Asn Phe Lys Ile Tyr Val Leu Gln Glu Leu Asp Asn Pro Gly Ala Lys
 100 105 110
 Arg Ile Leu Glu Leu Asp Gln Phe Lys Gly Gln Gln Gly Gln Lys Arg
 115 120 125
 Phe Gln Asp Met Met Gly His Gly Ser Asp Tyr Ser Leu Ser Glu Val
 130 135 140
 Leu Trp Val Cys Ala Asn Leu Phe Ser Asp Val Gln Phe Lys Met Ser
 145 150 155 160
 His Lys Arg Ile Met Leu Phe Thr Asn Glu Asp Asn Pro His Gly Asn
 165 170 175
 Asp Ser Ala Lys Ala Ser Arg Ala Arg Thr Lys Ala Gly Asp Leu Arg
 180 185 190
 Asp Thr Gly Ile Phe Leu Asp Leu His Leu Lys Lys Pro Gly Gly Phe
 195 200 205
 Asp Ile Ser Leu Phe Tyr Arg Asp Ile Ile Ser Ile Ala Glu Asp Glu
 210 215 220
 Asp Leu Arg Val His Phe Glu Glu Ser Ser Lys Leu Glu Asp Leu Leu
 225 230 235 240
 Arg Lys Val Arg Ala Lys Glu Thr Arg Lys Arg Ala Leu Ser Arg Leu
 245 250 255
 Lys Leu Lys Leu Asn Lys Asp Ile Val Ile Ser Val Gly Ile Tyr Asn
 260 265 270
 Leu Val Gln Lys Ala Leu Lys Pro Pro Pro Ile Lys Leu Tyr Arg Glu
 275 280 285
 Thr Asn Glu Pro Val Lys Thr Lys Thr Arg Thr Phe Asn Thr Ser Thr
 290 295 300
 Gly Gly Leu Leu Leu Pro Ser Asp Thr Lys Arg Ser Gln Ile Tyr Gly
 305 310 315 320
 Ser Arg Gln Ile Ile Leu Glu Lys Glu Glu Thr Glu Glu Leu Lys Arg
 325 330 335
 Phe Asp Asp Pro Gly Leu Met Leu Met Gly Phe Lys Pro Leu Val Leu
 340 345 350

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Leu Lys Lys His His Tyr Leu Arg Pro Ser Leu Phe Val Tyr Pro Glu
 355 360 365
 Glu Ser Leu Val Ile Gly Ser Ser Thr Leu Phe Ser Ala Leu Leu Ile
 370 375 380
 Lys Cys Leu Glu Lys Glu Val Ala Ala Leu Cys Arg Tyr Thr Pro Arg
 385 390 395 400
 Arg Asn Ile Pro Pro Tyr Phe Val Ala Leu Val Pro Gln Glu Glu Glu
 405 410 415
 Leu Asp Asp Gln Lys Ile Gln Val Thr Pro Pro Gly Phe Gln Leu Val
 420 425 430
 Phe Leu Pro Phe Ala Asp Asp Lys Arg Lys Met Pro Phe Thr Glu Lys
 435 440 445
 Ile Met Ala Thr Pro Glu Gln Val Gly Lys Met Lys Ala Ile Val Glu
 450 455 460
 Lys Leu Arg Phe Thr Tyr Arg Ser Asp Ser Phe Glu Asn Pro Val Leu
 465 470 475 480
 Gln Gln His Phe Arg Asn Leu Glu Ala Leu Ala Leu Asp Leu Met Glu
 485 490 495
 Pro Glu Gln Ala Val Asp Leu Thr Leu Pro Lys Val Glu Ala Met Asn
 500 505 510
 Lys Arg Leu Gly Ser Leu Val Asp Glu Phe Lys Glu Leu Val Tyr Pro
 515 520 525
 Pro Asp Tyr Asn Pro Glu Gly Lys Val Thr Lys Arg Lys His Asp Asn
 530 535 540
 Glu Gly Ser Gly Ser Lys Arg Pro Lys Val Glu Tyr Ser Glu Glu Glu
 545 550 555 560
 Leu Lys Thr His Ile Ser Lys Gly Thr Leu Gly Lys Phe Thr Val Pro
 565 570 575
 Leu Lys Glu Ala Cys Arg Ala Tyr Gly Leu Lys Ser Gly Leu Lys Lys
 580 585 590
 Gln Glu Leu Leu Glu Ala Leu Thr Lys His Phe Gln Asp
 595 600 605

<210> SEQ ID NO 50
 <211> LENGTH: 482
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 50

Met Val Arg Ser Gly Asn Lys Ala Ala Trp Leu Cys Met Asp Val Gly
 1 5 10 15
 Phe Thr Met Ser Asn Ser Ile Pro Gly Ile Glu Ser Pro Phe Glu Gln
 20 25 30
 Ala Lys Lys Val Ile Thr Met Phe Val Gln Arg Gln Val Phe Ala Glu
 35 40 45
 Asn Lys Asp Glu Ile Ala Leu Val Leu Phe Gly Thr Asp Gly Thr Asp
 50 55 60
 Asn Pro Leu Ser Gly Gly Asp Gln Tyr Gln Asn Ile Thr Val His Arg
 65 70 75 80
 His Leu Met Leu Pro Asp Phe Asp Leu Leu Glu Asp Ile Glu Ser Lys
 85 90 95
 Ile Gln Pro Gly Ser Gln Gln Ala Asp Phe Leu Asp Ala Leu Ile Val
 100 105 110

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Ser Met Asp Val Ile Gln His Glu Thr Ile Gly Lys Lys Phe Glu Lys
    115                      120                      125

Arg His Ile Glu Ile Phe Thr Asp Leu Ser Ser Arg Phe Ser Lys Ser
    130                      135                      140

Gln Leu Asp Ile Ile Ile His Ser Leu Lys Lys Cys Asp Ile Ser Glu
    145                      150                      155                      160

Arg His Ser Ile His Trp Pro Cys Arg Leu Thr Ile Gly Ser Asn Leu
    165                      170                      175

Ser Ile Arg Ile Ala Ala Tyr Lys Ser Ile Leu Gln Glu Arg Val Lys
    180                      185                      190

Lys Thr Thr Trp Asp Ala Lys Thr Leu Lys Lys Glu Asp Ile Gln Lys
    195                      200                      205

Glu Thr Val Tyr Cys Leu Asn Asp Asp Asp Glu Thr Glu Val Leu Lys
    210                      215                      220

Glu Asp Ile Ile Gln Gly Phe Arg Tyr Gly Ser Asp Ile Val Pro Phe
    225                      230                      235                      240

Ser Lys Val Asp Glu Glu Gln Met Lys Tyr Lys Ser Glu Gly Lys Cys
    245                      250                      255

Phe Ser Val Leu Gly Phe Cys Lys Ser Ser Gln Val Gln Arg Arg Phe
    260                      265                      270

Phe Met Gly Asn Gln Val Leu Lys Val Phe Ala Ala Arg Asp Asp Glu
    275                      280                      285

Ala Ala Ala Val Ala Leu Ser Ser Leu Ile His Ala Leu Asp Asp Leu
    290                      295                      300

Asp Ile Trp Ala Ile Val Arg Tyr Ala Tyr Asp Lys Arg Ala Asn Pro
    305                      310                      315                      320

Gln Val Gly Val Ala Phe Pro His Ile Lys His Asn Tyr Glu Cys Leu
    325                      330                      335

Val Tyr Val Gln Leu Pro Phe Met Glu Asp Leu Arg Gln Tyr Met Phe
    340                      345                      350

Ser Ser Leu Lys Asn Ser Lys Lys Tyr Ala Pro Thr Glu Ala Gln Leu
    355                      360                      365

Asn Ala Val Asp Ala Leu Ile Asp Ser Met Ser Leu Ala Lys Lys Asp
    370                      375                      380

Glu Lys Thr Asp Thr Leu Glu Asp Leu Phe Pro Thr Thr Lys Ile Pro
    385                      390                      395                      400

Asn Pro Arg Phe Gln Arg Leu Phe Gln Cys Leu Leu His Arg Ala Leu
    405                      410                      415

His Pro Arg Glu Pro Leu Pro Pro Ile Gln Gln His Ile Trp Asn Met
    420                      425                      430

Leu Asn Pro Pro Ala Glu Val Thr Thr Lys Ser Gln Ile Pro Leu Ser
    435                      440                      445

Lys Ile Lys Thr Leu Phe Pro Leu Ile Glu Ala Lys Lys Lys Asp Gln
    450                      455                      460

Val Thr Ala Gln Glu Ile Phe Gln Asp Asn His Glu Asp Gly Pro Thr
    465                      470                      475                      480

Ala Lys

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<210> SEQ ID NO 51

<211> LENGTH: 10

<212> TYPE: DNA

<213> ORGANISM: Methanobacterium thermoautotrophicum

<400> SEQUENCE: 51

-continued

aatttttga

10

<210> SEQ ID NO 52
 <211> LENGTH: 83
 <212> TYPE: PRT
 <213> ORGANISM: Methanobacterium thermoautotrophicum

<400> SEQUENCE: 52

Gly Ser Val Ile Asp Val Ser Ser Gln Arg Val Asn Val Gln Arg Pro
 1 5 10 15
 Leu Asp Ala Leu Gly Asn Ser Leu Asn Ser Pro Val Ile Ile Lys Leu
 20 25 30
 Lys Gly Asp Arg Glu Phe Arg Gly Val Leu Lys Ser Phe Asp Leu His
 35 40 45
 Met Asn Leu Val Leu Asn Asp Ala Glu Glu Leu Glu Asp Gly Glu Val
 50 55 60
 Thr Arg Arg Leu Gly Thr Val Leu Ile Arg Gly Asp Asn Ile Val Tyr
 65 70 75 80

Ile Ser Pro

<210> SEQ ID NO 53
 <211> LENGTH: 25
 <212> TYPE: DNA
 <213> ORGANISM: Bacteriophage MS2

<400> SEQUENCE: 53

gcgcacatga ggatcaccca tgtgc

25

<210> SEQ ID NO 54
 <211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Bacteriophage MS2

<400> SEQUENCE: 54

Met Ala Ser Asn Phe Thr Gln Phe Val Leu Val Asp Asn Gly Gly Thr
 1 5 10 15
 Gly Asp Val Thr Val Ala Pro Ser Asn Phe Ala Asn Gly Ile Ala Glu
 20 25 30
 Ile Ser Ser Asn Ser Arg Ser Gln Ala Tyr Lys Val Thr Cys Ser Val
 35 40 45
 Arg Gln Ser Ser Ala Gln Asn Arg Lys Tyr Thr Ile Lys Val Glu Val
 50 55 60
 Pro Lys Gly Ala Trp Arg Ser Tyr Leu Asn Met Glu Leu Thr Ile Pro
 65 70 75 80
 Ile Phe Ala Thr Asn Ser Asp Cys Glu Leu Ile Val Lys Ala Met Gln
 85 90 95
 Gly Leu Leu Lys Asp Gly Asn Pro Ile Pro Ser Ala Ile Ala Ala Asn
 100 105 110
 Ser Gly Ile Tyr
 115

<210> SEQ ID NO 55
 <211> LENGTH: 26
 <212> TYPE: DNA
 <213> ORGANISM: Bacteriophage PP7

<400> SEQUENCE: 55

ataaggagtt tatatggaaa ccctta

26

-continued

<210> SEQ ID NO 56
 <211> LENGTH: 127
 <212> TYPE: PRT
 <213> ORGANISM: Bacteriophage PP7

<400> SEQUENCE: 56

```

Met Ser Lys Thr Ile Val Leu Ser Val Gly Glu Ala Thr Arg Thr Leu
 1              5              10              15

Thr Glu Ile Gln Ser Thr Ala Asp Arg Gln Ile Phe Glu Glu Lys Val
          20          25          30

Gly Pro Leu Val Gly Arg Leu Arg Leu Thr Ala Ser Leu Arg Gln Asn
          35          40          45

Gly Ala Lys Thr Ala Tyr Arg Val Asn Leu Lys Leu Asp Gln Ala Asp
          50          55          60

Trp Asp Cys Ser Thr Ser Val Cys Gly Glu Leu Pro Lys Val Arg Tyr
65          70          75          80

Thr Gln Val Trp Ser His Asp Val Thr Ile Val Ala Asn Ser Thr Glu
          85          90          95

Ala Ser Arg Lys Ser Leu Tyr Asp Leu Thr Lys Ser Leu Val Ala Thr
          100          105          110

Ser Gln Val Glu Asp Leu Val Val Asn Leu Val Pro Leu Gly Arg
          115          120          125

```

<210> SEQ ID NO 57
 <211> LENGTH: 19
 <212> TYPE: DNA
 <213> ORGANISM: Shigella phage

<400> SEQUENCE: 57

ctgaatgcct gcgagcatc

19

<210> SEQ ID NO 58
 <211> LENGTH: 62
 <212> TYPE: PRT
 <213> ORGANISM: Shigella phage

<400> SEQUENCE: 58

```

Met Lys Ser Ile Arg Cys Lys Asn Cys Asn Lys Leu Leu Phe Lys Ala
 1              5              10              15

Asp Ser Phe Asp His Ile Glu Ile Arg Cys Pro Arg Cys Lys Arg His
          20          25          30

Ile Ile Met Leu Asn Ala Cys Glu His Pro Thr Glu Lys His Cys Gly
          35          40          45

Lys Arg Glu Lys Ile Thr His Ser Asp Glu Thr Val Arg Tyr
          50          55          60

```

<210> SEQ ID NO 59
 <211> LENGTH: 1367
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 59

```

Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val Gly
 1              5              10              15

Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe Lys
          20          25          30

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Val	Leu	Gly	Asn	Thr	Asp	Arg	His	Ser	Ile	Lys	Lys	Asn	Leu	Ile	Gly
	35						40					45			
Ala	Leu	Leu	Phe	Asp	Ser	Gly	Glu	Thr	Ala	Glu	Ala	Thr	Arg	Leu	Lys
	50					55					60				
Arg	Thr	Ala	Arg	Arg	Arg	Tyr	Thr	Arg	Arg	Lys	Asn	Arg	Ile	Cys	Tyr
65					70					75				80	
Leu	Gln	Glu	Ile	Phe	Ser	Asn	Glu	Met	Ala	Lys	Val	Asp	Asp	Ser	Phe
			85						90					95	
Phe	His	Arg	Leu	Glu	Glu	Ser	Phe	Leu	Val	Glu	Glu	Asp	Lys	Lys	His
			100					105					110		
Glu	Arg	His	Pro	Ile	Phe	Gly	Asn	Ile	Val	Asp	Glu	Val	Ala	Tyr	His
						120					125				
Glu	Lys	Tyr	Pro	Thr	Ile	Tyr	His	Leu	Arg	Lys	Lys	Leu	Val	Asp	Ser
	130					135					140				
Thr	Asp	Lys	Ala	Asp	Leu	Arg	Leu	Ile	Tyr	Leu	Ala	Leu	Ala	His	Met
145					150					155				160	
Ile	Lys	Phe	Arg	Gly	His	Phe	Leu	Ile	Glu	Gly	Asp	Leu	Asn	Pro	Asp
				165					170					175	
Asn	Ser	Asp	Val	Asp	Lys	Leu	Phe	Ile	Gln	Leu	Val	Gln	Thr	Tyr	Asn
			180					185					190		
Gln	Leu	Phe	Glu	Glu	Asn	Pro	Ile	Asn	Ala	Ser	Gly	Val	Asp	Ala	Lys
			195				200					205			
Ala	Ile	Leu	Ser	Ala	Arg	Leu	Ser	Lys	Ser	Arg	Arg	Leu	Glu	Asn	Leu
	210					215					220				
Ile	Ala	Gln	Leu	Pro	Gly	Glu	Lys	Lys	Asn	Gly	Leu	Phe	Gly	Asn	Leu
225					230					235					240
Ile	Ala	Leu	Ser	Leu	Gly	Leu	Thr	Pro	Asn	Phe	Lys	Ser	Asn	Phe	Asp
				245					250					255	
Leu	Ala	Glu	Asp	Ala	Lys	Leu	Gln	Leu	Ser	Lys	Asp	Thr	Tyr	Asp	Asp
			260					265					270		
Asp	Leu	Asp	Asn	Leu	Leu	Ala	Gln	Ile	Gly	Asp	Gln	Tyr	Ala	Asp	Leu
	275						280					285			
Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala	Ile	Leu	Leu	Ser	Asp	Ile
	290					295					300				
Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala	Pro	Leu	Ser	Ala	Ser	Met
305					310					315					320
Ile	Lys	Arg	Tyr	Asp	Glu	His	His	Gln	Asp	Leu	Thr	Leu	Leu	Lys	Ala
				325					330					335	
Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr	Lys	Glu	Ile	Phe	Phe	Asp
				340				345					350		
Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Ile	Asp	Gly	Gly	Ala	Ser	Gln
		355					360					365			
Glu	Glu	Phe	Tyr	Lys	Phe	Ile	Lys	Pro	Ile	Leu	Glu	Lys	Met	Asp	Gly
	370					375					380				
Thr	Glu														

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450	455	460
Thr Arg Lys Ser Glu Glu Thr Ile Thr Pro Trp Asn Phe Glu Glu Val		
465	470	475 480
Val Asp Lys Gly Ala Ser Ala Gln Ser Phe Ile Glu Arg Met Thr Asn		
	485	490 495
Phe Asp Lys Asn Leu Pro Asn Glu Lys Val Leu Pro Lys His Ser Leu		
	500	505 510
Leu Tyr Glu Tyr Phe Thr Val Tyr Asn Glu Leu Thr Lys Val Lys Tyr		
	515	520 525
Val Thr Glu Gly Met Arg Lys Pro Ala Phe Leu Ser Gly Glu Gln Lys		
	530	535 540
Lys Ala Ile Val Asp Leu Leu Phe Lys Thr Asn Arg Lys Val Thr Val		
545	550	555 560
Lys Gln Leu Lys Glu Asp Tyr Phe Lys Lys Ile Glu Cys Phe Asp Ser		
	565	570 575
Val Glu Ile Ser Gly Val Glu Asp Arg Phe Asn Ala Ser Leu Gly Thr		
	580	585 590
Tyr His Asp Leu Leu Lys Ile Ile Lys Asp Lys Asp Phe Leu Asp Asn		
	595	600 605
Glu Glu Asn Glu Asp Ile Leu Glu Asp Ile Val Leu Thr Leu Thr Leu		
	610	615 620
Phe Glu Asp Arg Glu Met Ile Glu Glu Arg Leu Lys Thr Tyr Ala His		
625	630	635 640
Leu Phe Asp Asp Lys Val Met Lys Gln Leu Lys Arg Arg Arg Tyr Thr		
	645	650 655
Gly Trp Gly Arg Leu Ser Arg Lys Leu Ile Asn Gly Ile Arg Asp Lys		
	660	665 670
Gln Ser Gly Lys Thr Ile Leu Asp Phe Leu Lys Ser Asp Gly Phe Ala		
	675	680 685
Asn Arg Asn Phe Met Gln Leu Ile His Asp Asp Ser Leu Thr Phe Lys		
	690	695 700
Glu Asp Ile Gln Lys Ala Gln Val Ser Gly Gln Gly Asp Ser Leu His		
705	710	715 720
Glu His Ile Ala Asn Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly Ile		
	725	730 735
Leu Gln Thr Val Lys Val Val Asp Glu Leu Val Lys Val Met Gly Arg		
	740	745 750
His Lys Pro Glu Asn Ile Val Ile Glu Met Ala Arg Glu Asn Gln Thr		
	755	760 765
Thr Gln Lys Gly Gln Lys Asn Ser Arg Glu Arg Met Lys Arg Ile Glu		
	770	775 780
Glu Gly Ile Lys Glu Leu Gly Ser Gln Ile Leu Lys Glu His Pro Val		
785	790	795 800
Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu Gln		
	805	810 815
Asn Gly Arg Asp Met Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg Leu		
	820	825 830
Ser Asp Tyr Asp Val Asp His Ile Val Pro Gln Ser Phe Leu Lys Asp		
	835	840 845
Asp Ser Ile Asp Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg Gly		
	850	855 860
Lys Ser Asp Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys Asn		
865	870	875 880

-continued

Tyr Trp Arg Gln Leu Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys Phe
 885 890 895

Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp Lys
 900 905 910

Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr Lys
 915 920 925

His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp Glu
 930 935 940

Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr Leu Lys Ser Lys
 945 950 955 960

Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg Glu
 965 970 975

Ile Asn Asn Tyr His His Ala His Asp Ala Tyr Leu Asn Ala Val Val
 980 985 990

Gly Thr Ala Leu Ile Lys Lys Tyr Pro Lys Leu Glu Ser Glu Phe Val
 995 1000 1005

Tyr Gly Asp Tyr Lys Val Tyr Asp Val Arg Lys Met Ile Ala Lys
 1010 1015 1020

Ser Glu Gln Glu Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe Tyr
 1025 1030 1035

Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala Asn
 1040 1045 1050

Gly Glu Ile Arg Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu Thr
 1055 1060 1065

Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr Val Arg
 1070 1075 1080

Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys Thr Glu
 1085 1090 1095

Val Gln Thr Gly Gly Phe Ser Lys Glu Ser Ile Leu Pro Lys Arg
 1100 1105 1110

Asn Ser Asp Lys Leu Ile Ala Arg Lys Lys Asp Trp Asp Pro Lys
 1115 1120 1125

Lys Tyr Gly Gly Phe Asp Ser Pro Thr Val Ala Tyr Ser Val Leu
 1130 1135 1140

Val Val Ala Lys Val Glu Lys Gly Lys Ser Lys Lys Leu Lys Ser
 1145 1150 1155

Val Lys Glu Leu Leu Gly Ile Thr Ile Met Glu Arg Ser Ser Phe
 1160 1165 1170

Glu Lys Asn Pro Ile Asp Phe Leu Glu Ala Lys Gly Tyr Lys Glu
 1175 1180 1185

Val Lys Lys Asp Leu Ile Ile Lys Leu Pro Lys Tyr Ser Leu Phe
 1190 1195 1200

Glu Leu Glu Asn Gly Arg Lys Arg Met Leu Ala Ser Ala Gly Glu
 1205 1210 1215

Leu Gln Lys Gly Asn Glu Leu Ala Leu Pro Ser Lys Tyr Val Asn
 1220 1225 1230

Phe Leu Tyr Leu Ala Ser His Tyr Glu Lys Leu Lys Gly Ser Pro
 1235 1240 1245

Glu Asp Asn Glu Gln Lys Gln Leu Phe Val Glu Gln His Lys His
 1250 1255 1260

Tyr Leu Asp Glu Ile Ile Glu Gln Ile Ser Glu Phe Ser Lys Arg
 1265 1270 1275

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Val	Ile	Leu	Ala	Asp	Ala	Asn	Leu	Asp	Lys	Val	Leu	Ser	Ala	Tyr
1280						1285					1290			
Asn	Lys	His	Arg	Asp	Lys	Pro	Ile	Arg	Glu	Gln	Ala	Glu	Asn	Ile
1295						1300					1305			
Ile	His	Leu	Phe	Thr	Leu	Thr	Asn	Leu	Gly	Ala	Pro	Ala	Ala	Phe
1310						1315					1320			
Lys	Tyr	Phe	Asp	Thr	Thr	Ile	Asp	Arg	Lys	Arg	Tyr	Thr	Ser	Thr
1325						1330					1335			
Lys	Glu	Val	Leu	Asp	Ala	Thr	Leu	Ile	His	Gln	Ser	Ile	Thr	Gly
1340						1345					1350			
Leu	Tyr	Glu	Thr	Arg	Ile	Asp	Leu	Ser	Gln	Leu	Gly	Gly	Asp	
1355						1360					1365			

<210> SEQ ID NO 60
 <211> LENGTH: 1367
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 60

Asp	Lys	Lys	Tyr	Ser	Ile	Gly	Leu	Ala	Ile	Gly	Thr	Asn	Ser	Val	Gly
1				5					10					15	
Trp	Ala	Val	Ile	Thr	Asp	Glu	Tyr	Lys	Val	Pro	Ser	Lys	Lys	Phe	Lys
			20					25					30		
Val	Leu	Gly	Asn	Thr	Asp	Arg	His	Ser	Ile	Lys	Lys	Asn	Leu	Ile	Gly
		35					40					45			
Ala	Leu	Leu	Phe	Asp	Ser	Gly	Glu	Thr	Ala	Glu	Ala	Thr	Arg	Leu	Lys
		50				55					60				
Arg	Thr	Ala	Arg	Arg	Arg	Tyr	Thr	Arg	Arg	Lys	Asn	Arg	Ile	Cys	Tyr
65				70						75				80	
Leu	Gln	Glu	Ile	Phe	Ser	Asn	Glu	Met	Ala	Lys	Val	Asp	Asp	Ser	Phe
			85					90						95	
Phe	His	Arg	Leu	Glu	Glu	Ser	Phe	Leu	Val	Glu	Glu	Asp	Lys	Lys	His
			100					105					110		
Glu	Arg	His	Pro	Ile	Phe	Gly	Asn	Ile	Val	Asp	Glu	Val	Ala	Tyr	His
		115				120					125				
Glu	Lys	Tyr	Pro	Thr	Ile	Tyr	His	Leu	Arg	Lys	Lys	Leu	Val	Asp	Ser
	130					135					140				
Thr	Asp	Lys	Ala	Asp	Leu	Arg	Leu	Ile	Tyr	Leu	Ala	Leu	Ala	His	Met
145					150					155				160	
Ile	Lys	Phe	Arg	Gly	His	Phe	Leu	Ile	Glu	Gly	Asp	Leu	Asn	Pro	Asp
			165					170						175	
Asn	Ser	Asp	Val	Asp	Lys	Leu	Phe	Ile	Gln	Leu	Val	Gln	Thr	Tyr	Asn
		180						185					190		
Gln	Leu	Phe	Glu	Glu	Asn	Pro	Ile	Asn	Ala	Ser	Gly	Val	Asp	Ala	Lys
		195				200						205			
Ala	Ile	Leu	Ser	Ala	Arg	Leu	Ser	Lys	Ser	Arg	Arg	Leu	Glu	Asn	Leu
	210					215					220				
Ile	Ala	Gln	Leu	Pro	Gly	Glu	Lys	Lys	Asn	Gly	Leu	Phe	Gly	Asn	Leu
225					230					235				240	
Ile	Ala	Leu	Ser	Leu	Gly	Leu	Thr	Pro	Asn	Phe	Lys	Ser	Asn	Phe	Asp
			245					250						255	
Leu	Ala	Glu	Asp	Ala	Lys	Leu	Gln	Leu	Ser	Lys	Asp	Thr	Tyr	Asp	Asp
		260					265						270		

Asp	Leu	Asn	Leu	Leu	Ala	Gln	Ile	Gly	Asp	Gln	Tyr	Ala	Asp	Leu
275						280					285			
Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala	Ile	Leu	Leu	Ser	Ile
290						295					300			
Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala	Pro	Leu	Ser	Ala	Met
305					310					315				320
Ile	Lys	Arg	Tyr	Asp	Glu	His	His	Gln	Asp	Leu	Thr	Leu	Leu	Ala
				325					330					335
Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr	Lys	Glu	Ile	Phe	Asp
			340					345					350	
Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Ile	Asp	Gly	Gly	Ala	Gln
		355					360					365		
Glu	Glu	Phe	Tyr	Lys	Phe	Ile	Lys	Pro	Ile	Leu	Glu	Lys	Met	Gly
						375					380			
Thr	Glu	Glu	Leu	Leu	Val	Lys	Leu	Asn	Arg	Glu	Asp	Leu	Leu	Lys
385					390					395				400
Gln	Arg	Thr	Phe	Asp	Asn	Gly	Ser	Ile	Pro	His	Gln	Ile	His	Gly
				405					410					
Glu	Leu	His	Ala	Ile	Leu	Arg	Arg	Gln	Glu	Asp	Phe	Tyr	Pro	Leu
			420					425					430	
Lys	Asp	Asn	Arg	Glu	Lys	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Pro
		435					440					445		
Tyr	Tyr	Val	Gly	Pro	Leu	Ala	Arg	Gly	Asn	Ser	Arg	Phe	Ala	Met
		450				455					460			
Thr	Arg	Lys	Ser	Glu	Glu	Thr	Ile	Thr	Pro	Trp	Asn	Phe	Glu	Val
465					470					475				480
Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Asn
				485					490					
Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Leu
			500					505					510	
Leu	Tyr	Glu	Tyr	Phe	Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Tyr
		515					520					525		
Val	Thr	Glu	Gly	Met	Arg	Lys	Pro	Ala	Phe	Leu	Ser	Gly	Glu	Lys
		530				535					540			
Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Val
545					550					555				560
Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Ser
				565					570					575
Val	Glu	Ile	Ser	Gly	Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Thr
			580					585					590	
Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asn
		595					600					605		
Glu	Glu	Asn	Glu	Asp	Ile	Leu	Glu	Asp	Ile	Val	Leu	Thr	Leu	Leu
					615						620			
Phe	Glu	Asp	Arg	Glu	Met	Ile	Glu	Glu	Arg	Leu	Lys	Thr	Tyr	His
625					630					635				640
Leu	Phe	Asp	Asp	Lys	Val	Met	Lys	Gln	Leu	Lys	Arg			

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690	695	700
Glu Asp Ile Gln Lys Ala Gln Val Ser Gly Gln Gly Asp Ser Leu His		
705	710	715 720
Glu His Ile Ala Asn Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly Ile		
	725	730 735
Leu Gln Thr Val Lys Val Val Asp Glu Leu Val Lys Val Met Gly Arg		
	740	745 750
His Lys Pro Glu Asn Ile Val Ile Glu Met Ala Arg Glu Asn Gln Thr		
	755	760 765
Thr Gln Lys Gly Gln Lys Asn Ser Arg Glu Arg Met Lys Arg Ile Glu		
	770 775	780
Glu Gly Ile Lys Glu Leu Gly Ser Gln Ile Leu Lys Glu His Pro Val		
785	790	795 800
Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu Gln		
	805	810 815
Asn Gly Arg Asp Met Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg Leu		
	820	825 830
Ser Asp Tyr Asp Val Asp His Ile Val Pro Gln Ser Phe Leu Ala Asp		
	835	840 845
Asp Ser Ile Asp Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg Gly		
	850 855	860
Lys Ser Asp Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys Asn		
865	870	875 880
Tyr Trp Arg Gln Leu Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys Phe		
	885	890 895
Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp Lys		
	900	905 910
Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr Lys		
	915	920 925
His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp Glu		
	930 935	940
Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr Leu Lys Ser Lys		
945	950	955 960
Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg Glu		
	965	970 975
Ile Asn Asn Tyr His His Ala His Asp Ala Tyr Leu Asn Ala Val Val		
	980	985 990
Gly Thr Ala Leu Ile Lys Lys Tyr Pro Ala Leu Glu Ser Glu Phe Val		
	995	1000 1005
Tyr Gly Asp Tyr Lys Val Tyr Asp Val Arg Lys Met Ile Ala Lys		
1010	1015	1020
Ser Glu Gln Glu Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe Tyr		
1025	1030	1035
Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala Asn		
1040	1045	1050
Gly Glu Ile Arg Lys Ala Pro Leu Ile Glu Thr Asn Gly Glu Thr		
1055	1060	1065
Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr Val Arg		
1070	1075	1080
Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys Thr Glu		
1085	1090	1095
Val Gln Thr Gly Gly Phe Ser Lys Glu Ser Ile Leu Pro Lys Arg		
1100	1105	1110

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Asn Ser	Asp Lys Leu Ile	Ala	Arg Lys Lys Asp	Trp	Asp Pro Lys
1115		1120		1125	
Lys Tyr	Gly Gly Phe Asp	Ser	Pro Thr Val Ala	Tyr	Ser Val Leu
1130		1135		1140	
Val Val	Ala Lys Val Glu	Lys	Gly Lys Ser Lys	Lys	Leu Lys Ser
1145		1150		1155	
Val Lys	Glu Leu Leu Gly	Ile	Thr Ile Met Glu	Arg	Ser Ser Phe
1160		1165		1170	
Glu Lys	Asn Pro Ile Asp	Phe	Leu Glu Ala Lys	Gly	Tyr Lys Glu
1175		1180		1185	
Val Lys	Lys Asp Leu Ile	Ile	Lys Leu Pro Lys	Tyr	Ser Leu Phe
1190		1195		1200	
Glu Leu	Glu Asn Gly Arg	Lys	Arg Met Leu Ala	Ser	Ala Gly Glu
1205		1210		1215	
Leu Gln	Lys Gly Asn Glu	Leu	Ala Leu Pro Ser	Lys	Tyr Val Asn
1220		1225		1230	
Phe Leu	Tyr Leu Ala Ser	His	Tyr Glu Lys Leu	Lys	Gly Ser Pro
1235		1240		1245	
Glu Asp	Asn Glu Gln Lys	Gln	Leu Phe Val Glu	Gln	His Lys His
1250		1255		1260	
Tyr Leu	Asp Glu Ile Ile	Glu	Gln Ile Ser Glu	Phe	Ser Lys Arg
1265		1270		1275	
Val Ile	Leu Ala Asp Ala	Asn	Leu Asp Lys Val	Leu	Ser Ala Tyr
1280		1285		1290	
Asn Lys	His Arg Asp Lys	Pro	Ile Arg Glu Gln	Ala	Glu Asn Ile
1295		1300		1305	
Ile His	Leu Phe Thr Leu	Thr	Asn Leu Gly Ala	Pro	Ala Ala Phe
1310		1315		1320	
Lys Tyr	Phe Asp Thr Thr	Ile	Asp Arg Lys Arg	Tyr	Thr Ser Thr
1325		1330		1335	
Lys Glu	Val Leu Asp Ala	Thr	Leu Ile His Gln	Ser	Ile Thr Gly
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<210> SEQ ID NO 61

<211> LENGTH: 4101

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

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<210> SEQ ID NO 62

<211> LENGTH: 4101

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 62

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<210> SEQ ID NO 63

<211> LENGTH: 4092

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 63

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<210> SEQ ID NO 64

<211> LENGTH: 4101

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

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<210> SEQ ID NO 65

<211> LENGTH: 4101

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 65

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<210> SEQ ID NO 66
 <211> LENGTH: 4092
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 66

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<210> SEQ ID NO 67

<211> LENGTH: 4101

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

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<210> SEQ ID NO 68

<211> LENGTH: 4101

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 68

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<210> SEQ ID NO 69

<211> LENGTH: 4101

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

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ttgtcacaac tgggtgggga t 4101

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<210> SEQ ID NO 70

<211> LENGTH: 3307

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 70

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cgagcatcac caggatctca ccttgcctaa ggcccttggt cggcagcagc tccctgagaa 240
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ttctgccag tccttcattg agcggatgac taacttcgac aagaacctgc ccaacgagaa 720
ggtgctgcc aagcacagcc tgctctacga atactttact gtgtacaatg agctgacgaa 780
ggtgaagtac gtgacagagg gtagcgga gcccgtttc ctgagcggcg agcaaaaaaa 840

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ggagaacatc	atacactgtg	tcacctcac	gaacctcggg	gctcccgag	ccttcaaata	3180
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<210> SEQ ID NO 71
 <211> LENGTH: 4101
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 71

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 acacgcttaa aacgcacagc taggaggaga tatacacggc gcaagaaccg aatatgctac 240
 ttacaggaga tattctccaa tgagatggcg aagggtggacg actctttctt ccatcggtctt 300
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 aacgcctctg ggggtggacgc taaagctatc ctctcagccc gcctgtcaaa gtcgaggagg 660
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 ctgtcggata tactgcggt gaacaccgag atcactaaag cacctctctc cgccagcatg 960
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gccgccttca aatacttcga cactaccatc gacaggaagc gctacacatc tacgaaggaa	4020
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<210> SEQ ID NO 72

<211> LENGTH: 427

<212> TYPE: PRT

<213> ORGANISM: Triticum aestivum

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<400> SEQUENCE: 72

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Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Thr Lys Asn Pro Thr Lys
 1           5           10           15

Thr Leu Ala Ala Ile Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
      20           25           30

Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
      35           40           45

Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Ser Ala Pro His
      50           55           60

Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val
      65           70           75           80

Leu Leu Gly Ser Leu Ala Gly Phe Leu Leu Ser Leu Ala Phe Leu Tyr
      85           90           95

Ala Ile Arg Val Ala Val Leu His Ala Gly Asn Ala Pro Ala Val Ala
      100          105          110

Arg Gly Pro Val Ser Phe Thr Pro Gln Ile Ser Pro Lys Ser Leu Gln
      115          120          125

Gly Ala Leu Pro Ser Ala Arg Pro Leu Ala Arg Gly Pro Arg Gly Thr
      130          135          140

Tyr His Lys Leu Asp Leu Asp Gly Asp Leu Thr Val Ala Val Lys Val
      145          150          155          160

Leu Asp Leu Ala Ala Ala Ser Arg Ala Glu Ala Ser Pro Ser Pro Ser
      165          170          175

Arg Pro Ala Asn Gly Ser Asn Ser Lys Ser Asp Met Arg Arg Val Gln
      180          185          190

Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Leu Asn Val Met Ser
      195          200          205

Leu Lys Ala Tyr Val Arg Asp Ala Asp Arg Leu Ser Leu Val Tyr Asp
      210          215          220

Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val Arg Ser
      225          230          235          240

Gln Gln Val Ser Leu Ser Trp Asp Thr Arg Asn Arg Ile Ala Ala Gly
      245          250          255

Val Ala Lys Gly Leu Arg His Leu His Phe Glu Cys Asn Pro Arg Ile
      260          265          270

Leu His Cys Asn Leu Lys Pro Ser Asn Val Met Leu Glu Glu Gly Phe
      275          280          285

Glu Pro Val Leu Ala Asp Cys Gly Val Ala Arg Leu Ile Asp Ser Gly
      290          295          300

Ser Pro Asp Pro Gln Ser Ser Gly Ser Leu Tyr Thr Ala Pro Glu Cys
      305          310          315          320

Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Cys Asp Val Tyr Ala Phe Gly
      325          330          335

Met Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Ser Asp Pro Phe
      340          345          350

Phe Ser Gly Glu Ser Gly Arg Gly Ser Leu Ala Arg Trp Leu Arg His
      355          360          365

Met Gln His Ser Gly Asp Met Lys Glu Ala Leu Asp Ser Ser Ile Val
      370          375          380

Gly Glu Glu Val Asp Glu Glu Glu Met Val Met Ala Ile Arg Ile Ala
      385          390          395          400

Ile Val Cys Leu Ser Glu Leu Pro Ala Asp Arg Pro Ser Ser Asp Glu

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405	410	415
Leu Val Ala Met Leu Ala Gln Leu His Ser Phe		
420	425	
 <210> SEQ ID NO 73		
<211> LENGTH: 427		
<212> TYPE: PRT		
<213> ORGANISM: Triticum dicoccoides		
 <400> SEQUENCE: 73		
Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Thr Lys Asn Pro Thr Lys		
1	5	10
Thr Leu Ala Ala Ile Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro		
20	25	30
Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr		
35	40	45
Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Ser Ala Pro His		
50	55	60
Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val		
65	70	75
Leu Leu Gly Ser Leu Ala Gly Phe Leu Leu Ser Leu Ala Phe Leu Tyr		
85	90	95
Ala Ile Arg Val Ala Val Leu His Ala Gly Asn Ala Pro Ala Val Ala		
100	105	110
Arg Gly Pro Val Ser Phe Thr Pro Gln Ile Ser Pro Lys Ser Leu Gln		
115	120	125
Cys Ala Leu Pro Ser Ala Arg Pro Leu Ala Arg Gly Pro Arg Gly Thr		
130	135	140
Tyr His Lys Leu Asp Leu Asp Gly Asp Leu Thr Val Ala Val Lys Val		
145	150	155
Leu Asp Leu Ala Ala Ala Ser Arg Ala Glu Ala Ser Pro Ser Pro Ser		
165	170	175
Arg Pro Ala Asn Gly Ser Asn Ser Lys Ser Asp Met Arg Arg Val Gln		
180	185	190
Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Leu Asn Val Met Ser		
195	200	205
Leu Lys Ala Tyr Val Arg Asp Ala Asp Arg Leu Ser Leu Val Tyr Asp		
210	215	220
Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val Arg Ser		
225	230	235
Gln Gln Val Ser Leu Ser Trp Asp Thr Arg Asn Arg Ile Ala Ala Gly		
245	250	255
Val Ala Lys Gly Leu Arg His Leu His Phe Glu Cys Asn Pro Arg Ile		
260	265	270
Leu His Cys Asn Leu Lys Pro Ser Asn Val Met Leu Glu Glu Gly Phe		
275	280	285
Glu Pro Val Leu Ala Asp Cys Gly Val Ala Arg Leu Ile Asp Ser Gly		
290	295	300
Ser Pro Asp Pro Glu Ser Ser Gly Ser Leu Tyr Thr Ala Pro Glu Cys		
305	310	315
Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Cys Asp Val Tyr Ala Phe Gly		
325	330	335
Met Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Ser Asp Pro Phe		
340	345	350

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Phe Ser Gly Glu Ser Gly Arg Gly Ser Leu Ala Arg Trp Leu Arg His
   355                               360                               365

Met Gln His Ser Gly Asp Met Lys Glu Ala Leu Asp Ser Ser Ile Val
   370                               375                               380

Gly Glu Glu Val Asp Glu Glu Glu Met Val Met Ala Val Arg Ile Ala
   385                               390                               395                               400

Ile Val Cys Leu Ser Glu Leu Pro Ala Asp Arg Pro Ser Ser Asp Glu
           405                               410                               415

Leu Val Ala Met Leu Ala Gln Leu His Ser Phe
           420                               425

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<210> SEQ ID NO 74
<211> LENGTH: 427
<212> TYPE: PRT
<213> ORGANISM: Triticum aestivum

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<400> SEQUENCE: 74

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Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Lys Asn Pro Thr Lys
 1          5          10          15

Thr Leu Ala Ala Thr Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
 20          25          30

Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
 35          40          45

Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Arg Ala Pro His
 50          55          60

Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val
 65          70          75          80

Leu Leu Gly Ser Leu Ala Gly Phe Leu Leu Ser Leu Ala Phe Leu Tyr
 85          90          95

Ala Ile Arg Val Ala Val Leu His Ala Gly Asn Ala Pro Ala Val Ala
100          105          110

Arg Gly Pro Val Ser Phe Thr Pro Gln Ile Ser Pro Lys Ser Leu Gln
115          120          125

Cys Ala Leu Pro Ser Ala Arg Pro Leu Ala Arg Gly Pro Arg Gly Thr
130          135          140

Tyr His Lys Leu Asp Leu Asp Gly Asp Val Thr Val Ala Val Lys Val
145          150          155          160

Leu Asp Leu Ala Ala Ala Ser Arg Ala Ser Pro Ser Pro Ser Pro Ser
165          170          175

Pro Ser Arg Pro Ala Ser Gly Ser Lys Ser Asp Met Arg Arg Val Gln
180          185          190

Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Leu Asn Val Met Ser
195          200          205

Leu Lys Ala Tyr Val Arg Asp Ala Asp Arg Leu Ser Leu Val Tyr Asp
210          215          220

Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val Arg Ser
225          230          235          240

Gln Gln Val Ser Leu Ser Trp Asp Thr Arg Asn Arg Ile Ala Ala Gly
245          250          255

Val Ala Lys Gly Leu Arg His Leu His Phe Glu Cys Asn Pro Arg Ile
260          265          270

Leu His Cys Asn Leu Lys Pro Ser Asn Val Met Leu Glu Glu Gly Phe
275          280          285

Glu Pro Val Leu Ala Asp Cys Gly Val Ala Arg Leu Leu Asp Ser Gly
290          295          300

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Ser Pro Asp Pro Glu Ser Ser Gly Ser Leu Tyr Thr Ala Pro Glu Cys
 305 310 315 320
 Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Cys Asp Val Tyr Ala Phe Gly
 325 330 335
 Met Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Ala Asp Leu Phe
 340 345 350
 Phe Ser Gly Glu Ser Gly Arg Gly Ser Leu Ala Arg Trp Leu Arg His
 355 360 365
 Met Gln His Ser Gly Asp Met Lys Glu Ala Leu Asp Ser Ser Ile Val
 370 375 380
 Gly Glu Glu Val Asp Glu Glu Glu Met Val Met Ala Val Arg Val Ala
 385 390 395 400
 Ile Val Cys Leu Ser Glu Leu Pro Ala Asp Arg Pro Ser Ser Asp Glu
 405 410 415
 Leu Val Ala Met Leu Ala Gln Leu His Ser Phe
 420 425

<210> SEQ ID NO 75
 <211> LENGTH: 429
 <212> TYPE: PRT
 <213> ORGANISM: Aegilops tauschii

<400> SEQUENCE: 75

Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Ala Lys Asn Pro Thr Lys
 1 5 10 15
 Thr Leu Ala Ala Thr Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
 20 25 30
 Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
 35 40 45
 Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Arg Ala Pro His
 50 55 60
 Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val
 65 70 75 80
 Leu Leu Gly Ser Leu Ala Gly Phe Leu Leu Ser Leu Ala Phe Leu Tyr
 85 90 95
 Ala Ile Arg Val Ala Val Leu His Ala Gly Asn Ala Pro Ala Val Ala
 100 105 110
 Arg Gly Pro Val Ser Phe Thr Pro Gln Ile Ser Pro Lys Ser Leu Gln
 115 120 125
 Cys Ala Leu Pro Ser Ala Arg Pro Leu Ala Arg Gly Pro Arg Gly Thr
 130 135 140
 Tyr His Lys Leu Asp Leu Asp Gly Asp Val Thr Val Ala Val Lys Val
 145 150 155 160
 Leu Asp Leu Ala Ala Ala Ser Arg Ala Ser Pro Ser Pro Ser Pro Ser
 165 170 175
 Pro Ser Pro Ser Arg Pro Ala Ser Gly Ser Lys Ser Asp Met Arg Arg
 180 185 190
 Val Gln Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Leu Asn Val
 195 200 205
 Met Ser Leu Lys Ala Tyr Val Arg Asp Ala Asp Arg Leu Ser Leu Val
 210 215 220
 Tyr Asp Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val
 225 230 235 240
 Arg Ser Gln Gln Val Ser Leu Ser Trp Asp Thr Arg Asn Arg Ile Ala

245										250					255				
Ala	Gly	Val	Ala	Lys	Gly	Leu	Arg	His	Leu	His	Phe	Glu	Cys	Asn	Pro				
			260					265					270						
Arg	Ile	Leu	His	Cys	Asn	Leu	Lys	Pro	Ser	Asn	Val	Met	Leu	Glu	Glu				
			275					280					285						
Gly	Phe	Glu	Pro	Val	Leu	Ala	Asp	Cys	Gly	Val	Ala	Arg	Leu	Leu	Asp				
			290		295						300								
Ser	Gly	Ser	Pro	Asp	Pro	Glu	Ser	Ser	Gly	Ser	Leu	Tyr	Thr	Ala	Pro				
305			310						315			320							
Glu	Cys	Tyr	Gln	Ser	Ser	Arg	Tyr	Thr	Asp	Lys	Cys	Asp	Val	Tyr	Ala				
			325					330					335						
Phe	Gly	Met	Ile	Leu	Gly	Val	Leu	Leu	Thr	Gly	Lys	Asp	Pro	Ala	Asp				
			340					345					350						
Leu	Phe	Phe	Ser	Gly	Glu	Ser	Gly	Arg	Gly	Ser	Leu	Ala	Arg	Trp	Leu				
355						360						365							
Arg	His	Met	Gln	His	Ser	Gly	Asp	Met	Lys	Glu	Ala	Leu	Asp	Ser	Ser				
370						375					380								
Ile	Val	Gly	Glu	Glu	Val	Asp	Glu	Glu	Glu	Met	Val	Met	Ala	Val	Arg				
385			390						395			400							
Val	Ala	Ile	Val	Cys	Leu	Ser	Glu	Leu	Pro	Ala	Asp	Arg	Pro	Ser	Ser				
			405					410						415					
Asp	Glu	Leu	Val	Ala	Met	Leu	Ala	Gln	Leu	His	Ser	Phe							
			420					425											

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<210> SEQ ID NO 76
<211> LENGTH: 428
<212> TYPE: PRT
<213> ORGANISM: Triticum aestivum
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<400> SEQUENCE: 76

Met	Ala	Thr	Gly	Ala	Gly	Thr	Ala	Ala	Ala	Ala	Ala	Lys	Asn	Pro	Thr
1			5						10					15	
Lys	Thr	Leu	Ala	Thr	Leu	Leu	Leu	Leu	Val	Leu	Leu	Leu	Ala	Asp	Leu
		20					25						30		
Pro	Leu	Cys	Ala	Ser	Gln	Pro	Pro	Leu	His	Ser	Gln	Pro	Leu	Pro	Ala
		35				40						45			
Thr	Gln	Ser	Pro	Ala	Ala	Pro	Leu	Pro	Pro	Pro	Gln	Pro	Arg	Gly	Pro
	50					55					60				
Arg	Ala	Gln	Ala	Gly	Gly	Ala	Ala	Arg	Leu	Arg	Arg	Ile	Ala	Leu	Gly
65				70						75				80	
Val	Leu	Leu	Gly	Ser	Leu	Ala	Gly	Phe	Ile	Leu	Ser	Leu	Ala	Phe	Leu
			85						90					95	
Tyr	Ala	Ile	Arg	Val	Ala	Val	Leu	His	Ala	Gly	Asn	Ala	Pro	Ala	Val
		100					105					110			
Ala	Arg	Gly	Pro	Val	Ser	Phe	Thr	Pro	Gln	Ile	Ser	Pro	Lys	Ser	Leu
		115					120					125			
Gln	Cys	Ala	Leu	Pro	Ser	Ala	Arg	Pro	Leu	Ala	Arg	Gly	Pro	Arg	Gly
	130					135					140				
Thr	Tyr	His	Lys	Leu	Asp	Leu	Asp	Gly	Asp	Leu	Thr	Val	Ala	Val	Lys
145				150					155					160	
Val	Leu	Asp	Leu	Ala	Ala	Ala	Ala	Gly	Ala	Ser	Pro	Ser	Pro	Ser	Pro
			165					170						175	
Ser	Pro	Ser	Arg	Pro	Ala	Ser	Gly	Ser	Lys	Ser	Asp	Met	Arg	Arg	Val
			180				185					190			

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Gln Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Leu Asn Val Met
 195 200 205
 Ser Leu Lys Ala Tyr Val Arg Asp Ala Asp Arg Leu Ser Leu Val Tyr
 210 215 220
 Asp Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val Arg
 225 230 235 240
 Ser Gln Gln Val Ser Leu Ser Trp Asp Thr Arg Asn Arg Ile Ala Ala
 245 250 255
 Gly Val Ala Lys Gly Leu Arg His Leu His Phe Glu Cys Asn Pro Arg
 260 265 270
 Ile Leu His Cys Asn Leu Lys Pro Ser Asn Val Met Leu Glu Glu Gly
 275 280 285
 Phe Glu Pro Val Leu Ala Asp Cys Gly Val Ala Arg Leu Leu Asp Ser
 290 295 300
 Gly Ser Pro Asp Pro Glu Ser Ser Gly Ser Leu Tyr Ala Ala Pro Glu
 305 310 315 320
 Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Cys Asp Val Tyr Ala Phe
 325 330 335
 Gly Met Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Ala Asp Pro
 340 345 350
 Phe Phe Ser Gly Glu Ser Gly Arg Gly Ser Leu Ala Arg Trp Leu Arg
 355 360 365
 His Met Gln His Ser Gly Asp Thr Lys Glu Ala Leu Asp Ser Ser Ile
 370 375 380
 Val Gly Glu Glu Val Asp Glu Glu Glu Met Val Met Ala Val Arg Val
 385 390 395 400
 Ala Ile Val Cys Leu Ser Glu Leu Pro Ala Asp Arg Pro Ser Ser Asp
 405 410 415
 Glu Leu Val Ala Met Leu Ala Gln Leu His Ser Phe
 420 425

<210> SEQ ID NO 77

<211> LENGTH: 428

<212> TYPE: PRT

<213> ORGANISM: Triticum dicoccoides

<400> SEQUENCE: 77

Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Ala Lys Asn Pro Thr
 1 5 10 15
 Lys Thr Leu Ala Thr Leu Leu Leu Val Leu Leu Leu Ala Asp Leu
 20 25 30
 Pro Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Ser Ala
 35 40 45
 Thr Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Arg Pro Pro
 50 55 60
 Arg Ala Gln Ala Gly Gly Ala Ala His Leu Arg Arg Ile Ala Leu Gly
 65 70 75 80
 Val Leu Leu Gly Ser Leu Ala Gly Phe Leu Leu Ser Leu Ala Phe Leu
 85 90 95
 Tyr Ala Ile Arg Val Ala Val Leu His Ala Gly Asn Ala Pro Ala Val
 100 105 110
 Ala Arg Gly Pro Val Ser Phe Thr Pro Gln Ile Ser Pro Lys Ser Leu
 115 120 125
 Gln Cys Ala Leu Pro Ser Ala Arg Pro Leu Ala Arg Gly Pro Arg Gly
 130 135 140

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Thr Tyr His Lys Leu Asp Leu Asp Gly Asp Leu Thr Val Ala Val Lys
 145 150 155 160
 Val Leu Asp Leu Ala Ala Ala Ala Arg Ala Ser Pro Ser Pro Ser Pro
 165 170 175
 Ser Pro Ser Arg Pro Ala Ser Gly Ser Lys Ser Asp Met Arg Arg Val
 180 185 190
 Gln Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Leu Asn Val Met
 195 200 205
 Ser Leu Lys Ala Tyr Val Arg Asp Ala Asp Arg Leu Ser Leu Val Tyr
 210 215 220
 Asp Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val Arg
 225 230 235 240
 Ser Gln Gln Val Ser Leu Ser Trp Asp Thr Arg Asn Arg Ile Ala Ala
 245 250 255
 Gly Val Ala Lys Gly Leu Arg His Leu His Phe Glu Cys Asn Pro Arg
 260 265 270
 Ile Leu His Cys Asn Leu Lys Pro Ser Asn Val Met Leu Glu Glu Gly
 275 280 285
 Phe Glu Pro Val Leu Ala Asp Cys Gly Val Ala Arg Leu Ile Asp Ser
 290 295 300
 Gly Leu Pro Asp Pro Glu Ser Ser Gly Ser Leu Tyr Ala Ala Pro Glu
 305 310 315 320
 Cys Tyr Gln Ser Asn Arg Tyr Thr Asp Lys Cys Asp Val Tyr Ala Phe
 325 330 335
 Gly Met Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Ala Asp Pro
 340 345 350
 Phe Phe Ser Gly Glu Ser Gly Arg Gly Ser Leu Ala Arg Trp Leu Arg
 355 360 365
 His Met Gln His Ser Gly Asp Thr Lys Glu Ala Leu Asp Ser Ser Ile
 370 375 380
 Val Gly Glu Glu Val Asp Glu Glu Glu Met Val Met Ala Val Arg Val
 385 390 395 400
 Ala Ile Val Cys Leu Ser Glu Leu Pro Ala Asp Arg Pro Ser Ser Asp
 405 410 415
 Glu Leu Val Ala Met Leu Ala Gln His His Ser Phe
 420 425

<210> SEQ ID NO 78

<211> LENGTH: 420

<212> TYPE: PRT

<213> ORGANISM: Brachypodium distachyon

<400> SEQUENCE: 78

Met Ala Arg Gly Gly Gly Val Val Val Gly Lys Lys Asn Pro Thr Lys
 1 5 10 15
 Thr Leu Ala Ala Ile Leu Leu Leu Leu Leu Leu Ala Val Phe Pro
 20 25 30
 Arg Pro Ala Ala Ser Gln Pro Leu His Ser Glu Pro Met Ser Thr Gln
 35 40 45
 Gln Ser Pro Pro Pro Pro Gln Gln Gln Ser Lys Ile Pro His Ala Gln
 50 55 60
 Pro Gly Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val Leu
 65 70 75 80
 Phe Gly Ser Leu Thr Gly Phe Leu Leu Ser Leu Ala Leu Leu Tyr Ala

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85				90				95							
Ile	Arg	Leu	Thr	Val	Leu	His	Ala	Arg	Asp	Ala	Pro	Ala	Ile	Ala	Lys
		100							105				110		
Gly	Pro	Val	Ser	Phe	Thr	Pro	Gln	Ile	Ser	Pro	Lys	Asn	Leu	Leu	Cys
		115					120					125			
Ala	Leu	Pro	Ala	Ala	Gln	Pro	Leu	Ala	His	Gly	Thr	Tyr	His	Lys	Leu
		130				135					140				
Ala	Leu	Asp	Gly	Gly	Leu	Thr	Val	Ala	Val	Lys	Arg	Leu	Glu	Ser	Val
145					150					155					160
Ala	Ala	Asn	Arg	Pro	Glu	Ala	Ser	Pro	Ser	Ala	Pro	Pro	Pro	Asn	Gly
			165						170					175	
Ser	Lys	Ser	Asp	Met	Arg	Arg	Val	Gln	Arg	Gln	Leu	Glu	Val	Leu	Ala
			180					185					190		
Arg	Val	Arg	His	Gln	Ser	Val	Met	Ser	Leu	Lys	Ala	Tyr	Val	Arg	Glu
		195					200					205			
Val	Asp	Arg	Leu	Ser	Leu	Val	Tyr	Asn	Phe	Val	Pro	Gly	Gly	Ser	Leu
	210					215					220				
Glu	Asp	Val	Met	Lys	Arg	Val	Arg	Ser	Gln	Gln	Leu	Ser	Leu	Asp	Trp
225					230					235				240	
Asp	Thr	Arg	Ser	Arg	Ile	Ala	Ala	Gly	Val	Ala	Lys	Gly	Leu	Arg	Tyr
			245						250					255	
Leu	His	Phe	Glu	Cys	Cys	Pro	Ser	Ile	Leu	His	Cys	Asn	Leu	Lys	Pro
		260						265					270		
Ser	Asn	Val	Met	Leu	Asp	Glu	Gly	Phe	Glu	Pro	Ile	Leu	Ala	Asp	Cys
		275					280					285			
Gly	Val	Ala	Ser	Leu	Ile	Asp	Ser	Gly	Pro	Ala	Asp	Pro	Glu	Ser	Phe
	290					295					300				
Gly	Ser	Val	Tyr	Ala	Ala	Pro	Glu	Phe	Tyr	Gln	Asn	Ser	Arg	Tyr	Thr
305					310					315				320	
Asp	Lys	Cys	Asp	Val	Tyr	Ala	Phe	Gly	Met	Ile	Leu	Gly	Val	Leu	Leu
			325						330					335	
Thr	Gly	Arg	Asp	Pro	Lys	Asp	Pro	Phe	Phe	Ser	Gly	Glu	Ala	Gly	Arg
			340					345					350		
Gly	Gly	Leu	Ala	Arg	Trp	Leu	Arg	His	Met	Gln	His	Ser	Gly	Glu	Ala
		355				360					365				
Lys	Glu	Ala	Leu	Asp	Ser	Ser	Ile	Leu	Gly	Glu	Glu	Val	Cys	Glu	Glu
	370					375					380				
Glu	Met	Leu	Met	Ala	Ile	Arg	Val	Ala	Ile	Val	Cys	Leu	Ser	Asp	Leu
385					390					395				400	
Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu	Val	Ala	Met	Leu	Ala	Gln
			405						410					415	
Leu	His	Ser	Leu												
			420												

<210> SEQ ID NO 79

<211> LENGTH: 421

<212> TYPE: PRT

<213> ORGANISM: Oryza brachyantha

<400> SEQUENCE: 79

Met	Glu	Glu	Gly	Gly	Gly	Met	Met	Ala	Ser	Lys	Asn	Pro	Thr	Lys	Thr
1				5						10				15	

Leu	Ala	Ile	Leu	Leu	Leu	Val	Leu	Val	Phe	Phe	Leu	Leu	Ser	Leu	Cys
			20					25					30		

Ser	Leu	Ala	Ala	Ser	Gln	Pro	Leu	His	Ser	Glu	Pro	Met	Ser	Thr	Ala
35						40						45			
Glu	Tyr	Ser	Pro	Pro	Pro	Ser	Pro	Pro	Pro	Pro	Gln	Ser	Lys	Ile	Pro
50						55						60			
His	Ala	Gln	Ala	Gly	Gly	Ala	Ala	Arg	Leu	Arg	Arg	Ile	Val	Leu	Gly
65			70						75			80			
Val	Leu	Phe	Gly	Ser	Leu	Thr	Gly	Phe	Leu	Leu	Ala	Leu	Ala	Phe	Leu
			85						90			95			
Tyr	Ala	Ile	Arg	Val	Ala	Ile	Leu	His	Ala	Lys	Tyr	Ala	Pro	Ala	Ile
			100			105						110			
Ile	Lys	Gly	Pro	Val	Ser	Phe	Thr	Pro	Gln	Ile	Ser	Pro	Lys	Asn	Leu
115						120						125			
Gln	Ser	Ala	Leu	Pro	Ser	Ala	Gln	Pro	Leu	Thr	His	Gly	Pro	Asn	Gly
130						135						140			
Lys	Tyr	Tyr	Lys	Leu	Val	Leu	Asp	Asn	Asp	Val	Thr	Val	Ala	Val	Lys
145			150						155			160			
Arg	Leu	Glu	Ala	Ala	Ser	Arg	Pro	Glu	Ala	Ser	Pro	Ser	Met	Pro	Asn
			165						170			175			
Val	Ser	Lys	Ser	Asp	Met	Arg	Arg	Val	Gln	Arg	Gln	Leu	Glu	Leu	Leu
			180			185						190			
Ala	Arg	Val	Arg	His	Gln	Asn	Val	Met	Ala	Leu	Lys	Ala	Tyr	Val	Arg
195						200						205			
Glu	Ala	Asp	Arg	Leu	Ser	Leu	Ala	Tyr	Asp	Phe	Val	Pro	Gly	Gly	Ser
210						215						220			
Leu	Glu	Asp	Met	Met	Lys	Arg	Val	Arg	Ser	Gln	Gln	Val	Asn	Leu	Asn
225			230						235			240			
Trp	Asp	Ala	Arg	Asn	Arg	Ile	Ala	Ile	Gly	Val	Ala	Lys	Gly	Leu	Arg
			245						250			255			
Tyr	Leu	His	Phe	Glu	Cys	Asn	Pro	Arg	Ile	Leu	His	Ser	Asn	Leu	Lys
			260			265						270			
Pro	Ser	Asn	Val	Met	Leu	Asp	Glu	Gly	Phe	Glu	Pro	Ser	Leu	Ala	Asp
275						280						285			
Cys	Gly	Val	Ser	Arg	Leu	Ile	Ala	Ser	Gly	Ser	Ala	Asp	Pro	Glu	Leu
290						295						300			
Ala	Asn	Ser	Leu	Tyr	Ser	Ala	Pro	Glu	Cys	Tyr	Gln	Ser	Ser	Arg	Tyr
305			310						315			320			
Thr	Asp	Lys	Ser	Asp	Val	Tyr	Ser	Phe	Gly	Met	Ile	Leu	Gly	Val	Leu
			325						330			335			
Leu	Thr	Gly	Arg	Asp	Pro	Ala	Asp	Gln	Phe	Phe	Ser	Gly	Glu	Thr	Gly
			340			345						350			
Arg	Gly	Gly	Leu	Ala	Arg	Trp	Leu	Arg	His	Met	Gln	Gln	Ser	Gly	Asp
355						360						365			
Ala	Lys	Asp	Ala	Leu	Asp	Ser	Ser	Ile	Leu	Gly	Glu	Glu	Gly	Glu	Glu
370						375						380			
Asp	Glu	Met	Val	Met	Ala	Val	Arg	Val	Ala	Ile	Ile	Cys	Leu	Ser	Asp
385			390						395			400			
Leu	Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu	Val	Ala	Met	Leu	Ile
			405						410			415			
Gln	Leu	His	Ser	Phe											
			420												

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<210> SEQ ID NO 80
<211> LENGTH: 414
<212> TYPE: PRT
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<213> ORGANISM: *Oryza punctata*

<400> SEQUENCE: 80

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Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Ala Val His
 1          5          10          15

Leu Leu Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Ser Ser Gln Pro
 20          25          30

Leu His Ser Glu Pro Met Ser Thr Thr Thr Gln Ser Pro Pro Pro Pro
 35          40          45

Gln Ser Lys Ile Pro His Ala Gln Ala Gly Gly Ala Ala Arg Leu Arg
 50          55          60

Arg Ile Val Leu Gly Val Leu Phe Gly Ser Leu Thr Gly Phe Leu Leu
 65          70          75          80

Ser Leu Ala Phe Leu Tyr Ala Ile Arg Val Ala Ile Leu His Ala Lys
 85          90          95

Tyr Ala Pro Ala Ile Val Lys Gly Pro Val Ser Phe Thr Pro Gln Ile
 100         105         110

Ser Pro Lys Asn Leu His Ser Ala Leu Pro Ser Ala Gln Pro Leu Ala
 115         120         125

His Gly Pro Asn Gly Lys Tyr Tyr Lys Leu Val Leu Asp Asn Asp Val
 130         135         140

Thr Val Ala Val Lys Arg Leu Glu Gly Val Ala Ala Ala Ala Ser Arg
 145         150         155         160

Pro Glu Ala Ser Pro Ser Met Ser Asn Val Ser Lys Ser Asp Met Arg
 165         170         175

Arg Leu Gln Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Gln Asn
 180         185         190

Val Met Gly Leu Lys Ala Tyr Val Arg Glu Ala Asp Arg Leu Ser Leu
 195         200         205

Ala Tyr Asp Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg
 210         215         220

Val Arg Ser Gln Gln Val Asn Leu Asn Trp Asp Ala Arg Asn Arg Ile
 225         230         235         240

Ala Ile Gly Val Ala Lys Gly Leu Arg Tyr Leu His Phe Glu Cys Ser
 245         250         255

Pro Arg Ile Leu His Cys Asn Leu Lys Pro Ser Asn Val Met Leu Asp
 260         265         270

Glu Gly Phe Glu Pro Arg Leu Ala Asp Cys Gly Val Ser Arg Leu Ile
 275         280         285

Ala Ser Gly Ser Ala Asp Arg Glu Leu Ala Ser Ser Leu Tyr Ser Ala
 290         295         300

Pro Glu Cys Tyr Gln Thr Ser Arg Tyr Thr Asp Lys Ser Asp Val Tyr
 305         310         315         320

Ser Phe Gly Met Ile Leu Gly Val Leu Leu Ser Gly Arg Asp Pro Thr
 325         330         335

Asp Gln Phe Phe Ser Gly Glu Thr Gly Arg Gly Gly Leu Ala Arg Trp
 340         345         350

Leu Arg His Met Gln Gln Ser Gly Asp Ala Lys Asp Ala Leu Asp Ser
 355         360         365

Ser Ile Leu Gly Glu Glu Gly Glu Glu Asp Glu Met Val Met Ala Val
 370         375         380

Arg Val Ala Ile Ile Cys Leu Ser Asp Leu Pro Ala Asp Arg Pro Ser
 385         390         395         400

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Ser Asp Glu Leu Val Ala Met Leu Thr Gln Leu His Ser Leu
405 410

<210> SEQ ID NO 81
 <211> LENGTH: 413
 <212> TYPE: PRT
 <213> ORGANISM: Oryza meridionalis

<400> SEQUENCE: 81

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu
1 5 10 15
 Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His
20 25 30
 Ser Glu Pro Met Ser Thr Thr Ala Thr Thr Gln Ser Ala Pro Pro Pro
35 40 45
 Pro Pro Pro Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala Ala
50 55 60
 Arg Leu Arg Arg Ile Val Leu Gly Val Leu Phe Gly Ser Leu Thr Gly
65 70 75 80
 Phe Leu Leu Ser Leu Ala Phe Leu Tyr Ala Ile Arg Val Ala Ile Leu
85 90 95
 His Ala Lys Tyr Ala Pro Ala Ile Val Arg Gly Pro Val Ser Phe Thr
100 105 110
 Pro Gln Ile Ser Pro Lys Asn Leu His Ser Ala Leu Pro Ser Ala Gln
115 120 125
 Pro Leu Ala His Gly Pro Asn Gly Lys Tyr Tyr Lys Leu Val Leu Asp
130 135 140
 Asn Asp Val Thr Val Ala Val Lys Arg Leu Glu Ala Ser Arg Pro Glu
145 150 155 160
 Ala Ser Pro Ser Ser Met Pro Asn Val Ser Lys Ser Asp Met Arg Arg
165 170 175
 Val Gln Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Gln Asn Val
180 185 190
 Met Gly Leu Lys Ala Tyr Val Arg Glu Ala Asp Arg Leu Ser Leu Ala
195 200 205
 Tyr Asp Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val
210 215 220
 Arg Ser Gln Gln Val Asn Leu Asn Trp Asp Ala Arg Asn Arg Ile Ala
225 230 235 240
 Ile Gly Val Ala Lys Gly Leu Arg His Leu His Phe Glu Cys Thr Pro
245 250 255
 Arg Ile Leu His Cys Ser Leu Lys Pro Ser Asn Val Met Leu Asp Glu
260 265 270
 Gly Phe Glu Pro Arg Leu Ala Asp Cys Gly Val Ser Arg Leu Ile Ala
275 280 285
 Ser Gly Ser Thr Asp Pro Glu Leu Ala Ser Ser Leu Tyr Ser Ala Pro
290 295 300
 Glu Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Ser Asp Val Tyr Ser
305 310 315 320
 Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr Asp
325 330 335
 His Phe Phe Ser Gly Glu Thr Gly Arg Gly Gly Leu Ala Arg Trp Leu
340 345 350
 Arg His Met Gln Gln Ser Gly Asp Ala Lys Asp Ala Leu Asp Ser Ser
355 360 365

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Val Leu Gly Glu Glu Gly Glu Glu Asp Glu Met Val Met Ala Val Arg
 370 375 380

Val Ala Ile Ile Cys Leu Ser Asp Leu Pro Ala Asp Arg Pro Ser Ser
 385 390 395 400

Asp Glu Leu Val Ala Met Leu Thr Gln Leu His Ser Leu
 405 410

<210> SEQ ID NO 82
 <211> LENGTH: 410
 <212> TYPE: PRT
 <213> ORGANISM: Oryza sativa Indica Group

<400> SEQUENCE: 82

Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu Leu Val
 1 5 10 15

Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His Ser Glu Pro Met
 20 25 30

Ser Thr Thr Thr Thr Ala Thr Gln Pro Ala Pro Pro Pro Pro Pro
 35 40 45

Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala Ala Arg Leu Arg
 50 55 60

Arg Ile Val Leu Gly Val Leu Phe Gly Ser Leu Thr Gly Phe Leu Leu
 65 70 75 80

Ser Leu Ala Phe Leu Tyr Ala Ile Arg Val Ala Ile Leu His Ala Lys
 85 90 95

Tyr Ser Pro Ala Ile Val Arg Gly Pro Val Ser Phe Thr Pro Gln Ile
 100 105 110

Ser Pro Lys Asn Leu His Ser Ala Leu Pro Ser Ala Gln Pro Leu Ala
 115 120 125

His Gly Pro Asn Gly Lys Tyr Tyr Lys Leu Val Leu Asp Asn Asp Val
 130 135 140

Thr Val Ala Val Lys Arg Leu Glu Ala Ser Arg Pro Glu Ala Ser Pro
 145 150 155 160

Ser Ser Met Pro Asn Val Ser Lys Ser Asp Met Arg Arg Val Gln Arg
 165 170 175

Gln Leu Glu Leu Leu Ala Arg Val Arg His Gln Asn Val Met Gly Leu
 180 185 190

Lys Ala Tyr Val Arg Glu Ala Asp Arg Leu Ser Leu Ala Tyr Asp Phe
 195 200 205

Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val Arg Ser Leu
 210 215 220

Gln Val Asn Leu Asn Trp Asp Ala Arg Asn Arg Ile Ala Ile Gly Val
 225 230 235 240

Ala Lys Gly Leu Arg Tyr Leu His Phe Glu Cys Thr Pro Arg Ile Leu
 245 250 255

His Cys Ser Leu Lys Pro Ser Asn Val Met Leu Asp Glu Gly Phe Glu
 260 265 270

Pro Arg Leu Ala Asp Cys Gly Val Ser Arg Leu Ile Ala Ser Gly Ser
 275 280 285

Ala Asp Pro Glu Leu Ala Ser Ser Leu Tyr Ser Ala Pro Glu Cys Tyr
 290 295 300

Gln Ser Ser Arg Tyr Thr Asp Lys Ser Asp Val Tyr Ser Phe Gly Met
 305 310 315 320

Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr Asp His Phe Phe

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325				330				335							
Ser	Gly	Glu	Thr	Gly	Arg	Gly	Gly	Leu	Ala	Arg	Trp	Leu	Arg	His	Met
			340						345				350		
Gln	Gln	Ser	Gly	Asp	Ala	Lys	Asp	Ala	Leu	Asp	Ser	Ser	Val	Leu	Gly
		355					360						365		
Glu	Glu	Gly	Glu	Glu	Asp	Glu	Met	Val	Met	Ala	Val	Arg	Val	Ala	Ile
		370				375					380				
Ile	Cys	Leu	Ser	Asp	Leu	Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu
385					390					395					400
Val	Pro	Met	Leu	Thr	Gln	Leu	His	Ser	Leu						
			405						410						
<210> SEQ ID NO 83															
<211> LENGTH: 414															
<212> TYPE: PRT															
<213> ORGANISM: Oryza nivara															
<400> SEQUENCE: 83															
Met	Glu	Met	Met	Ala	Ser	Lys	Asn	Pro	Thr	Lys	Thr	Leu	Leu	Leu	Leu
1				5					10					15	
Leu	Leu	Leu	Val	Phe	Phe	Leu	Ser	Leu	Ala	Cys	Ser	Gln	Pro	Leu	His
			20						25				30		
Ser	Glu	Pro	Met	Ser	Thr	Thr	Thr	Thr	Ala	Thr	Gln	Pro	Ala	Pro	Pro
		35					40					45			
Pro	Pro	Pro	Pro	Gln	Ser	Lys	Ile	Pro	His	Ala	Glu	Ala	Gly	Gly	Ala
		50				55					60				
Ala	Arg	Leu	Arg	Arg	Ile	Val	Leu	Gly	Val	Leu	Phe	Gly	Ser	Leu	Thr
65					70					75				80	
Gly	Phe	Leu	Leu	Ser	Leu	Ala	Phe	Leu	Tyr	Ala	Ile	Arg	Val	Ala	Ile
			85						90					95	
Leu	His	Ala	Lys	Tyr	Ser	Pro	Ala	Ile	Val	Arg	Gly	Pro	Val	Ser	Phe
			100						105				110		
Thr	Pro	Gln	Ile	Ser	Pro	Lys	Asn	Leu	His	Ser	Ala	Leu	Pro	Ser	Ala
			115				120						125		
Gln	Pro	Leu	Ala	His	Gly	Pro	Asn	Gly	Lys	Tyr	Tyr	Lys	Leu	Val	Leu
		130				135					140				
Asp	Asn	Asp	Val	Thr	Val	Ala	Val	Lys	Arg	Leu	Glu	Ala	Ser	Arg	Pro
145					150					155				160	
Glu	Ala	Ser	Pro	Ser	Ser	Met	Pro	Asn	Val	Ser	Lys	Ser	Asp	Met	Arg
			165						170					175	
Arg	Val	Gln	Arg	Gln	Leu	Glu	Leu	Leu	Ala	Arg	Val	Arg	His	Gln	Asn
			180				185						190		
Val	Met	Gly	Leu	Lys	Ala	Tyr	Val	Arg	Glu	Ala	Asp	Arg	Leu	Ser	Leu
		195					200				205				
Ala	Tyr	Asp	Phe	Val	Pro	Gly	Gly	Ser	Leu	Glu	Asp	Val	Met	Lys	Arg
		210				215					220				
Val	Arg	Ser	Leu	Gln	Val	Asn	Leu	Asn	Trp	Asp	Ala	Arg	Asn	Arg	Ile
225					230					235				240	
Ala	Ile	Gly	Val	Ala	Lys	Gly	Leu	Arg	Tyr	Leu	His	Phe	Glu	Cys	Thr
			245						250					255	
Pro	Arg	Ile	Leu	His	Cys	Ser	Leu	Lys	Pro	Ser	Asn	Val	Met	Leu	Asp
			260						265					270	
Glu	Gly	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Cys	Gly	Val	Ser	Arg	Leu	Ile
		275					280							285	

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Ala Ser Gly Ser Ala Asp Pro Glu Leu Ala Ser Ser Leu Tyr Ser Ala
 290 295 300

Pro Glu Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Ser Asp Val Tyr
 305 310 315 320

Ser Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr
 325 330 335

Asp His Phe Phe Ser Gly Glu Thr Gly Arg Gly Gly Leu Ala Arg Trp
 340 345 350

Leu Arg His Met Gln Gln Ser Gly Asp Ala Lys Asp Ala Leu Asp Ser
 355 360 365

Ser Val Leu Gly Glu Glu Gly Glu Glu Asp Glu Met Val Met Ala Val
 370 375 380

Arg Val Ala Ile Ile Cys Leu Ser Asp Leu Pro Ala Asp Arg Pro Ser
 385 390 395 400

Ser Asp Glu Leu Val Pro Met Leu Thr Gln Leu His Ser Leu
 405 410

<210> SEQ ID NO 84
 <211> LENGTH: 442
 <212> TYPE: PRT
 <213> ORGANISM: Oryza rufipogon

<400> SEQUENCE: 84

Met Pro Thr Leu Ala Ala Ala Val Ala Pro Leu Leu Ser Ser Pro Leu
 1 5 10 15

Leu Cys Ser Pro Arg Ala Ile Ala Ser Pro Leu Ser Met Glu Met Met
 20 25 30

Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu Leu Leu Val
 35 40 45

Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His Ser Glu Pro Met
 50 55 60

Ser Thr Thr Thr Thr Ala Thr Gln Pro Ala Pro Pro Pro Pro Pro
 65 70 75 80

Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala Ala Arg Leu Arg
 85 90 95

Arg Ile Val Leu Gly Val Leu Phe Gly Ser Leu Thr Gly Phe Leu Leu
 100 105 110

Ser Leu Ala Phe Leu Tyr Ala Ile Arg Val Ala Ile Leu His Ala Lys
 115 120 125

Tyr Ala Pro Ala Ile Val Arg Gly Pro Val Ser Phe Thr Pro Gln Ile
 130 135 140

Ser Pro Lys Asn Leu His Ser Ala Leu Pro Ser Ala Gln Pro Leu Ala
 145 150 155 160

His Gly Pro Asn Gly Lys Tyr Tyr Lys Leu Val Leu Asp Asn Asp Val
 165 170 175

Thr Val Ala Val Lys Arg Leu Glu Ala Ser Arg Pro Glu Ala Ser Pro
 180 185 190

Ser Ser Met Pro Asn Val Ser Lys Ser Asp Met Arg Arg Val Gln Arg
 195 200 205

Gln Leu Glu Leu Leu Ala Arg Val Arg His Gln Asn Val Met Gly Leu
 210 215 220

Lys Ala Tyr Val Arg Glu Ala Asp Arg Leu Ser Leu Ala Tyr Asp Phe
 225 230 235 240

Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg Val Arg Ser Leu
 245 250 255

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Gln Val Asn Leu Asn Trp Asp Ala Arg Asn Arg Ile Ala Ile Gly Val
 260 265 270
 Ala Lys Gly Leu Arg Tyr Leu His Phe Glu Cys Thr Pro Arg Ile Leu
 275 280 285
 His Cys Ser Leu Lys Pro Ser Asn Val Met Leu Asp Glu Gly Phe Glu
 290 295 300
 Pro Arg Leu Ala Asp Cys Gly Val Ser Arg Leu Ile Ala Ser Gly Ser
 305 310 315 320
 Ala Asp Pro Glu Leu Ala Ser Ser Leu Tyr Ser Ala Pro Glu Cys Tyr
 325 330 335
 Gln Ser Ser Arg Tyr Thr Asp Lys Ser Asp Val Tyr Ser Phe Gly Met
 340 345 350
 Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr Asp His Phe Phe
 355 360 365
 Ser Gly Glu Thr Gly Arg Gly Gly Leu Ala Arg Trp Leu Arg His Met
 370 375 380
 Gln Gln Ser Gly Asp Ala Lys Asp Ala Leu Asp Ser Ser Val Leu Gly
 385 390 395 400
 Glu Glu Gly Glu Glu Asp Glu Met Val Met Ala Val Arg Val Ala Ile
 405 410 415
 Ile Cys Leu Ser Asp Leu Pro Ala Asp Arg Pro Ser Ser Asp Glu Leu
 420 425 430
 Val Pro Met Leu Thr Gln Leu His Ser Leu
 435 440

<210> SEQ ID NO 85
 <211> LENGTH: 416
 <212> TYPE: PRT
 <213> ORGANISM: Oryza barthii

<400> SEQUENCE: 85

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr His Leu Leu Leu
 1 5 10 15
 Leu Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu
 20 25 30
 His Ser Glu Pro Met Ser Thr Thr Thr Thr Ala Thr Gln Pro Thr
 35 40 45
 Pro Pro Pro Pro Pro Pro Gln Ser Lys Ile Pro His Ala Glu Ala Gly
 50 55 60
 Gly Ala Ala Arg Leu Arg Arg Ile Val Leu Gly Val Leu Phe Gly Ser
 65 70 75 80
 Leu Thr Gly Phe Leu Leu Ser Leu Ala Phe Leu Tyr Ala Ile Arg Val
 85 90 95
 Ala Ile Leu His Ala Lys Tyr Ala Pro Ala Ile Val Arg Gly Pro Val
 100 105 110
 Ser Phe Thr Pro Gln Ile Ser Pro Lys Asn Leu His Ser Ala Leu Pro
 115 120 125
 Ser Ala Gln Pro Leu Ala His Gly Pro Asn Gly Lys Tyr Tyr Lys Leu
 130 135 140
 Val Leu Asp Asn Asp Val Thr Val Ala Val Lys Arg Leu Glu Ala Ser
 145 150 155 160
 Arg Pro Glu Ala Ser Pro Ser Ser Met Pro Asn Val Ser Lys Ser Asp
 165 170 175
 Met Arg Arg Val Gln Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His

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180					185					190					
Gln	Asn	Val	Met	Gly	Leu	Lys	Ala	Tyr	Val	Arg	Glu	Ala	Asp	Arg	Leu
	195						200					205			
Ser	Leu	Ala	Tyr	Asp	Phe	Val	Pro	Gly	Gly	Ser	Leu	Glu	Asp	Val	Met
	210					215					220				
Lys	Arg	Val	Arg	Ser	Leu	Gln	Val	Asn	Leu	Asn	Trp	Asp	Ala	Arg	Asn
	225					230					235				240
Arg	Ile	Ala	Ile	Gly	Val	Ala	Lys	Gly	Leu	Arg	Tyr	Leu	His	Phe	Glu
				245					250					255	
Cys	Thr	Pro	Arg	Ile	Leu	His	Cys	Ser	Leu	Lys	Pro	Ser	Asn	Val	Met
			260					265					270		
Leu	Asp	Glu	Gly	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Cys	Gly	Val	Ser	Arg
	275						280					285			
Leu	Ile	Ala	Ser	Gly	Ser	Ala	Asp	Pro	Glu	Leu	Ala	Ser	Ser	Leu	Tyr
	290					295					300				
Ser	Ala	Pro	Glu	Cys	Tyr	Gln	Ser	Ser	Arg	Tyr	Thr	Asp	Lys	Ser	Asp
	305					310					315				320
Val	Tyr	Ser	Phe	Gly	Met	Ile	Leu	Gly	Val	Leu	Leu	Thr	Gly	Arg	Asp
				325					330					335	
Pro	Thr	Asp	His	Phe	Phe	Ser	Gly	Glu	Thr	Gly	Arg	Gly	Gly	Leu	Ala
			340					345					350		
Arg	Trp	Leu	Arg	His	Met	Gln	Gln	Ser	Gly	Asp	Ala	Lys	Asp	Ala	Leu
	355					360						365			
Asp	Ser	Ser	Val	Leu	Gly	Glu	Glu	Gly	Glu	Glu	Asp	Glu	Met	Val	Met
	370					375					380				
Ala	Val	Arg	Val	Ala	Ile	Ile	Cys	Leu	Ser	Asp	Leu	Pro	Ala	Asp	Arg
	385					390					395				400
Pro	Ser	Ser	Asp	Glu	Leu	Val	Pro	Met	Leu	Thr	Gln	Leu	His	Ser	Leu
			405						410					415	
<210> SEQ ID NO 86															
<211> LENGTH: 414															
<212> TYPE: PRT															
<213> ORGANISM: Oryza sativa Japonica Group															
<400> SEQUENCE: 86															
Met	Glu	Met	Met	Ala	Ser	Lys	Asn	Pro	Thr	Lys	Thr	Leu	Leu	Leu	Leu
1				5					10					15	
Leu	Leu	Leu	Val	Phe	Phe	Leu	Ser	Leu	Ala	Cys	Ser	Gln	Pro	Leu	His
			20					25					30		
Ser	Glu	Pro	Met	Ser	Thr	Thr	Thr	Thr	Ala	Thr	Gln	Pro	Ala	Pro	Pro
	35						40				45				
Pro	Pro	Pro	Pro	Gln	Ser	Lys	Ile	Pro	His	Ala	Glu	Ala	Gly	Gly	Ala
	50					55					60				
Ala	Arg	Leu	Arg	Arg	Ile	Val	Leu	Gly	Val	Leu	Phe	Gly	Ser	Leu	Thr
	65					70					75			80	
Gly	Phe	Leu	Leu	Ser	Leu	Ala	Phe	Leu	Tyr	Ala	Ile	Arg	Val	Ala	Ile
				85					90					95	
Leu	His	Ala	Lys	Tyr	Ala	Pro	Ala	Ile	Val	Arg	Gly	Pro	Val	Ser	Phe
			100					105					110		
Thr	Pro	Gln	Ile	Ser	Pro	Lys	Asn	Leu	His	Ser	Ala	Leu	Pro	Ser	Ala
			115				120					125			
Gln	Pro	Leu	Ala	His	Gly	Pro	Asn	Gly	Lys	Tyr	Tyr	Lys	Leu	Val	Leu
	130					135						140			

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Asp	Asn	Asp	Val	Thr	Val	Ala	Val	Lys	Arg	Leu	Glu	Ala	Ser	Arg	Pro
145					150					155					160
Glu	Ala	Ser	Pro	Ser	Ser	Met	Pro	Asn	Val	Ser	Lys	Ser	Asp	Met	Arg
			165						170					175	
Arg	Val	Gln	Arg	Gln	Leu	Glu	Leu	Leu	Ala	Arg	Val	Arg	His	Gln	Asn
			180					185					190		
Val	Met	Gly	Leu	Lys	Ala	Tyr	Val	Arg	Glu	Ala	Asp	Arg	Leu	Ser	Leu
		195					200					205			
Ala	Tyr	Asp	Phe	Val	Pro	Gly	Gly	Ser	Leu	Glu	Asp	Val	Met	Lys	Arg
	210					215					220				
Val	Arg	Ser	Leu	Gln	Val	Asn	Leu	Asn	Trp	Asp	Ala	Arg	Asn	Arg	Ile
	225				230					235					240
Ala	Ile	Gly	Val	Ala	Lys	Gly	Leu	Arg	Tyr	Leu	His	Phe	Glu	Cys	Thr
			245						250					255	
Pro	Arg	Ile	Leu	His	Cys	Ser	Leu	Lys	Pro	Ser	Asn	Val	Met	Leu	Asp
			260					265					270		
Glu	Asp	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Cys	Gly	Val	Ser	Arg	Leu	Ile
		275					280					285			
Ala	Ser	Gly	Ser	Ala	Asp	Pro	Glu	Leu	Ala	Ser	Ser	Leu	Tyr	Ser	Ala
	290					295					300				
Pro	Glu	Cys	Tyr	Gln	Ser	Ser	Arg	Tyr	Thr	Asp	Lys	Ser	Asp	Val	Tyr
	305				310					315					320
Ser	Phe	Gly	Met	Ile	Leu	Gly	Val	Leu	Leu	Thr	Gly	Arg	Asp	Pro	Thr
			325					330						335	
Asp	His	Phe	Phe	Ser	Gly	Glu	Thr	Gly	Arg	Gly	Gly	Leu	Ala	Arg	Trp
			340					345					350		
Leu	Arg	His	Met	Gln	Gln	Ser	Gly	Asp	Ala	Lys	Asp	Ala	Leu	Asp	Ser
		355					360					365			
Ser	Val	Leu	Gly	Glu	Glu	Gly	Glu	Glu	Asp	Glu	Met	Val	Met	Ala	Val
	370					375					380				
Arg	Val	Ala	Ile	Ile	Cys	Leu	Ser	Asp	Leu	Pro	Ala	Asp	Arg	Pro	Ser
	385				390					395					400
Ser	Asp	Glu	Leu	Val	Pro	Met	Leu	Thr	Gln	Leu	His	Ser	Leu		
			405						410						

<210> SEQ ID NO 87

<211> LENGTH: 414

<212> TYPE: PRT

<213> ORGANISM: Oryza glumipatula

<400> SEQUENCE: 87

Met	Glu	Met	Met	Ala	Ser	Lys	Asn	Pro	Thr	Lys	Thr	Leu	Leu	Leu	Leu
1			5					10						15	
Leu	Leu	Leu	Val	Phe	Phe	Leu	Ser	Leu	Ala	Cys	Ser	Gln	Pro	Leu	His
			20				25					30			
Ser	Glu	Pro	Met	Ser	Thr	Thr	Thr	Thr	Ala	Thr	Gln	Pro	Ala	Pro	Pro
	35					40					45				
Pro	Pro	Pro	Pro	Gln	Ser	Lys	Ile	Pro	His	Ala	Glu	Ala	Gly	Gly	Ala
	50				55					60					
Ala	Arg	Leu	Arg	Arg	Ile	Val	Leu	Gly	Val	Leu	Phe	Gly	Ser	Leu	Thr
	65				70				75					80	
Gly	Phe	Leu	Leu	Ser	Leu	Ala	Phe	Leu	Tyr	Ala	Ile	Arg	Val	Ala	Ile
			85					90						95	
Leu	His	Ala	Lys	Tyr	Ala	Pro	Ala	Ile	Val	Arg	Gly	Pro	Val	Ser	Phe
			100					105						110	

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Thr Pro Gln Ile Ser Pro Lys Asn Leu His Ser Ala Leu Pro Ser Ala
 115 120 125
 Gln Pro Leu Ala His Gly Pro Asn Gly Lys Tyr Tyr Lys Leu Val Leu
 130 135 140
 Asp Asn Asp Val Thr Val Ala Val Lys Arg Leu Glu Ala Ser Arg Pro
 145 150 155 160
 Glu Ala Ser Pro Ser Ser Met Pro Asn Val Ser Lys Ser Asp Met Arg
 165 170 175
 Arg Val Gln Arg Gln Leu Glu Leu Leu Ala Arg Val Arg His Gln Asn
 180 185 190
 Val Met Gly Leu Lys Ala Tyr Val Arg Glu Ala Asp Arg Leu Ser Leu
 195 200 205
 Ala Tyr Asp Phe Val Pro Gly Gly Ser Leu Glu Asp Val Met Lys Arg
 210 215 220
 Val Arg Ser Leu Gln Val Asn Leu Asn Trp Asp Ala Arg Asn Arg Ile
 225 230 235 240
 Ala Ile Gly Val Ala Lys Gly Leu Arg Tyr Leu His Phe Glu Cys Thr
 245 250 255
 Pro Arg Ile Leu His Cys Ser Leu Lys Pro Ser Asn Val Met Leu Asp
 260 265 270
 Glu Gly Phe Glu Pro Arg Leu Ala Asp Cys Gly Val Ser Arg Leu Ile
 275 280 285
 Ala Ser Gly Ser Ala Asp Pro Glu Leu Ala Ser Ser Leu Tyr Ser Ala
 290 295 300
 Pro Glu Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Ser Asp Val Tyr
 305 310 315 320
 Ser Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr
 325 330 335
 Asp His Phe Phe Ser Gly Glu Thr Gly Arg Gly Gly Leu Ala Arg Trp
 340 345 350
 Leu Arg His Met Gln Gln Ser Gly Asp Ala Lys Asp Ala Leu Asp Ser
 355 360 365
 Ser Val Leu Gly Glu Glu Gly Glu Glu Asp Glu Met Val Met Ala Val
 370 375 380
 Arg Val Ala Ile Ile Cys Leu Ser Asp Leu Pro Ala Asp Arg Pro Ser
 385 390 395 400
 Ser Asp Glu Leu Val Pro Met Leu Thr Gln Leu His Ser Leu
 405 410

<210> SEQ ID NO 88

<211> LENGTH: 433

<212> TYPE: PRT

<213> ORGANISM: Panicum hallii FIL2

<400> SEQUENCE: 88

Met Leu Val Phe Pro Ser Lys Phe Gly Ser Cys Ile Met Glu Gln Ala
 1 5 10 15
 Val Arg Met Thr Ser Lys Asn Pro Thr Lys Thr Pro Pro Leu Leu Ala
 20 25 30
 Thr Leu Leu Leu Leu Ala Phe Leu Ala Leu Cys Ala Pro Ala Ser
 35 40 45
 Ser Gln Pro Leu His Ser Glu Pro Met Pro Thr Gln Ser Pro Pro Pro
 50 55 60
 Ser Pro Thr Pro Pro Gln Ser Thr Ile Pro Arg Ala Pro Ala Gly Gly

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65	70	75	80
Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val Leu Phe Gly Ser Ile	85	90	95
Ser Gly Phe Leu Leu Ala Leu Ala Phe Leu Tyr Gly Ile Arg Val Ala	100	105	110
Ile Leu His Ala Lys Ser Ala Pro Ala Ile Val Lys Gly Pro Val Ser	115	120	125
Phe Thr Pro Gln Ile Ser Pro Lys Asn Leu Leu Ala Ala Leu Pro Ser	130	135	140
Ala Gln Pro Leu Ala His Gly Pro His Gly Lys Tyr Cys Lys Leu Ala	145	150	155
Leu Asp Asn Asp Leu Thr Val Ala Val Lys Arg Leu Glu Ala Ala Asn	165	170	175
Arg Pro Glu Glu Ala Ser Pro Ser Met Ser Pro Ser Thr Ser Lys Ser	180	185	190
Asp Met Arg Arg Val Gln Arg Gln Leu Glu Ala Leu Ala Arg Val Arg	195	200	205
His Gln Asn Val Met Thr Leu Lys Ala Tyr Val Arg Glu Ala Asp Arg	210	215	220
Leu Ser Leu Val Tyr Asp Phe Ile Pro Gly Gly Ser Leu Glu Glu Leu	225	230	235
Met Lys Arg Val Arg Ser Gln Gln Val Ser Leu Asn Trp Asp Ala Arg	245	250	255
Ser Arg Ile Val Val Gly Ile Ala Lys Gly Leu Arg His Leu His Phe	260	265	270
Glu Tyr Ser Pro Arg Ile Leu His Cys Asn Leu Lys Pro Ser Asn Val	275	280	285
Met Leu Asp Glu Gly Phe Glu Pro Ile Leu Thr Asp Cys Gly Ile Ala	290	295	300
Arg Leu Ile Ala Ala Gly Ser Gly Asp Pro Glu Leu Cys Ser Gly Leu	305	310	315
Tyr Ala Ala Pro Glu Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Ser	325	330	335
Asp Val Tyr Ala Leu Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg	340	345	350
Asp Pro Thr Asp Pro Phe Phe Ser Gly Glu Thr Gly Arg Gly Gly Leu	355	360	365
Pro Arg Trp Leu Arg His Met Gln Gln Ser Ala Asp Pro Lys Glu Ala	370	375	380
Leu Asp Ser Ser Ile Leu Gly Asp Glu Gly Glu Glu Glu Glu Met Phe	385	390	395
Met Ala Ile Arg Val Ala Ile Val Cys Leu Ser Asp Ser Pro Val Asp	405	410	415
Arg Pro Ser Ser Asp Glu Leu Val Ala Met Leu Thr Gln Leu His Ser	420	425	430
Leu			

<210> SEQ ID NO 89

<211> LENGTH: 433

<212> TYPE: PRT

<213> ORGANISM: Panicum hallii HAL2

<400> SEQUENCE: 89

Met Leu Val Phe Pro Ser Lys Phe Gly Ser Cys Ile Met Glu Gln Ala

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1	5	10	15
Val Arg Met Thr Ser Lys Asn Pro Thr Lys Thr Pro Pro Phe Leu Ala	20	25	30
Ile Leu Leu Leu Leu Leu Ala Phe Leu Ala Leu Cys Ala Pro Ala Ser	35	40	45
Ser Gln Pro Leu His Ser Glu Pro Met Ala Thr Gln Ser Pro Pro Pro	50	55	60
Ser Pro Thr Pro Pro Gln Ser Thr Ile Pro Arg Ala Gln Ala Gly Gly	65	70	75
Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val Leu Phe Gly Ser Ile	85	90	95
Ser Gly Phe Leu Leu Ala Leu Ala Phe Leu Tyr Gly Ile Arg Val Ala	100	105	110
Ile Leu His Ala Lys Ser Ala Pro Ala Ile Val Lys Gly Pro Val Ser	115	120	125
Phe Thr Pro Gln Ile Ser Pro Lys Asn Leu Leu Ala Ala Leu Pro Ser	130	135	140
Ala Gln Pro Leu Ala His Gly Pro His Gly Lys Tyr Cys Lys Leu Ala	145	150	155
Leu Asp Asn Asp Leu Thr Val Ala Val Lys Arg Leu Glu Ala Ala Asn	165	170	175
Arg Pro Glu Glu Ala Ser Pro Ser Met Ser Pro Ser Thr Ser Lys Ser	180	185	190
Asp Met Arg Arg Val Gln Arg Gln Leu Glu Ala Leu Ala Arg Val Arg	195	200	205
His Gln Asn Val Met Thr Leu Lys Ala Tyr Val Arg Glu Ala Asp Arg	210	215	220
Leu Ser Leu Val Tyr Asp Phe Ile Pro Trp Gly Ser Leu Glu Asp Val	225	230	235
Met Lys Arg Val Arg Ser Gln Gln Val Ser Leu Asn Trp Asp Ala Arg	245	250	255
Ser Arg Ile Val Val Gly Ile Ala Lys Gly Leu Arg His Leu His Phe	260	265	270
Glu Tyr Ser Pro Arg Ile Leu His Cys Asn Leu Lys Pro Ser Asn Val	275	280	285
Met Leu Asp Glu Gly Phe Glu Pro Ile Leu Thr Asp Cys Gly Ile Ala	290	295	300
Arg Leu Ile Ala Ala Gly Ser Gly Asp Pro Glu Leu Cys Ser Gly Leu	305	310	315
Tyr Ala Ala Pro Glu Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Ser	325	330	335
Asp Val Tyr Ala Leu Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg	340	345	350
Asp Pro Thr Asp Pro Phe Phe Ser Gly Glu Thr Gly Arg Gly Ser Leu	355	360	365
Pro Arg Trp Leu Arg His Met Gln Gln Ser Ala Asp Pro Lys Glu Ala	370	375	380
Leu Asp Ser Ser Ile Leu Gly Asp Glu Gly Glu Glu Glu Glu Met Phe	385	390	395
Met Ala Ile Arg Val Ala Ile Val Cys Leu Ser Asp Ser Pro Val Asp	405	410	415
Arg Pro Ser Ser Asp Glu Leu Val Ala Met Leu Thr Gln Leu His Ser	420	425	430

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Leu

<210> SEQ ID NO 90

<211> LENGTH: 417

<212> TYPE: PRT

<213> ORGANISM: Sorghum bicolor

<400> SEQUENCE: 90

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Met Val Val Gly Ala Arg Met Thr Ser Arg Asn Pro Thr Lys Thr Leu
1      5      10      15

Pro Leu Leu Ala Thr Thr Leu Val Leu Leu Leu Leu Ala Phe Leu Ala
20      25      30

Leu Cys Ala Pro Ala Ser Ser Gln Pro Leu His Ser Glu Pro Met Ala
35      40      45

Thr Gln Ser Pro Pro Pro Arg Thr Ser Ile Pro Arg Ala Gln Val Gly
50      55      60

Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val Leu Phe Gly Ser
65      70      75      80

Leu Ser Gly Phe Leu Leu Ala Leu Ala Phe Leu Tyr Ala Ile Arg Leu
85      90      95

Ala Ile Leu His Ala Lys Ser Thr Pro Ala Ile Ala Lys Gly Pro Val
100     105     110

Ser Phe Val Pro Gln Ile Ser Ala Lys Asn Leu Leu Ala Ala Leu Pro
115     120     125

Ala Ala Gln Pro Leu Ala His Gly Pro His Gly Lys Tyr Tyr Lys Leu
130     135     140

Ala Leu Asp Asn Asp Leu Thr Val Ala Val Lys Arg Leu Glu Ala Ala
145     150     155     160

Ser Arg Pro Glu Ala Ser Pro Ser Met Ser Pro Ser Ala Ser Lys Ser
165     170     175

Asp Met Arg Arg Val Gln Arg Gln Leu Glu Val Leu Ala Arg Val Arg
180     185     190

His Gln Asn Val Leu Ser Leu Lys Ala Tyr Val Arg Glu Pro Asp Arg
195     200     205

Leu Ser Leu Val Tyr Asp Phe Val Pro Gly Gly Ser Leu Glu Asp Val
210     215     220

Met Lys Arg Val Arg Ser Gln Gln Val Ser Leu Ser Trp Asp Ala Arg
225     230     235     240

Ser Arg Ile Ala Val Gly Ile Ala Lys Gly Leu Arg His Leu His Phe
245     250     255

Glu Cys Asn Pro Arg Ile Leu His Cys Asn Leu Lys Pro Ser Asn Val
260     265     270

Met Leu Asp Glu Gly Phe Glu Pro Ile Leu Thr Asp Cys Gly Val Ala
275     280     285

Arg Leu Ile Ala Ala Ser Ser Gly Asp Pro Glu Leu Cys Ser Gly Leu
290     295     300

Tyr Ala Ala Pro Glu Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Ser
305     310     315     320

Asp Val Tyr Ala Leu Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg
325     330     335

Asp Pro Thr Asp Ser Phe Phe Ser Gly Glu Ala Gly Gln Gly Gly Leu
340     345     350

Ala Arg Trp Leu Arg His Val Gln Gln Ser Ala Asp Pro Lys Glu Ala
355     360     365

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Leu Asp Ser Ser Ile Leu Gly Asp Glu Gly Glu Glu Glu Met Leu
 370 375 380

Met Ala Ile Arg Ile Ala Ile Val Cys Leu Ser Asp Ser Pro Ser Asp
 385 390 395 400

Arg Pro Ser Ser Asp Glu Leu Val Ala Met Leu Met Gln Leu His Ser
 405 410 415

Leu

<210> SEQ ID NO 91
 <211> LENGTH: 394
 <212> TYPE: PRT
 <213> ORGANISM: Brassica rapa

<400> SEQUENCE: 91

Met Glu Gln Arg Arg Arg Arg Asn Ile Asn Thr Leu Ser Leu Leu Leu
 1 5 10 15

Leu Phe Phe Leu Ala Phe Thr Ser Thr Thr Ala Thr Thr Ser Ser Cys
 20 25 30

Arg Arg Arg Ala Val Lys His Leu Ser Thr Thr Gln Ala Thr Pro Leu
 35 40 45

Gln Ser Arg Ile Thr Pro Lys Val Ile Val Leu Ser Ile Val Ser Gly
 50 55 60

Val Leu Thr Gly Leu Leu Ser Ala Leu Ala Leu Ala Phe Leu Val Arg
 65 70 75 80

Cys Ile Val Lys Tyr Met Lys Gln Thr Pro Ile Leu Lys Gly Pro Val
 85 90 95

Val Phe Ser Pro Lys Ile Thr Pro Lys Ser Leu His Ala Ala Leu Gly
 100 105 110

Asn Ser Ile His Leu Leu Gly Ser Asp Pro Asn Gly Lys Tyr His Lys
 115 120 125

Thr Val Leu Asp Asn Gly Leu Val Val Ala Val Lys Lys Leu Ser Tyr
 130 135 140

Ser Ser Pro Glu Gly Gly Ala Thr Ser Lys Ser Val Lys Arg Arg Leu
 145 150 155 160

Gln Lys Glu Leu Glu Leu Leu Ala Gly Leu Arg His Arg Asn Ile Met
 165 170 175

Ser Leu Arg Ala Tyr Val Arg Glu Ser Asn Glu Phe Ser Leu Val Tyr
 180 185 190

Asp Tyr Leu Pro Asn Gly Ser Leu Glu Asp Val Met Ser Lys Val Arg
 195 200 205

Gly Asn Glu Met Glu Leu Gly Trp Glu Val Arg Leu Arg Ile Ala Val
 210 215 220

Gly Ile Val Lys Gly Leu Gln Tyr Leu His Phe Gly Cys Glu Glu Gln
 225 230 235 240

Gln Ile Leu His Tyr Asn Leu Lys Pro Thr Asn Val Val Leu Asp Ser
 245 250 255

Glu Phe Glu Pro Arg Leu Thr Asp Cys Gly Leu Ala Lys Ile Met Leu
 260 265 270

Ser Ser Gln Thr Ala Val Val Ser Cys Tyr Ser Ala Pro Glu Ser Ser
 275 280 285

Lys Ser Asn Arg Tyr Thr Glu Lys Ser Asp Val Phe Ser Phe Gly Met
 290 295 300

Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Thr Gln Pro Phe Cys
 305 310 315 320

Val	Asn	Gly	Ala	Ser	Gly	Gly	Ser	Leu	Gly	Leu	Trp	Leu	Lys	His	Leu		
				325					330					335			
Gln	Gln	Ser	Gly	Glu	Ser	Arg	Glu	Ala	Leu	Asp	Glu	Ser	Ile	Leu	Gly		
				340					345					350			
Glu	Glu	Val	Glu	Glu	Asp	Glu	Met	Leu	Met	Ala	Leu	Arg	Ile	Thr	Ile		
				355					360					365			
Val	Cys	Leu	Ser	Asp	Phe	Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu		
				370					375					380			
Val	His	Met	Leu	Thr	Gln	Leu	His	Ser	Phe								
				385					390								
<210> SEQ ID NO 92																	
<211> LENGTH: 394																	
<212> TYPE: PRT																	
<213> ORGANISM: Brassica napus																	
<400> SEQUENCE: 92																	
Met	Glu	Gln	Arg	Arg	Arg	Arg	Asn	Ile	Asn	Thr	Leu	Ser	Leu	Leu	Leu		
1					5					10					15		
Leu	Phe	Phe	Leu	Ala	Phe	Thr	Ser	Thr	Thr	Ala	Thr	Thr	Ser	Ser	Cys		
				20					25					30			
Arg	Arg	Arg	Ala	Val	Lys	His	Leu	Ser	Thr	Thr	Gln	Ala	Thr	Pro	Leu		
				35					40					45			
Gln	Ser	Arg	Ile	Thr	Pro	Lys	Val	Ile	Val	Leu	Ser	Ile	Val	Ser	Gly		
				50					55					60			
Val	Leu	Thr	Gly	Leu	Leu	Ser	Ala	Leu	Ala	Leu	Ala	Phe	Leu	Val	Arg		
				65					70					75			
Cys	Ile	Val	Lys	Tyr	Met	Lys	Gln	Thr	Pro	Ile	Leu	Lys	Gly	Pro	Val		
				85					90					95			
Val	Phe	Ser	Pro	Lys	Ile	Thr	Pro	Lys	Ser	Leu	His	Ala	Ala	Leu	Gly		
				100					105					110			
Asn	Ser	Ile	His	Leu	Leu	Gly	Ser	Asp	Pro	Asn	Gly	Lys	Tyr	His	Lys		
				115					120					125			
Thr	Val	Leu	Asp	Asn	Gly	Leu	Val	Val	Ala	Val	Lys	Lys	Leu	Ser	Tyr		
				130					135					140			
Ser	Ser	Pro	Glu	Gly	Gly	Ala	Thr	Ser	Lys	Ser	Val	Lys	Arg	Arg	Leu		
				145					150					155			
Gln	Lys	Glu	Leu	Glu	Leu	Leu	Ala	Gly	Leu	Arg	His	Arg	Asn	Ile	Met		
				165					170					175			
Ser	Leu	Arg	Ala	Tyr	Val	Arg	Glu	Ser	Asn	Glu	Phe	Ser	Leu	Val	Tyr		
				180					185					190			
Asp	Tyr	Leu	Pro	Asn	Gly	Ser	Leu	Glu	Asp	Val	Met	Ser	Lys	Val	Arg		
				195					200					205			
Gly	Asn	Glu	Met	Glu	Leu	Gly	Trp	Glu	Val	Arg	Leu	Arg	Ile	Ala	Val		
				210					215					220			
Gly	Ile	Val	Lys	Gly	Leu	Gln	Tyr	Leu	His	Phe	Gly	Cys	Glu	Glu	Gln		
				225					230					235			
Gln	Ile	Leu	His	Tyr	Asn	Leu	Lys	Pro	Thr	Asn	Val	Val	Leu	Asp	Ser		
				245					250					255			
Glu	Phe	Glu	Pro	Arg	Leu	Thr	Asp	Cys	Gly	Leu	Ala	Lys	Ile	Met	Leu		
				260					265					270			
Ser	Ser	Gln	Thr	Ala	Val	Val	Ser	Cys	Tyr	Ser	Ala	Pro	Glu	Ser	Ser		
				275					280					285			
Lys	Ser	Asn	Arg	Tyr	Thr	Glu	Lys	Ser	Asp	Val	Phe	Ser	Phe	Gly	Met		

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290	295	300
Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Thr Gln Pro Phe Cys		
305	310	315 320
Val Asn Gly Ala Ser Gly Gly Ser Leu Gly Leu Trp Leu Lys His Leu		
	325	330 335
Gln Gln Ser Gly Glu Ser Arg Glu Ala Leu Asp Glu Ser Ile Leu Gly		
	340	345 350
Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala Leu Arg Ile Thr Ile		
	355	360 365
Val Cys Leu Ser Asp Phe Pro Ala Asp Arg Pro Ser Ser Asp Glu Leu		
	370	375 380
Val His Met Leu Thr Gln Leu His Ser Phe		
385	390	
<210> SEQ ID NO 93		
<211> LENGTH: 398		
<212> TYPE: PRT		
<213> ORGANISM: Brassica napus		
<400> SEQUENCE: 93		
Met Glu Gln Arg Gly Arg Lys Arg Tyr Cys Asn Thr Val Tyr Leu Leu		
1	5	10 15
Leu Phe Ile Phe Leu Val Phe Ser Ser Arg Thr Thr Ser Ser Ala Ser		
	20	25 30
Cys Arg Arg Arg Ala Val Lys His Leu Ser Thr Ala Pro Pro Ser Ser		
	35	40 45
Thr Pro Leu Glu Ser Arg Ile Thr Thr Lys Val Ile Ile Val Ser Ile		
	50	55 60
Val Ser Gly Val Leu Thr Gly Phe Phe Ser Ala Leu Ala Leu Ala Phe		
65	70	75 80
Leu Val Arg Cys Thr Val Lys Tyr Leu Lys Gln Thr Pro Ile Leu Lys		
	85	90 95
Gly Pro Val Val Phe Ser Pro Lys Ile Thr Pro Lys Ser Leu His Ala		
	100	105 110
Ala Leu Ala Asn Gly Ile Gln Leu Leu Gly Ser Asp Pro Asn Gly Lys		
	115	120 125
Tyr Tyr Lys Met Val Leu Asp Asn Gly Leu Val Val Ala Val Lys Arg		
	130	135 140
Leu Gly Ser Leu Glu Gly Asn Gly Gly Ser Pro Glu Ala Ala Thr Lys		
145	150	155 160
Ser Val Lys Arg Arg Leu Gln Lys Glu Leu Glu Leu Leu Ala Gly Leu		
	165	170 175
Arg Asp Arg Asn Leu Met Ser Leu Arg Ala Tyr Val Arg Glu Ser Asp		
	180	185 190
Glu Phe Ser Leu Val Tyr Asp Tyr Met Pro Asn Gly Ser Leu Asp Asp		
	195	200 205
Val Met Thr Lys Val Arg Ala Gln Glu Leu Glu Leu Gly Trp Glu Ile		
	210	215 220
Arg Leu Arg Val Ala Val Gly Ile Val Lys Gly Leu Gln Tyr Leu His		
225	230	235 240
Phe Ser Cys Glu Gln Gln Ile Leu His Tyr Asn Leu Lys Pro Thr Asn		
	245	250 255
Val Met Leu Asp Ser Glu Phe Glu Pro Arg Leu Ala Asp Cys Gly Leu		
	260	265 270

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Ala Lys Ile Ile Pro Thr Ser His Thr Ser Val Ser Cys Tyr Ser Ala
275 280 285

Pro Glu Ser Ser Gln Thr Asn Arg Tyr Thr Asp Lys Ser Asp Val Phe
290 295 300

Ser Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr
305 310 315 320

His Pro Phe Cys Ile Glu Gly Ala Thr Gly Gly Ser Leu Gly Gln Trp
325 330 335

Leu Lys His Leu Gln Gln Thr Gly Glu Val Arg Glu Gly Leu Asp Lys
340 345 350

Ser Ile Leu Gly Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala Leu
355 360 365

Arg Ile Thr Ile Ile Cys Leu Ser Asp Phe Pro Ala Asp Arg Pro Ser
370 375 380

Ser Asp Glu Leu Val His Met Leu Thr Gln Leu His Ser Phe
385 390 395

<210> SEQ ID NO 94

<211> LENGTH: 397

<212> TYPE: PRT

<213> ORGANISM: Brassica napus

<400> SEQUENCE: 94

Met Glu Gln Arg Arg Arg Lys Arg Tyr Cys Asn Thr Val Tyr Leu Leu
1 5 10 15

Leu Phe Ile Leu Leu Val Phe Ser Ser Ile Thr Thr Ser Ser Ala Ser
20 25 30

Cys Arg Arg Arg Ala Val Lys His Leu Ser Thr Ala Pro Ser Ser
35 40 45

Thr Pro Leu Glu Ser Lys Ile Thr Ser Lys Val Ile Ala Ile Ser Ile
50 55 60

Val Ser Gly Val Leu Thr Gly Leu Val Ser Ala Leu Ala Leu Phe
65 70 75 80

Leu Val Arg Cys Thr Val Lys Tyr Leu Lys Gln Thr Pro Ile Leu Lys
85 90 95

Gly Pro Val Val Phe Ser Pro Lys Ile Thr Pro Lys Ser Leu His Ala
100 105 110

Ala Leu Ala Asn Gly Ile Gln Leu Leu Gly Ser Asp Pro Asn Gly Lys
115 120 125

Tyr Tyr Lys Met Val Leu Asp Asn Gly Leu Val Val Ala Val Lys Arg
130 135 140

Leu Gly Ser Leu Glu Gly Asn Gly Gly Ser Pro Glu Ala Thr Lys Ser
145 150 155 160

Val Lys Arg Arg Leu Gln Lys Glu Leu Glu Leu Ala Gly Leu Arg
165 170 175

Asp Arg Asn Leu Met Ser Leu Arg Ala Tyr Val Arg Glu Ser Asp Glu
180 185 190

Phe Ser Leu Val Tyr Asp Tyr Met Pro Asn Gly Ser Leu Glu Asp Val
195 200 205

Met Thr Lys Val Arg Ala Glu Glu Leu Glu Leu Gly Trp Glu Ile Arg
210 215 220

Leu Arg Val Ala Val Gly Ile Val Lys Gly Leu Gln Tyr Leu His Phe
225 230 235 240

Ser Cys Glu Gln Gln Ile Leu His Tyr Asn Leu Lys Pro Ala Asn Val
245 250 255

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Met Leu Asp Ser Glu Phe Glu Pro Arg Leu Ala Asp Cys Gly Leu Ala
  260                               265               270

Lys Ile Ile Pro Ala Ser Gln Thr Ala Val Ser Cys Tyr Ser Ala Pro
  275                               280               285

Glu Ser Ser Gln Ile Asn Arg Tyr Thr Asp Lys Ser Asp Val Phe Ser
  290                               295               300

Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr Leu
  305                               310               315               320

Pro Phe Ser Ile Glu Gly Ala Ser Gly Gly Ser Leu Gly Gln Trp Leu
  325                               330               335

Lys His Leu Gln Gln Thr Gly Glu Val Arg Glu Ala Leu Asp Lys Ser
  340                               345               350

Ile Leu Gly Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala Leu Arg
  355                               360               365

Ile Thr Ile Ile Cys Leu Ser Asp Phe Pro Ala Asp Arg Pro Ser Ser
  370                               375               380

Asp Glu Leu Val His Met Leu Thr Gln Leu His Ser Phe
  385                               390               395

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<210> SEQ ID NO 95

<211> LENGTH: 395

<212> TYPE: PRT

<213> ORGANISM: Brassica rapa

<400> SEQUENCE: 95

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Met Glu Gln Arg Arg Arg Lys Arg Tyr Cys Asn Thr Val His Leu Leu
  1                               5               10               15

Leu Phe Ile Phe Leu Val Phe Ser Ser Arg Thr Ser Ala Ser Cys Arg
  20                               25               30

Arg Arg Ala Val Lys His Leu Ser Thr Ala Pro Pro Ser Ser Thr Pro
  35                               40               45

Leu Glu Ser Lys Ile Thr Ser Lys Val Ile Val Val Ser Ile Val Ser
  50                               55               60

Gly Val Leu Thr Gly Leu Val Ser Ala Leu Ala Leu Ala Phe Leu Val
  65                               70               75               80

Arg Ser Thr Val Lys Tyr Leu Lys Gln Thr Pro Ile Leu Lys Gly Pro
  85                               90               95

Val Val Phe Ser Pro Lys Ile Thr Pro Lys Ser Leu His Ala Ala Leu
  100                              105               110

Ala Asn Gly Ile Gln Leu Leu Gly Ser Asp Pro Asn Gly Lys Tyr Tyr
  115                              120               125

Lys Met Val Leu Asp Asn Gly Leu Val Val Ala Val Lys Arg Leu Gly
  130                              135               140

Ser Leu Glu Gly Asn Gly Gly Ser Pro Glu Ala Thr Lys Ser Val Lys
  145                              150               155               160

Arg Arg Leu Gln Lys Glu Leu Glu Leu Leu Ala Gly Leu Arg Asp Arg
  165                              170               175

Asn Leu Met Ser Leu Arg Ala Tyr Val Arg Glu Ser Asp Glu Phe Ser
  180                              185               190

Leu Val Tyr Asp Tyr Thr Pro Asn Gly Ser Leu Glu Asp Val Met Thr
  195                              200               205

Lys Val Arg Ala Gln Glu Leu Glu Leu Gly Trp Glu Ile Arg Leu Arg
  210                              215               220

Val Ala Val Gly Ile Val Lys Gly Leu Gln Tyr Leu His Phe Ser Cys

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225	230	235	240
Glu Gln Gln Ile Leu His Tyr Asn Leu Lys Pro Thr Asn Val Met Leu	245	250	255
Asp Ser Glu Phe Glu Pro Arg Leu Ala Asp Cys Gly Leu Ala Lys Ile	260	265	270
Ile Pro Ala Ser His Thr Ser Val Ser Cys Tyr Ser Ala Pro Glu Ser	275	280	285
Ser Gln Thr Asn Arg Tyr Thr Asp Lys Ser Asp Val Phe Ser Phe Gly	290	295	300
Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr Leu Pro Phe	305	310	315
Ser Ile Glu Gly Ala Ser Gly Gly Ser Leu Gly Gln Trp Leu Lys His	325	330	335
Leu Gln Gln Thr Gly Glu Val Arg Glu Ala Leu Asp Lys Ser Ile Leu	340	345	350
Gly Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala Leu Arg Ile Thr	355	360	365
Ile Ile Cys Leu Ser Asp Phe Pro Ala Asp Arg Pro Ser Ser Asp Glu	370	375	380
Leu Val His Met Leu Thr Gln Leu His Ser Phe	385	390	395

<210> SEQ ID NO 96

<211> LENGTH: 401

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 96

Met Lys Gln Arg Arg Arg Arg Asn Gly Cys Ser Ser Ser Asn Thr Ile	1	5	10	15
Ser Leu Leu Leu Leu Phe Phe Leu Val Phe Phe Ser Arg Thr Ser Thr	20	25	30	
Ser Thr Ser Cys Arg Arg Arg Thr Val Lys His Leu Ser Thr Thr Ser	35	40	45	
Thr Ser Ser Thr Pro Leu Glu Ser Arg Ile Thr Ser Lys Val Ile Val	50	55	60	
Ile Ser Ile Val Ser Gly Ile Leu Thr Gly Leu Val Ser Ala Leu Val	65	70	75	80
Leu Ala Phe Leu Val Arg Ser Ile Val Lys Phe Met Lys Gln Thr Pro	85	90	95	
Ile Leu Lys Gly Pro Val Val Phe Ser Pro Lys Ile Thr Pro Lys Ser	100	105	110	
Leu His Ala Ala Leu Ser Asn Gly Ile Gln Leu Leu Gly Ser Asp Leu	115	120	125	
Asn Gly Lys Tyr Tyr Lys Met Val Leu Asp Asn Gly Leu Val Val Ala	130	135	140	
Val Lys Arg Leu Gly Ser Leu Glu Gly Val Gly Ser Pro Glu Ser Ser	145	150	155	160
Ser Ser Lys Ser Val Lys Arg Arg Leu Gln Lys Glu Leu Glu Leu Leu	165	170	175	
Ala Gly Leu Arg His Arg Asn Leu Met Ser Leu Arg Ala Tyr Val Arg	180	185	190	
Glu Ser Asp Glu Phe Ser Leu Val Tyr Asp Tyr Met Pro Asn Gly Ser	195	200	205	

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Leu Glu Asp Val Met Asn Lys Val Arg Thr Lys Glu Val Glu Leu Gly
 210                215                220

Trp Glu Ile Arg Leu Arg Val Ala Val Gly Ile Val Lys Gly Leu Gln
 225                230                235                240

Tyr Leu His Phe Ser Cys Glu Thr Gln Ile Leu His Tyr Asn Leu Lys
    245                250                255

Pro Thr Asn Val Met Leu Asp Ser Glu Phe Glu Pro Arg Leu Ala Asp
    260                265                270

Cys Gly Leu Ala Lys Ile Met Pro Ser Ser His Thr Ala Val Ser Cys
    275                280                285

Tyr Ser Ala Pro Glu Ser Ser Gln Ser Asn Arg Tyr Thr Asp Lys Ser
    290                295                300

Asp Ile Phe Ser Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg
 305                310                315                320

Asp Pro Thr His Pro Phe Cys Glu Glu Ser Ala Ser Gly Gly Ser Leu
    325                330                335

Gly Gln Trp Leu Lys His Leu Gln Gln Ser Gly Glu Ala Arg Glu Ala
    340                345                350

Leu Asp Lys Thr Ile Leu Gly Glu Glu Val Glu Glu Asp Glu Met Leu
    355                360                365

Met Ala Leu Arg Ile Thr Ile Ile Cys Leu Ser Asp Phe Pro Ala Asp
    370                375                380

Arg Pro Ser Ser Asp Glu Leu Val His Met Leu Thr Gln Leu His Ser
 385                390                395                400

Phe

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<210> SEQ ID NO 97
<211> LENGTH: 402
<212> TYPE: PRT
<213> ORGANISM: Arabidopsis lyrata

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<400> SEQUENCE: 97

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Met Glu Gln Arg Arg Arg Arg Arg Asn Gly Cys Ser Ser Asn Thr
 1                5                10                15

Ile Ser Leu Leu Leu Phe Phe Phe Leu Val Phe Phe Ser Arg Thr Ser
    20                25                30

Thr Ser Thr Ser Cys Arg Arg Arg Thr Val Lys His Leu Ser Thr Thr
    35                40                45

Ser Thr Ser Ser Thr Pro Leu Glu Ser Arg Ile Thr Ser Lys Val Ile
    50                55                60

Val Val Ser Ile Val Ser Gly Ile Leu Thr Gly Leu Val Ser Ala Leu
    65                70                75                80

Val Leu Ala Phe Leu Val Arg Ser Ile Val Lys Tyr Met Lys Gln Thr
    85                90                95

Pro Ile Leu Lys Gly Pro Val Val Phe Ser Pro Lys Ile Thr Pro Lys
    100                105                110

Ser Leu His Ala Ala Leu Gly Asn Gly Ile Gln Leu Leu Gly Ser Asp
    115                120                125

Pro Asn Gly Lys Tyr Tyr Lys Met Val Leu Asp Asn Gly Leu Val Val
    130                135                140

Ala Val Lys Arg Leu Gly Ser Leu Glu Gly Ile Gly Ser Pro Glu Thr
    145                150                155                160

Asn Ser Ser Lys Ser Val Lys Arg Arg Leu Gln Lys Glu Leu Glu Leu
    165                170                175

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Leu Ala Gly Leu Arg His Arg Asn Leu Met Ser Leu Arg Ala Tyr Val
    180              185              190

Arg Glu Ser Asp Glu Phe Ser Leu Val Tyr Asp Tyr Met Pro Asn Gly
    195              200              205

Ser Leu Glu Asp Val Met His Lys Val Arg Thr Lys Glu Val Glu Leu
    210              215              220

Gly Trp Glu Ile Arg Leu Arg Val Ala Val Gly Ile Val Lys Gly Leu
    225              230              235              240

Gln Tyr Leu His Phe Ser Cys Glu Thr Gln Ile Leu His Tyr Asn Leu
    245              250              255

Lys Pro Thr Asn Val Met Leu Asp Ser Glu Phe Glu Pro Arg Leu Ala
    260              265              270

Asp Cys Gly Leu Ala Lys Ile Met Pro Ser Ser His Thr Ala Val Phe
    275              280              285

Cys Tyr Ser Ala Pro Glu Ser Ser Gln Ser Asn Arg Tyr Thr Asp Lys
    290              295              300

Ser Asp Ile Phe Ser Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly
    305              310              315              320

Arg Asp Pro Thr His Pro Phe Cys Glu Glu Ser Ala Ser Gly Gly Ser
    325              330              335

Leu Gly Gln Trp Leu Lys His Leu Gln Gln Ser Gly Glu Ala Arg Glu
    340              345              350

Ala Leu Asp Lys Thr Ile Leu Gly Glu Glu Val Glu Glu Asp Glu Met
    355              360              365

Leu Met Ala Leu Arg Ile Thr Ile Ile Cys Leu Ser Asp Phe Pro Ala
    370              375              380

Asp Arg Pro Ser Ser Asp Glu Leu Val His Met Leu Thr Gln Leu His
    385              390              395              400

Ser Phe

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<210> SEQ ID NO 98
<211> LENGTH: 394
<212> TYPE: PRT
<213> ORGANISM: Brassica napus

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<400> SEQUENCE: 98

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Met Glu Gln Arg Arg Arg Ser Asn Thr Ile Tyr Leu Leu Leu Leu Phe
 1              5              10              15

Phe Phe Leu Val Phe Thr Ser Arg Thr Ser Thr Ser Ser Ser Cys Arg
 20              25              30

Arg Arg Thr Val Lys His Leu Ser Thr Thr Pro Pro Ser Ser Thr Pro
 35              40              45

Leu Glu Ser Arg Ile Thr Ser Lys Val Ile Val Phe Ser Ile Val Ser
 50              55              60

Gly Ile Leu Thr Gly Leu Val Ser Ala Leu Val Leu Ala Phe Leu Val
 65              70              75              80

Arg Ser Ile Val Lys Tyr Met Lys Gln Thr Pro Ile Leu Lys Gly Pro
 85              90              95

Val Val Phe Ser Pro Glu Leu Thr Pro Lys Ser Leu His Gly Ala Leu
100              105              110

Gly Asn Gly Ile His Leu Leu Gly Ser Asp Pro Asn Gly Lys Tyr His
115              120              125

Lys Val Val Leu Asp Asn Gly Leu Val Val Ala Val Lys Arg Leu Cys
130              135              140

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Ser	Pro	Glu	Ala	Ser	Gly	Ser	Ser	Thr	Ser	Arg	Asn	Ser	Val	Lys	Arg
145					150					155					160
Arg	Leu	Gln	Lys	Glu	Leu	Glu	Leu	Leu	Ala	Glu	Ile	Arg	His	Arg	Asn
			165						170					175	
Leu	Met	Ser	Leu	Arg	Ala	Tyr	Val	Arg	Glu	Ser	Asn	Glu	Phe	Ser	Leu
			180					185					190		
Val	Tyr	Asp	Tyr	Met	Pro	Asn	Gly	Ser	Leu	Glu	Asp	Val	Met	Thr	Lys
		195					200					205			
Val	Arg	Gly	Lys	Glu	Leu	Glu	Leu	Gly	Trp	Glu	Val	Arg	Leu	Arg	Val
	210					215					220				
Ala	Val	Gly	Ile	Val	Gln	Gly	Leu	Gln	Tyr	Leu	His	Phe	Ser	Cys	Asp
225					230					235					240
His	Gln	Ile	Leu	His	Tyr	Asn	Leu	Lys	Pro	Thr	Asn	Val	Met	Leu	Asp
			245						250					255	
Ser	Glu	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Cys	Gly	Leu	Ala	Lys	Ile	Met
			260					265					270		
Ala	Ala	Ser	His	Thr	Gly	Val	Ser	Cys	Tyr	Ser	Ala	Pro	Glu	Ser	Ser
		275					280					285			
Gln	Ser	Asn	Arg	Tyr	Thr	Asp	Lys	Ser	Asp	Ile	Phe	Ser	Phe	Gly	Met
	290					295					300				
Ile	Leu	Gly	Leu	Leu	Leu	Thr	Gly	Arg	Asp	Pro	Thr	Gln	Pro	Phe	Cys
305					310					315					320
Val	Asp	Gly	Ala	Ser	Gly	Gly	Ser	Leu	Gly	Gln	Trp	Leu	Lys	His	Leu
			325						330					335	
Gln	Gln	Ser	Gly	Glu	Ala	Arg	Glu	Ala	Leu	Asp	Lys	Ser	Ile	Leu	Gly
			340					345					350		
Glu	Glu	Val	Lys	Glu	Asp	Glu	Met	Leu	Met	Ala	Leu	Arg	Ile	Thr	Ile
		355					360					365			
Ile	Cys	Leu	Ser	Asp	Phe	Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu
	370					375					380				
Val	His	Met	Leu	Thr	Gln	Leu	His	Ser	Phe						
385					390										

<210> SEQ ID NO 99

<211> LENGTH: 394

<212> TYPE: PRT

<213> ORGANISM: Brassica oleracea

<400> SEQUENCE: 99

Met	Glu	Gln	Arg	Arg	Arg	Ser	Asn	Thr	Ile	Tyr	Leu	Leu	Leu	Leu	Phe
1				5					10						15
Phe	Phe	Leu	Val	Phe	Thr	Ser	Arg	Thr	Asn	Thr	Ser	Ala	Ser	Cys	Arg
		20					25					30			
Arg	Arg	Thr	Val	Lys	His	Leu	Ser	Thr	Thr	Pro	Pro	Ser	Ser	Thr	Pro
		35					40					45			
Leu	Gln	Ser	Ser	Ile	Thr	Ser	Lys	Val	Met	Val	Phe	Ser	Ile	Ala	Ser
	50					55					60				
Gly	Ile	Leu	Ile	Gly	Leu	Val	Ser	Ala	Leu	Val	Leu	Ala	Phe	Leu	Val
65					70					75					80
Arg	Cys	Ile	Val	Lys	Tyr	Met	Gln	Gln	Thr	Pro	Ile	Leu	Lys	Gly	Pro
				85					90					95	
Val	Val	Phe	Ser	Pro	Glu	Leu	Thr	Pro	Lys	Ser	Leu	Tyr	Gly	Ala	Leu
			100					105					110		
Gly	Asn	Gly	Ile	Gln	Leu	Leu	Gly	Ser	Gly	Pro	Asn	Gly	Lys	Tyr	His
		115					120					125			

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Lys Val Val Leu Asp Asn Gly Leu Val Val Ala Val Lys Arg Leu Cys
 130 135 140
 Ser Pro Glu Ala Ser Gly Ser Ser Thr Ser Arg Asn Ser Val Lys Arg
 145 150 155 160
 Arg Leu Gln Lys Glu Leu Glu Leu Leu Ala Ile Leu Arg His Met Asn
 165 170 175
 Leu Met Ser Leu Arg Ala Tyr Val Leu Glu Ser Asn Glu Phe Ser Leu
 180 185 190
 Val Tyr Asp Tyr Met Pro Asn Gly Ser Leu Glu Asp Val Met Thr Lys
 195 200 205
 Val Arg Ala Lys Glu Leu Glu Leu Gly Trp Glu Val Arg Leu Arg Val
 210 215 220
 Ala Val Gly Ile Val Gln Gly Leu Gln Tyr Leu His Phe Ser Cys Asp
 225 230 235 240
 His Gln Val Leu His Tyr Asn Leu Lys Pro Thr Asn Val Met Leu Asp
 245 250 255
 Ser Glu Phe Glu Pro Arg Leu Ala Asp Cys Gly Leu Ala Lys Ile Met
 260 265 270
 Ala Ala Ser His Thr Gly Val Ser Cys Tyr Ser Ala Pro Glu Ser Ser
 275 280 285
 Gln Ser Asn Arg Tyr Thr Asp Lys Ser Asp Ile Phe Ser Phe Gly Met
 290 295 300
 Ile Leu Gly Leu Leu Leu Thr Gly Arg Asp Pro Thr Gln Pro Phe Cys
 305 310 315 320
 Val Asp Gly Ala Ser Gly Gly Ser Leu Gly Gln Trp Leu Lys His Leu
 325 330 335
 Gln Gln Ser Gly Glu Ala Arg Glu Ala Leu Asp Lys Ser Ile Leu Gly
 340 345 350
 Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala Leu Arg Ile Thr Ile
 355 360 365
 Ile Cys Leu Ser Asp Phe Pro Ala Asp Arg Pro Ser Ser Asp Glu Leu
 370 375 380
 Val His Met Leu Thr Gln Leu His Ser Phe
 385 390

<210> SEQ ID NO 100

<211> LENGTH: 394

<212> TYPE: PRT

<213> ORGANISM: Brassica napus

<400> SEQUENCE: 100

Met Glu Gln Arg Arg Arg Ser Asn Thr Ile Tyr Leu Leu Leu Leu Phe
 1 5 10 15
 Phe Phe Leu Val Phe Thr Ser Arg Thr Asn Thr Ser Ala Ser Cys Arg
 20 25 30
 Arg Arg Thr Val Lys His Leu Ser Thr Thr Pro Pro Ser Ser Thr Pro
 35 40 45
 Leu Gln Ser Ser Ile Thr Ser Lys Val Ile Val Phe Ser Ile Ala Ser
 50 55 60
 Gly Ile Leu Ile Gly Leu Val Ser Ala Leu Val Leu Ala Phe Leu Val
 65 70 75 80
 Arg Cys Ile Val Lys Tyr Met Gln Gln Thr Pro Ile Leu Lys Gly Pro
 85 90 95
 Val Val Phe Ser Pro Glu Leu Thr Pro Lys Ser Leu Tyr Gly Ala Leu

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100					105					110					
Gly	Asn	Gly	Ile	Gln	Leu	Leu	Gly	Ser	Gly	Pro	Asn	Gly	Lys	Tyr	His
	115						120					125			
Lys	Val	Val	Leu	Asp	Asn	Gly	Leu	Val	Val	Ala	Val	Lys	Arg	Leu	Cys
	130					135					140				
Ser	Pro	Glu	Ala	Ser	Gly	Ser	Ser	Thr	Ser	Arg	Asn	Ser	Val	Lys	Arg
145					150					155					160
Arg	Leu	Gln	Lys	Glu	Leu	Glu	Leu	Leu	Ala	Ile	Leu	Arg	His	Met	Asn
			165						170					175	
Leu	Met	Ser	Leu	Arg	Ala	Tyr	Val	Pro	Glu	Ser	Asn	Glu	Phe	Ser	Leu
			180					185					190		
Val	Tyr	Asp	Tyr	Met	Pro	Asn	Gly	Ser	Leu	Glu	Asp	Val	Met	Thr	Lys
	195						200					205			
Val	Arg	Ala	Lys	Glu	Leu	Glu	Leu	Gly	Trp	Glu	Val	Arg	Leu	Arg	Val
	210					215					220				
Ala	Val	Gly	Ile	Val	Gln	Gly	Leu	Gln	Tyr	Leu	His	Phe	Ser	Cys	Asp
225					230					235					240
His	Gln	Val	Leu	His	Tyr	Asn	Leu	Lys	Pro	Thr	Asn	Val	Met	Leu	Asp
			245						250					255	
Ser	Glu	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Cys	Gly	Leu	Ala	Lys	Ile	Met
			260					265					270		
Ala	Ala	Ser	His	Thr	Gly	Leu	Ser	Cys	Tyr	Ser	Ala	Pro	Glu	Ser	Ser
		275					280					285			
Gln	Ser	Asn	Arg	Tyr	Thr	Asp	Lys	Ser	Asp	Ile	Phe	Ser	Phe	Gly	Met
	290					295					300				
Ile	Leu	Gly	Leu	Leu	Leu	Thr	Gly	Arg	Asp	Pro	Thr	Gln	Pro	Phe	Cys
305					310					315					320
Val	Asp	Gly	Ala	Ser	Gly	Gly	Ser	Leu	Gly	Gln	Trp	Leu	Lys	His	Leu
			325						330					335	
Gln	Gln	Ser	Gly	Glu	Ala	Arg	Glu	Ala	Leu	Asp	Lys	Ser	Ile	Leu	Gly
			340					345					350		
Glu	Glu	Val	Glu	Glu	Asp	Glu	Met	Leu	Met	Ala	Leu	Arg	Ile	Thr	Ile
		355					360					365			
Ile	Cys	Leu	Ser	Asp	Phe	Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu
	370					375					380				
Val	His	Met	Leu	Thr	Gln	Ile	His	Ser	Phe						
385					390										

<210> SEQ ID NO 101

<211> LENGTH: 413

<212> TYPE: PRT

<213> ORGANISM: Trifolium pratense

<400> SEQUENCE: 101

Met	Gly	Leu	Ile	Cys	Gly	Cys	Gly	Phe	Val	Phe	Cys	Val	Leu	Met	Phe
1				5					10					15	
Lys	Lys	Arg	Gln	Thr	Phe	Phe	Tyr	Leu	Ala	Lys	Glu	Leu	Leu	Val	Phe
			20					25					30		
Gln	Pro	Leu	Val	Leu	Leu	Leu	Phe	Leu	Phe	Ser	Leu	His	His	Asn	Thr
		35					40					45			
Val	Gln	Cys	Glu	Gly	Arg	Leu	Ser	Lys	Asn	Ile	Ser	Ser	Glu	Thr	Ser
	50						55				60				
Ser	Gln	Ser	Asp	Tyr	Lys	Asp	Asn	Pro	Arg	Lys	Val	Ile	Val	Ser	Ile
65					70					75					80

Val	Leu	Gly	Ala	Val	Thr	Gly	Leu	Ile	Gly	Ser	Val	Leu	Phe	Ala	Phe	
				85					90					95		
Val	Ile	Arg	Cys	Ile	Val	Arg	Tyr	Leu	Asn	Arg	Thr	Pro	Ile	Leu	Lys	
				100					105					110		
Gly	Pro	Val	Ile	Phe	Ser	Pro	Lys	Ile	Ala	Ser	Lys	Thr	Leu	Gln	Leu	
				115					120					125		
Ala	Leu	Ser	Lys	Glu	Asn	His	Leu	Leu	Gly	Ser	Ser	Pro	Asn	Gly	Lys	
				130					135					140		
Tyr	Tyr	Lys	Thr	Val	Leu	Glu	Asn	Gly	Leu	Thr	Ile	Ala	Val	Lys	Arg	
				145					150					155		
Leu	Thr	Pro	Phe	Glu	Ser	Asn	Ser	Pro	Glu	Ser	Arg	Arg	Lys	Ser	Val	
				165					170					175		
Lys	Arg	Gln	Ile	Gln	Met	Glu	Leu	Glu	Leu	Leu	Ala	Ser	Leu	Arg	His	
				180					185					190		
Arg	Asn	Leu	Met	Ser	Leu	Arg	Ala	Tyr	Val	Arg	Gln	Asn	Asp	Gly	Phe	
				195					200					205		
Ser	Leu	Val	Tyr	Asp	Tyr	Val	Ser	Thr	Gly	Ser	Leu	Ala	Asp	Val	Met	
				210					215					220		
Asn	Arg	Val	Arg	Glu	Asn	Glu	Leu	Gln	Ile	Gly	Trp	Glu	Val	Arg	Leu	
				225					230					235		
Arg	Ile	Ala	Val	Gly	Val	Val	Lys	Gly	Leu	Gln	Tyr	Leu	His	Phe	Thr	
				245					250					255		
Cys	Val	Pro	Gln	Ile	Leu	His	Phe	Asn	Leu	Lys	Pro	Thr	Asn	Val	Met	
				260					265					270		
Leu	Asp	Ala	Glu	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Tyr	Gly	Leu	Ala	Lys	
				275					280					285		
Leu	Leu	Pro	Asn	Leu	Asp	Arg	Gly	Ile	Ser	Leu	Asn	Thr	Pro	Pro	Glu	
				290					295					300		
Cys	Leu	His	Asn	Arg	Ser	Arg	Tyr	Thr	Glu	Lys	Ser	Asp	Ile	Phe	Gly	
				305					310					315		
Phe	Gly	Met	Ile	Leu	Gly	Val	Leu	Leu	Thr	Gly	Lys	Asp	Ala	Ala	Asp	
				325					330					335		
Pro	Phe	Phe	Gly	Glu	Ala	Ser	Arg	Gly	Gly	Asn	Leu	Gly	Cys	Trp	Leu	
				340					345					350		
Arg	His	Leu	Lys	Gln	Ala	Gly	Glu	Glu	Arg	Glu	Ala	Leu	Asp	Lys	Ser	
				355					360					365		
Ile	Leu	Gly	Glu	Gln	Gly	Glu	Glu	Asp	Glu	Met	Leu	Met	Ala	Val	Gly	
				370					375					380		
Ile	Ala	Ala	Ala	Cys	Leu	Ser	Asp	Met	Pro	Ala	Asp	Arg	Pro	Ser	Ser	
				385					390					395		
Asp	Glu	Leu	Val	His	Met	Leu	Thr	Gln	Leu	Asn	Ser	Phe				
				405					410							

<400> SEQUENCE: 102

Met Gly Leu Ile Cys Gly Cys Gly Phe Val Phe Cys Val Phe Met Phe
1 5 10 15

Lys Lys Arg His Thr Pro Phe Ser Leu Ala Arg Lys Phe Leu Ser Phe
20 25 30

Gln Pro Phe Val Leu Leu Phe Leu Phe Ser Leu His His Asp Thr
35 40 45

Val	Gln	Cys	Gln	Gly	Arg	Leu	Ser	Lys	His	Met	Ser	Ser	Glu	Pro	Ser
50						55				60					
Ser	Ser	Thr	Ser	Glu	Tyr	Lys	Asp	Asp	Pro	Arg	Lys	Ile	Ile	Ile	Ser
65					70					75					80
Ile	Val	Leu	Gly	Ala	Val	Thr	Gly	Leu	Ile	Gly	Ser	Val	Leu	Phe	Ala
			85						90					95	
Phe	Val	Ile	Arg	Cys	Val	Val	Trp	Tyr	Leu	Asn	Arg	Thr	Pro	Ile	Leu
			100					105					110		
Lys	Gly	Pro	Val	Ile	Phe	Ser	Pro	Lys	Ile	Ala	Ser	Lys	Thr	Leu	Gln
		115					120					125			
Val	Ala	Leu	Thr	Lys	Glu	Asn	His	Met	Leu	Gly	Ser	Ser	Pro	Asn	Gly
						135						140			
Lys	Tyr	Tyr	Lys	Thr	Val	Leu	Glu	Asn	Gly	Leu	Ile	Ile	Ala	Val	Lys
145					150					155					160
Arg	Leu	Thr	Pro	Phe	Glu	Ser	Asn	Ser	Pro	Glu	Ser	Arg	Lys	Lys	Ser
				165					170					175	
Val	Lys	Arg	Gln	Ile	Gln	Met	Glu	Leu	Glu	Leu	Leu	Ala	Ser	Leu	Arg
			180					185					190		
His	Arg	Asn	Leu	Met	Ser	Leu	Arg	Ala	Tyr	Val	Arg	Glu	Asn	Asp	Gly
		195					200					205			
Phe	Ser	Leu	Val	Tyr	Asp	Tyr	Val	Ser	Thr	Gly	Ser	Leu	Ala	Asp	Val
		210				215					220				
Met	Asn	Arg	Val	Arg	Glu	Asn	Glu	Leu	Gln	Ile	Gly	Trp	Glu	Ala	Arg
225					230					235					240
Leu	Arg	Ile	Ala	Val	Gly	Val	Val	Lys	Gly	Leu	Gln	Tyr	Leu	His	Phe
				245					250					255	
Thr	Cys	Val	Pro	Gln	Ile	Leu	His	Phe	Asn	Leu	Lys	Pro	Thr	Asn	Val
			260					265					270		
Met	Leu	Asp	Ala	Glu	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Tyr	Gly	Leu	Ala
		275					280					285			
Lys	Leu	Leu	Pro	Asn	Leu	Asp	Arg	Gly	Ile	Ser	Val	Asn	Ile	Pro	Pro
		290				295					300				
Glu	Cys	Ser	His	Asn	Cys	Arg	Tyr	Thr	Glu	Lys	Ser	Asp	Ile	Phe	Ser
305					310					315					320
Phe	Gly	Met	Ile	Leu	Gly	Val	Leu	Leu	Thr	Gly	Lys	Asp	Pro	Thr	Asp
				325					330					335	
Pro	Phe	Phe	Gly	Glu	Ala	Ser	Arg	Gly	Gly	Ser	Leu	Gly	Cys	Trp	Leu
			340					345					350		
Arg	His	Leu	Lys	Gln	Ala	Gly	Glu	Glu	Arg	Glu	Ala	Leu	Asp	Lys	Ser
			355				360					365			
Ile	Leu	Gly	Glu	Gln	Gly	Glu	Glu	Asp	Glu	Met	Leu	Met	Ala	Val	Gly
					375						380				
Ile	Ala	Ala	Ala	Cys	Leu	Ser	Asp	Met	Pro	Ala	Glu	Arg	Pro	Ser	Ser
385					390					395					400
Asp	Glu	Leu	Val	His	Met</										

Met Phe Lys Arg Arg His Thr Leu Cys Ser Leu Leu Arg Glu Leu Phe

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1	5	10	15
Leu Leu Leu	Leu Phe Ser	Leu His His Ser Thr Val	Gln Cys Gln Gly
	20	25	30
Arg Leu Ser	Lys His Val Ser	Ser Ala Pro Tyr Ser	Pro Ser Glu Tyr
	35	40	45
Lys Asp Asp	Leu Arg Arg	Ile Ile Ile Ser Ile Val	Leu Gly Gly Leu
	50	55	60
Thr Gly Leu	Val Gly Ser Val	Phe Phe Ala Phe Val	Val Arg Cys Val
	65	70	80
Val Arg Tyr	Leu Asn Arg Thr	Pro Ile Leu Lys Gly	Pro Val Ile Phe
	85	90	95
Ser Pro Lys	Ile Ala Pro Lys	Thr Leu Glu Ser Ala	Leu Ala Lys Glu
	100	105	110
Asn His Leu	Leu Gly Ser Ser	Pro Asn Gly Lys Tyr	Tyr Lys Thr Val
	115	120	125
Leu Asp Asn	Gly Leu Thr Ile	Ala Val Lys Arg Leu	Thr Pro Phe Glu
	130	135	140
Ser Asn Ser	Pro Glu Met Arg	Arg Lys Ser Val Lys	Arg Gln Ile Gln
	145	150	155
Lys Glu Leu	Glu Leu Leu Ala	Ser Leu Arg His Arg	Asn Leu Met Ser
	165	170	175
Leu Arg Ala	Tyr Val Arg Glu	Ala Asp Arg Phe Ser	Leu Val Tyr Asp
	180	185	190
Tyr Val Ser	Thr Gly Ser Leu	Ala Asp Leu Met Asn	Arg Val Arg Glu
	195	200	205
Asn Glu Leu	Gln Leu Gly Trp	Glu Val Arg Leu Arg	Ile Ala Val Gly
	210	215	220
Ile Val Lys	Gly Leu Gln Tyr	Leu His Phe Thr Cys	Val Pro Gln Ile
	225	230	235
Leu His Cys	Asn Leu Lys Pro	Thr Asn Val Met Leu	Asp Ala Glu Phe
	245	250	255
Glu Pro Arg	Leu Ala Asp Tyr	Gly Leu Ser Lys Leu	Leu Pro Asn Leu
	260	265	270
Asp Arg Gly	Thr His Gly Asn	Asn Pro Ser Gln Ser	Leu His Asn Cys
	275	280	285
Ser Arg Tyr	Thr Asp Lys Ser	Asp Ile Phe Ser Phe	Gly Met Ile Leu
	290	295	300
Gly Thr Leu	Leu Thr Gly Lys	Asp Pro Thr Asp Pro	Phe Phe Gly Glu
	305	310	315
Ala Ala Ser	Gly Gly Ser Leu	Gly Ser Trp Leu	Gln His Leu Arg
	325	330	335
Ala Gly Glu	Ala Arg Glu Ala	Leu Asp Lys Ser Ile	Leu Gly Glu Glu
	340	345	350
Gly Glu Glu	Asp Glu Met Leu	Met Ala Val Arg Ile	Ala Ala Ala Cys
	355	360	365
Leu Ser Asp	Ile Pro Ala Asp	Arg Pro Ser Ser Asp	Glu Leu Val His
	370	375	380
Met Leu Thr	Gln Leu His Ser	Phe	
	385	390	

<210> SEQ ID NO 104

<211> LENGTH: 406

<212> TYPE: PRT

<213> ORGANISM: Phaseolus vulgaris

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<400> SEQUENCE: 104

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Met Phe Arg Lys Arg His Ile Ala Ser Ser Leu Ala Arg Glu Leu Leu
 1           5           10           15

Ala Leu Gln Pro Leu Leu Ile Leu Phe Leu Phe Ser Leu His His His
 20           25           30

Thr Val Gln Cys Gln Gly Arg Leu Ser Lys His Val Ser Ser Glu Pro
 35           40           45

Pro Ser Pro Ser Arg Thr Ser Pro Pro Ser Ser Ser Gly Tyr Arg Asp
 50           55           60

Asp Pro Lys Lys Ile Ile Leu Ser Met Val Leu Gly Ala Val Thr Gly
 65           70           75           80

Leu Val Cys Ser Val Leu Phe Ala Leu Val Val Arg Cys Val Val Gln
 85           90           95

Tyr Leu Asn Arg Thr Pro Ile Leu Lys Gly Pro Val Ile Phe Ser Pro
100           105           110

Lys Ile Ala Pro Lys Thr Leu Gln Leu Ala Leu Ala Lys Glu Asn His
115           120           125

Leu Leu Gly Ser Ser Pro Asn Gly Lys Tyr Tyr Lys Thr Val Leu Asp
130           135           140

Asn Gly Leu Thr Ile Ala Val Lys Arg Leu Thr Pro Phe Glu Ser Asn
145           150           155           160

Ser Pro Glu Ala Arg Lys Asn Ser Val Lys Arg Gln Ile Gln Thr Glu
165           170           175

Leu Glu Leu Leu Ala Ser Leu Arg His Arg Asn Leu Met Ser Leu Arg
180           185           190

Ala Tyr Val Arg Glu Pro Asp Gly Phe Ser Leu Val Tyr Asp Tyr Val
195           200           205

Ser Thr Gly Ser Leu Ala Asp Val Met Asn Lys Val Trp Glu Asn Glu
210           215           220

Leu Pro Phe Gly Trp Glu Val Arg Leu Arg Ile Ala Val Gly Val Val
225           230           235           240

Lys Gly Leu Gln Tyr Leu His Phe Thr Cys Ala Pro Gln Ile Leu His
245           250           255

Tyr Ser Leu Lys Pro Thr Asn Val Met Leu Asp Ala Asp Phe Glu Pro
260           265           270

Arg Leu Ala Asp Tyr Gly Leu Ala Lys Leu Leu Pro Asn Leu Asp Arg
275           280           285

Gly Thr Ser Leu Tyr Thr Pro Pro Glu Cys Phe His Asn Cys Ser Arg
290           295           300

Tyr Thr Asp Lys Ser Asp Ile Phe Ser Phe Gly Met Ile Leu Gly Val
305           310           315           320

Leu Leu Thr Gly Lys Asp Pro Thr Asp Pro Phe Phe Gly Glu Ala Ala
325           330           335

Ser Gly Gly Ser Leu Gly Cys Trp Leu Arg His Leu Gln Gln Ala Gly
340           345           350

Glu Ala Arg Asp Ala Leu Asp Lys Ser Met Leu Gly Glu Glu Gly Glu
355           360           365

Glu Asp Glu Met Leu Met Ala Val Arg Ile Ala Ala Ala Cys Leu Ser
370           375           380

Asp Met Pro Ala Asp Arg Pro Ser Ser Asp Glu Leu Val His Met Leu
385           390           395           400

Thr Gln Leu His Ser Phe

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405

<210> SEQ ID NO 105

<211> LENGTH: 429

<212> TYPE: PRT

<213> ORGANISM: *Vigna radiata*

<400> SEQUENCE: 105

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Met Ile His Ala Ser Phe Trp Ser Phe Arg Asp Trp Phe Leu Ala Leu
 1             5             10             15

Phe Val Cys Cys Ser Leu Met Phe Arg Lys Arg Asn Ile Ala Ser Ser
      20             25             30

Leu Ala Arg Glu Leu Leu Ala Leu Gln Pro Leu Phe Ile Ile Phe Leu
      35             40             45

Phe Ser Leu His His Asn Thr Val Gln Cys Gln Gly Arg Leu Ser Lys
 50             55             60

His Val Ser Ser Glu Pro Pro Ser Pro Ser Arg Pro Ser Pro Pro Ser
 65             70             75             80

Ser Ser Gly Tyr Arg Asp Asp Pro Lys Lys Ile Ile Leu Ser Leu Val
      85             90             95

Leu Gly Ala Val Thr Gly Leu Val Cys Ser Val Phe Phe Ala Leu Val
      100            105            110

Val Arg Cys Val Val Gln Tyr Leu Asn Arg Thr Pro Ile Leu Lys Gly
      115            120            125

Pro Val Ile Phe Ser Pro Lys Ile Ala Pro Lys Thr Leu Gln Leu Ala
      130            135            140

Leu Ala Lys Glu Asn His Leu Leu Gly Ser Ser Pro Asn Gly Lys Tyr
      145            150            155            160

Tyr Lys Thr Val Leu Asp Asn Gly Leu Thr Ile Ala Val Lys Arg Leu
      165            170            175

Thr Pro Phe Gly Ser Ser Ser Pro Glu Ala Lys Lys Lys Ser Val Lys
      180            185            190

Arg Gln Ile Gln Thr Glu Leu Glu Leu Leu Ala Ser Leu Arg His Arg
      195            200            205

Asn Leu Met Ser Leu Arg Ala Tyr Val Arg Glu Pro Asp Gly Phe Ser
      210            215            220

Leu Val Tyr Asp Tyr Val Ser Thr Gly Ser Leu Ala Asp Val Met Asn
      225            230            235            240

Arg Val Trp Glu Asn Glu Leu Ala Phe Gly Trp Glu Val Arg Leu Arg
      245            250            255

Ile Ala Val Gly Val Val Lys Gly Leu Gln Tyr Leu His Phe Thr Cys
      260            265            270

Ala Pro Gln Ile Leu Leu His Tyr Asn Leu Lys Pro Thr Asn Val Met
      275            280            285

Leu Asp Gly Asp Phe Glu Pro Arg Leu Ala Asp Tyr Gly Leu Ala Lys
      290            295            300

Leu Leu Pro Asn Leu Asp Arg Gly Thr Ser Leu Tyr Thr Pro Pro Glu
      305            310            315            320

Cys Phe His Asn Cys Ser Arg Tyr Thr Asp Lys Ser Asp Ile Phe Ser
      325            330            335

Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Lys Asp Pro Thr Asp
      340            345            350

Pro Phe Phe Gly Glu Ala Ala Ser Gly Gly Ser Leu Gly Gly Trp Leu
      355            360            365

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Arg His Leu Gln Gln Ala Gly Asp Ala Arg Asp Ala Leu Asp Lys Ser
 370                               375                               380

Met Leu Gly Glu Glu Gly Glu Glu Asp Glu Met Leu Met Ala Val Arg
385                               390                               395                               400

Ile Ala Ala Ala Cys Leu Ser Asp Met Pro Ala Asp Arg Pro Ser Ser
      405                               410                               415

Asp Glu Leu Val His Met Leu Thr Gln Leu His Ser Phe
 420                               425

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<210> SEQ ID NO 106

<211> LENGTH: 408

<212> TYPE: PRT

<213> ORGANISM: Glycine max

<400> SEQUENCE: 106

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Met Phe Arg Lys Arg His Ile Leu Ser Ser Leu Ala Arg Glu Leu Leu
 1           5           10           15

Ala Leu Gln Pro Leu Phe Leu Leu Phe Leu Phe Ser Leu His His Asn
      20           25           30

Thr Val Gln Cys Gln Gly Arg Leu Ser Lys His Val Ser Ser Glu Pro
 35           40           45

Pro Ser Pro Ser Arg Pro Ser Ser Ala Ala Pro Ser Ser Ser Gly Tyr
 50           55           60

Lys Asp Asp Pro Arg Lys Ile Ile Leu Ser Met Val Leu Gly Ala Val
65           70           75           80

Thr Gly Leu Val Cys Ser Val Leu Phe Ala Leu Val Val Arg Cys Val
      85           90           95

Val Gln Tyr Leu Asn Arg Thr Pro Ile Leu Lys Gly Pro Val Ile Phe
 100           105           110

Ser Pro Lys Ile Ala Ser Lys Thr Leu Gln Ser Ala Leu Ala Lys Glu
 115           120           125

Asn His Leu Leu Gly Ser Ser Pro Asn Gly Lys Tyr Tyr Lys Thr Met
 130           135           140

Leu Asp Asn Gly Leu Thr Ile Ala Val Lys Arg Leu Thr Pro Phe Glu
145           150           155           160

Ser Asn Ser Pro Glu Ala Lys Arg Lys Ser Val Lys Arg Gln Ile Gln
 165           170           175

Thr Glu Leu Glu Leu Leu Ala Ser Leu Arg Asn Arg Asn Leu Met Ser
 180           185           190

Leu Arg Ala Tyr Val Arg Glu Pro Asp Gly Phe Ser Leu Val Tyr Asp
 195           200           205

Tyr Ala Ser Thr Gly Ser Leu Ala Asp Val Leu Asn Arg Val Arg Glu
 210           215           220

Asn Glu Leu Pro Phe Gly Trp Glu Val Arg Leu Arg Ile Ala Val Gly
225           230           235           240

Val Val Lys Gly Leu Gln Tyr Leu His Phe Thr Cys Val Pro Gln Ile
 245           250           255

Leu His Tyr Asn Leu Lys Pro Thr Asn Val Met Leu Asp Ala Glu Phe
 260           265           270

Glu Pro Arg Leu Ala Asp Tyr Gly Leu Ala Lys Leu Leu Pro Asn Leu
 275           280           285

Asp Arg Gly Ser Ser Leu Tyr Thr Pro Pro Glu Cys Phe His Asn Cys
 290           295           300

Ser Arg Tyr Thr Asp Lys Ser Asp Ile Phe Ser Phe Gly Ile Ile Leu
305           310           315           320

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Met 1	Phe	Arg	Lys	Arg 5	His	Thr	Leu	Ser	Ser 10	Leu	Ala	Arg	Glu	Leu 15	Leu
Ala	Phe	Gln	Pro 20	Leu	Phe	Leu	Leu	Phe 25	Leu	Phe	Ser	Leu	His 30	His	Asn
Thr	Met	Gln	Cys 35	Gln	Gly	Arg	Leu 40	Ser	Lys	His	Val	Ser 45	Ser	Glu	Pro
Pro	Ser 50	Pro	Ser	Arg	Ser	Thr 55	Pro	Ser	Pro	Pro	Ser 60	Ser	Ser	Gly	Tyr
Lys 65	Asp	Asp	Pro	Arg	Lys 70	Ile	Ile	Leu	Ser	Met 75	Val	Leu	Gly	Ala	Val 80
Thr	Gly	Leu	Val	Ser 85	Ser	Ala	Leu	Phe	Ala 90	Leu	Val	Val	Arg	Cys 95	Val
Val	Gln	Tyr	Leu 100	Asn	Arg	Thr	Pro	Ile 105	Leu	Lys	Gly	Pro	Val 110	Ile	Phe
Ser	Pro	Lys 115	Ile	Ala	Pro	Met	Thr 120	Leu	Gln	Ser	Ala	Leu 125	Ala	Lys	Glu
Asn	His 130	Leu	Leu	Gly	Ser	Ser 135	Pro	Asn	Gly	Lys	Tyr 140	Tyr	Lys	Thr	Val
Leu 145	Asp	Asn	Gly	Leu	Thr 150	Ile	Ala	Val	Lys	Arg 155	Leu	Thr	Pro	Phe	Glu 160
Ser	Asn	Ser	Pro	Glu 165	Ala	Lys	Arg	Lys	Ser 170	Val	Lys	Arg	Gln	Ile 175	Gln
Thr	Glu	Leu	Glu 180	Leu	Leu	Ala	Ser	Leu 185	Arg	His	Arg	Asn	Leu 190	Met	Ser
Leu	Arg 195	Ala	Tyr	Val	Arg	Glu	Pro 200	Asp	Gly	Phe	Ser	Leu 205	Val	Tyr	Asp
Tyr 210	Val	Ser	Thr	Gly	Ser	Leu 215	Ala	Asp	Val	Leu	Ser 220	Lys	Val	Arg	Glu
Asn 225	Glu	Leu	Pro	Phe 230	Gly	Trp	Glu	Val	Arg	Leu 235	Arg	Ile	Ala	Val	Gly 240
Val	Val	Lys	Gly 245	Leu	Gln	Tyr	Leu	His	Phe 250	Thr	Cys	Val	Pro	Gln 255	Ile
Leu	His	Tyr	Asn 260	Leu	Lys	Pro	Thr	Asn 265	Val	Met	Leu	Asp	Ala	Glu	Phe
Glu	Pro	Arg	Leu	Ala	Asp	Tyr	Gly	Leu	Ala	Lys	Leu	Leu	Pro	Asn	Leu

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275	280	285
Asp Gly Gly Ser Ser Leu Tyr Thr Pro Pro Glu Cys Phe His Asn Cys		
290	295	300
Ser Arg Tyr Thr Asp Lys Ser Asp Ile Phe Ser Phe Gly Met Ile Leu		
305	310	315
Gly Val Leu Leu Thr Gly Lys Asp Pro Thr Asp Pro Phe Phe Gly Glu		
	325	330
Ala Ala Ser Gly Gly Ser Leu Gly Cys Trp Leu Arg His Leu Gln Gln		
	340	345
Ala Gly Glu Ala His Glu Ala Leu Asp Lys Ser Met Leu Gly Glu Glu		
	355	360
Gly Glu Glu Asp Glu Met Leu Met Ala Val Arg Ile Ala Ala Ala Cys		
	370	375
Leu Ser Asp Met Pro Ala Asp Arg Pro Ser Ser Asp Glu Leu Val His		
385	390	395
Met Leu Thr Gln Leu His Ser Phe		
	405	

<210> SEQ ID NO 108
 <211> LENGTH: 399
 <212> TYPE: PRT
 <213> ORGANISM: Actinidia chinensis

<400> SEQUENCE: 108

Met Leu Arg Phe Val Val Ser Met Glu Lys Thr Ser Cys Val Asn Leu
1 5 10 15
Glu Leu Val Leu Leu Leu Ile Ile Cys Leu Cys Asn Thr Ser Val Gln
20 25 30
Cys Gln Thr Arg Pro Ile Asn His Ile Ser Ala Glu Pro Pro Ser Pro
35 40 45
Ser Arg Ala Pro Glu Leu Lys Asn Lys Phe Gln Lys Ile Ile Leu Ser
50 55 60
Ile Leu Phe Gly Ile Val Thr Gly Leu Ile Cys Ser Phe Leu Phe Ala
65 70 75 80
Phe Thr Val Arg Cys Phe Val Arg Tyr Ile Ser Arg Thr Pro Ile Leu
85 90 95
Lys Gly Pro Val Ile Phe Ser Pro Lys Ile Asp Pro Lys Thr Leu Gln
100 105 110
Leu Ala Leu Ala Asn Glu Asn Gln Leu Leu Gly Ser Ser Pro Asn Gly
115 120 125
Lys Tyr Tyr Lys Thr Val Leu Asp Asn Gly Leu Thr Ile Ala Val Lys
130 135 140
Arg Leu Glu Pro Phe Glu Asn Gly Ser Pro Gln Ala Gln Ile Lys Ser
145 150 155 160
Val Lys Arg Arg Met Gln Gln Glu Leu Glu Met Leu Ala Ser Leu Arg
165 170 175
His Arg Asn Leu Met Ser Leu Arg Ala Phe Val Arg Glu Ser Asp Gly
180 185 190
Leu Cys Leu Val Tyr Asp Tyr Met His Ile Gly Ser Leu Glu Asp Ala
195 200 205
Met Lys Arg Val Arg Glu Asn Gln Leu Glu Leu Arg Trp Glu Pro Arg
210 215 220
Leu Arg Ile Ala Val Gly Ile Val Lys Gly Leu Gln Phe Leu His Phe
225 230 235 240

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Ser Cys Asn Pro Ser Arg Val Leu His Tyr Asn Leu Lys Pro Ser Asn
      245                      250                      255

Val Met Leu Asp Ala Glu Phe Glu Pro Arg Leu Ala Asp Cys Gly Leu
      260                      265                      270

Ala Lys Met Val Met Pro Asn Leu Asp Arg Ala Pro Ser Gly Tyr Asp
      275                      280                      285

Ala Pro Glu Cys Phe Gln Ser Cys Arg Tyr Thr Asp Lys Ser Asp Val
      290                      295                      300

Phe Ser Phe Gly Val Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro
305                      310                      315                      320

Met Asp Pro Phe Phe Glu Glu Ser Ala Ser Gly Gly Ser Leu Gly Arg
      325                      330                      335

Trp Leu Arg His Leu Gln His Ala Gly Glu Ala Arg Glu Ala Leu Asp
      340                      345                      350

Lys Ser Ile Ile Gly Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala
      355                      360                      365

Val Arg Ile Ser Val Val Cys Gln Ser Asp Leu Pro Ala Asp Arg Pro
      370                      375                      380

Ser Ser Asp Glu Leu Val Ser Met Leu Thr Gln Leu Asn Ser Phe
385                      390                      395

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<210> SEQ ID NO 109

<211> LENGTH: 399

<212> TYPE: PRT

<213> ORGANISM: Actinidia chinensis

<400> SEQUENCE: 109

```

Met Leu Arg Val Val Ala Ser Met Glu Lys Thr Asn Cys Val Thr Leu
1      5      10      15

Glu Leu Val Leu Leu Leu Ile Phe Cys Leu Cys Asn Thr Ser Val Gln
20     25     30

Cys Gln Thr Arg Pro Ile Asn His Ile Ser Ala Glu Pro Pro Ser Pro
35     40     45

Ser Arg Ala Pro Glu Leu Lys Asn Lys Phe Gln Lys Ile Ile Leu Ser
50     55     60

Ile Leu Leu Gly Ile Val Thr Gly Leu Ile Cys Ser Ile Leu Phe Ala
65     70     75     80

Phe Thr Val Arg Cys Phe Ile Arg Tyr Ile Ser Arg Thr Pro Ile Leu
85     90     95

Lys Gly Pro Val Ile Phe Ser Pro Lys Ile Asp Pro Lys Thr Leu Gln
100    105    110

Leu Ala Leu Ala Asn Glu Asn Gln Leu Leu Gly Ser Ser Pro Asn Gly
115    120    125

Lys Tyr Tyr Lys Thr Met Leu Asp Asn Gly Leu Thr Ile Ala Val Lys
130    135    140

Arg Leu Glu Pro Phe Glu Asn Gly Ser Pro Gln Ala Gln Ser Lys Ser
145    150    155    160

Val Lys Arg Arg Met Gln Gln Glu Leu Glu Met Leu Ala Ser Leu Arg
165    170    175

His Arg Asn Leu Met Ser Leu Arg Ala Phe Val Arg Glu Ser Asp Arg
180    185    190

Leu Cys Leu Val Tyr Asp Tyr Met His Ile Gly Ser Leu Glu Asp Val
195    200    205

Met Lys Arg Val Arg Glu Asn Gln Leu Glu Leu Arg Trp Glu Leu Arg
210    215    220

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Leu Arg Ile Ala Val Gly Ile Val Lys Gly Leu Gln Phe Leu His Phe
225          230          235          240

Ser Cys Asn Pro Thr Arg Phe Leu His Tyr Asn Leu Lys Pro Ser Asn
          245          250          255

Val Met Leu Asp Ala Glu Phe Glu Pro Arg Leu Ala Asp Cys Gly Leu
          260          265          270

Ala Lys Leu Val Met Pro Asn Leu Asp Arg Ala Pro Ser Gly Tyr Asp
          275          280          285

Ala Pro Glu Cys Phe Gln Asn Cys Arg Tyr Thr Asp Lys Ser Asp Val
          290          295          300

Phe Ser Phe Gly Val Ile Leu Gly Val Leu Leu Met Gly Arg Asp Pro
305          310          315          320

Met Asp Pro Phe Phe Glu Glu Ser Ala Ser Gly Gly Ser Phe Gly Arg
          325          330          335

Trp Leu Arg His Leu Gln His Ala Gly Glu Ala Arg Glu Ala Leu Asp
          340          345          350

Lys Arg Ile Ile Gly Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala
          355          360          365

Val Arg Ile Ser Val Val Cys Gln Ser Asp Leu Pro Ala Asp Arg Pro
          370          375          380

Ser Ser Asp Glu Leu Val Ser Met Leu Thr Gln Leu Asn Ser Phe
385          390          395

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<210> SEQ ID NO 110

<211> LENGTH: 390

<212> TYPE: PRT

<213> ORGANISM: Prunus persica

<400> SEQUENCE: 110

```

Met Asp Arg Arg Arg Ser Thr Ser Lys Phe Asn Ser Arg Leu Thr Arg
1          5          10          15

Leu Leu Leu Leu Leu Phe Ile Phe Cys Phe Arg Gln Glu Thr Val Gln
          20          25          30

Cys Gln Glu Thr Glu Pro Pro Ser Ser Ser Lys Gln Pro His Phe Lys
          35          40          45

Asn Gln Leu Gln Arg Ile Ile Leu Ser Ile Val Leu Gly Val Val Thr
          50          55          60

Gly Leu Ile Cys Ala Leu Leu Thr Ala Leu Leu Val Arg Cys Phe Leu
65          70          75          80

Arg Tyr Ile Ser Arg Thr Pro Ile Leu Lys Gly Pro Val Ile Phe Ser
          85          90          95

Pro Lys Ile Asp Pro Lys Thr Leu Gln Leu Ala Leu Thr Asn Glu Asn
          100          105          110

Gln Leu Leu Gly Ser Ser Pro Asn Gly Lys Tyr Tyr Arg Thr Val Leu
          115          120          125

Asp Asn Gly Leu Ile Ile Ala Val Lys Gln Leu Gly Pro Phe Ser Glu
          130          135          140

Cys Ser Ser Pro Glu Ser Gln Ser Lys Ser Val Lys Arg Arg Ile Gln
          145          150          155          160

Gln Glu Leu Glu Val Leu Ala Gly Leu Arg His Arg His Leu Met Ser
          165          170          175

Leu Arg Ala Tyr Val Arg Glu His Asp Arg Phe Ser Leu Val Tyr Asp
          180          185          190

Phe Val Pro Asn Gly Ser Leu Glu Asp Ala Met Asn Arg Val Arg Ala

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195	200	205
Asn Gln Leu Gln Leu Gly Trp	Glu Val Arg Leu Arg	Val Ala Val Gly
210	215	220
Val Ile Lys Gly Leu Gln Tyr	Leu His Ser Tyr	Val Pro Gln Ile Met
225	230	235
His Tyr Asn Leu Lys Pro Thr	Asn Val Met Leu Asp	Ser Glu Phe Glu
245	250	255
Pro Arg Leu Gly Asp Tyr Gly	Leu Ala Lys Leu Thr	Pro Tyr Leu Asp
260	265	270
Gly Ala Thr Ser Gly Tyr Ser	Ala Pro Glu Cys Phe	Gln Asn Gly Arg
275	280	285
Tyr Ser Asp Lys Ser Asp Ile	Phe Ser Phe Gly Met	Ile Leu Gly Val
290	295	300
Leu Leu Thr Gly Arg Asp Pro	Thr Asp Ala Phe Phe	Gly Glu Ala Ala
305	310	315
Ser Gly Gly Ser Leu Gly Arg	Trp Leu Arg His Leu	Gln Gln Ala Gly
325	330	335
Glu Ala Arg Glu Ala Leu Asp	Lys Ser Ile Ile Gly	Glu Glu Gly Glu
340	345	350
Glu Asp Glu Met Leu Met Ala	Val Arg Ile Ala Val	Val Cys Leu Ser
355	360	365
Asp Leu Pro Ala Glu Arg Pro	Ser Ser Asp Glu Leu	Val His Met Leu
370	375	380
Thr Gln Leu His Ser Phe		
385	390	
<210> SEQ ID NO 111		
<211> LENGTH: 390		
<212> TYPE: PRT		
<213> ORGANISM: Vitis vinifera		
<400> SEQUENCE: 111		
Met Glu Lys Thr Ser Arg Glu Ala Ile Ser Leu Leu Phe Leu Leu Ile		
1	5	10
Phe Ile His Phe Tyr Asn Val Lys Cys Gln Glu Asn Ile Ile Asn Arg		
20	25	30
Val Ser Thr Asp Ser Ser Pro Ser Pro Ser Arg Glu Ala Glu Phe Lys		
35	40	45
Ser Gly Leu Lys Lys Ile Ile Leu Ser Ile Val Phe Gly Val Leu Thr		
50	55	60
Gly Leu Leu Ser Ala Ala Leu Cys Ala Gly Ile Val His Cys Val Val		
65	70	75
Arg Tyr Met Asn Arg Thr Pro Ile Leu Lys Gly Pro Val Val Phe Ser		
85	90	95
Pro Lys Ile Ala Pro Lys Thr Val Gln Ser Ala Leu Asp Ser Glu Asn		
100	105	110
Gln Leu Leu Gly Ser Ser Pro Asn Gly Lys Tyr Tyr Lys Thr Ala Leu		
115	120	125
Asp Asn Gly Leu Ile Ile Ala Val Lys Arg Leu Glu Pro Phe Glu Thr		
130	135	140
Gly Ser Pro Glu Thr Gln Ser Lys Ser Val Lys Arg Arg Ile Gln Gln		
145	150	155
Glu Leu Glu Met Leu Ala Asn Leu Arg His Arg Asn Leu Met Ser Leu		
165	170	175

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Arg	Ala	Tyr	Val	Arg	Asp	Ser	Asp	Arg	Phe	Ser	Leu	Val	Tyr	Asp	Tyr
			180					185					190		
Val	Pro	Thr	Gly	Ser	Leu	Glu	Asp	Ala	Met	Asn	Arg	Val	Arg	Glu	Asn
		195					200					205			
Gln	Leu	Gln	Leu	Ser	Trp	Glu	Val	Arg	Leu	Arg	Ile	Ala	Val	Gly	Val
	210					215					220				
Ile	Lys	Gly	Leu	Gln	Tyr	Leu	His	Phe	Ser	Cys	Val	Pro	Arg	Ile	Leu
225					230					235					240
His	Tyr	Asn	Leu	Lys	Pro	Thr	Asn	Val	Met	Leu	Asp	Ala	Glu	Phe	Glu
			245						250					255	
Pro	Arg	Leu	Ala	Asp	Cys	Gly	Leu	Ala	Lys	Leu	Met	Pro	Asn	Leu	Asp
			260					265					270		
Arg	Ala	Ala	Ser	Gly	Tyr	Ser	Ala	Pro	Glu	Cys	Phe	Gln	Asn	Cys	Arg
		275					280					285			
Tyr	Thr	Glu	Lys	Ser	Asp	Val	Phe	Ser	Phe	Gly	Val	Ile	Leu	Gly	Val
	290					295					300				
Leu	Leu	Thr	Gly	Arg	Asp	Pro	Met	Asp	Pro	Phe	Phe	Ser	Glu	Ala	Ala
305					310					315					320
Gly	Gly	Gly	Ser	Leu	Gly	Arg	Trp	Leu	Arg	His	Leu	Gln	Gln	Ala	Gly
			325					330						335	
Glu	Ala	Arg	Glu	Ala	Leu	Asp	Lys	Ser	Ile	Leu	Gly	Glu	Glu	Val	Glu
			340				345						350		
Glu	Asp	Glu	Met	Leu	Met	Ala	Val	Arg	Ile	Ala	Val	Val	Cys	Leu	Ser
		355				360						365			
Asp	Leu	Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu	Val	Pro	Met	Leu
	370					375					380				
Thr	Gln	Leu	His	Ser	Phe										
385					390										

<210> SEQ ID NO 112

<211> LENGTH: 392

<212> TYPE: PRT

<213> ORGANISM: Gossypium raimondii

<400> SEQUENCE: 112

Met	Ala	Lys	Met	Ser	Gly	Cys	Ile	Val	Thr	Val	Ala	Leu	Leu	Leu	Leu
1			5					10					15		
Leu	Leu	Leu	Cys	Leu	His	Gln	Thr	Pro	Val	Glu	Cys	Lys	Glu	Arg	Leu
		20					25					30			
Ile	Arg	Gln	Leu	Ser	Ser	Gln	Pro	Ser	Ser	Ala	Thr	Lys	Pro	Gln	Glu
	35					40					45				
Phe	Lys	Ile	Gly	Phe	Lys	Arg	Val	Ile	Leu	Ser	Ile	Val	Leu	Gly	Ile
	50				55					60					
Leu	Thr	Gly	Leu	Ile	Gly	Ala	Ile	Leu	Phe	Ala	Leu	Leu	Ile	Lys	Ile
65			70					75						80	
Ala	Val	Gln	Tyr	Ile	Asn	Gln	Thr	Pro	Phe	Leu	Lys	Gly	Pro	Val	Ile
			85					90					95		
Phe	Ser	Pro	Lys	Ile	Ser	Ser	Lys	Thr	Leu	Gln	Ser	Ala	Leu	Ala	Asn
		100					105						110		
Glu	Asn	Gln	Leu	Leu	Gly	Ser	Ser	Ser	Asn	Gly	Lys	Tyr	Tyr	Lys	Thr
		115				120						125			
Val	Leu	Asp	Asn	Gly	Leu	Thr	Val	Ala	Val	Lys	Val	Leu	Glu	Pro	Phe
	130					135					140				
Asp	Ser	Gly	Ser	Pro	Glu	Met	Gln	Ser	Lys	Ser	Val	Lys	Arg	Arg	Ile
145					150					155					160

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Gln Gln Glu Leu Glu Val Leu Ala Ser Leu Arg His Arg His Leu Arg
 165 170 175
 Ser Leu Arg Ala Tyr Val Arg Glu Ser Asp Arg Phe Ser Leu Val Tyr
 180 185 190
 Asp Tyr Met Pro Met Gly Ser Leu Glu Asp Ala Met Asn Gly Val Arg
 195 200 205
 Glu Asn His Leu Glu Leu Arg Trp Asp Val Arg Leu Arg Ile Ala Val
 210 215 220
 Gly Val Ile Lys Gly Leu Gln Tyr Leu His Phe Thr Cys Val Pro Gln
 225 230 235 240
 Ile Leu His Tyr Asn Leu Lys Pro Thr Asn Val Met Leu Asp Ala Glu
 245 250 255
 Phe Glu Pro Arg Leu Ala Asp Cys Gly Leu Ala Lys Leu Met Pro Asn
 260 265 270
 Ile Asp Arg Ala Thr Ser Gly Tyr Cys Ala Pro Glu Cys Leu Gln Asn
 275 280 285
 Cys Arg Tyr Thr Asp Lys Ser Asp Ile Phe Ser Phe Gly Met Ile Leu
 290 295 300
 Gly Val Leu Leu Thr Gly Arg Tyr Pro Thr Asp Ser Phe Phe Arg Glu
 305 310 315 320
 Ala Val Ser Gly Gly Ser Leu Gly Gln Trp Leu Arg His Leu Gln Gln
 325 330 335
 Ala Gly Glu Ala His Glu Ala Leu Asp Lys Ser Ile Leu Gly Glu Glu
 340 345 350
 Val Glu Glu Asp Glu Met Leu Met Ala Val Arg Ile Ala Val Val Cys
 355 360 365
 Leu Ser Asp Ser Pro Asp Asp Arg Pro Ser Ser Asp Glu Leu Val Thr
 370 375 380
 Met Leu Thr Gln Leu His Ser Phe
 385 390

<210> SEQ ID NO 113

<211> LENGTH: 390

<212> TYPE: PRT

<213> ORGANISM: *Corchorus capsularis*

<400> SEQUENCE: 113

Met Gly Lys Gly Arg Arg Cys Thr Leu Ser Leu Leu Phe Leu Leu Leu
 1 5 10 15
 Leu Cys Phe His Gln Thr Thr Val His Cys Lys Glu Arg Leu Ile Arg
 20 25 30
 His Leu Ser Ser Gln Pro Pro Ser Pro Ser Lys Pro Gln Glu Phe Lys
 35 40 45
 Ile Gly Leu Lys Arg Ile Ile Leu Ser Ile Val Leu Gly Ile Leu Thr
 50 55 60
 Gly Leu Ala Gly Ala Leu Val Val Ala Leu Phe Ile Lys Phe Ala Val
 65 70 75 80
 Gln Tyr Met Asn Gln Thr Pro Ile Leu Lys Gly Pro Val Ile Phe Ser
 85 90 95
 Pro Lys Ile Ser Ala Lys Thr Leu Gln Ser Ala Leu Ala Asn Glu Asn
 100 105 110
 Gln Leu Leu Gly Ser Ser Ser Asn Gly Lys Tyr Tyr Lys Thr Val Leu
 115 120 125
 Asp Asn Gly Leu Thr Val Ala Val Lys Val Leu Glu Pro Phe Asp Asn

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130	135	140
Gly Ser Pro Glu Arg Gln Ser Lys Ser Val Lys Arg Arg Ile Gln Gln		
145	150	155 160
Glu Leu Glu Val Leu Ala Ser Leu Arg His Arg His Leu Met Ser Leu		
165	170	175
Arg Ala Tyr Val Arg Glu Ser Asp Arg Phe Ser Leu Val Tyr Asp Tyr		
180	185	190
Met Pro Thr Gly Ser Leu Glu Asp Ala Met Asn Arg Val Arg Gly Asn		
195	200	205
Gln Leu Gln Leu Gly Trp Asp Val Arg Leu Arg Ile Ala Val Gly Val		
210	215	220
Ile Lys Gly Leu Gln Tyr Leu His Phe Thr Cys Ile Pro Gln Ile Leu		
225	230	235 240
His Tyr Asn Leu Lys Pro Thr Asn Ile Met Leu Asp Ala Glu Phe Glu		
245	250	255
Pro Arg Leu Ala Asp Cys Gly Leu Ala Lys Leu Met Pro Asn Ile Asp		
260	265	270
Arg Ala Thr Ser Gly Tyr Ser Ala Pro Glu Cys Phe Gln Asn Cys Arg		
275	280	285
Tyr Thr Asp Lys Ser Asp Ile Phe Ser Phe Gly Met Ile Leu Gly Val		
290	295	300
Leu Leu Thr Gly Arg Asp Pro Ala Asp Pro Phe Phe Gly Glu Ala Ala		
305	310	315 320
Ser Gly Gly Ser Leu Gly Gln Trp Leu Arg His Leu Gln Gln Ala Gly		
325	330	335
Glu Ala Arg Glu Ala Leu Asp Lys Ser Ile Leu Gly Glu Glu Val Glu		
340	345	350
Glu Asp Glu Met Leu Met Ala Val Arg Ile Ala Val Val Cys Leu Ser		
355	360	365
Asp Leu Pro Ala Asp Arg Pro Ser Ser Asp Glu Leu Val Pro Met Leu		
370	375	380
Thr Gln Leu His Ser Phe		
385	390	

<210> SEQ ID NO 114

<211> LENGTH: 400

<212> TYPE: PRT

<213> ORGANISM: Theobroma cacao Matina 1-6

<400> SEQUENCE: 114

Met Phe Val Cys Gly Phe Val Ala Ser Met Glu Lys Arg Ser Cys Thr		
1	5	10 15
Leu Ser Leu Leu Leu Leu Leu Leu Leu Leu Cys Phe His Leu Thr		
20	25	30
Thr Val Gln Cys Glu Gly Arg Leu Ile Arg Tyr Leu Ser Ser Gln Pro		
35	40	45
Pro Ser Pro Ser Thr Pro Gln Glu Phe Lys Ile Gly Phe Lys Arg Ile		
50	55	60
Val Leu Ser Ile Ala Leu Gly Ile Leu Thr Gly Leu Ile Gly Ala Ile		
65	70	75 80
Leu Phe Ala Leu Leu Ile Lys Phe Ala Val Gln Tyr Met Asn Gln Thr		
85	90	95
Pro Ile Leu Lys Gly Pro Val Ile Phe Ser Pro Lys Ile Ser Ala Lys		
100	105	110

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Thr Leu Gln Ser Ala Leu Ser Thr Glu Asn Gln Leu Leu Gly Ser Ser
 115 120 125
 Ser Asn Gly Lys Tyr Tyr Lys Thr Val Leu Asp Asn Gly Leu Thr Val
 130 135 140
 Ala Val Lys Val Leu Glu Pro Phe Asp Asn Gly Ser Pro Glu Arg His
 145 150 155 160
 Ser Lys Ser Val Lys Arg Arg Ile Gln Gln Glu Leu Glu Ile Leu Ala
 165 170 175
 Ser Leu Arg His Arg His Leu Met Ser Leu Arg Ala Tyr Val Arg Glu
 180 185 190
 Ser Asp Arg Phe Ser Leu Val Tyr Asp Tyr Met Pro Thr Gly Ser Leu
 195 200 205
 Glu Asp Ala Met Asn Arg Val Arg Gly Asn Gln Leu Gln Leu Gly Trp
 210 215 220
 Asp Val Arg Leu Arg Ile Ala Val Gly Val Ile Lys Gly Leu Gln Tyr
 225 230 235 240
 Leu His Phe Thr Cys Ile Pro Gln Ile Leu His Tyr Asn Leu Lys Pro
 245 250 255
 Thr Asn Val Met Leu Asp Ala Glu Leu Glu Pro Arg Val Ala Asp Cys
 260 265 270
 Gly Leu Ala Lys Leu Met Pro Asn Ile Asp Arg Ala Thr Ser Gly Tyr
 275 280 285
 Gly Ala Pro Glu Cys Phe Glu Asn Cys Arg Tyr Thr Asp Lys Ser Asp
 290 295 300
 Ile Phe Ser Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp
 305 310 315 320
 Pro Thr Asp Pro Phe Phe Gly Glu Ala Ser Ser Gly Gly Ser Leu Arg
 325 330 335
 Gln Trp Leu Arg His Leu Gln Gln Ala Gly Glu Ala Arg Glu Ala Leu
 340 345 350
 Asp Lys Ser Ile Leu Gly Glu Glu Val Glu Glu Asp Glu Met Leu Met
 355 360 365
 Ala Val Arg Ile Ala Val Val Cys Leu Ser Asp Leu Pro Ala Asp Arg
 370 375 380
 Pro Ser Ser Asp Glu Leu Val Pro Met Leu Thr Gln Leu His Ser Phe
 385 390 395 400

<210> SEQ ID NO 115

<211> LENGTH: 398

<212> TYPE: PRT

<213> ORGANISM: Manihot esculenta

<400> SEQUENCE: 115

Met Glu Lys Arg Arg Tyr Ser Ser Gln Leu Thr Ser Lys Val Thr Ala
 1 5 10 15
 Leu Leu Leu Leu Leu Ser Phe Cys Phe His Pro Ile Val Val Gln Cys
 20 25 30
 Gln Glu Ser Asn Ile Lys Met Thr Arg His Leu Ser Ser Glu Thr Pro
 35 40 45
 Pro Ser Arg Ser Pro Gln Phe His Thr Gly Leu Lys Arg Ile Leu Leu
 50 55 60
 Ser Ile Gly Leu Gly Val Leu Thr Gly Leu Thr Gly Ala Val Leu Cys
 65 70 75 80
 Ala Cys Val Val Arg Phe Phe Val Arg Tyr Met Asn Arg Thr Pro Ile
 85 90 95

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Leu Lys Gly Pro Val Thr Phe Ser Pro Asn Ile Ala Pro Lys Thr Leu
    100                      105                      110

Gln Ser Ala Leu Ala Ser Glu Asn Gln Leu Leu Gly Ser Ser Ser Asn
    115                      120                      125

Gly Lys Tyr Tyr Arg Lys Val Leu Asp Asn Asp Leu Thr Val Ala Val
    130                      135                      140

Lys Arg Leu Glu Pro Phe Glu Asn Gly Ser Pro Glu Arg Gln Ser Lys
    145                      150                      155                      160

Ala Val Lys Arg Arg Ile Gln Gln Glu Leu Glu Arg Leu Ala Ser Leu
    165                      170                      175

Arg His Arg Asn Leu Met Ser Leu Arg Ala Tyr Val Arg Glu Ser Asp
    180                      185                      190

Arg Phe Ser Leu Val Tyr Asp Tyr Val Pro Thr Gly Ser Leu Glu Asp
    195                      200                      205

Ala Met Asn Arg Val Arg Asp Asn Gln Leu Gln Leu Gly Trp Glu Val
    210                      215                      220

Arg Leu Arg Ile Ala Val Gly Val Val Lys Gly Leu Arg Tyr Leu His
    225                      230                      235                      240

Phe Glu Cys Val Pro Gln Ile Leu His Tyr Asn Leu Lys Pro Thr Asn
    245                      250                      255

Val Met Leu Asp Ala Glu Phe Glu Pro Arg Leu Ala Asp Ser Gly Leu
    260                      265                      270

Ala Lys Leu Met Pro Asn Leu Asp Arg Thr Thr Ser Gly Tyr Ser Ala
    275                      280                      285

Pro Glu Cys Phe Gln Asn Cys Arg Tyr Thr Glu Lys Ser Asp Ile Phe
    290                      295                      300

Ser Phe Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr
    305                      310                      315                      320

Asp Pro Phe Phe Gly Glu Ala Ala Ser Gly Gly Ser Leu Gly His Trp
    325                      330                      335

Leu Arg His Leu Gln His Ala Gly Glu Ala Arg Glu Ala Leu Asp Lys
    340                      345                      350

Ser Ile Leu Gly Glu Glu Gly Glu Glu Asp Glu Met Leu Met Ala Val
    355                      360                      365

Arg Ile Ala Val Val Cys Leu Ser Asp Leu Pro Ala Asp Arg Pro Ser
    370                      375                      380

Ser Asp Glu Leu Val Leu Met Leu Ser Gln Leu His Ser Phe
    385                      390                      395

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<210> SEQ ID NO 116

<211> LENGTH: 394

<212> TYPE: PRT

<213> ORGANISM: Populus trichocarpa

<400> SEQUENCE: 116

```

Met Gly Lys Arg Arg Tyr Ser Thr Gln Ile Ser Asn Asn Lys Asp Thr
  1                      5                      10                      15

Ser Ser Leu Leu Phe Leu Phe Ile Leu Cys Leu Tyr Tyr Thr Thr Val
    20                      25                      30

Gln Cys Gln Glu Ser Ser Lys Val Thr Pro Pro Phe Pro Ser Thr Pro
    35                      40                      45

Thr His Ser Lys Asn Gly Leu Lys Arg Ile Leu Val Ser Ile Phe Leu
    50                      55                      60

Gly Val Leu Thr Gly Leu Thr Gly Ala Val Val Phe Ala Phe Val Val

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65	70	75	80
Arg Phe Leu Val Arg Tyr Met Lys Arg Thr Pro Ile Leu Lys Gly Pro			
	85	90	95
Val Ile Phe Ser Pro Lys Ile Thr Pro Lys Ser Leu Gln Ser Ala Leu			
	100	105	110
Glu Asn Glu Asn Gln Leu Leu Gly Ser Ser Ser Asn Gly Lys Tyr Tyr			
	115	120	125
Arg Thr Val Leu Asp Asn Gly Leu Thr Ile Ala Val Lys Arg Phe Glu			
	130	135	140
Pro Phe Glu Ile Gly Ser Pro Glu Arg Gln Ser Lys Ser Val Lys Arg			
	145	150	155
Arg Ile Gln Gln Glu Leu Glu Met Leu Ala Ser Leu Arg His Arg Asn			
	165	170	175
Leu Met Ser Leu Arg Ala Tyr Val Arg Glu Pro Asp Arg Phe Ser Leu			
	180	185	190
Val Tyr Asp Cys Val Pro Thr Gly Ser Leu Glu Asp Ala Met Asn Arg			
	195	200	205
Val Arg Glu Asn Glu Leu Gln Leu Gly Trp Glu Val Arg Leu Arg Ile			
	210	215	220
Ala Val Gly Val Ile Lys Gly Leu Arg Tyr Leu His Phe Asp Cys Val			
	225	230	235
Pro Gln Ile Leu His Tyr Asn Leu Lys Pro Arg Asn Val Ile Leu Asp			
	245	250	255
Ala Glu Phe Glu Pro Arg Leu Ala Asp Phe Gly Leu Ala Lys Leu Thr			
	260	265	270
Pro Asn Leu Asp Arg Ala Thr Ser Gly Tyr Ser Ala Pro Glu Cys Phe			
	275	280	285
Gln Asn Cys Arg Tyr Thr Asp Lys Ser Asp Val Phe Ser Phe Gly Met			
	290	295	300
Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr Asp Pro Phe Phe			
	305	310	315
Arg Glu Thr Ala Ser Gly Gly Ser Leu Gly Pro Trp Leu Arg His Leu			
	325	330	335
Gln Gln Ala Gly Glu Ala Arg Glu Ala Leu Asp Lys Ser Ile Leu Gly			
	340	345	350
Glu Glu Val Glu Glu Asp Glu Met Leu Met Ala Val Arg Ile Ala Val			
	355	360	365
Val Cys Leu Ser Asp Met Pro Ala Asp Arg Pro Ser Ser Asp Glu Leu			
	370	375	380
Val Pro Met Leu Ser Gln Leu His Ser Phe			
	385	390	

<210> SEQ ID NO 117

<211> LENGTH: 394

<212> TYPE: PRT

<213> ORGANISM: Populus trichocarpa

<400> SEQUENCE: 117

Met Glu Lys Arg Arg Tyr Ser Leu Arg Val Ser Asn Ser Lys Val Thr			
1	5	10	15
Ala Ser Leu Leu Phe Leu Phe Val Leu Cys Leu Tyr Tyr Ala Ser Val			
	20	25	30
Gln Cys Gln Glu Ser Ser Lys Val Thr Pro Pro Ser Pro Ser Thr Pro			
	35	40	45

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Thr	Gln	Ser	Lys	Asn	Gly	Leu	Lys	Arg	Ile	Leu	Val	Ser	Ile	Phe	Leu
50						55				60					
Gly	Val	Leu	Thr	Gly	Leu	Ile	Gly	Ala	Val	Val	Phe	Ala	Phe	Val	Val
65					70					75					80
Arg	Phe	Leu	Val	Arg	Tyr	Met	Lys	Arg	Thr	Pro	Ile	Leu	Lys	Gly	Pro
				85					90					95	
Val	Ile	Phe	Ser	Pro	Lys	Ile	Thr	Pro	Lys	Ser	Leu	Gln	Ser	Ala	Leu
			100					105					110		
Glu	Asn	Glu	Asn	Gln	Leu	Leu	Gly	Ser	Ser	Ser	Asn	Gly	Lys	Tyr	Tyr
		115					120					125			
Arg	Thr	Ala	Leu	Asp	Asn	Gly	Leu	Thr	Ile	Ala	Val	Lys	Arg	Phe	Glu
	130					135					140				
Pro	Phe	Glu	Ile	Gly	Ser	Pro	Glu	Arg	Gln	Ser	Lys	Ser	Val	Lys	Arg
145					150					155					160
Arg	Ile	Gln	Gln	Glu	Leu	Glu	Met	Leu	Ala	Ser	Leu	Arg	His	Arg	Asn
				165					170					175	
Leu	Met	Ser	Leu	Arg	Ala	Tyr	Val	Arg	Glu	Pro	Asp	Arg	Phe	Ser	Leu
			180					185					190		
Val	Tyr	Asp	Tyr	Val	Pro	Thr	Gly	Ser	Leu	Glu	Asp	Ala	Met	Asn	Arg
		195					200					205			
Val	Arg	Glu	Asn	Glu	Leu	Gln	Leu	Gly	Trp	Glu	Val	Arg	Leu	Arg	Ile
	210					215					220				
Ala	Val	Gly	Val	Ile	Lys	Gly	Leu	Arg	Tyr	Leu	His	Phe	Glu	Cys	Ala
225					230					235					240
Pro	Gln	Ile	Leu	His	Tyr	Asn	Leu	Lys	Pro	Arg	Asn	Val	Ile	Leu	Asp
				245					250					255	
Ala	Glu	Phe	Glu	Pro	Arg	Leu	Ala	Asp	Phe	Gly	Leu	Ala	Lys	Leu	Thr
			260					265					270		
Pro	Asn	Leu	Asp	Arg	Ala	Thr	Ser	Gly	Tyr	Ser	Ala	Pro	Glu	Cys	Phe
		275					280					285			
Gln	Asp	Cys	Arg	Tyr	Ser	Asp	Lys	Ser	Asp	Val	Phe	Ser	Phe	Gly	Met
	290					295					300				
Ile	Leu	Gly	Val	Leu	Leu	Thr	Gly	Arg	Asp	Pro	Thr	Asp	Pro	Phe	Phe
305					310					315					320
Gly	Glu	Thr	Ala	Ser	Gly	Gly	Ser	Leu	Gly	Arg	Trp	Leu	Arg	His	Leu
				325					330					335	
Gln	Gln	Ala	Gly	Glu	Ala	Arg	Glu	Ala	Leu	Asp	Lys	Ser	Leu	Leu	Gly
			340					345					350		
Glu	Glu	Val	Glu	Glu	Asp	Glu	Met	Leu	Met	Ala	Val	Arg	Ile	Ala	Val
		355					360					365			
Val	Cys	Gln	Ser	Glu	Met	Pro	Ala	Asp	Arg	Pro	Ser	Ser	Asp	Glu	Leu
	370					375					380				
Val	Pro	Met	Leu	Ser	Gln	Leu	His	Ser	Phe						
385					390										

<210> SEQ ID NO 118

<211> LENGTH: 414

<212> TYPE: PRT

<213> ORGANISM: Zea mays

<400> SEQUENCE: 118

Met	Thr	Ser	Arg	Asn	Pro	Thr	Lys	Thr	Leu	Ser	Val	Leu	Thr	Thr	Leu
1				5					10					15	
Leu	Leu	Leu	Leu	Glu	Phe	Leu	Ala	Leu	Cys	Ala	Pro	Ala	Ser	Ala	Gln
			20					25					30		

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Pro Leu His Ser Glu Pro Met Ala Thr Gln Ser Pro Pro Pro Pro Leu
   35                               40               45

Pro Pro Gly Ser Thr Ile Pro Arg Ala Gln Ala Gly Gly Ala Ala Arg
   50                               55               60

Leu Arg Arg Ile Ala Leu Gly Val Leu Phe Gly Ser Leu Ser Gly Phe
   65                               70               75               80

Leu Leu Ala Leu Ala Phe Leu Tyr Ala Ile Arg Leu Ala Ile Leu His
                        85               90               95

Ala Lys Ser Thr Pro Ala Ile Ala Lys Gly Pro Val Ser Phe Ala Pro
      100               105               110

Gln Ile Ser Ala Lys Asn Leu Leu Ala Ala Leu Pro Ala Ala Gln Pro
      115               120               125

Leu Ala His Gly Pro His Gly Lys Tyr Tyr Lys Leu Ala Leu Asp Asn
      130               135               140

Asp Leu Thr Val Ala Val Lys Arg Leu Glu Ala Ala Ser Arg Pro Glu
      145               150               155               160

Ala Ser Pro Ser Met Ser Pro Ser Ala Ser Lys Ser Asp Met Arg Arg
      165               170               175

Val Gln Arg Gln Leu Glu Ala Leu Ala Arg Val Arg His Gln Asn Val
      180               185               190

Leu Ser Leu Lys Ala Tyr Val Arg Glu Ala Asp Arg Leu Ser Leu Val
      195               200               205

Tyr Asp Phe Val His Gly Gly Ser Leu Glu Asp Leu Met Lys Arg Val
      210               215               220

Arg Ser Gln Gln Val Ser Leu Gly Trp Asp Ala Arg Ser Arg Ile Ala
      225               230               235               240

Val Gly Ile Ala Lys Gly Leu Arg His Leu His Phe Glu Cys Asn Pro
      245               250               255

Arg Ile Val His Cys Asn Leu Lys Pro Ser Asn Val Met Leu Asp Glu
      260               265               270

Gly Leu Glu Pro Leu Leu Ala Asp Cys Gly Val Ala Arg Leu Ile Ala
      275               280               285

Ala Gly Ser Gly Asp Pro Glu Leu Cys Thr Gly Leu Tyr Ala Ala Pro
      290               295               300

Glu Cys Tyr Gln Ser Ser Arg Tyr Thr Asp Lys Ser Asp Val Tyr Ala
      305               310               315               320

Leu Gly Met Ile Leu Gly Val Leu Leu Thr Gly Arg Asp Pro Thr Asp
      325               330               335

Ser Phe Phe Ser Gly Glu Ser Thr Gly Gln Gly Gly Leu Pro Arg Trp
      340               345               350

Leu Arg His Thr Gln Gln Ser Ala Asp Pro Lys Glu Thr Leu Asp Ser
      355               360               365

Ser Ile Leu Gly Asp Glu Gly Glu Glu Glu Glu Met Leu Met Ala Ile
      370               375               380

Arg Val Ala Ile Val Cys Leu Ser Asp Ser Pro Thr Asp Arg Pro Ser
      385               390               395               400

Ser Asp Glu Leu Val Ala Met Leu Thr Gln Leu His Ser Leu
      405               410

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<210> SEQ ID NO 119
<211> LENGTH: 44
<212> TYPE: PRT
<213> ORGANISM: Zea mays

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-continued

<400> SEQUENCE: 119

Leu Ala Leu Cys Ala Pro Ala Ser Ala Gln Pro Leu His Ser Glu Pro
 1 5 10 15
 Met Ala Thr Gln Ser Pro Pro Pro Pro Leu Pro Pro Gly Ser Thr Ile
 20 25 30
 Pro Arg Ala Gln Ala Gly Gly Ala Ala Arg Leu Arg
 35 40

<210> SEQ ID NO 120

<211> LENGTH: 64

<212> TYPE: PRT

<213> ORGANISM: Zea mays

<400> SEQUENCE: 120

Leu Thr Thr Leu Leu Leu Leu Leu Glu Phe Leu Ala Leu Cys Ala Pro
 1 5 10 15
 Ala Ser Ala Gln Pro Leu His Ser Glu Pro Met Ala Thr Gln Ser Pro
 20 25 30
 Pro Pro Pro Leu Pro Pro Gly Ser Thr Ile Pro Arg Ala Gln Ala Gly
 35 40 45
 Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly Val Leu Phe Gly Ser
 50 55 60

<210> SEQ ID NO 121

<211> LENGTH: 84

<212> TYPE: PRT

<213> ORGANISM: Zea mays

<400> SEQUENCE: 121

Ser Arg Asn Pro Thr Lys Thr Leu Ser Val Leu Thr Thr Leu Leu Leu
 1 5 10 15
 Leu Leu Glu Phe Leu Ala Leu Cys Ala Pro Ala Ser Ala Gln Pro Leu
 20 25 30
 His Ser Glu Pro Met Ala Thr Gln Ser Pro Pro Pro Pro Leu Pro Pro
 35 40 45
 Gly Ser Thr Ile Pro Arg Ala Gln Ala Gly Gly Ala Ala Arg Leu Arg
 50 55 60
 Arg Ile Ala Leu Gly Val Leu Phe Gly Ser Leu Ser Gly Phe Leu Leu
 65 70 75 80
 Ala Leu Ala Phe

<210> SEQ ID NO 122

<211> LENGTH: 4433

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 122

gcgcgcgcgc cgccggcgcg ggacggggcg gaggagaatg tttggctggt ccgggtccgg 60
 tcgcacggca tgggtgtgga cgcgagccgc ctgcgagggc cgaggggaga gggatgcgtc 120
 aagaaagctc tcggttttta gcttgccttg ggcgcccaca ccaatgccac actgctctgc 180
 tctccgagta acttattgcg agcagcgccc caggtgaccg agcgagcggg ccggcgaggg 240
 ctcagctttg gataattctt gctccctctt gtcttttcgg tttgaagctt tctggaggga 300
 gaagggagca cgcagctcgg cgccagttag agagaggatt agaggagcga gaggaaggga 360
 ggagatcatg cttgcagagc gtgctccgtg atgcggtgag cgccgctaac ctcggttgat 420
 ttcttggtgg gaggcgacaa acgcgtgcgt gagccgccgc ttgcttgga ggtggtggtg 480

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agaccaggcg	tgtccttcg	cacttgctc	tgtttgagtt	cgttgctgc	aggtaactcct	540
cacgagctcc	acgcgcgcg	cgccttgggc	gaetgccttt	tcgattcgtc	ttctttgatg	600
catattttct	ctttctgtag	gtaaatatca	cgagcaccat	gctcgagttc	ttgtagctcc	660
gcagcttctc	cgttctttgc	atcctttgca	acttgctctc	tgttgttttc	ctctgatttc	720
catggaggaa	agatgctggg	tgtgccagt	atgtgggcgt	ggagcaggca	acaattggct	780
cgtctggac	attgatcgat	tctccctat	tctcgccaa	ggaattttat	tcgtttctcc	840
tagcaagtcg	tgtatgcttg	ttctgaggag	gtaggagcta	ggatgacgag	caggaaacct	900
acaaaaacc	tgtcggctct	gacctctc	ctgctgctgc	ttgagttcct	cgcgctgtgc	960
gcgccgcct	cgcgccagcc	gctccactcg	gagcccatgg	ctacgcagtc	ccgcgcgcg	1020
cgcgtgcgc	cgggggtccac	gattcccccg	gcgcaggccg	gcggcgcgc	acgcctccgc	1080
cgcattgcgc	tcggggctct	cttcggctcc	ctctccggtt	tcctctcgc	gctcgccttc	1140
ctctacgcc	tccgcctcgc	cactctccac	gccaagagca	cgccgcgat	cgccaagggc	1200
cccgctctct	tcgccccgca	aatctccgc	aagaacctcc	tggccgcgct	ccccgcgcg	1260
cagccgctcg	cccatgggcc	ccatggcaag	tactacaagc	tggcactcga	caacgaacct	1320
accgtcgcgc	tcaagcgct	cgaggcgcc	agccgccccg	aggcctcgc	gtccatgtcc	1380
ccgagcgcgt	ccaagtcgga	catgaggagg	gtgcagaggc	agcttgaggc	acttgctcgg	1440
gtcaggcacc	agaatgtgtt	gtcgttgaag	gcatacgttc	gcgaggcgga	ccgcctctcg	1500
ctcgtgtacg	acttcgtcca	tgggggcagc	ctcaggatc	tgatgaagag	ggtaaggctc	1560
cagcaagtca	gcctcgggtg	ggatgcgagg	agcaggattg	ctgttgggat	tgccaaggga	1620
ctgaggcact	tgcattttga	gtgcaatccg	aggatagtcc	actgcaacct	caagccgtcc	1680
aatgtgatgc	tagacgaagg	tttgaggcca	ttattggcag	attgtggtgt	tgcaaggctg	1740
attgcagcag	gttcagggtga	tccagaattg	tgcactggcc	tctacgtgc	tccagaatgc	1800
tatcaaagca	gcaggtaaca	tcaacgtgct	agcattttca	gttctttttt	ctgaatttaa	1860
agaattagct	aaattctcga	aatctcagc	attctggtaa	ccctgtggga	taggtaggaa	1920
tttacaggaa	acaatccaat	tcttgcatga	gtattgaaaa	aaacacattc	taagggtctc	1980
tggcttattg	ttttgatact	tgatcgtata	aatgaaacct	tcagatctac	caatgtattt	2040
ctcatgattg	tcactcttga	agtgaagaag	ggaatctaaa	tctagtttgg	gtcagaattt	2100
tctagtatta	aattatatca	actaaatgtt	acttatatga	cattcttgat	gaatggtgat	2160
catatatcat	tcccgcctcg	tgaaaggata	cccaatgaca	agaagatatg	tcgtagactt	2220
tgtttgagtt	ccttgctatg	catgtatata	atctttagaa	acagatggct	ggcgcgtata	2280
tgttcgcatt	tttgatatac	aaatctcaac	tgttctaata	aaattctcgg	accatggaaa	2340
tcccaaagg	gtttcgtgca	ttatagaagc	atggagatat	aagaaaggaa	ttgtagaaat	2400
ctatgacgta	tatagagggt	cctttaatta	ttttcgtttt	gctttgtcta	gcttgttgac	2460
attgtggaca	aactatatgt	tgtagtgact	ttaccgatac	attctctgta	tttgacttct	2520
gcctgggttc	ggagtgtatg	cacaagcatc	tgagtattgt	tttagaataa	ctagtggaa	2580
tttgatatgt	ctctggtttg	gcaaacacaga	aattaagctt	aagacttcca	ctttacatag	2640
gatgggggtt	tagctcaatt	gattgtgtgg	gagactaaga	ggtttttagt	catgcacacc	2700
caacctggtt	ggttaaagcc	ctcttcagct	tgagtttgag	tgtttttccc	ctcttaataga	2760
ttaatgatca	tgaaaacttt	tattcaaaaa	aaaatcatca	aaactgccta	ccctcctag	2820

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tttggatatct ttttgaact tccattatat atatttttat taaaaaaatt cttaaccac	2880
cagataccag tggtttaaag gcgcgcgcgcgc ctccctaggcg tocaagagct caagcaatct	2940
gaggcgcctt aagaaccatt gtagatagta tacatccatt gacttggtga tgctaccatc	3000
cagtttttcc aagtactac tagctagtgg tcttctttgt cattaaactt gccacaagag	3060
taactgagaa ggcataaagc ataaaaatga tgatcatccc tagggcttag tgaaagtatc	3120
taacttttcc ttttggcgac atgagaatct ttacttttgc agaggacctc aattttgcgc	3180
gcctttgatg ctcccgcttt tcataaattc ctggcgtgag ccgtgagggc atgcagaata	3240
ccggcccgct ccctttttat gtccaaaaat gctattgcta cccttttttc attcaggcgcg	3300
accttcata tgcccccct caaagaataa atccctttgag gattggcatt ctgattccat	3360
cgttcatatt tctcctggca aaccagcta tgctatgtca gtagattttt agagggtttg	3420
cttcagtcatt tgtagtggtg ggtttttagt ttacttgac agttgcaccc cttgagagga	3480
agaagaagta tactgattgt ttccaacaaa acggatcttc tttctgtac aagtactgta	3540
gttgatctat ccatgtaatt ctttctttga tgatttgaga ctcaactgtt tgggtgtacg	3600
tacaggtaga cggataagag tgatgtgtac gcccttgca tgatccctggg cgtgctgctc	3660
actgggagag acccgacaga ctctttcttc tcgggagaga gtaccgggca gggcggcctc	3720
ccaagatggc tccggcacac gcagcagtc ccggaccga aagaaacgt ggatagcagc	3780
atccctgggg acgagggaga agaggaggag atgctgatgg ccattcgggt tgccattgtc	3840
tgccctcctg actcgccaac cgaccggcca tccagcgacg agctcgtgc catgctaacg	3900
cagctccaca gcttatgatt aatcaccata tatcgcaaat ttcttgagcg gctgccgctg	3960
cgcattggct taaaaccatg ggggggcgat catgttgat cagtgcctcc tgtttcacag	4020
ctagcacggg ggagtgtgca gcagatctca ttgtgtggtg cttgtgagtc acaagcgctg	4080
aaatggtcga caagggaact gacagttcaa ggccatatag gacaggcgca gtcgctgtgt	4140
tccctgtacg attcatcagt gatgtagtag ctaggtagaa tacaagatgg tggttgagct	4200
tcctgggttg ttttgcgcgt gggctgtgaa tatgtcttg tagaaaaaa gctctgtgcc	4260
gtggatctga tctgttggtg ttagctgtag cctgtgtata gcctgtagaa agacgtgctt	4320
gtgctgtgca ctgagaccca tcattgtacac cccccggct cactccaat ttgcctgaga	4380
gcagagagac tatgatgccg agtatattaac aggtgctggt ggccaacggt ttc	4433

<210> SEQ ID NO 123
 <211> LENGTH: 133
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 123

tcgcgctgtg cgcgcgcgcgc tccgcccagc cgctccactc ggagcccatg gctacgcagt	60
ccccgcgcgc gccgctgccg ccgggggtcca cgattccccg ggcgcaaggc ggcggcgccg	120
cacgcctccg ccg	133

<210> SEQ ID NO 124
 <211> LENGTH: 93
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 124

tccgcccagc cgctccactc ggagcccatg gctacgcagt cccccgcgc gccgctgccg	60
ccgggggtcca cgattccccg ggcgcaaggc ggc	93

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<210> SEQ ID NO 125
<211> LENGTH: 78
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 125

cagccgctcc actcggagcc catggctacg cagtccccgc cgccgccgct gccgccgggg 60
tccacgattc cccgggcg 78

<210> SEQ ID NO 126
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial sequences
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 126

ttcctcgcgc tgtgcgcgcc cgcctcc 27

<210> SEQ ID NO 127
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 127

tccgcccagc cgtccactc ggagccc 27

<210> SEQ ID NO 128
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 128

tccactcgga gcccatggct acgcagt 27

<210> SEQ ID NO 129
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Artificial sequence

<400> SEQUENCE: 129

tccccgccgc cgcgctgcc gccgggg 27

<210> SEQ ID NO 130
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 130

tccacgattc cccgggcgca ggccggc 27

<210> SEQ ID NO 131
<211> LENGTH: 66
<212> TYPE: PRT
<213> ORGANISM: Arabidopsis thaliana

-continued

<400> SEQUENCE: 131

```

Met Lys Gln Arg Arg Arg Arg Asn Gly Cys Ser Ser Ser Asn Thr Ile
1           5           10           15
Ser Leu Leu Leu Leu Phe Phe Leu Val Phe Phe Ser Arg Thr Ser Thr
          20           25           30
Ser Thr Ser Cys Arg Arg Arg Thr Val Lys His Leu Ser Thr Thr Ser
          35           40           45
Thr Ser Ser Thr Pro Leu Glu Ser Arg Ile Thr Ser Lys Val Ile Val
          50           55           60
Ile Ser
65

```

<210> SEQ ID NO 132

<211> LENGTH: 71

<212> TYPE: PRT

<213> ORGANISM: Zea mays

<400> SEQUENCE: 132

```

Met Thr Ser Arg Asn Pro Thr Lys Thr Leu Ser Val Leu Thr Thr Leu
1           5           10           15
Leu Leu Leu Leu Glu Phe Leu Ala Leu Cys Ala Pro Ala Ser Ala Gln
          20           25           30
Pro Leu His Ser Glu Pro Met Ala Thr Gln Ser Pro Pro Pro Pro Leu
          35           40           45
Pro Pro Gly Ser Thr Ile Pro Arg Ala Gln Ala Gly Gly Ala Ala Arg
          50           55           60
Leu Arg Arg Ile Ala Leu Gly
65           70

```

<210> SEQ ID NO 133

<211> LENGTH: 79

<212> TYPE: PRT

<213> ORGANISM: Triticum aestivum

<400> SEQUENCE: 133

```

Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Thr Lys Asn Pro Thr Lys
1           5           10           15
Thr Leu Ala Ala Ile Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
          20           25           30
Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
          35           40           45
Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Ser Ala Pro His
          50           55           60
Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65           70           75

```

<210> SEQ ID NO 134

<211> LENGTH: 79

<212> TYPE: PRT

<213> ORGANISM: Triticum dicoccoides

<400> SEQUENCE: 134

```

Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Thr Lys Asn Pro Thr Lys
1           5           10           15
Thr Leu Ala Ala Ile Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
          20           25           30
Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
          35           40           45

```

-continued

Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Ser Ala Pro His
50 55 60

Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65 70 75

<210> SEQ ID NO 135

<211> LENGTH: 79

<212> TYPE: PRT

<213> ORGANISM: Triticum aestivum

<400> SEQUENCE: 135

Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Thr Lys Asn Pro Thr Lys
1 5 10 15

Thr Leu Ala Ala Thr Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
20 25 30

Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
35 40 45

Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Arg Ala Pro His
50 55 60

Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65 70 75

<210> SEQ ID NO 136

<211> LENGTH: 79

<212> TYPE: PRT

<213> ORGANISM: Aegilops tauschii

<400> SEQUENCE: 136

Met Ala Pro Gly Ala Gly Thr Ala Ala Ala Thr Lys Asn Pro Thr Lys
1 5 10 15

Thr Leu Ala Ala Thr Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
20 25 30

Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
35 40 45

Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Arg Ala Pro His
50 55 60

Ala Arg Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65 70 75

<210> SEQ ID NO 137

<211> LENGTH: 79

<212> TYPE: PRT

<213> ORGANISM: Triticum aestivum

<400> SEQUENCE: 137

Ala Thr Gly Ala Gly Thr Ala Ala Ala Ala Ala Lys Asn Pro Thr Lys
1 5 10 15

Thr Leu Ala Thr Leu Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
20 25 30

Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Pro Ala Thr
35 40 45

Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Arg Gly Pro Arg
50 55 60

Ala Gln Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65 70 75

<210> SEQ ID NO 138

<211> LENGTH: 79

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Triticum dicoccoides

<400> SEQUENCE: 138

Ala Pro Gly Ala Gly Thr Ala Ala Ala Ala Lys Asn Pro Thr Lys
1          5          10          15
Thr Leu Ala Thr Leu Leu Leu Leu Val Leu Leu Leu Ala Asp Leu Pro
          20          25          30
Leu Cys Ala Ser Gln Pro Pro Leu His Ser Gln Pro Leu Ser Ala Thr
          35          40          45
Gln Ser Pro Ala Ala Pro Leu Pro Pro Pro Gln Pro Arg Pro Pro Arg
          50          55          60
Ala Gln Ala Gly Gly Ala Ala His Leu Arg Arg Ile Ala Leu Gly
65          70          75

```

```

<210> SEQ ID NO 139
<211> LENGTH: 78
<212> TYPE: PRT
<213> ORGANISM: Brachypodium distachyon

<400> SEQUENCE: 139

Met Ala Arg Gly Gly Gly Val Val Val Gly Lys Lys Asn Pro Thr Lys
1          5          10          15
Thr Leu Ala Ala Ile Leu Leu Leu Leu Leu Leu Ala Val Phe Pro
          20          25          30
Arg Pro Ala Ala Ser Gln Pro Leu His Ser Glu Pro Met Ser Thr Gln
          35          40          45
Gln Ser Pro Pro Pro Pro Gln Gln Gln Ser Lys Ile Pro His Ala Gln
          50          55          60
Pro Gly Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65          70          75

```

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<210> SEQ ID NO 140
<211> LENGTH: 80
<212> TYPE: PRT
<213> ORGANISM: Oryza brachyantha

<400> SEQUENCE: 140

Met Glu Glu Gly Gly Gly Met Met Ala Ser Lys Asn Pro Thr Lys Thr
1          5          10          15
Leu Ala Ile Leu Leu Leu Val Leu Val Phe Phe Leu Leu Ser Leu Cys
          20          25          30
Ser Leu Ala Ala Ser Gln Pro Leu His Ser Glu Pro Met Ser Thr Ala
          35          40          45
Glu Tyr Ser Pro Pro Pro Ser Pro Pro Pro Pro Gln Ser Lys Ile Pro
          50          55          60
His Ala Gln Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Val Leu Gly
65          70          75          80

```

```

<210> SEQ ID NO 141
<211> LENGTH: 69
<212> TYPE: PRT
<213> ORGANISM: Oryza punctata

<400> SEQUENCE: 141

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Ala Val His
1          5          10          15
Leu Leu Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Ser Ser Gln Pro
          20          25          30

```

-continued

Leu His Ser Glu Pro Met Ser Thr Thr Thr Gln Ser Pro Pro Pro Pro
 35 40 45

Gln Ser Lys Ile Pro His Ala Gln Ala Gly Gly Ala Ala Arg Leu Arg
 50 55 60

Arg Ile Val Leu Gly
 65

<210> SEQ ID NO 142
 <211> LENGTH: 72
 <212> TYPE: PRT
 <213> ORGANISM: Oryza meridionalis

<400> SEQUENCE: 142

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu
 1 5 10 15

Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His
 20 25 30

Ser Glu Pro Met Ser Thr Thr Ala Thr Thr Gln Ser Ala Pro Pro Pro
 35 40 45

Pro Pro Pro Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala Ala
 50 55 60

Arg Leu Arg Arg Ile Val Leu Gly
 65 70

<210> SEQ ID NO 143
 <211> LENGTH: 69
 <212> TYPE: PRT
 <213> ORGANISM: Oryza sativa Indica Group

<400> SEQUENCE: 143

Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu Leu Val
 1 5 10 15

Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His Ser Glu Pro Met
 20 25 30

Ser Thr Thr Thr Thr Ala Thr Thr Thr Ala Thr Gln Pro Ala Pro Pro Pro Pro
 35 40 45

Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala Ala Arg Leu Arg
 50 55 60

Arg Ile Val Leu Gly
 65

<210> SEQ ID NO 144
 <211> LENGTH: 73
 <212> TYPE: PRT
 <213> ORGANISM: Oryza nivara

<400> SEQUENCE: 144

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu
 1 5 10 15

Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His
 20 25 30

Ser Glu Pro Met Ser Thr Thr Thr Thr Ala Thr Gln Pro Ala Pro Pro
 35 40 45

Pro Pro Pro Pro Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala
 50 55 60

Ala Arg Leu Arg Arg Ile Val Leu Gly
 65 70

-continued

<210> SEQ ID NO 145
 <211> LENGTH: 85
 <212> TYPE: PRT
 <213> ORGANISM: Oryza rufipogon

<400> SEQUENCE: 145

```

Leu Cys Ser Pro Arg Ala Ile Ala Ser Pro Leu Ser Met Glu Met Met
1           5           10           15
Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu Leu Leu Val
          20           25           30
Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His Ser Glu Pro Met
          35           40           45
Ser Thr Thr Thr Thr Ala Thr Gln Pro Ala Pro Pro Pro Pro Pro
          50           55           60
Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala Ala Arg Leu Arg
65           70           75           80
Arg Ile Val Leu Gly
          85

```

<210> SEQ ID NO 146
 <211> LENGTH: 75
 <212> TYPE: PRT
 <213> ORGANISM: Oryza barthii

<400> SEQUENCE: 146

```

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr His Leu Leu Leu
1           5           10           15
Leu Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu
          20           25           30
His Ser Glu Pro Met Ser Thr Thr Thr Thr Thr Ala Thr Gln Pro Thr
          35           40           45
Pro Pro Pro Pro Pro Pro Gln Ser Lys Ile Pro His Ala Glu Ala Gly
          50           55           60
Gly Ala Ala Arg Leu Arg Arg Ile Val Leu Gly
65           70           75

```

<210> SEQ ID NO 147
 <211> LENGTH: 73
 <212> TYPE: PRT
 <213> ORGANISM: Oryza sativa Japonica Group

<400> SEQUENCE: 147

```

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu
1           5           10           15
Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His
          20           25           30
Ser Glu Pro Met Ser Thr Thr Thr Thr Ala Thr Gln Pro Ala Pro Pro
          35           40           45
Pro Pro Pro Pro Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala
          50           55           60
Ala Arg Leu Arg Arg Ile Val Leu Gly
65           70

```

<210> SEQ ID NO 148
 <211> LENGTH: 73
 <212> TYPE: PRT
 <213> ORGANISM: Oryza glumipatula

<400> SEQUENCE: 148

-continued

```

Met Glu Met Met Ala Ser Lys Asn Pro Thr Lys Thr Leu Leu Leu Leu
1          5          10          15

Leu Leu Leu Val Phe Phe Leu Ser Leu Ala Cys Ser Gln Pro Leu His
20          25          30

Ser Glu Pro Met Ser Thr Thr Thr Thr Ala Thr Gln Pro Ala Pro Pro
35          40          45

Pro Pro Pro Pro Gln Ser Lys Ile Pro His Ala Glu Ala Gly Gly Ala
50          55          60

Ala Arg Leu Arg Arg Ile Val Leu Gly
65          70

```

```

<210> SEQ ID NO 149
<211> LENGTH: 72
<212> TYPE: PRT
<213> ORGANISM: Setaria italica

```

```

<400> SEQUENCE: 149

```

```

Met Thr Ser Lys Asn Pro Thr Lys Thr Pro Pro Leu Leu Ala Thr Leu
1          5          10          15

Leu Leu Leu Leu Ala Phe Leu Ala Leu Cys Ala Pro Ala Ala Ser Gln
20          25          30

Pro Leu His Ser Glu Pro Met Ala Thr Gln Ser Pro Pro Pro Ser Pro
35          40          45

Ala Pro Pro Gln Ser Arg Ile Pro Arg Ala Gln Val Gly Gly Ala Ala
50          55          60

Arg Leu Arg Arg Ile Ala Leu Gly
65          70

```

```

<210> SEQ ID NO 150
<211> LENGTH: 79
<212> TYPE: PRT
<213> ORGANISM: Panicum hallii FIL2

```

```

<400> SEQUENCE: 150

```

```

Ile Met Glu Gln Ala Val Arg Met Thr Ser Lys Asn Pro Thr Lys Thr
1          5          10          15

Pro Pro Leu Leu Ala Thr Leu Leu Leu Leu Ala Phe Leu Ala Leu
20          25          30

Cys Ala Pro Ala Ser Ser Gln Pro Leu His Ser Glu Pro Met Pro Thr
35          40          45

Gln Ser Pro Pro Pro Ser Pro Thr Pro Pro Gln Ser Thr Ile Pro Arg
50          55          60

Ala Pro Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65          70          75

```

```

<210> SEQ ID NO 151
<211> LENGTH: 79
<212> TYPE: PRT
<213> ORGANISM: Panicum hallii HAL2

```

```

<400> SEQUENCE: 151

```

```

Ile Met Glu Gln Ala Val Arg Met Thr Ser Lys Asn Pro Thr Lys Thr
1          5          10          15

Pro Pro Phe Leu Ala Ile Leu Leu Leu Leu Ala Phe Leu Ala Leu
20          25          30

Cys Ala Pro Ala Ser Ser Gln Pro Leu His Ser Glu Pro Met Ala Thr
35          40          45

```

-continued

Gln Ser Pro Pro Pro Ser Pro Thr Pro Pro Gln Ser Thr Ile Pro Arg
50 55 60

Ala Gln Ala Gly Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65 70 75

<210> SEQ ID NO 152

<211> LENGTH: 75

<212> TYPE: PRT

<213> ORGANISM: Sorghum bicolor

<400> SEQUENCE: 152

Met Val Val Gly Ala Arg Met Thr Ser Arg Asn Pro Thr Lys Thr Leu
1 5 10 15

Pro Leu Leu Ala Thr Thr Leu Val Leu Leu Leu Ala Phe Leu Ala
20 25 30

Leu Cys Ala Pro Ala Ser Ser Gln Pro Leu His Ser Glu Pro Met Ala
35 40 45

Thr Gln Ser Pro Pro Pro Arg Thr Ser Ile Pro Arg Ala Gln Val Gly
50 55 60

Gly Ala Ala Arg Leu Arg Arg Ile Ala Leu Gly
65 70 75

<210> SEQ ID NO 153

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Brassica rapa

<400> SEQUENCE: 153

Met Glu Gln Arg Arg Arg Arg Asn Ile Asn Thr Leu Ser Leu Leu Leu
1 5 10 15

Leu Phe Phe Leu Ala Phe Thr Ser Thr Thr Ala Thr Thr Ser Ser Cys
20 25 30

Arg Arg Arg Ala Val Lys His Leu Ser Thr Thr Gln Ala Thr Pro Leu
35 40 45

Gln Ser Arg Ile Thr Pro Lys Val Ile Val Leu Ser
50 55 60

<210> SEQ ID NO 154

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Brassica napus

<400> SEQUENCE: 154

Met Glu Gln Arg Arg Arg Arg Asn Ile Asn Thr Leu Ser Leu Leu Leu
1 5 10 15

Leu Phe Phe Leu Ala Phe Thr Ser Thr Thr Ala Thr Thr Ser Ser Cys
20 25 30

Arg Arg Arg Ala Val Lys His Leu Ser Thr Thr Gln Ala Thr Pro Leu
35 40 45

Gln Ser Arg Ile Thr Pro Lys Val Ile Val Leu Ser
50 55 60

<210> SEQ ID NO 155

<211> LENGTH: 63

<212> TYPE: PRT

<213> ORGANISM: Brassica napus

<400> SEQUENCE: 155

Met Glu Gln Arg Gly Arg Lys Arg Tyr Cys Asn Thr Val Tyr Leu Leu
1 5 10 15

-continued

Leu Phe Ile Phe Leu Val Phe Ser Ser Arg Thr Thr Ser Ser Ala Ser
20 25 30

Cys Arg Arg Arg Ala Val Lys His Leu Ser Thr Ala Pro Pro Ser Ser
35 40 45

Thr Pro Leu Glu Ser Arg Ile Thr Thr Lys Val Ile Ile Val Ser
50 55 60

<210> SEQ ID NO 156

<211> LENGTH: 63

<212> TYPE: PRT

<213> ORGANISM: Brassica napus

<400> SEQUENCE: 156

Met Glu Gln Arg Arg Arg Lys Arg Tyr Cys Asn Thr Val Tyr Leu Leu
1 5 10 15

Leu Phe Ile Leu Leu Val Phe Ser Ser Ile Thr Thr Ser Ser Ala Ser
20 25 30

Cys Arg Arg Arg Ala Val Lys His Leu Ser Thr Ala Pro Pro Ser Ser
35 40 45

Thr Pro Leu Glu Ser Lys Ile Thr Ser Lys Val Ile Ala Ile Ser
50 55 60

<210> SEQ ID NO 157

<211> LENGTH: 61

<212> TYPE: PRT

<213> ORGANISM: Brassica rapa

<400> SEQUENCE: 157

Met Glu Gln Arg Arg Arg Lys Arg Tyr Cys Asn Thr Val His Leu Leu
1 5 10 15

Leu Phe Ile Phe Leu Val Phe Ser Ser Arg Thr Ser Ala Ser Cys Arg
20 25 30

Arg Arg Ala Val Lys His Leu Ser Thr Ala Pro Pro Ser Ser Thr Pro
35 40 45

Leu Glu Ser Lys Ile Thr Ser Lys Val Ile Val Val Ser
50 55 60

<210> SEQ ID NO 158

<211> LENGTH: 67

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 158

Met Glu Gln Arg Arg Arg Arg Arg Arg Asn Gly Cys Ser Ser Asn Thr
1 5 10 15

Ile Ser Leu Leu Leu Phe Phe Phe Leu Val Phe Phe Ser Arg Thr Ser
20 25 30

Thr Ser Thr Ser Cys Arg Arg Arg Thr Val Lys His Leu Ser Thr Thr
35 40 45

Ser Thr Ser Ser Thr Pro Leu Glu Ser Arg Ile Thr Ser Lys Val Ile
50 55 60

Val Val Ser
65

<210> SEQ ID NO 159

<211> LENGTH: 61

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis lyrata

-continued

<400> SEQUENCE: 159

```

Met Glu Gln Arg Arg Arg Ser Asn Thr Ile Tyr Leu Leu Leu Leu Phe
1           5           10           15
Phe Phe Leu Val Phe Thr Ser Arg Thr Ser Thr Ser Ser Ser Cys Arg
20           25           30
Arg Arg Thr Val Lys His Leu Ser Thr Thr Pro Pro Ser Ser Thr Pro
35           40           45
Leu Glu Ser Arg Ile Thr Ser Lys Val Ile Val Phe Ser
50           55           60

```

<210> SEQ ID NO 160

<211> LENGTH: 61

<212> TYPE: PRT

<213> ORGANISM: Brassica napus

<400> SEQUENCE: 160

```

Met Glu Gln Arg Arg Arg Ser Asn Thr Ile Tyr Leu Leu Leu Leu Phe
1           5           10           15
Phe Phe Leu Val Phe Thr Ser Arg Thr Asn Thr Ser Ala Ser Cys Arg
20           25           30
Arg Arg Thr Val Lys His Leu Ser Thr Thr Pro Pro Ser Ser Thr Pro
35           40           45
Leu Gln Ser Ser Ile Thr Ser Lys Val Met Val Phe Ser
50           55           60

```

<210> SEQ ID NO 161

<211> LENGTH: 61

<212> TYPE: PRT

<213> ORGANISM: Brassica oleracea

<400> SEQUENCE: 161

```

Met Glu Gln Arg Arg Arg Ser Asn Thr Ile Tyr Leu Leu Leu Leu Phe
1           5           10           15
Phe Phe Leu Val Phe Thr Ser Arg Thr Asn Thr Ser Ala Ser Cys Arg
20           25           30
Arg Arg Thr Val Lys His Leu Ser Thr Thr Pro Pro Ser Ser Thr Pro
35           40           45
Leu Gln Ser Ser Ile Thr Ser Lys Val Ile Val Phe Ser
50           55           60

```

<210> SEQ ID NO 162

<211> LENGTH: 74

<212> TYPE: PRT

<213> ORGANISM: Trifolium pratense

<400> SEQUENCE: 162

```

Gly Cys Gly Phe Val Phe Cys Val Leu Met Phe Lys Lys Arg Gln Thr
1           5           10           15
Phe Phe Tyr Leu Ala Lys Glu Leu Leu Val Phe Gln Pro Leu Val Leu
20           25           30
Leu Leu Phe Leu Phe Ser Leu His His Asn Thr Val Gln Cys Glu Gly
35           40           45
Arg Leu Ser Lys Asn Ile Ser Ser Glu Thr Ser Ser Gln Ser Asp Tyr
50           55           60
Lys Asp Asn Pro Arg Lys Val Ile Val Ser
65           70

```

<210> SEQ ID NO 163

-continued

<211> LENGTH: 75
 <212> TYPE: PRT
 <213> ORGANISM: *Medicago truncatula*

<400> SEQUENCE: 163

Gly Cys Gly Phe Val Phe Cys Val Phe Met Phe Lys Lys Arg His Thr
 1 5 10 15
 Pro Phe Ser Leu Ala Arg Lys Phe Leu Ser Phe Gln Pro Phe Val Leu
 20 25 30
 Leu Leu Phe Leu Phe Ser Leu His His Asp Thr Val Gln Cys Gln Gly
 35 40 45
 Arg Leu Ser Lys His Met Ser Ser Glu Pro Ser Ser Ser Thr Ser Glu
 50 55 60
 Tyr Lys Asp Asp Pro Arg Lys Ile Ile Ile Ser
 65 70 75

<210> SEQ ID NO 164
 <211> LENGTH: 58
 <212> TYPE: PRT
 <213> ORGANISM: *Lupinus angustifolius*

<400> SEQUENCE: 164

Met Phe Lys Arg Arg His Thr Leu Cys Ser Leu Leu Arg Glu Leu Phe
 1 5 10 15
 Leu Leu Leu Leu Phe Ser Leu His His Ser Thr Val Gln Cys Gln Gly
 20 25 30
 Arg Leu Ser Lys His Val Ser Ser Ala Pro Tyr Ser Pro Ser Glu Tyr
 35 40 45
 Lys Asp Asp Leu Arg Arg Ile Ile Ile Ser
 50 55

<210> SEQ ID NO 165
 <211> LENGTH: 72
 <212> TYPE: PRT
 <213> ORGANISM: *Phaseolus vulgaris*

<400> SEQUENCE: 165

Met Phe Arg Lys Arg His Ile Ala Ser Ser Leu Ala Arg Glu Leu Leu
 1 5 10 15
 Ala Leu Gln Pro Leu Leu Ile Leu Phe Leu Phe Ser Leu His His His
 20 25 30
 Thr Val Gln Cys Gln Gly Arg Leu Ser Lys His Val Ser Ser Glu Pro
 35 40 45
 Pro Ser Pro Ser Arg Thr Ser Pro Pro Ser Ser Ser Gly Tyr Arg Asp
 50 55 60
 Asp Pro Lys Lys Ile Ile Leu Ser
 65 70

<210> SEQ ID NO 166
 <211> LENGTH: 81
 <212> TYPE: PRT
 <213> ORGANISM: *Vigna radiata*

<400> SEQUENCE: 166

Leu Ala Leu Phe Val Cys Cys Ser Leu Met Phe Arg Lys Arg Asn Ile
 1 5 10 15
 Ala Ser Ser Leu Ala Arg Glu Leu Leu Ala Leu Gln Pro Leu Phe Ile
 20 25 30
 Ile Phe Leu Phe Ser Leu His His Asn Thr Val Gln Cys Gln Gly Arg

-continued

35	40	45
Leu Ser Lys His Val Ser	Ser Glu Pro Pro Ser	Pro Ser Arg Pro Ser
50	55	60
Pro Pro Ser Ser Ser Gly	Tyr Arg Asp Asp	Pro Lys Lys Ile Ile Leu
65	70	75 80

Ser

<210> SEQ ID NO 167
 <211> LENGTH: 74
 <212> TYPE: PRT
 <213> ORGANISM: Glycine max

<400> SEQUENCE: 167

Met Phe Arg Lys Arg His Ile	Leu Ser Ser Leu Ala Arg Glu Leu Leu
1	5 10 15
Ala Leu Gln Pro Leu Phe Leu	Leu Phe Ser Leu His His Asn
20	25 30
Thr Val Gln Cys Gln Gly Arg	Leu Ser Lys His Val Ser Ser Glu Pro
35	40 45
Pro Ser Pro Ser Arg Pro Ser	Ser Ala Ala Pro Ser Ser Ser Gly Tyr
50	55 60
Lys Asp Asp Pro Arg Lys Ile	Ile Leu Ser
65	70

<210> SEQ ID NO 168
 <211> LENGTH: 74
 <212> TYPE: PRT
 <213> ORGANISM: Glycine max

<400> SEQUENCE: 168

Met Phe Arg Lys Arg His Thr	Leu Ser Ser Leu Ala Arg Glu Leu Leu
1	5 10 15
Ala Phe Gln Pro Leu Phe Leu	Leu Phe Ser Leu His His Asn
20	25 30
Thr Met Gln Cys Gln Gly Arg	Leu Ser Lys His Val Ser Ser Glu Pro
35	40 45
Pro Ser Pro Ser Arg Ser Thr	Pro Ser Pro Ser Ser Ser Gly Tyr
50	55 60
Lys Asp Asp Pro Arg Lys Ile	Ile Leu Ser
65	70

<210> SEQ ID NO 169
 <211> LENGTH: 64
 <212> TYPE: PRT
 <213> ORGANISM: Actinidia chinensis

<400> SEQUENCE: 169

Met Leu Arg Phe Val Val Ser	Met Glu Lys Thr Ser Cys Val Asn Leu
1	5 10 15
Glu Leu Val Leu Leu Leu Ile	Ile Cys Leu Cys Asn Thr Ser Val Gln
20	25 30
Cys Gln Thr Arg Pro Ile Asn	His Ile Ser Ala Glu Pro Pro Ser Pro
35	40 45
Ser Arg Ala Pro Glu Leu Lys	Asn Lys Phe Gln Lys Ile Ile Leu Ser
50	55 60

<210> SEQ ID NO 170
 <211> LENGTH: 64

-continued

<212> TYPE: PRT

<213> ORGANISM: *Actinidia chinensis*

<400> SEQUENCE: 170

```

Met Leu Arg Val Val Ala Ser Met Glu Lys Thr Asn Cys Val Thr Leu
1           5           10           15
Glu Leu Val Leu Leu Leu Ile Phe Cys Leu Cys Asn Thr Ser Val Gln
                20           25           30
Cys Gln Thr Arg Pro Ile Asn His Ile Ser Ala Glu Pro Pro Ser Pro
                35           40           45
Ser Arg Ala Pro Glu Leu Lys Asn Lys Phe Gln Lys Ile Ile Leu Ser
                50           55           60

```

<210> SEQ ID NO 171

<211> LENGTH: 57

<212> TYPE: PRT

<213> ORGANISM: *Prunus persica*

<400> SEQUENCE: 171

```

Met Asp Arg Arg Arg Ser Thr Ser Lys Phe Asn Ser Arg Leu Thr Arg
1           5           10           15
Leu Leu Leu Leu Leu Phe Ile Phe Cys Phe Arg Gln Glu Thr Val Gln
                20           25           30
Cys Gln Glu Thr Glu Pro Pro Ser Ser Ser Lys Gln Pro His Phe Lys
                35           40           45
Asn Gln Leu Gln Arg Ile Ile Leu Ser
                50           55

```

<210> SEQ ID NO 172

<211> LENGTH: 57

<212> TYPE: PRT

<213> ORGANISM: *Vitis vinifera*

<400> SEQUENCE: 172

```

Met Glu Lys Thr Ser Arg Glu Ala Ile Ser Leu Leu Phe Leu Leu Ile
1           5           10           15
Phe Ile His Phe Tyr Asn Val Lys Cys Gln Glu Asn Ile Ile Asn Arg
                20           25           30
Val Ser Thr Asp Ser Ser Pro Ser Pro Ser Arg Glu Ala Glu Phe Lys
                35           40           45
Ser Gly Leu Lys Lys Ile Ile Leu Ser
                50           55

```

<210> SEQ ID NO 173

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Gossypium raimondii*

<400> SEQUENCE: 173

```

Met Ala Lys Met Ser Gly Cys Ile Val Thr Val Ala Leu Leu Leu Leu
1           5           10           15
Leu Leu Leu Cys Leu His Gln Thr Pro Val Glu Cys Lys Glu Arg Leu
                20           25           30
Ile Arg Gln Leu Ser Ser Gln Pro Ser Ser Ala Thr Lys Pro Gln Glu
                35           40           45
Phe Lys Ile Gly Phe Lys Arg Val Ile Leu Ser
                50           55

```

<210> SEQ ID NO 174

-continued

<211> LENGTH: 57
 <212> TYPE: PRT
 <213> ORGANISM: *Gorchorus capsularis*

<400> SEQUENCE: 174

Met Gly Lys Gly Arg Arg Cys Thr Leu Ser Leu Leu Phe Leu Leu Leu
 1 5 10 15
 Leu Cys Phe His Gln Thr Thr Val His Cys Lys Glu Arg Leu Ile Arg
 20 25 30
 His Leu Ser Ser Gln Pro Pro Ser Pro Ser Lys Pro Gln Glu Phe Lys
 35 40 45
 Ile Gly Leu Lys Arg Ile Ile Leu Ser
 50 55

<210> SEQ ID NO 175
 <211> LENGTH: 67
 <212> TYPE: PRT
 <213> ORGANISM: *Theobroma cacao* Matina 16

<400> SEQUENCE: 175

Met Phe Val Cys Gly Phe Val Ala Ser Met Glu Lys Arg Ser Cys Thr
 1 5 10 15
 Leu Ser Leu Leu Leu Leu Leu Leu Leu Leu Cys Phe His Leu Thr
 20 25 30
 Thr Val Gln Cys Glu Gly Arg Leu Ile Arg Tyr Leu Ser Ser Gln Pro
 35 40 45
 Pro Ser Pro Ser Thr Pro Gln Glu Phe Lys Ile Gly Phe Lys Arg Ile
 50 55 60
 Val Leu Ser
 65

<210> SEQ ID NO 176
 <211> LENGTH: 65
 <212> TYPE: PRT
 <213> ORGANISM: *Manihot esculenta*

<400> SEQUENCE: 176

Met Glu Lys Arg Arg Tyr Ser Ser Gln Leu Thr Ser Lys Val Thr Ala
 1 5 10 15
 Leu Leu Leu Leu Leu Ser Phe Cys Phe His Pro Ile Val Val Gln Cys
 20 25 30
 Gln Glu Ser Asn Ile Lys Met Thr Arg His Leu Ser Ser Glu Thr Pro
 35 40 45
 Pro Ser Arg Ser Pro Gln Phe His Thr Gly Leu Lys Arg Ile Leu Leu
 50 55 60
 Ser
 65

<210> SEQ ID NO 177
 <211> LENGTH: 61
 <212> TYPE: PRT
 <213> ORGANISM: *Populus trichocarpa*

<400> SEQUENCE: 177

Met Gly Lys Arg Arg Tyr Ser Thr Gln Ile Ser Asn Asn Lys Asp Thr
 1 5 10 15
 Ser Ser Leu Leu Phe Leu Phe Ile Leu Cys Leu Tyr Tyr Thr Thr Val
 20 25 30
 Gln Cys Gln Glu Ser Ser Lys Val Thr Pro Pro Phe Pro Ser Thr Pro

-continued

35	40	45	
Thr His Ser Lys Asn Gly	Leu Lys Arg Ile Leu	Val Ser	
50	55	60	

<210> SEQ ID NO 178
 <211> LENGTH: 61
 <212> TYPE: PRT
 <213> ORGANISM: Populus trichocarpa
 <400> SEQUENCE: 178

Met Glu Lys Arg Arg Tyr Ser Leu Arg Val Ser Asn Ser Lys Val Thr	
1 5 10 15	

Ala Ser Leu Leu Phe Leu Phe Val Leu Cys Leu Tyr Tyr Ala Ser Val	
20 25 30	

Gln Cys Gln Glu Ser Ser Lys Val Thr Pro Pro Ser Pro Ser Thr Pro	
35 40 45	

Thr Gln Ser Lys Asn Gly Leu Lys Arg Ile Leu Val Ser	
50 55 60	

<210> SEQ ID NO 179
 <211> LENGTH: 200
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays
 <400> SEQUENCE: 179

acaaaaaccc tgtcggctct gaccactctc ctgctgctgc ttgagttcct cgcgctgtgc	60
gcgcgccgct cgcgccagcc gctccactcg gagcccatgg ctacgcagtc ccgcgccg	120
ccgctgccgc cgggggtccac gattcccccg gcgcaggccg gcggcgccgc acgcctccgc	180
cgcattgcgc tcggggctct	200

<210> SEQ ID NO 180
 <211> LENGTH: 197
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polynucleotide
 <400> SEQUENCE: 180

acaaaaaccc tgtcggctct gaccactctc ctgctgctgc ttgagttcct cgcgctgtgc	60
gcgcctccg ccagccgct cactcggag cccatggcta cgcagtcgcc gccgcgccg	120
ctgccgccg ggtccacgat tccccggcg caggccggcg gcgcgcacg cctccgccgc	180
attgcgctcg gggtct	197

<210> SEQ ID NO 181
 <211> LENGTH: 194
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polynucleotide
 <400> SEQUENCE: 181

acaaaaaccc tgtcggctct gaccactctc ctgctgctgc ttgagttcct cgcgctgtgc	60
gcgcgccgct cgcgccagcc gctccactcg gagcccatgc agtccccgcc gccgcgctg	120
ccgcgggggt ccacgattcc cggggcgag gccggcgcg ccgcacgct ccgccgatt	180
gcgctcgggg tcct	194

<210> SEQ ID NO 182

-continued

<211> LENGTH: 194
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 182

accaaaacc	tgctgggtcct gaccactctc ctgctgctgc ttgagttcct cgcgctgtgc	60
gcgcgccgcc	agcgcgtcca ctggagccc atggctacgc agtccccgcc gcgcgcgtg	120
ccgcgggggt	ccacgattcc ccgggcgcag gccggcggcg ccgcacgcct ccgcgcatt	180
gcgctcgggg	tcct	194

<210> SEQ ID NO 183
<211> LENGTH: 194
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 183

accaaaacc	tgctgggtcct gaccactctc ctgctgctgc ttgagttcct cgcgctgtgc	60
gcgcgccgct	ccgcccagcc gctccacccc atggctacgc agtccccgcc gcgcgcgtg	120
ccgcgggggt	ccacgattcc ccgggcgcag gccggcggcg ccgcacgcct ccgcgcatt	180
gcgctcgggg	tcct	194

<210> SEQ ID NO 184
<211> LENGTH: 191
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 184

accaaaacc	tgctgggtcct gaccactctc ctgctgctgc ttgagttcct cgcgctgtgc	60
gcgcgccgct	ccgcccagcc gctccactcg gagcccatgt cccgcgcgc gcgcgtgccg	120
ccgggggtcca	cgattccccg ggcgcaggcc ggcggcgcgc cagcctccg ccgcattgcg	180
ctcgggggtcc	t	191

<210> SEQ ID NO 185
<211> LENGTH: 191
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 185

accaaaacc	tgctgggtcct gaccactctc ctgctgctgc ttgagttcct cgcgctgtgc	60
gcgcgccgct	ccgcccagcc gctccactcg gagcccatgg ctacgcagtc cccgcgcgcg	120
ccgcgggtcca	cgattccccg ggcgcaggcc ggcggcgcgc cagcctccg ccgcattgcg	180
ctcgggggtcc	t	191

<210> SEQ ID NO 186
<211> LENGTH: 191
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 186

-continued

```

<210> SEQ ID NO 191
<211> LENGTH: 155
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 191

accaaacc tgcggtct gaccactct cagcgctcc actcggagcc catggctacg      60
cagtcctccg cgcgcgcgt gccgcgggg tccacgattc cccgggcgca ggccggcgcc      120
gccgcacgcc tccgcgcgt tgcgctcggg gtctct                                155

```

That which is claimed is:

1. A corn plant or plant part thereof comprising at least one non-natural mutation in an endogenous CORYNE (CRN) gene that encodes a wild type CRN protein, wherein the wild type CRN protein comprises a sequence having at least 95% identity to SEQ ID NO:118 and the mutation is an in-frame deletion of one or more codons that results in the deletion of one or more amino acid residues located from position 23 to position 66 with reference to amino acid position numbering of SEQ ID NO:118, or is an out-of-frame deletion that results in a premature stop codon and a truncated CRN protein, wherein the out-of-frame deletion is located in a region of the endogenous CRN gene that encodes position 23 to position 66 with reference to amino acid position numbering of SEQ ID NO:118, wherein the corn plant exhibits increased kernel row number without fasciation compared to a control corn plant devoid of the at least one non-natural mutation.

2. The corn plant or part thereof of claim 1, wherein the at least one non-natural mutation is a base deletion of at least 1 base pair to 132 base pairs.

3. The corn plant or part thereof of claim 1, wherein the at least one non-natural mutation is located in a nucleic acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 123-125.

4. The corn plant or part thereof of claim 1, wherein the endogenous CRN gene comprises a sequence having at least 95% sequence identity to SEQ ID NO: 122.

5. The corn plant or part thereof of claim 1, wherein the at least one non-natural mutation results in a mutated CRN protein that is encoded by a nucleotide sequence having at least 95% sequence identity to any one of SEQ ID NOs: 180-191.

6. The corn plant or part thereof of claim 1, wherein the mutated CRN gene comprises a sequence having at least 90% sequence identity to the nucleotide sequence of any one of SEQ ID NOs: 180-191.

7. A method for editing a specific site in the genome of a corn plant cell, the method comprising: cleaving, in a site specific manner, a target site within an endogenous CORYNE (CRN) gene in the corn plant cell, wherein the endogenous CRN gene encodes a wild type CRN protein having at least 95% sequence identity to the amino acid sequence of SEQ ID NO:118, thereby generating an edit in the endogenous CRN gene of the corn plant cell and producing a corn plant cell comprising the edit in the endogenous CRN gene, wherein the edit is an in-frame deletion of one or more codons that results in the deletion of one or more amino acid residues located from position 23 to position 66 with reference amino acid position numbering of SEQ ID NO: 118, or is an out-of-frame deletion that results in a premature stop codon and a truncated CRN protein, wherein the out-of-frame deletion is located in a region of the endogenous CRN gene that encodes position 23 to position 66 with reference to amino acid position numbering of SEQ ID NO:118.

8. The method of claim 7, further comprising regenerating a corn plant from the corn plant cell comprising the edit in the endogenous CRN gene to produce a corn plant comprising the edit in its endogenous CRN gene, wherein the corn plant exhibits increased kernel row number without fasciation compared to a control corn plant devoid of the edit.

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